WellSky® Blood Centers

Mobile PC Guide

Software Version: 2026 R1

Release Date: November 2025

WBCGMPC2026R1_01

How to Contact Us

WellSky Corporation
11300 Switzer Road
Overland Park, KS 66210

Customer Support: 866.714.9499

Fax: 630.684.0462

Web: http://www.wellsky.com

Disclaimer

WellSky Corporation provides the WellSky Blood Centers application or licensed software to the customer after extensive planning, designing, and validation of the system. Validation involves a range of hardware and data that necessarily is limited. As such, WellSky's in-house validation of the system can only assure that the system performs its functions and routines as designed by WellSky. WellSky cannot represent validation of all functioning possibilities and related consequences under ultimate conditions of use. Any warranty or guarantee, explicit or implied, pertains to the level of quality and functionality of the system's software, as planned, designed and validated by WellSky.

The user facility, not WellSky, is responsible for the control, management, and supervision of the facility's use of the Blood Centers application or licensed software, including, but not limited to, insuring that proper controls are in place to validate data, outcomes, and results obtained through its use. The user facility must run sufficient validation tests with appropriate frequency on the software and hardware, and on any modifications or supplements thereof as made available by WellSky, to be certain that the facility's use of the Blood Centers system, in the configuration and under the conditions at the facility, meets the facility's requirements in terms of policy, protocol, assumptions, and expectations, regardless of the participation or assistance of WellSky personnel in the form of suggested/published validation procedures and/or verbal discussions regarding Blood Centers systems capabilities, functions, performance, or validation.

WellSky may use data submitted by client and processed through the cloud services for interoperability and transparency between providers, payers, and patients for care coordination; continuity of care; and performance, payment, and claims related activities, including patient matching and claims data sharing with mutual patients of other providers and payers, making the matching patient information visible to these providers and payers. WellSky may aggregate the matching data from providers and payers for the creation, comparison, expansion, amendment, and maintenance of consumer or patient health records necessary for the deployment of features and functionality of the cloud services.

WellSky Corporation
Proprietary and confidential

Revision History

Software Release & Revision Date	Document Number	Content Affected	Description of Change
2026 R1 November 2025	WBCGMPC2026R1_01	All	Updated the document to reflect the current release version and release date.
2026 R1 November 2025	WBCGMPC2026R1_01	5.6 One-time Initialization	Previously Section 5.7. Moved and renumbered as 5.6. No other changes.
2026 R1 November 2025	WBCGMPC2026R1_01	5.7 Oracle Wallet Setup for Mobile PC Server	New section added as part of Mobile PC Master Server set up.
2026 R1 November 2025	WBCGMPC2026R1_01	5.8 IIS Installation	Previously Section 5.6. Moved and renumbered as 5.8. No other changes.
2026 R1 November 2025	WBCGMPC2026R1_01	5.8.1 Microsoft .NET Core Installation	Updated the Microsoft.NET Core to .NET 8.0.20.
2026 R1 November 2025	WBCGMPC2026R1_01	5.9 Google Chrome Setup for HTTP	Previously Section 5.8. Moved and renumbered as 5.9. No other changes.

Contents

Chapter 1 - Mobile PC System Overview 1
1.1 Introduction 2
1.2 User ID and Password 3
1.3 Mobile PC Identification and Tracking 3
1.4 Update Facilities 3
1.5 Mobile PC Update Log 3
1.6 Update Frequency 3
1.7 Mobile PC Storage 3
1.8 Mobile Donor PC Upload/Download Configuration Overview 4
1.9 Mobile PC Wireless Configuration Overview 6
1.9.1 Wireless Description 6
1.9.2 Computers 14
1.9.3 Wireless and Cellular Quality of Service 15
1.9.4 Security Measures 16
1.9.5 Electromagnetic Compatibility 17
1.9.6 Other Important Information Regarding Use of Wireless Devices 19
Chapter 2 - Mobile Donor Database Overview 23
2.1 Database Description 24
2.2 Mobile Donor Upload/Download Configuration Overview 24
Chapter 3 - Mobile PC Handling Guidelines 25
Chapter 4 - Mobile Donor Client Environment 27
4.1 Mobile Donor Collections Lookup Laptop 28
4.1.1 Features 28
4.1.2 Hardware 28
4.1.3 Software 28
4.2 Mobile Donor Collections Printers 29
4.3 Mobile Donor Upload/Download Laptop 29
4.3.1 Hardware 29
4.3.2 Software 29
4.4 Mobile Donor Laptop Browser Clients 30

WellSky Corporation
Proprietary and confidential

4.4.1 Hardware 30
4.4.2 Software 30
Chapter 5 - Set Up Mobile PC Master Server 31
5.1 Introduction 32
5.2 Software Preparation 32
5.3 Configure TCP/IP Protocol 32
5.4 Configure Windows Domain 33
5.4.1 Configure Microsoft TCP/IP Client 33
5.4.2 Download and Install File Archive Utility 33
5.5 Install Oracle Database 21c Express Edition 34
5.6 One-time Initialization 41
5.7 Oracle Wallet Setup for Mobile PC Server 44
5.8 IIS Installation 47
5.8.1 Microsoft .NET Core Installation 56
5.8.2 Deploy the BC Onsite Application on the Application Server 60
5.8.3 HTTPS Setup for IIS Server – Windows 68
5.8.4 Deploy the BC Donor Card Application on the Application Server 77
5.8.5 Deploy the BC Survey Application on the Application Server 79
5.9 Google Chrome Setup for HTTP 80
Chapter 6 - Mobile PC Download, Extract, and Upload Process in PUSH Method 85
6.1 Mobile PC Download Process in PUSH Method 86
6.1.1 Set Up the Check_download Process in Windows Task Scheduler 86
6.1.2 Download Data 91
6.2 Mobile PC Data Extract Process in PUSH Method 96
6.2.1 Introduction 96
6.2.2 Process 96
6.2.3 Extract Data to File 96
6.2.4 Set Up the Extract Queue Process in Windows Task Scheduler 97
6.3 Create Data Extract File 103
6.4 Upload Data to the Main Server from the Mobile PC Server 107
6.4.1 Set Up the Upload Archive Process in Windows Task Scheduler 107

6.4.2 Upload Data to the Server from the Mobile PC 112
Chapter 7 - Mobile PC Download, Extract, and Upload Process in Script Method 119
7.1 Mobile PC Download Process in Script Method 120
7.1.1 WinSCP (SFTP) Install Process 120
7.1.2 Automating The Process Through Task Scheduler 131
7.2 Mobile PC Data Extract Process in Script Method 139
7.2.1 Introduction 139
7.2.2 Process 139
7.2.3 Extract Data to File 139
7.2.4 Set Up the Extract Queue Process in Windows Task Scheduler 140
7.3 Extract Data Using Batch Script 146
7.4 Upload Data to Server 147
7.5 Copy the Extracted Files into Database Server - File Copy Method 148
Appendix A - Recreate the Audit Sequence 149
Appendix B - Mobile PC Server Setup in Compress Mode 151

WellSky Corporation
Proprietary and confidential

1

Chapter 1 - Mobile PC System Overview

1.1 Introduction 2
1.2 User ID and Password 3
1.3 Mobile PC Identification and Tracking 3
1.4 Update Facilities 3
1.5 Mobile PC Update Log 3
1.6 Update Frequency 3
1.7 Mobile PC Storage 3
1.8 Mobile Donor PC Upload/Download Configuration Overview 4
1.9 Mobile PC Wireless Configuration Overview 6

1.1 Introduction

This guide details the procedures to set up, configure, and update mobile donor collections computers used with Donor. To better comply with requirements of regulatory agencies, Blood Centers Customers are expected to incorporate the requirements, and might include the recommendations, set out in this guide.

There are two possible configurations for the Mobile Donor PCs:

- Mobile Donor Collections Lookup configuration (read-only) – where personnel use mobile PCs to look up donor history and status, and to print donor cards. The Donor data replication downloads data from the main server in a read-only mode to the mobile donor laptop or server.
- Mobile Donor Upload/Download configuration – where the Donor data replication downloads data from the main server (Donor Master Database) in a read/write mode to the Mobile Donor laptop server. After a mobile drive, the data collected is then uploaded from the Mobile Donor laptop server to the Donor Master Database.

The Mobile PC provides access to only the On-Site module of Donor. Specific functions include the following:

- Register donations and deferrals
- Perform health history and physical findings
- Link or unlink a unit or deferral
- Input bleed time and phlebotomy data
- Modify a donor scan ID
- Print donor cards from LOOKQ or other onsite screens (for example, WELCOME and HHPF)
- Walk a donor out
- Use query screens within the On-Site module (for example OSTATUSQ and ENTRYQ) to query registered donors in the drive and see the status of the donor registration
- Extract and encrypt data from the API tables

Functions that are included in the Blood Centers application but are *not* available from a Mobile PC include the following:

- Recruitment – used to create, modify, and manage mobile blood drives and manage drive sponsors
- Collection Staff Services – used to assign and manage collection staff
- Donor Enlistment – used to create and manage lists of current and potential donors
- Donor Center Management – used to schedule donors, create and modify drives, and manage donor centers
- Production Control and Transportation – used to schedule and manage vehicles and transportation staff
- Donor Profile – used to add donors to the database, place and remove donor attributes and interdictions
- HLA/RBC – used to enter and define HLA antigens, cross-reactive groups, and matching grades, and RBC antigen groups
- Donor Notification – used to define and manage notification letters to donors

WellSky Corporation
Proprietary and confidential

- Donor Communication – used to generate donor address labels and ID cards, and to manage donor communications
- Special Donations – used to manage special donations and maintain physician lists

These functions are described in detail in the WellSky Blood Centers User Guide.

1.2 User ID and Password

Access to the Blood Centers system through mobile PCs is controlled by user ID and password. Issue and control of user IDs and passwords are the responsibility of each customer's Systems Management personnel in compliance with the customer's policies and practices.

1.3 Mobile PC Identification and Tracking

Proper identification of each mobile PC, and tracking of the system and data currency on each mobile PC, is critical to meeting regulatory requirements. Each customer needs to establish proper policies and procedures to assure accurate tracking and proper update of mobile PCs used for donor deferral.

1.4 Update Facilities

The typical customer mobile PC update facility provides network connections and PC network cards for updating mobile PCs. The facility can use network hubs to add extra connections, but take care not to overload the hubs.

1.5 Mobile PC Update Log

Each customer facility should maintain accurate and timely records of updates to document that the donor deferral database is up-to-date. Customer Management should periodically review the update log to ensure that established procedures and policies are being followed.

1.6 Update Frequency

PCs used for donor deferral should be updated at least weekly. The mobile PC database should be updated in compliance with a schedule defined in accordance with customer policies and practices.

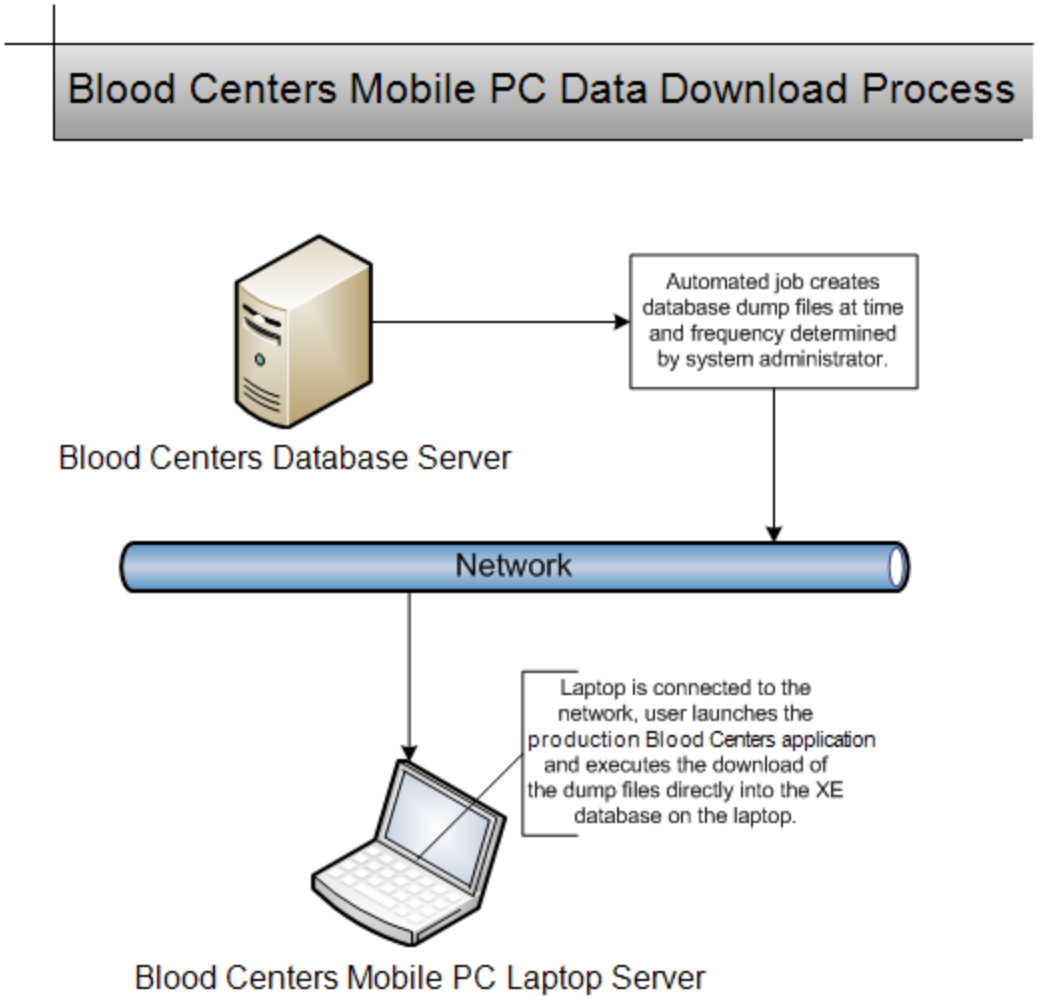
1.7 Mobile PC Storage

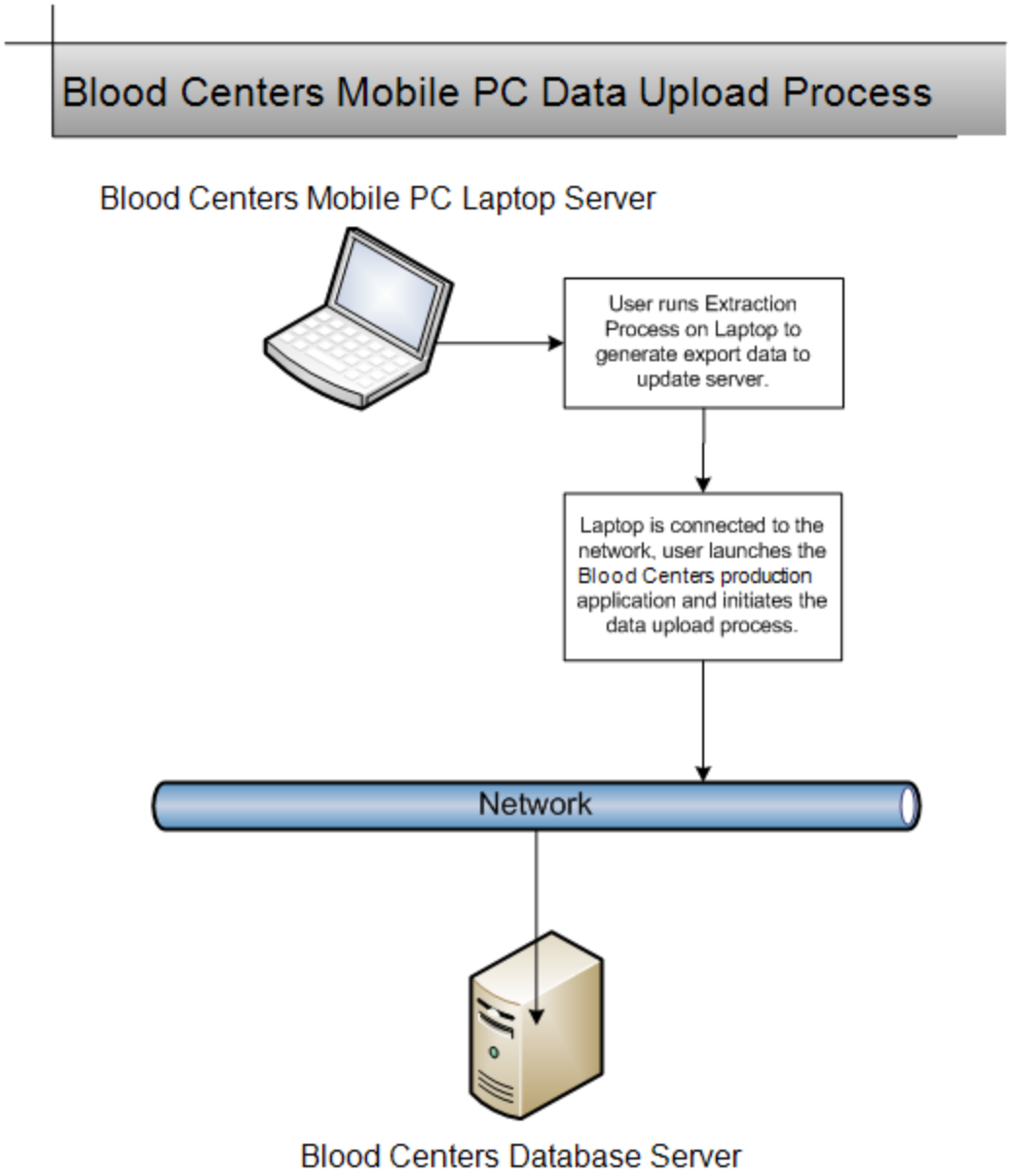
Data stored in the mobile PCs is sensitive and confidential. Mobile PCs are expensive pieces of fragile electronic equipment. Customers should take all reasonable precautions to ensure the safety and security of their data and equipment.

1.8 Mobile Donor PC Upload/Download Configuration Overview

The Mobile Donor Upload/Download configuration has the On-Site module and other modules which are required to run the application loaded on the Master Laptop. The Oracle Application Server is installed to allow additional Client Laptops to access the On-Site module through an Internet browser. Adobe Reader is used for viewing the reports. There may be multiple Master Laptops with this configuration at each drive.

The following diagrams illustrate the connectivity of the Mobile Donor Upload and Download configuration:




Blood Centers Mobile PC Data Upload Process
Blood Centers Mobile PC Laptop Server
User runs Extraction Process on Laptop to generate export data to update server.
Laptop is connected to the network, user launches the Blood Centers production application and initiates the data upload process.
Network
Blood Centers Database Server

1.9 Mobile PC Wireless Configuration Overview

Wireless networks offer the flexibility of setting up a network infrastructure without the need for cables, etc. to connect the computers. Donor is designed and tested to perform on either a traditionally wired network or a wireless network with the same response, data accuracy and integrity. The following should be considered when using Donor on a wireless network:

1.9.1 Wireless Description

Donor was tested from RF wireless connections to an appropriately configured and secured wireless network (WLAN) within a blood collection facility, simulated drive location, and mobile collection bus.

WellSky has evaluated the performance of Donor on three distinct wireless network configurations.

- Wireless Local Area Network (WLAN) using 802.11n network topology
- Wireless Local Area Network (WLAN) using 802.11g network topology
- Connected to a server WAN link using cellular network air cards

NOTE – WellSky has not qualified and does not recommend using a WLAN configuration together with cellular network air cards at the same drive.

Wireless Local Area Network (WLAN) using 802.11n and 802.11g Network Topology

WellSky used a WLAN utilizing 802.11n and 802.11g network topology. In this configuration, all computers were connected to a LAN using 802.11n and 802.11g wireless technology. Most of the testing performed was done on a wireless network consisting of a master laptop with two client laptops representing a typical setup for a blood drive. Additionally, a wireless network consisting of a master laptop and six client laptops was tested to verify that Donor would operate correctly on a wireless network with a maximum expected number of laptops.

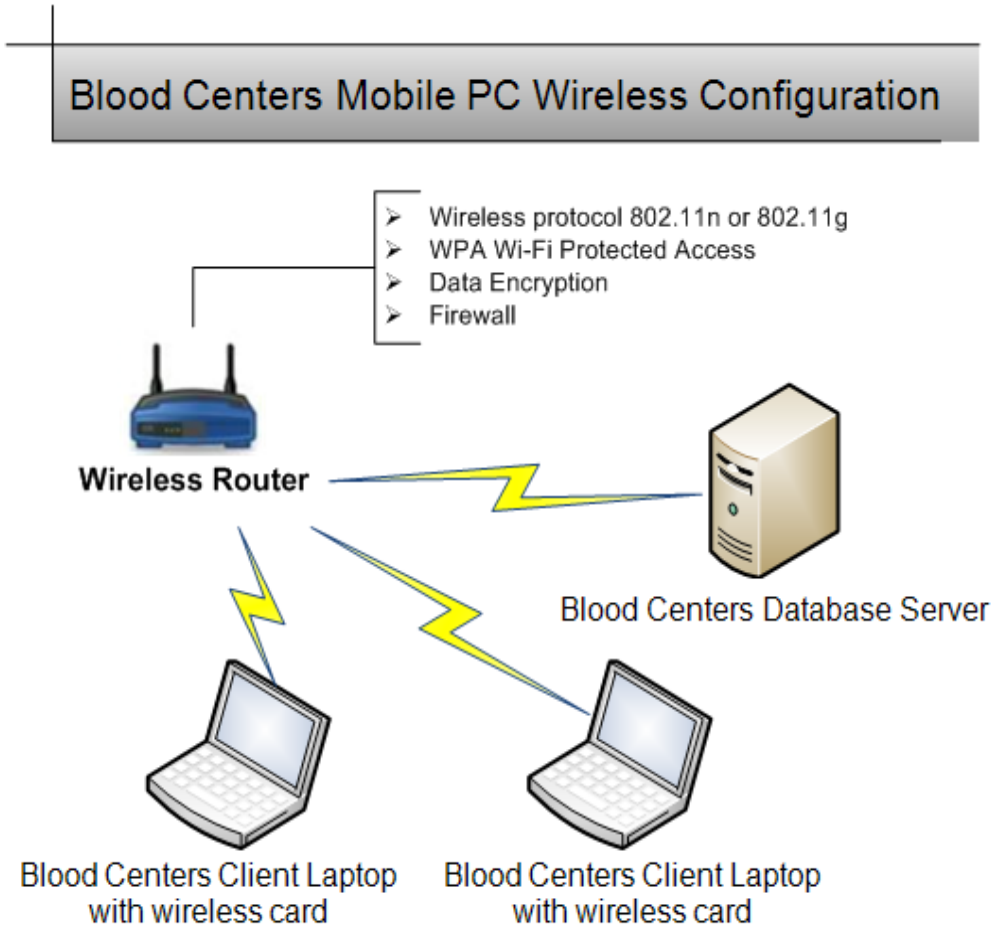
Summary of WLAN Essential Operating Requirements

- Use only 802.11n or 802.11g protocols
- Do not use WLAN and cellular network cards at he same drive
- Select an operating location and position network components to avoid sources of electromagnetic interference
- Position all stations within 75 feet of the wireless router
- Maintain a separation distance of three feet between active network components
- Apply WLAN network security according to recommendations
- Do not operate a WLAN station above two percent packet loss

WellSky Corporation
Proprietary and confidential

- Do not exceed one master laptop and six client stations one a single WLAN network
- Follow all of your equipment manufacturer's guidelines

Example 802.11 Network Configuration



Wireless 802.11n and 802.11g Standard

The 802.11 family includes over-the-air modulation techniques that use the same basic protocol. WellSky has qualified 802.11n and 802.11g wireless networks for Blood Centers. Because of this choice of frequency band, 802.11n and 802.11g equipment can suffer interference from microwave ovens, cordless telephones, and Bluetooth devices.

NOTE – In the following table, MIMO stands for Multiple Input/Multiple Output Streams.

802.11 Protocol	Freq. (GHz)	Bandwidth (MHz)	Data Rate per Stream (MBPS)	Allowable MIMO*	Approx. Indoor Range (Ft)	Approx. Outdoor Range (Ft)	Modulation
g	2.4	20	6, 9, 12, 18, 24, 36, 48, 54	1	125	460	OFDM DSSS
n	2.4/5	20	7.2, 14.4, 21.7, 28.9, 43.3, 57.8, 65, 72.2	4	230	820	OFDM
		40	15, 30, 45, 60, 90, 120, 135, 150				

Wireless 802.11n and 802.11g Range of Operation

Donor is a Blood Establishment Computer Software (BECS) software product that communicates between computers in a computer network. Donor is designed to operate on commercial off-the-shelf (COTS) computer equipment and peripherals. WellSky has tested the Donor software product in a wireless network environment, using industry standard 802.11g and 802.11n wireless networks. It is the responsibility of the customer to follow your equipment manufacturer's specific guidelines regarding separation distances, EMC considerations, etc., and to validate the equipment as suitable for use with Blood Centers in the customer's operating environment.

Examples of manufacturer's requirements for equipment used in testing by WellSky are listed below. The equipment utilized during the testing performed on Donor was labeled on the data plates on the equipment or in the literature accompanying the hardware as FCC compliant to the standards for commercial off-the-shelf (COTS) electronic equipment.

Table 1-1: COTS 802.11 Equipment Specifications

Equipment	Separation Distance from Manufacturer Documentation
Cisco 1800 Wireless G Router	A minimum 20 cm (7.9 in.) co-located separation distance from other FCC approved indoor/outdoor antennas used with the access point. The co-located 2.4-GHz and 5-GHz antennas support a minimum separation distance of 10 cm (3.9 in.) and are compliant with the applicable FCC RF exposure limit when transmitting simultaneously.
Netgear WNR2000 Wireless N Router	This equipment should be installed and operated with a minimum distance of 20 cm between the radiator and your body. The antennas used for this transmitter must be installed to provide a separation distance of at least 20 cm from all persons and must not be co-located or operating in conjunction with any other antenna or transmitter.
Intel 5100 Wireless network card	In order to comply with FCC RF exposure limits, low gain integrated antennas should be located at a minimum distance of 20 cm (8 inches) or more from the body of all persons. High-gain, wall-mount, or mast-mount antennas are designed to be professionally installed and should be located at a minimum distance of 30 cm (12 inches) or more from the body of all persons. Please contact your professional installer, VAR, or antenna manufacturer for proper installation requirements. To comply with the FCC and ANSI C95.1 RF exposure limits, it is recommended for the adapter installed in a desktop or portable computer, that the antenna for this device be installed so as to provide a separation distance of at least 20 cm (8 inches) from all persons. It is recommended that the user limit exposure time if the antenna is positioned closer than 20 cm (8 inches). Intel® PRO/Wireless LAN products are not designed for use with high-gain directional antennas. Use of such antennas with these products in a manner other than as described in the previous section titled, "Antenna Use" is illegal.

WellSky Corporation
Proprietary and confidential

In addition to your manufacturer's recommendations regarding equipment setup, WellSky recommends the following derived from our own system testing and field experience:

1. Do not plan to use wireless communication near known sources of electromagnetic interference, including but not limited to:
    - Heavy machinery with large motors
    - Large compressors (e.g. industrial refrigerators)
    - X-ray machines or any other type of radiology equipment
    - Microwave emitters such as microwave ovens and other microwave devices
    - Electric generators
    - Rooms with walls that block radio signals such as basements and reinforced walls
    - Rooms that do not allow enough room to properly deploy the network equipment for the blood drive
    - 2.4 MHz cordless telephones that use wireless network frequencies
    - Radar equipment
    - Locations outside the service area of your broadband provider
    - Known dead zones
2. Locate your wireless router centrally to your wireless network and elevated for best results.
3. Maintain a separation distance of three feet between operational units for best results.
4. Keep operational units within 75 feet of the wireless router for best results.
5. Do not allow phones or other hand-held electronic devices to be laid on top of or next to routers or workstations.
6. Advise staff and visitors to turn off their personal electronic equipment while in the donation area.
7. The WLAN should include a maximum of one master laptop and six client laptops or six laptops connected to your center server through the WLAN. Configurations exceeding this number must be validated prior to use to ensure that they operate properly.

Wireless Cellular WAN Connection

Client laptops located at a blood drive are connected directly to the central server located in the Blood Center through a WAN link using cellular network air cards. This wireless functionality can only be used where a cellular signal can be obtained at the mobile drive. Connections are made through a VPN to access the main database in the central server to verify donor eligibility and to update donation information.

Wireless Cellular Specifications

WellSky has tested cellular broadband communication using the following specifications for commercial off-the-shelf (COTS) equipment.

Table 1-2: COTS Cellular Equipment Specifications

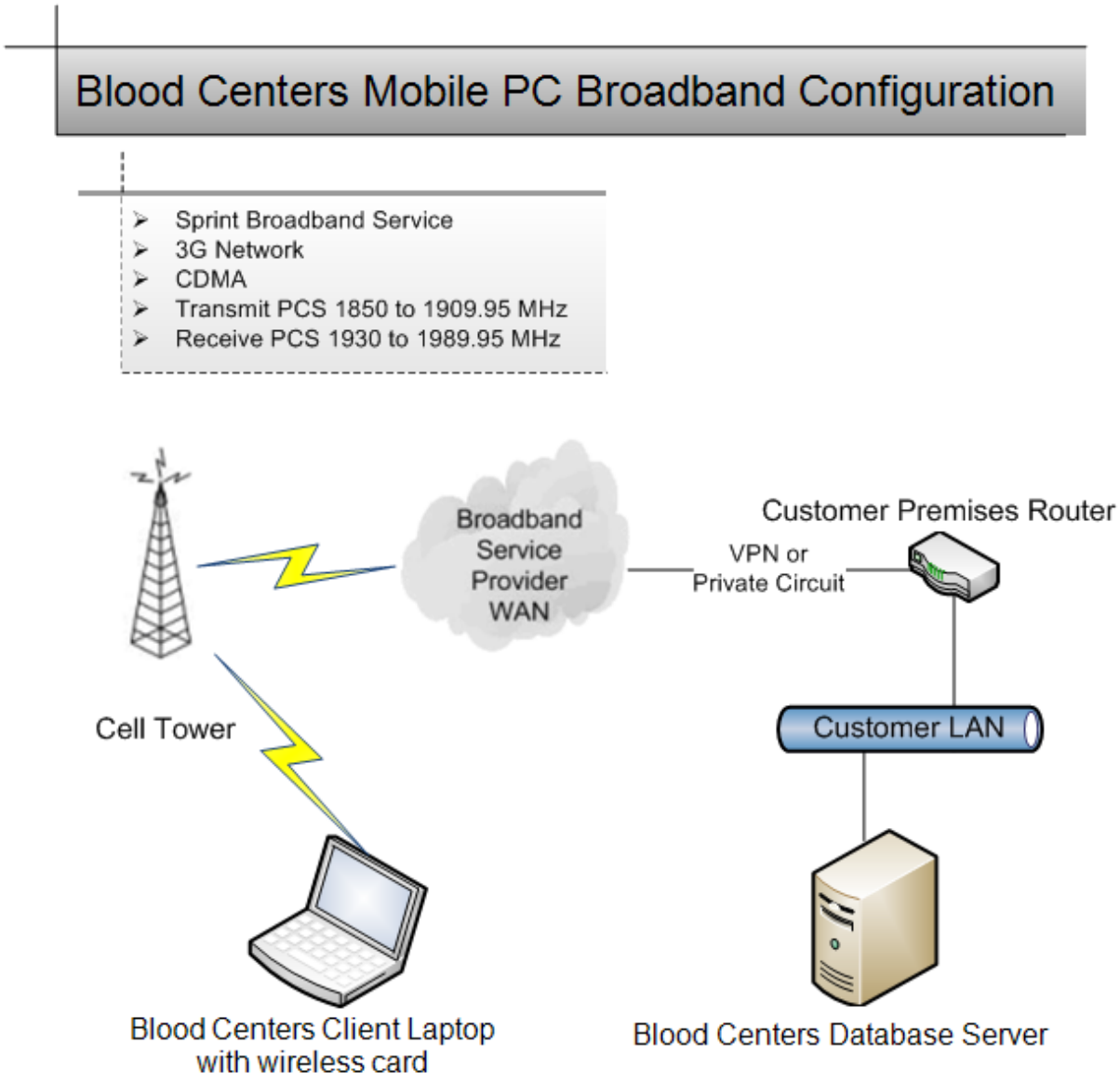
Device	RF Frequency	Modulation	Data Rate/ Data Flow
Sprint U301 USB Air card	CDMA 800/1900 MHz	CDMA	Up to 2 MBPS (3G)
Sprint Mobile Broadband Card by Sierra Wireless Air Card 597E	CDMA 824 to 849 MHz	CDMA	Up to 2 MBPS (3G)

Summary of Cellular Workstation Essential Operating Requirements

- Do not use WLAN and cellular network cards at the same drive.
- Select an operating location and position network components to avoid sources of electromagnetic interference.
- Maintain a separation distance of three feet between active network components.
- Use VPN network security protocols.
- Do not operate a cellular workstation above two percent packet loss.
- Follow all of your equipment manufacturer's guidelines.

WellSky Corporation
Proprietary and confidential

Example Cellular Network Configuration



Wireless Cellular Range of Operation

Donor is a Blood Establishment Computer Software (BECS) software product that communicates between computers in a computer network. Donor is designed to operate on commercial off-the-shelf (COTS) computer equipment and peripherals. WellSky has tested the Donor software product with cellular connectivity to the internet using 3G and 4G cellular technology. It is the responsibility of the customer to follow your manufacturer's specific guidelines regarding separation distances, EMC considerations, etc., and to validate the equipment as suitable for use with Blood Centers in the customer's operating environment.

Examples of manufacturer's requirements for equipment used in testing by WellSky are listed below. The equipment utilized during the testing performed on Donor was labeled on the data plates on the equipment or in the literature accompanying the hardware as FCC compliant to the standards for off-the-shelf electronic equipment.

Table 1-3: Example COTS Wireless Cellular Equipment Specifications

Equipment	Separation Distance from Manufacturer Documentation
Sprint U301 USB Aircard	This modem must be 5 mm (0.2 inches) or more from the user and bystanders during operation to satisfy FCC (regulatory) RF exposure requirements. It is approved for use with normal-size notebook computers only (typically with 12-inch or larger display screens). For notebook computers with a display screen smaller than 12 inches or if the orientation of the USB slot does not allow for at least 5 mm between the modem and the user or bystanders, use the provided extension cable to ensure a safe distance between the modem antenna and users or bystanders. To meet FCC requirements, use only the supplied USB extension cable; other cables should not be used.
Sprint Mobile Broadband Card by Sierra Wireless AirCard 597E	At least 20 cm separation distance between the antenna and the user's body must be maintained at all times.

In addition to your manufacturer's recommendations regarding equipment setup, WellSky recommends the following derived from our own system testing and field experience when using cellular equipment:

1. Do not plan to use wireless communication near known sources of electromagnetic interference, including but not limited to:
    - Heavy machinery with large motors
    - Large compressors (e.g. industrial refrigerators)
    - X-ray machines or any other type of radiology equipment
    - Microwave emitters such as microwave ovens and other microwave devices
    - Electric generators
    - Rooms with walls that block radio signals such as basements and reinforced walls
    - Rooms that do not allow enough room to properly deploy the network equipment for the blood drive
    - 2.4 MHz cordless telephones that use wireless network frequencies
    - Radar equipment
    - Locations outside the service area of your broadband provider
    - Known dead zones
2. Maintain a separation distance of three feet between operational units for best results.
3. Do not allow phones or other hand-held electronic devices to be laid on top of or next to the cellular equipped workstation.
4. Advise staff and visitors to turn off their personal electronic equipment while in the donation area.

WellSky Corporation
Proprietary and confidential

Electromagnetic Compatibility

Customers should be aware that electronic equipment, including computers and peripherals, has the potential for causing interference with other electronic equipment present in the environment where the equipment operates. Although no specific interference was noted in the testing performed by WellSky on Donor, customers must be aware of the potential for interference as part of the installation and validation process.

Recognizing Electromagnetic Interference

Blood Centers wireless communication was validated to a standard of less than 2% packet loss on a wireless network. You can easily verify your wireless setup to the same standard.

1. From any client station on your wireless network, ping the laptop server on the network. If ping reports packet loss of more than two percent, then that station does not meet the standard to continue operation on the wireless network.
2. Watch for these other warning signs of packet loss.
    - Response from the laptop server is slow.
    - The client hangs while communicating with the laptop server.
    - System reports wireless connection lost.

Procedures for Dealing with Lost or Problem Wireless Connections

If you lose or are unable to establish a connection to your wireless network or become aware that your wireless network is interfering with other equipment in the area, consider the following mitigation and strategies as appropriate for your situation and in accordance with your facility's Standard Operating Procedures (SOPs).

1. Reboot the workstation to reestablish your connection.
2. Use another workstation.
3. Relocate or change the orientation of the workstation.
4. Increase the distance from other nearby electronic components.
5. Move closer to the wireless router.
6. Change your power source.
7. Reorient your wireless antenna.
8. Connect to the host directly using a wired connection instead of a wireless connection.
9. Reset your wireless router if multiple stations are affected.
10. If no wireless network can be established, then continue drive operations on the laptop server alone if practical to do so.
11. Invoke the manual screening and draw procedures for your facility.
12. Fall back to backup copies of the screening database on the remote workstation or other options provided by your facility's configuration and its SOPs.

13. Discontinue the drive if that is your facility's SOP or appropriate for the situation (e.g. drive is almost over).
14. Seek expert advice from the manufacturer of the equipment and other wireless networking experts.

1.9.2 Computers

The following describes the equipment and configurations that were tested for wireless connection and transmission of donor data and information at the specified Quality of Service:

Servers

Mobile Donor Laptop

- Lenovo T510
- Dual Core i5 with Hyper-threading Technology @ 2.67 GHz - 64-bit
- 16 GB RAM Memory
- 250GB Hard Disk
- 10/100/1000 Ethernet Network Interface
- Intel 5100 wireless network card

Clients

- Lenovo T500
- 2.66 GHz Intel Core 2 Duo CPU
- 16 GB RAM memory
- 150 GB hard disk
- Windows 11, 32-bit or 64-bit
- 10/100/1000 Ethernet Network Interface
- Intel 5100 wireless network card

Network Devices

- Cisco 1811W, Security router with dual 10/100 WAN ports, 8-port 10/100BASE-T switch, V.92 analog modem backup, Cisco IOS Advanced IP Services, 32 MB Flash, 128 MB DRAM, and integrated FCC-compliant 802.11n and 802.11g wireless capability
- Sprint U301 USB Device 3G/4G Mobile Broadband, Sprint CDMA frequency of 800 MHz/1.9 GHz and Sprint WiMAX frequency band 2.5 GHz
- Netgear WNR2000
- Sierra Wireless Compass 597

WellSky Corporation
Proprietary and confidential

1.9.3 Wireless and Cellular Quality of Service

To achieve the same level of wireless quality of service that was demonstrated during internal WellSky testing any system installed at a user site should demonstrate the following Quality of Service:

Wireless

- Maintain the network according to IEEE 802.11n or 802.11g standards with encryption.
- The wireless access security measures outlined below should be applied.
- The wireless devices should be at least three feet and no more than 75 feet from the wireless access point to maintain connectivity.
- The error rate of data transfer must be no more than 2% packet loss.
- During user testing at WellSky, typical devices that could coexist with the wireless network during a drive were tested. These included a tube sealer, cell phone, laptop computer, printer, microwave oven, and apheresis equipment. The system was also tested in a mobile bus while operating. No interference with the wireless network was detected. Users should be aware that other electronic devices, especially those with wireless capabilities could potentially interfere with wireless data transmission.
  - Personal electronics including cell phones, portable phones, music players, Bluetooth devices should be kept at least six inches from laptops that are being used during a blood drive. As a general policy, recommend that blood drive personnel instruct donors to power-off any electronic devices in their possession while in the blood drive area.
  - Common lunch or break-room equipment including microwave ovens, refrigerators and coffee makers should be kept at least six feet from laptops that are being used during a blood drive.
  - Blood processing equipment commonly found in the blood donation sites including aphaeresis equipment and heat sealing devices should be kept at least three feet from laptops being used during a blood drive.
- An alternative backup system such as a wired network or manual paper records should be available at the donation site to be used if the wireless network is not working properly.

Cellular

- The cellular network should be maintained on a 3G network.
- The wireless access security measures outlined below should be applied.
- The wireless devices should be at least 3 feet and no more than 75 feet from the wireless access point to maintain connectivity.
- The error rate of data transfer must be no more than 2% packet loss.
- Typical devices that we believe could co-exist with the wireless network at the blood center or during a remote blood drive include tube sealer, cell phone/PDA, laptop computers, printers, microwave ovens and aphaeresis equipment. Therefore, we tested the Donor software operating on a wireless network against each of the identified devices. The system was also tested on a mobile bus while the bus was in operation.
  - Personal electronics including cell phones, portable phones, music players, Bluetooth devices should be kept at least six inches from laptops that are being used during a blood drive. As a general policy,

recommend that blood drive personnel instruct donors to power-off any electronic devices in their possession while in the blood drive area.

- Common lunch or break-room equipment including microwave ovens, refrigerators and coffee makers should be kept at least six feet from laptops that are being used during a blood drive.
- Blood processing equipment commonly found in the blood donation sites including aphaeresis equipment and heat sealing devices should be kept at least three feet from laptops being used during a blood drive.

1.9.4 Security Measures

These security measures are recommended for all wireless access point devices when using the Blood Centers application. These security measures are general guidelines; refer to the owner's manual of the wireless access point to implement specific settings.

Wireless

- The wireless access point administration will be configured with a password. The password must be changed from the default provided by the manufacturer. The recommended minimum password length is eight characters. The password should include lower case alphabetic, upper case alphabetic, numeric and special characters. At least one of each type is required.
- The default wireless Service Set Identifier (SSID) or network name should be changed. The name should have a length of at least eight characters and require both alphabetic and numeric characters. Additionally, broadcast of the SSID should be disabled.
- The wireless devices (routers and network cards) should be secured using encryption technology. It is recommended to use theWPA2 authentication level. The encryption should be AES (Advanced Encryption Standard). The WPA2 shared key should be a minimum of fifteen characters. These should include lower case alphabetic, upper case alphabetic, numeric and special characters. At least one of each type should be required.
- To prevent unauthorized computers from communicating with the wireless access points, it is recommended to use the Media Access Control (MAC) filtering. MAC filtering allows only authorized equipment to connect to the wireless access point. This prevents unauthorized equipment from connecting to a wireless network.

Cellular

Connections made over the cellular network must be made over a virtual private network (VPN). This will ensure that the traffic will be encrypted while being transmitted over the public internet.

Risks and Mitigation

a. Risk of failure of the wireless network is that individual client stations cannot be used for screening on the drive. WellSky recommends that this risk be covered by client operating procedures since failure of an individual laptop is always an attendant risk of any remote blood drive and must be planned for regardless of whether wireless communication is utilized or not. In the case of a drive that plans to use wireless communication, additional mitigations are available since multiple laptops provide redundancy in the case of a single laptop or wireless network failure including the option of using the database server itself to continue screening when no wireless network can be established.

WellSky Corporation
Proprietary and confidential

b. Risk of failure of the wireless connection is that the remote workstation cannot connect to the central database host for direct interaction with the system. WellSky recommends that this risk must also be covered by client operating procedures since failure of an individual laptop is always an attendant risk of any remote blood drive and must be planned for regardless of whether wireless communication is utilized or not. Mitigations for a failed remote workstation (or its wireless connection) range from full manual procedures, deployment of backup workstations, call-in screening procedures, up to cancelling the drive. Failure of the wireless connection alone allows the additional option of falling back to screening and registration from a backup local copy of the database on the workstation if the facility so chooses to configure their remote workstations.

1.9.5 Electromagnetic Compatibility

Users should be aware that electronic equipment, including computers and peripherals, has the potential for causing interference with other electronic equipment present in the environment where the equipment operates. Although no specific interference was noted in the testing performed by WellSky on Donor, users should be aware of the potential for interference as part of the installation and validation process.

When setting up your equipment for a blood drive where you plan to use wireless communication, consider the following guidelines.

Selecting a Location

"Location, location, location." It is true in real estate and for wireless networks. Do not plan to use wireless communication near known sources of electromagnetic interference, including but not limited to the following:

- Heavy machinery with large motors
- Large compressors (e.g. industrial refrigerators)
- X-ray machines or any other type of radiology equipment
- Microwave emitters such as microwave ovens and other microwave devices
- Electric generators
- Rooms with walls that block radio signals such as basements and reinforced walls
- Rooms that do not allow enough room to properly deploy the network equipment for the blood drive
- 2.4 MHz cordless telephones that use wireless network frequencies
- Radar equipment
- Locations outside the service area of your broadband provider
- Known dead zones

Setup

Observe the following guidelines in setting up your wireless network:

a. Position equipment in accordance with WellSky guidelines and the specific recommendations of your commercial off-the-shelf wireless network equipment manufacturer.

b. WellSky-recommended guidelines include:
    i. Locate your wireless router centrally to your wireless network and elevated for best results.
    ii. Maintain a separation distance of three feet between operational units for best results.
    iii. Do not overextend your wireless network. Keep operational units within 75 feet of the wireless router for best results.
    iv. Do not allow phones or other hand-held electronic devices to be laid on top of or next to routers or workstations.
    v. Advise staff and visitors to turn off their personal electronic equipment while in the donation area.

Recognizing Electromagnetic Interference

Blood Centers wireless communication was validated to a standard of less than 2% packet loss on a wireless network. You can easily verify your wireless setup to the same standard, as follows:

a. From any client station on your wireless network, ping the laptop server on the network. If ping reports packet loss of more than two percent, then that station does not meet the standard to continue operation on the wireless network.
b. Watch for these other warning signs of packet loss.
    i. Response from the laptop server is slow.
    ii. The client hangs while communicating with the laptop server.
    iii. System reports wireless connection lost

Mitigation Strategies

If you lose or are unable to establish a connection to your wireless network or become aware that your wireless network is interfering with other equipment in the area, consider the following mitigation and strategies as appropriate for your situation and in accordance with your facility's Standard Operating Procedures (SOPs).

a. Reboot the workstation to reestablish your connection.
b. Use another workstation.
c. Relocate or change the orientation of the workstation.
d. Increase the distance from other nearby electronic components.
e. Move closer to the wireless router.
f. Change your power source.
g. Reorient your wireless antenna.
h. Connect to the host directly using a wired connection instead of a wireless connection.
i. Reset your wireless router if multiple stations are affected.
j. If no wireless network can be established, then continue drive operations on the laptop server alone if practical to do so.

WellSky Corporation
Proprietary and confidential

k. Invoke the manual screening and draw procedures for your facility.

l. Fall back to backup copies of the screening database on the remote workstation or other options provided by your facility's configuration and its SOPs.

m. Discontinue the drive if that is your facility's SOP or appropriate for the situation (e.g. drive is almost over).

n. Seek expert advice from the manufacturer of the equipment and other wireless networking experts.

1.9.6 Other Important Information Regarding Use of Wireless Devices

During WellSky's internal and external validation testing of Donor several wireless devices were used to evaluate the product. Wireless devices are required by the FCC to be tested for the potential for interference with other devices that are located in the same general area as the wireless device. Additionally, devices are tested for the potential effect on persons that are using the devices or are in the general vicinity of these devices when they are being used. This information is usually provided in the labeling shipped with the device or may be available on the equipment supplier's Web site. When setting up Donor for use in a wireless network, you should be aware of such information and to utilize the any cautions and warnings contained therein in the planning for your network. Some examples of the information provided by the manufacturer of the devices used by WellSky during testing are summarized in the following:

Device	Separation Distance from Manufacturer Documentation
Cisco 1800 Wireless G Router	A minimum 20 cm (7.9 in.) co-located separation distance from other FCC approved indoor/outdoor antennas used with the access point. The co-located 2.4-GHz and 5-GHz antennas support a minimum separation distance of 10 cm (3.9 in.) and are compliant with the applicable FCC RF exposure limit when transmitting simultaneously.
Netgear WNR2000 Wireless N Router	This equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body. The antennas used for this transmitter must be installed to provide a separation distance of at least 20 cm from all persons and must not be co-located or operating in conjunction with any other antenna or transmitter.
Intel 5100 Wireless network card	In order to comply with FCC RF exposure limits, low gain integrated antennas should be located at a minimum distance of 20 cm (8 inches) or more from the body of all persons. High-gain, wall-mount, or mast-mount antennas are designed to be professionally installed and should be located at a minimum distance of 30 cm (12 inches) or more from the body of all persons. Please contact your professional installer, VAR, or antenna manufacturer for proper installation requirements. To comply with the FCC and ANSI C95.1 RF exposure limits, it is recommended for the adapter installed in a desktop or portable computer, that the antenna for this device be installed so as to provide a separation distance of at least 20 cm (8 inches) from all persons. It is recommended that the user limit exposure time if the antenna is positioned closer than 20 cm (8 inches). Intel® PRO/Wireless LAN products are not designed for use with high-gain directional antennas. Use of such antennas with these products in a manner other than as described in the previous section titled, "Antenna Use" is illegal.

Device	Separation Distance from Manufacturer Documentation
Sprint U301 USB Aircard	This modem must be 5 mm (0.2 inches) or more from the user and bystanders during operation to satisfy FCC (regulatory) RF exposure requirements. It is approved for use with normal-size notebook computers only (typically with 12-inch or larger display screens). For notebook computers with a display screen smaller than 12 inches or if the orientation of the USB slot does not allow for at least 5 mm between the modem and the user or bystanders, use the provided extension cable to ensure a safe distance between the modem antenna and users or bystanders. To meet FCC requirements, use only the supplied USB extension cable; other cables should not be used.
Sprint Mobile Broadband Card by Sierra WirelessAirCard 597E	At least 20 cm separation distance between the antenna and the user's body must be maintained at all times.

WellSky Corporation
Proprietary and confidential

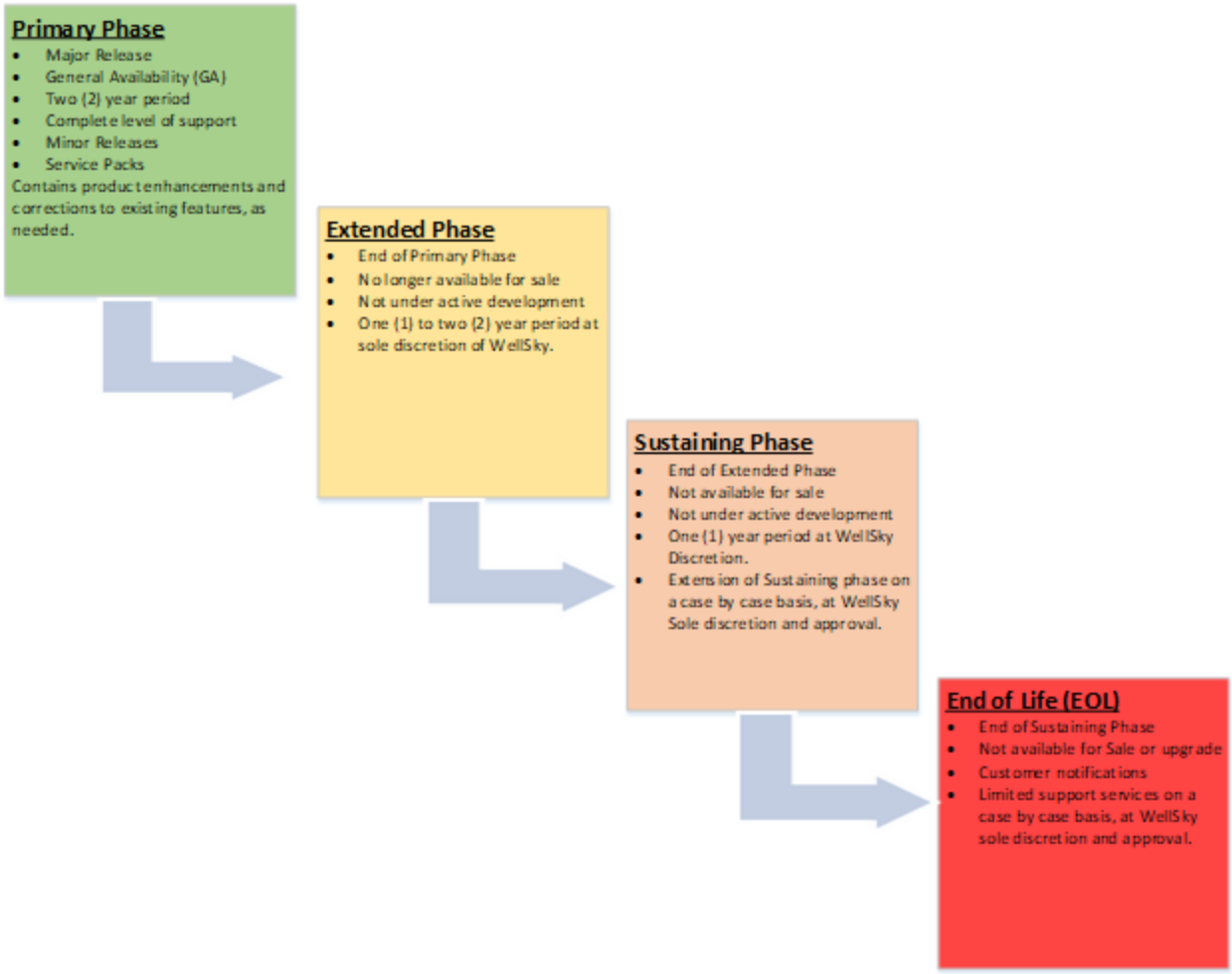
Solution Life Cycle

The WellSky® Solution Life Cycle describes the sequence of phases our solutions go through from General Availability (GA) through maturity and decline. The value of identifying the phases and applying timeframes, i.e. actual dates to each phase, allows you to plan your upgrades and migrations without any gaps in support.

To help realize the maximum value from WellSky's applications, customers should install the latest version of your WellSky solution. The latest versions allow customer(s) to benefit from the latest functionality and feature enhancements, are designed to handle modern workloads, and provide access to our most complete technical support and engineering assistance and resources.

WellSky strives to make each version easier to upgrade, and we also have a professional services team to assist with upgrade plans as well. WellSky understands that business constraints may delay customers from upgrading, and that certain customers may need to have the flexibility either to upgrade to new versions or decide to remain on an older version for a while longer. WellSky, at its discretion, may provide customers with additional support offerings that can extend certain limited features of maintenance/support coverage until the customer is able to upgrade.

Milestone	Estimated Schedules
Major Release	Every two (2) years, at a minimum
Minor Release	At least one during the life cycle
Service Packs	One (1) – two (2) per life cycle, according to need during the life cycle


Primary Phase
Major Release
General Availability (GA)
Two (2) year period
Complete level of support
Minor Releases
Service Packs
Contains product enhancements and corrections to existing features, as needed.
Extended Phase
End of Primary Phase
No longer available for sale
Not under active development
One (1) to two (2) year period at sole discretion of WellSky.
Sustaining Phase
End of Extended Phase
Not available for sale
Not under active development
One (1) year period at WellSky Discretion.
Extension of Sustaining phase on a case by case basis, at WellSky Sole discretion and approval.
End of Life (EOL)
End of Sustaining Phase
Not available for Sale or upgrade
Customer notifications
Limited support services on a case by case basis, at WellSky sole discretion and approval.

2

Chapter 2 - Mobile Donor Database Overview

2.1 Database Description 24
2.2 Mobile Donor Upload/Download Configuration Overview 24

2.1 Database Description

The mobile donor database contains a subset of information stored in the Donor production database on the primary host. The Oracle database manager on the mobile system uses SQL*Net to synchronize the mobile donor database with the production database on the primary host. The process transfers all donor deferrals to the mobile donor database. The update process also transfers donation data for at least the past year and the donor's gallon donation status.

The two configurations for the Mobile Donor PC Database are as follows:

1. The **Mobile Donor Collections Lookup configuration (read-only)** is for lookup of existing donor information (name, deferrals, donation history for past year, and gallon status) that currently resides in the Donor Master Database. This configuration supports printing of donor cards.
2. The **Mobile Donor Upload/Download configuration** provides complete donor information to identify uniqueness of donor, eligibility, deferrals (interdictions) and documents the donation (or deferral) process as currently done within the On-Site Registration module. This includes, but is not limited to donor registration, donor questionnaire, health history, physical finding, phlebotomy, unit (or deferral) attachment, and comments.

2.2 Mobile Donor Upload/Download Configuration Overview

The Mobile Donor Upload/Download configuration replicates donor data from the Donor database server to the Master Laptops in read/write mode for the registration and donation (or deferral) process of donors. The data downloaded contains complete donor information to identify uniqueness of donor, eligibility, deferrals (interdictions), and document the donation (or deferral) process as currently done within the On-Site Registration module. This includes, but is not limited to, donor registration, donor questionnaire, health history, physical finding, phlebotomy, unit (or deferral) identification, and comments.

At a predetermined time during or after a mobile drive, the Master Laptop provides updates to the Donor Master Database in the form of an extract file.



Important! Observe the following when selecting the DBA role in the master database:

- If any user has the DBA role on the master database, the system will not add this user in the laptop database during the replication process.
- Only the LAPTOP user has DBA privileges on the laptop server; other users have only connect privileges on the laptop server even though these users may have the LT_DONOR role on the master database.
- If the DBA role is granted for other user in the master database, the system will not add that user. If the DBA role is granted in the middle, then the system will drop that user in the LAPTOP database during replication.

WellSky Corporation
Proprietary and confidential

3

Chapter 3 - Mobile PC Handling Guidelines

Observe these do's and don'ts to protect equipment and data.

Do:

- Transport mobile PCs in hard-shell cases.
- Close cases securely before transport.
- Open and use the unit only on a secure, flat surface.
- Keep all liquids away from the equipment.
- Operate the mobile PC at room temperature.
- Use the mobile PC only for approved business purposes.
- Notify your company's mobile PC support personnel of any problems with the mobile PC.

Do Not:

- Do not put the mobile PC on an unstable surface.
- Do not drop the equipment.
- Do not spill any liquid on the mobile PC.
- Do not unplug any peripheral attached to the mobile PC.
- Do not leave the mobile PC unattended at any time.
- Do not lend the mobile PC to anyone.
- Do not allow anyone other than authorized personnel to use the mobile PC.
- Do not attempt to load software or change the machine configuration.

This Page Intentionally Left Blank

4

Chapter 4 - Mobile Donor Client Environment

4.1 Mobile Donor Collections Lookup Laptop 28
4.2 Mobile Donor Collections Printers 29
4.3 Mobile Donor Upload/Download Laptop 29
4.4 Mobile Donor Laptop Browser Clients 30

4.1 Mobile Donor Collections Lookup Laptop

The Mobile Donor Collections Lookup client houses a subset of Donor screens for the lookup of donor information and to print donor cards. The database is read-only and no update is allowed.

4.1.1 Features

- Donor Management
- Local Donor Deferral Database (DDR)
- DHCP IP address resolution is recommended

4.1.2 Hardware

- Dell Latitude with 7490
- Intel Core i7-8650U @ 1.9GHz - 64-bit
- 32GB RAM Memory
- 250GB SSD

4.1.3 Software

- Microsoft Windows Server 2022 Standard
- Oracle Database 21c Express Edition Release (21.3.0.0)
- Google Chrome with latest updates
- Adobe Acrobat Reader DC

WellSky Corporation
Proprietary and confidential

4.2 Mobile Donor Collections Printers

Mobile donor card printing requires a portable color printer.

4.3 Mobile Donor Upload/Download Laptop

The Mobile Donor Upload/Download client includes the On-Site module and other modules which are required to run the application loaded on the Master Laptop.

4.3.1 Hardware

- Dell Latitude with 7490
- Intel Core i7-8650U @ 1.9GHz 2.11 GHz - 64-bit
- 32GB RAM Memory
- 250GB SSD

4.3.2 Software

- Microsoft Windows Server 2022 Standard
- Oracle Database 21c Express Edition Release (21.3.0.0)
- Google Chrome with latest updates
- Adobe Acrobat Reader DC

4.4 Mobile Donor Laptop Browser Clients

Additional laptop clients can be networked (see illustration in section 1.8) to access the On-Site module on the Master Laptop Server (**Mobile Donor Upload/Download configuration** only). The requirement for these laptops or PCs is as follows:

4.4.1 Hardware

- Dell Latitude with 7490
- Intel Core i7-8650U @ 1.9GHz 2.11 GHz - 64-bit
- 16 GB RAM memory
- 250 GB SSD

4.4.2 Software

- Windows 11 Enterprise
- Google Chrome with latest updates
- Adobe Acrobat Reader DC

WellSky Corporation
Proprietary and confidential

5

Chapter 5 - Set Up Mobile PC Master Server

5.1 Introduction 32
5.2 Software Preparation 32
5.3 Configure TCP/IP Protocol 32
5.4 Configure Windows Domain 33
5.5 Install Oracle Database 21c Express Edition 34
5.6 One-time Initialization 41
5.7 Oracle Wallet Setup for Mobile PC Server 44
5.8 IIS Installation 47
5.9 Google Chrome Setup for HTTP 80

5.1 Introduction

This section describes the setup process for all configurations of Mobile PCs that are used to replicate or display the donor information locally.

NOTE – The networked Mobile Donor Laptop Browser Clients described in section 4.4 do not require the additional Oracle software described in this section. They will access the On-Site module on the Master Laptop Server (Mobile Donor Upload/Download configuration only) via a wireless network using an Internet browser.

5.2 Software Preparation

Verify that the Mobile PC meets the minimum requirements for disk space, memory, CPU speed, and operating system, as described in chapter 4 for the configuration you are implementing.

1. Partition the Mobile PC hard drive to create a 200 GB partition labeled "c:\" drive.
2. If you will be using Blood Centers across multiple time zones and have the Time Zone feature enabled, set the time zone on the Mobile master laptop to the same time zone as the Blood Centers database and application servers.

5.3 Configure TCP/IP Protocol

The mobile PC must be configured for TCP/IP protocol, according to site specifications, to communicate with the Oracle host and to perform database synchronization. Database synchronization and Donor require DNS (Domain Name Services) to be operational for SQL*Net to function properly. DHCP address assignment is highly recommended to allow the mobile PC to be used on networks at other locations without resetting TCP/IP parameters.

When PCs are moved from one site to another, the operator may need to use the “ipconfig” program to release the old IP configuration and to get a new IP configuration at the new site.

WellSky Corporation
Proprietary and confidential

5.4 Configure Windows Domain

5.4.1 Configure Microsoft TCP/IP Client

Configure the Microsoft TCP/IP Client to allow the user to log onto the local network. This includes setting the Windows domain and WINS servers for the site.

The Replication Laptop name should be in the following format: hostname.domainname

Example:

repl.lifetrak.com (where repl is the hostname and lifetrak.com is the domain name)

5.4.2 Download and Install File Archive Utility

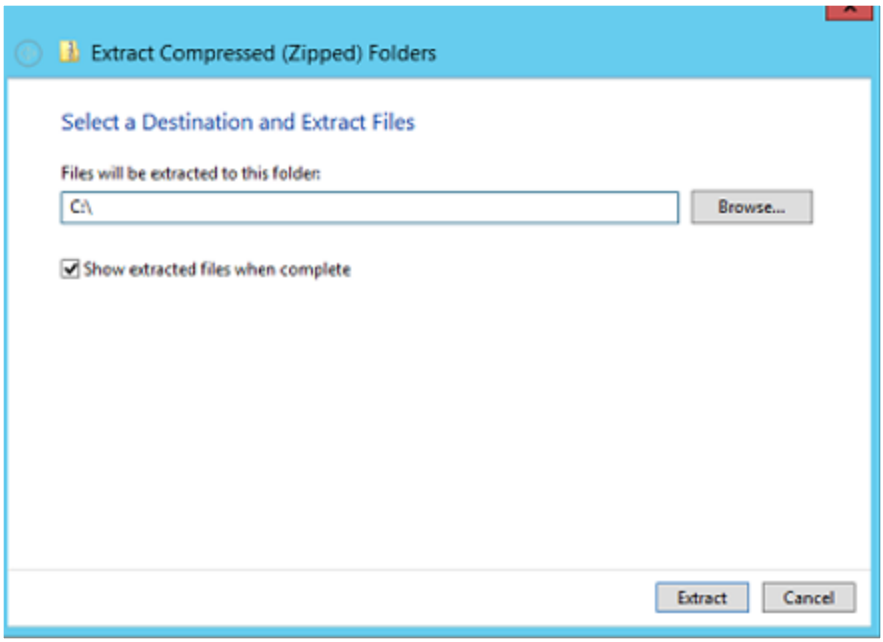
Download and install the 7zip (64-bit) software from https://www.7-zip.org.

5.5 Install Oracle Database 21c Express Edition

Follow these steps to complete the installation of the Oracle Database 21c Express Edition.

1. Insert the Blood Centers Software CD into the CD-ROM drive.
2. Copy the **BCMPC2026R1.zip** file from the CD to **C:**.
3. In Windows Explorer, right-click on **BCMPC2026R1.zip** and click **Extract All**. The Extract Compressed (Zipped) Folders window opens.

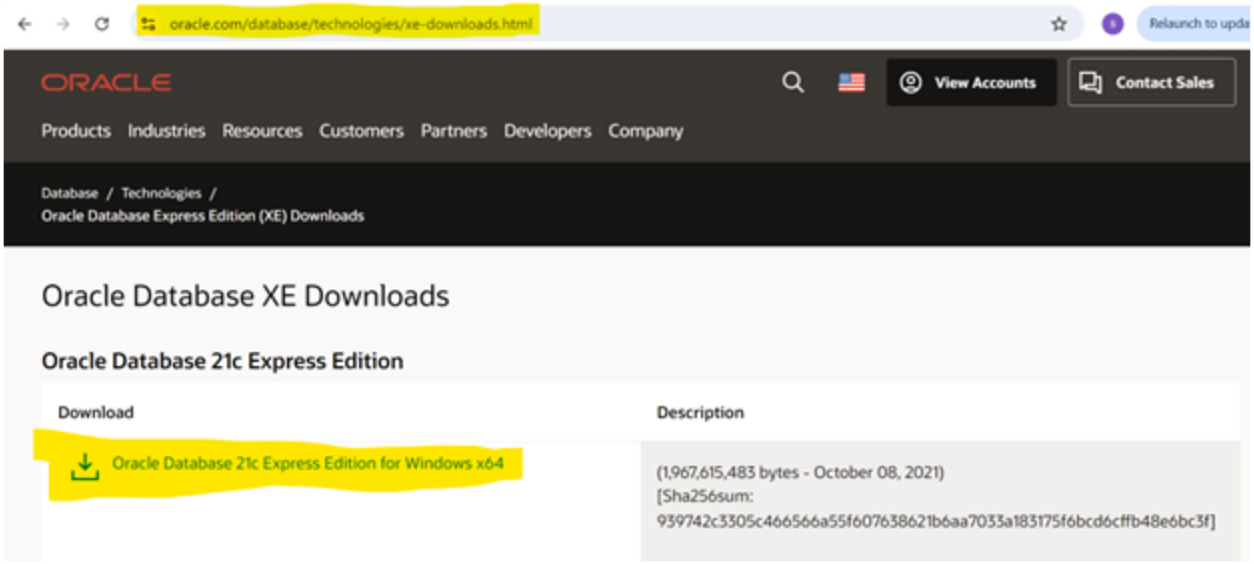
   Example:



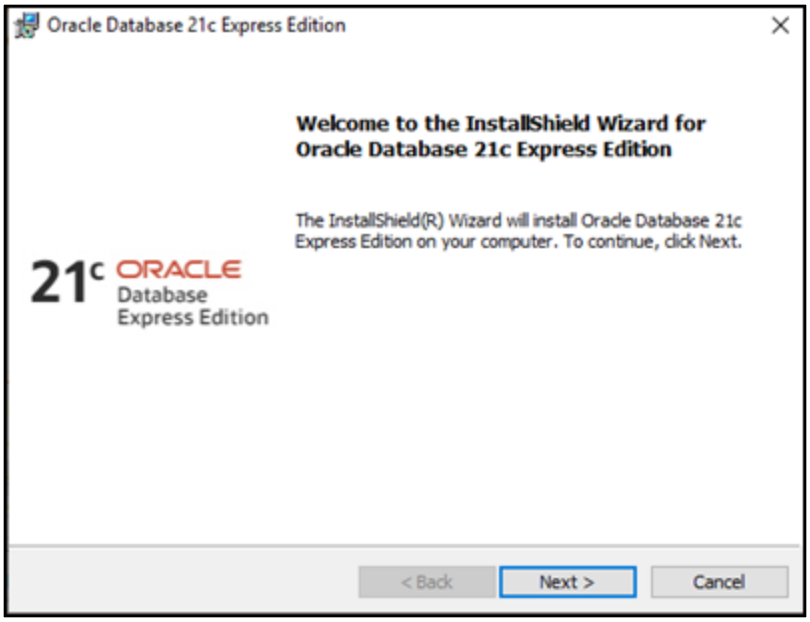
4. Accept the default location and click **Extract**.
5. Go to https://www.oracle.com/database/technologies/xe-downloads.html.

WellSky Corporation
Proprietary and confidential

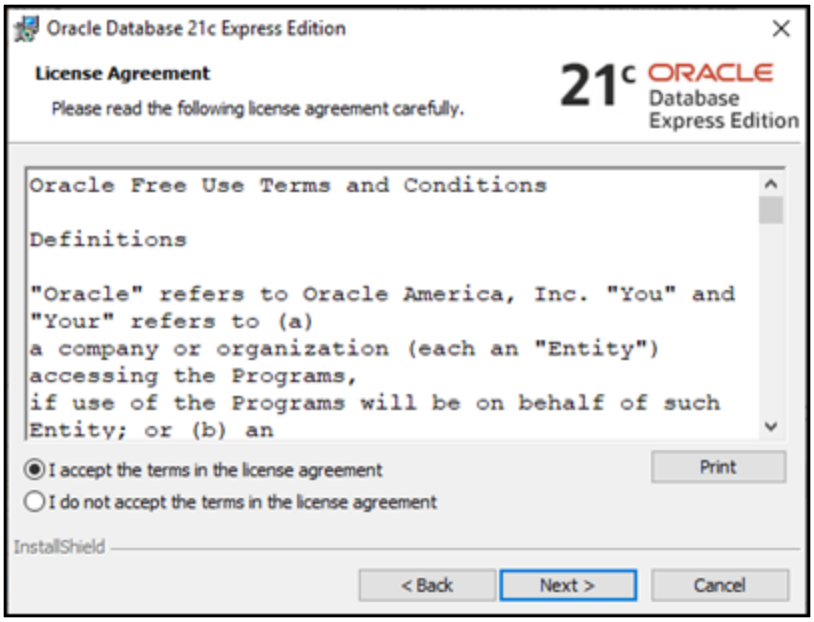
6. Click on **Oracle Database 21c Express Edition for Windows x64** to download.



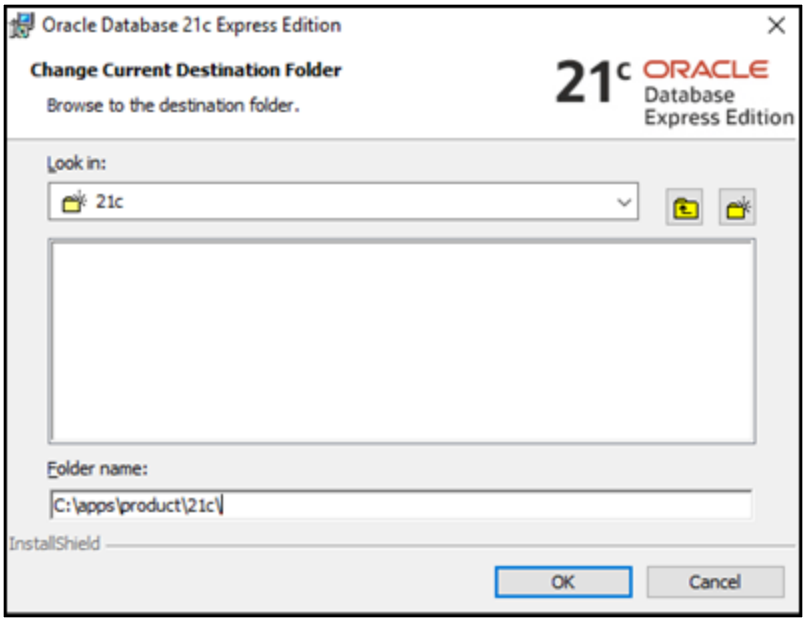
7. You will be prompted to log in to the Oracle Software Support link. Log in with Oracle Support credentials. The **OracleXE213_Win64.zip** file will download.
8. Save the file in the **C:\bcmpc\Software** folder.
9. Once the download completes, go back to Windows Explorer and unzip the downloaded **OracleXE213_Win64.zip** file.
10. Double-click the **Setup.exe** file in the download directory.



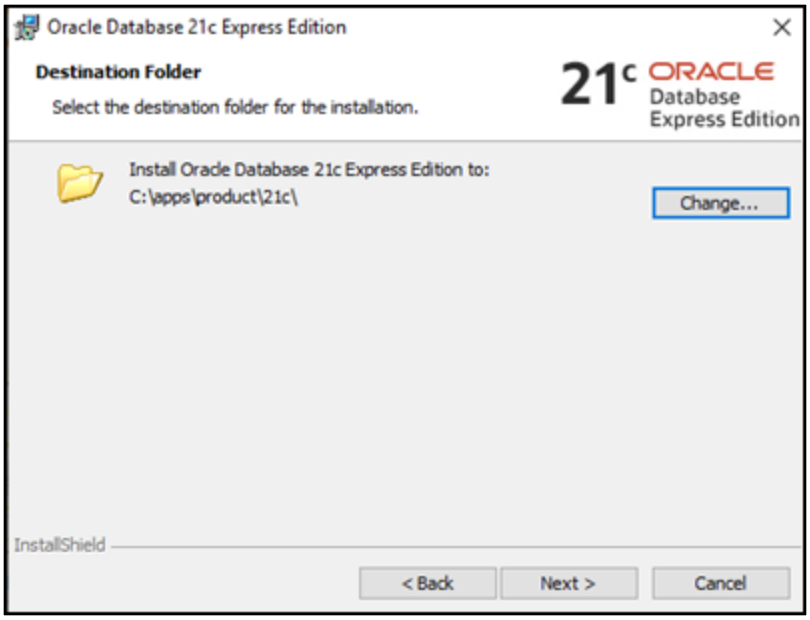
11. Click **Next**.



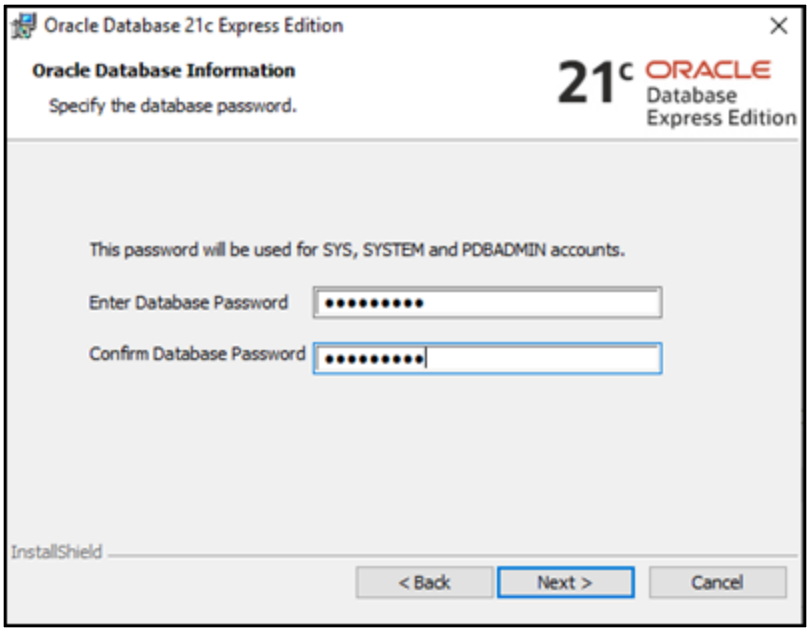
12. Select the **I accept the terms in the license agreement** option.
13. Click **Next**.
14. Click **Change...**, and specify the destination folder as: `C:\apps\product\21c\`.



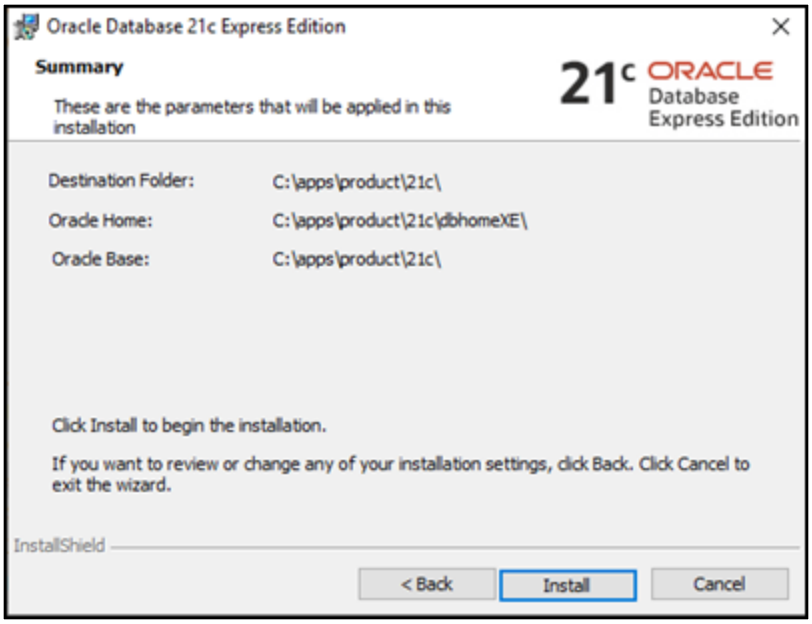
15. Click **OK**.



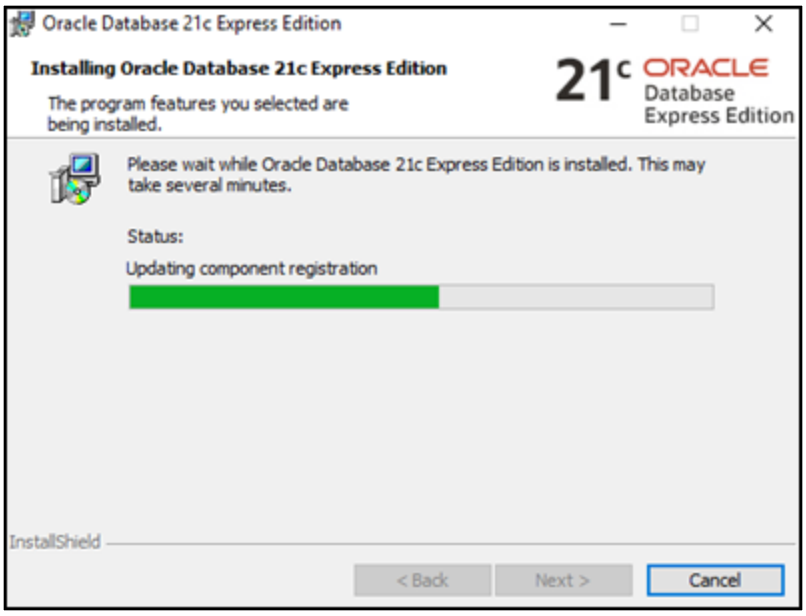
16. Click **Next**.
17. In the Enter Database Password field, enter the password as `Password1`. Enter it again in the Confirm Database Password field.

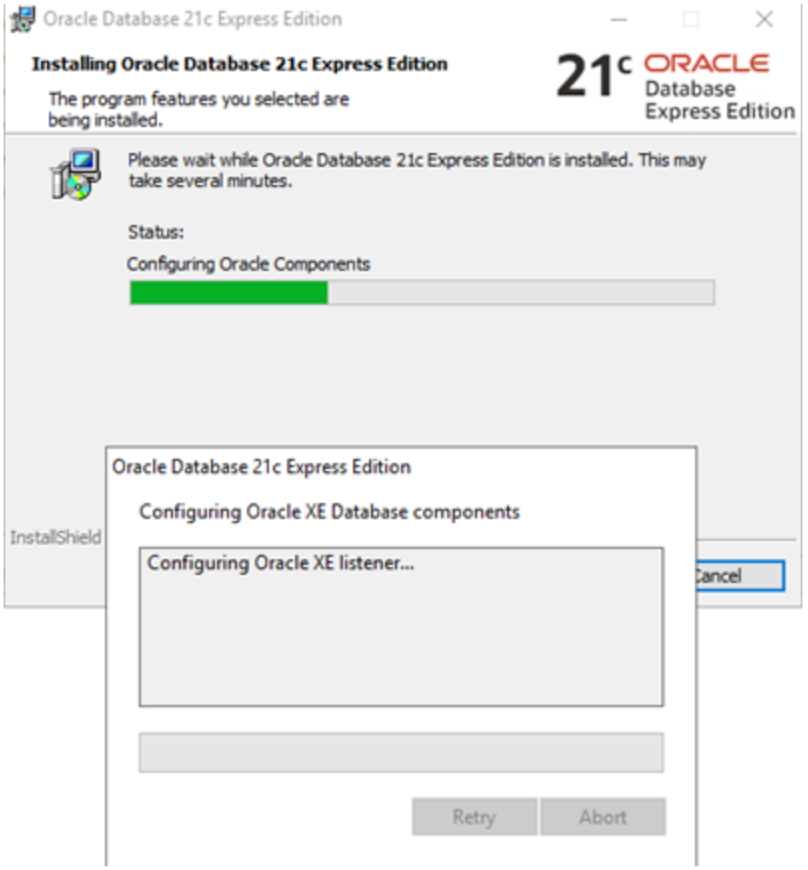


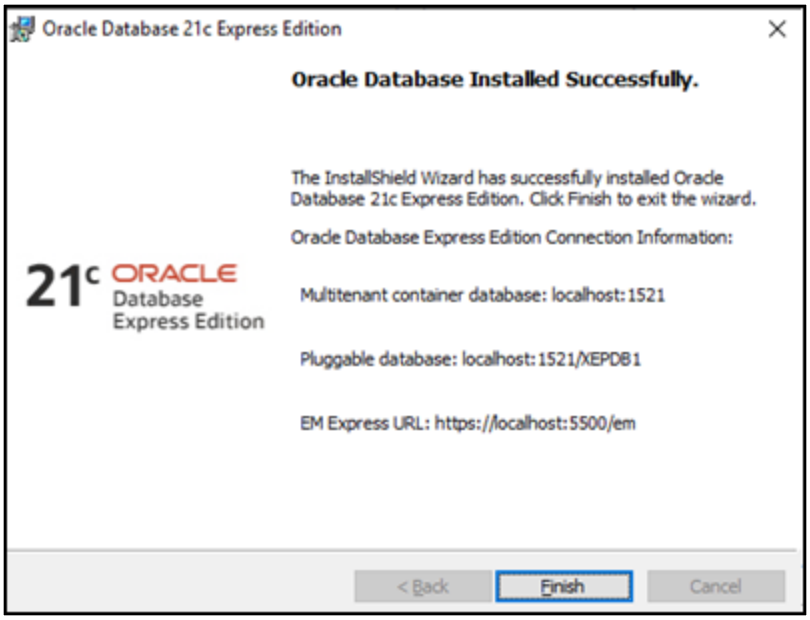
18. Click **Next**.



19. Click **Install**.

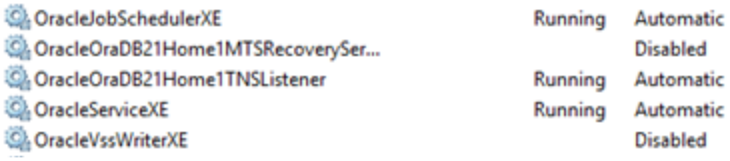






20. Click **Finish**.

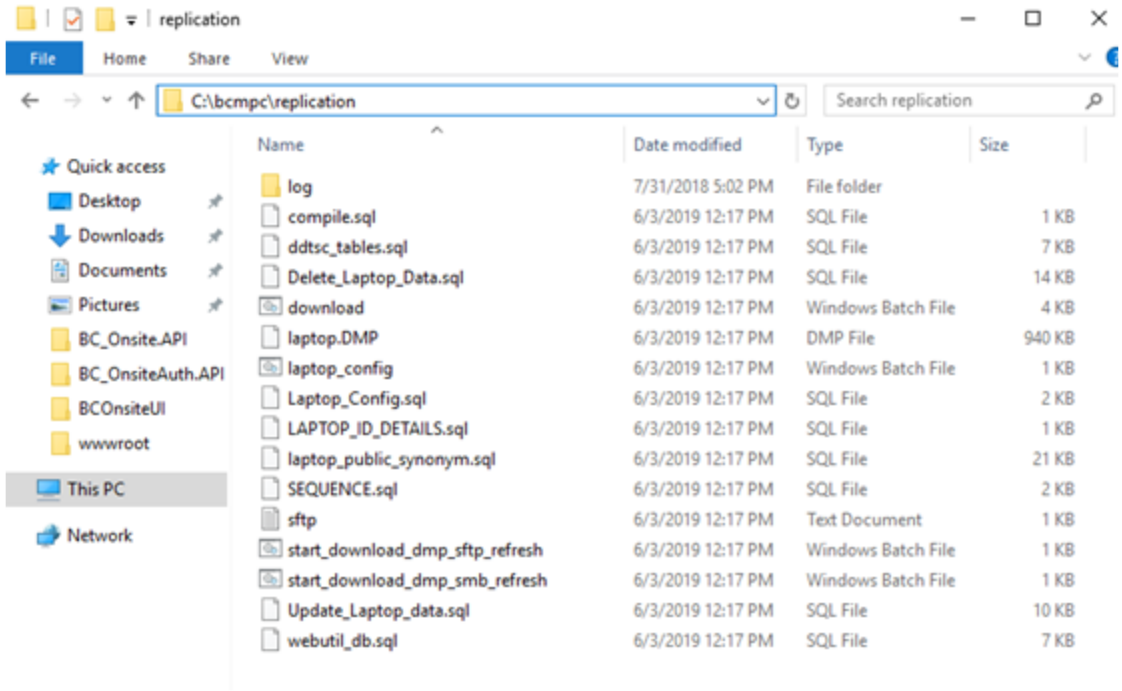
21. Make sure the following database services are running, and that they are configured for 'Automatic Startup Type'. You can verify them in Services.
    - OracleJobSchedulerXE
    - OracleOraDB21Home1TNSListener
    - OracleServiceXE



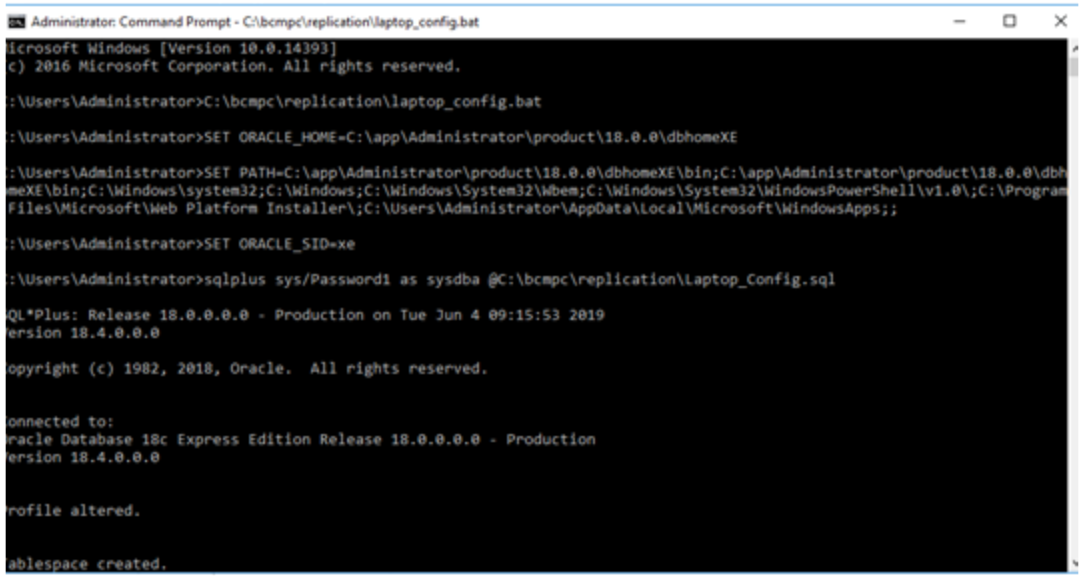
WellSky Corporation
Proprietary and confidential

5.6 One-time Initialization

1. In Windows Explorer, navigate to **C:\bcmpc\replication**.

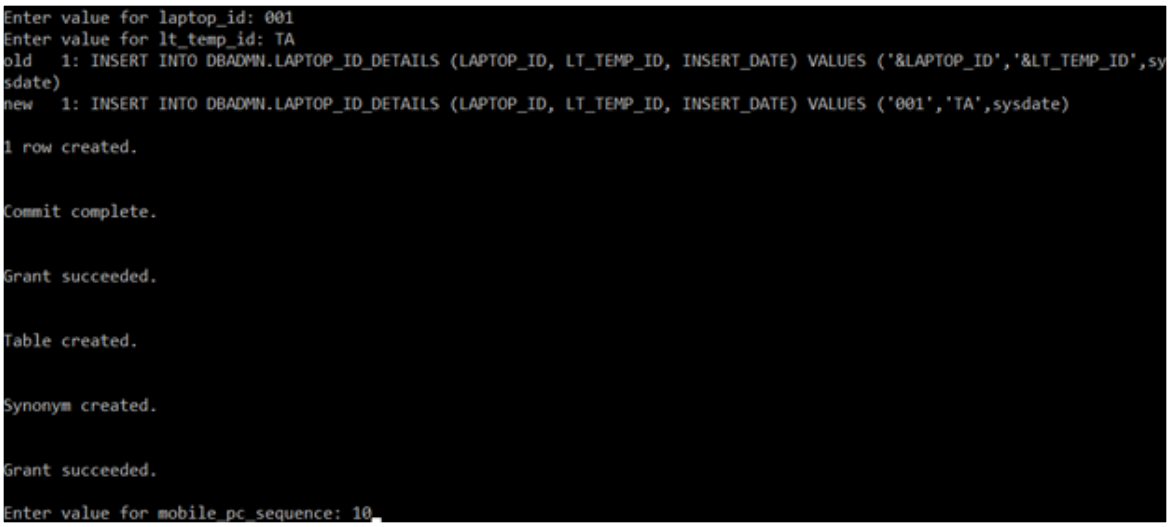


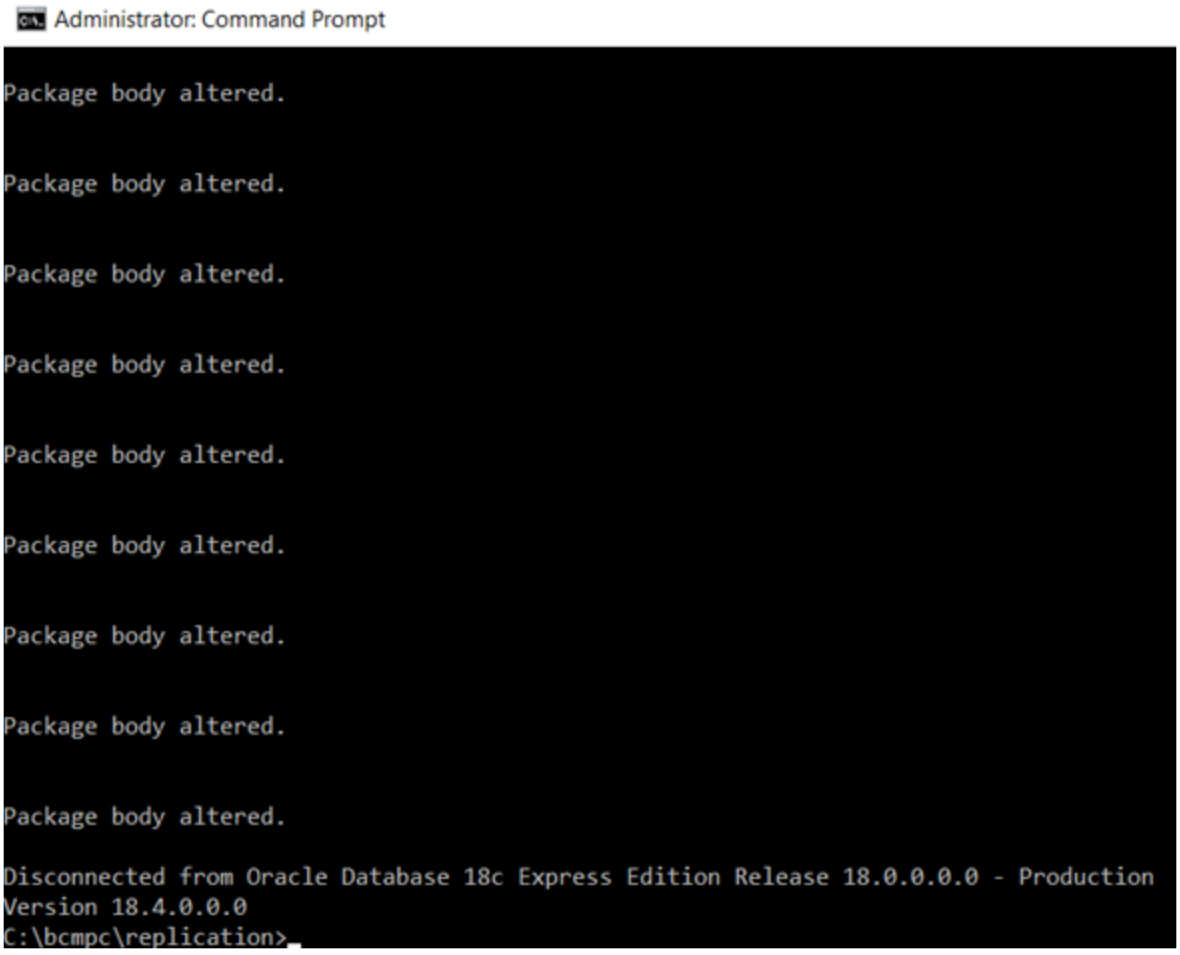
2. Open the command prompt, and run the **C:\bcmpc\replication\laptop_config.bat** script.



3. When prompted, enter the following information:
   a. **laptop_id** – Enter a 3-digit number unique to each Mobile PC server laptop. Example: 001
   b. **lt_temp_id** – Enter two letters that are unique to each Mobile PC server laptop. Example: TA

c. mobile_pc_sequence – Enter a 2-digit number that is unique to each Mobile PC server laptop. Example: 10





WellSky Corporation
Proprietary and confidential

d. Verify any ORA- errors in the following log files in C:\bcmpc\replication\log directory:

- Laptop_Config.log
- imp_laptop.log

4. Export the Mobile PC data tables in the main database server:

a. Copy the **C:\bcmpc\replication\mobile_dmp.tar** file to the Blood Centers database server's /home/bcorapp directory.

b. Untar the **mobile_dmp.tar** file using the following commands:

```
$ cd /home/bcorapp
$ tar -xvf mobile_dmp.tar
```

c. Update the **ORACLE_SID** value in the following two files, if needed:

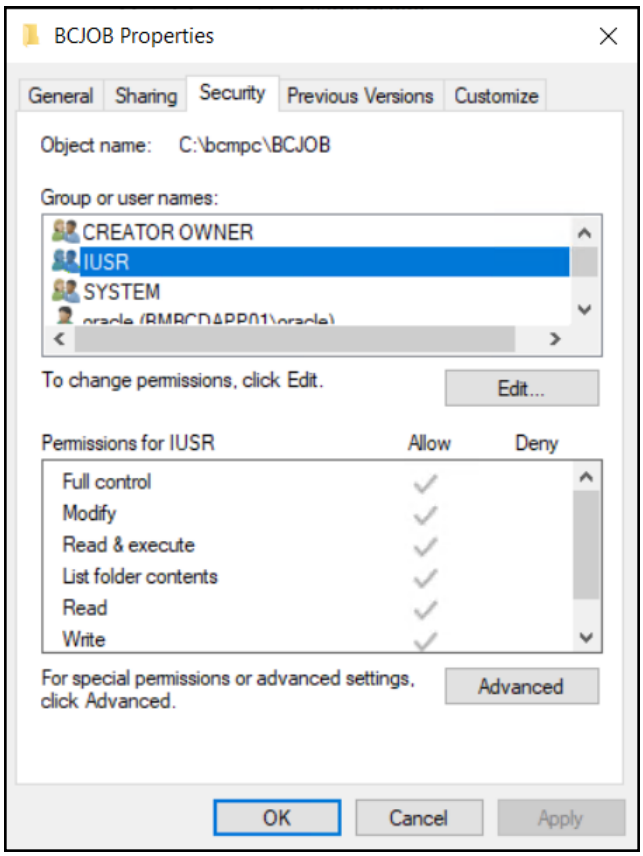
- **/home/bcorapp/mobile_dmp/export19c.sh**
- **/home/bcorapp/mobile_dmp/load_dump_files_in_database.sh**

d. If the mobile PC database is the Oracle 19c Edition, run the **/home/bcorapp/mobile_dmp/export19c.sh** to export the Mobile PC database tables.

> **NOTE –** You can set up these scripts in Linux CRON based on the Mobile PC database version.

5.7 Oracle Wallet Setup for Mobile PC Server

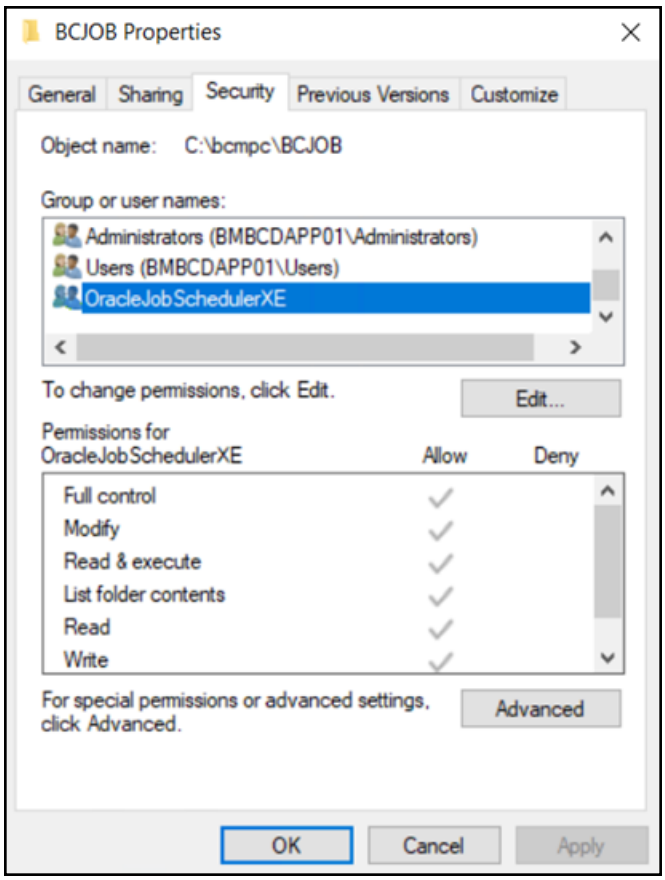
1. Right click on the **C:\bcmpc\BCJOB** directory.
    a. Click **Properties** > **EDIT** > **Add**.
    b. Enter **IUSR** then click **OK**.
    c. Select **IUSR**.
    d. In **Full Control**, select **Allow**, click **Apply**, then click **OK**.



NOTE – If Mobile PC Server does *not* have IUSR, recheck and configure this Wallet Setup section after installing IIS & dotnet software's (Section 5.8)

2. Right click on the **C:\bcmpc\BCJOB** directory.
    a. Click **Properties** > **EDIT** > **Add**.
    b. Enter **OracleJobSchedularXE** then click **OK**.
    c. Select **OracleJobSchedularXE**.
    d. In **Full Control**, select **Allow**, click **Apply**, then click **OK**.

WellSky Corporation
Proprietary and confidential



3. Verify same permissions are available for IUSR & OracleJobSchedularXE users under C:\bcmpc\BCJOB sub directories and files.
4. Backup the existing sqlnet.ora and tnsnames.ora files under the C:\apps\product\21c\dbhomeXE\network\admin directory to archive the directory.
5. Copy the sqlnet.ora and tnsnames.ora files from C:\bcmpc\Software directory C:\apps\product\21c\dbhomeXE\network\admin directory.
6. The sqlnet.ora file has a wallet location for XE database under C:\bcmpc\BCJOB\bcpstore.
7. The tnsnames.ora file has laptop and dbadmn tns aliases which are configured for wallet.
8. Verify these sqlnet.ora, tnanames.ora and appsettings.json files if any wallet changes needed.
    - This applies if mobile database is not an XE version and any custom database names).
9. For Mobile PC server database setup with default database user and passwords and wallet setups, if a database user's (laptop & dbadmn) passwords need to change, DBA must update wallet files as well using the below procedure and commands
    - `mkstore -wrl C:\bcmpc\BCJOB\bcpstore -modifyCredential LAPTOP_XE laptop idemerawlaptop`
        - In the above command the XE Database user laptop password is modified as `idemerawlaptop`.

- When prompted for the wallet password, enter **welcome1**.

```
C:\Users\oracle>mkstore -wrl C:\bcmpc\BCJOB\bcpstore -modifyCredential LAPTOP_XE laptop  idemerawlaptop
Oracle Secret Store Tool Release 21.0.0.0.0 - Production
Version 21.3.0.0.0
Copyright (c) 2004, 2021, Oracle and/or its affiliates. All rights reserved.

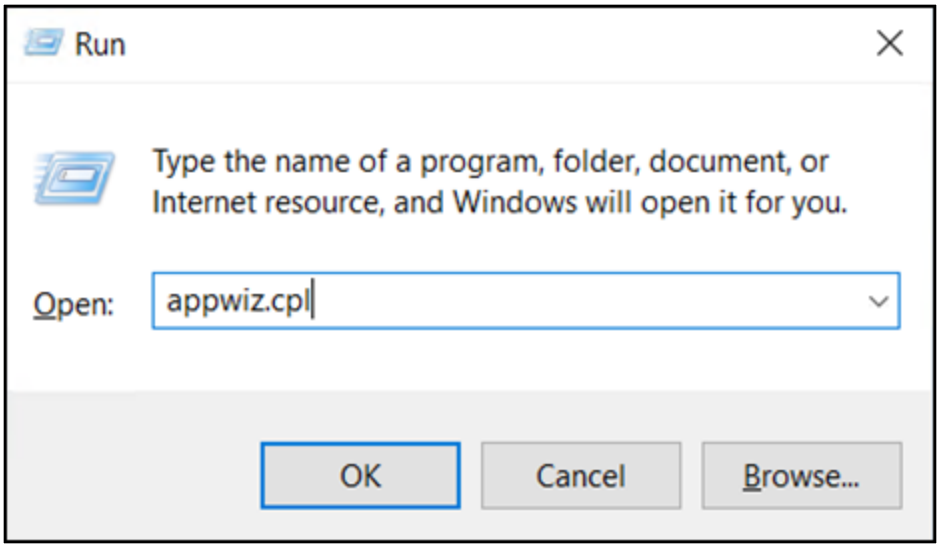
Enter wallet password:
```

10. Whenever oracle wallet files are updated with any changes like those above, update the **cwallet.sso** file from the **C:\bcmpc\BCJOB\bcpstore** directory to the **C:\bcmpc\BCJOB\onsite** directory.
    - This location is for Onsite application services.

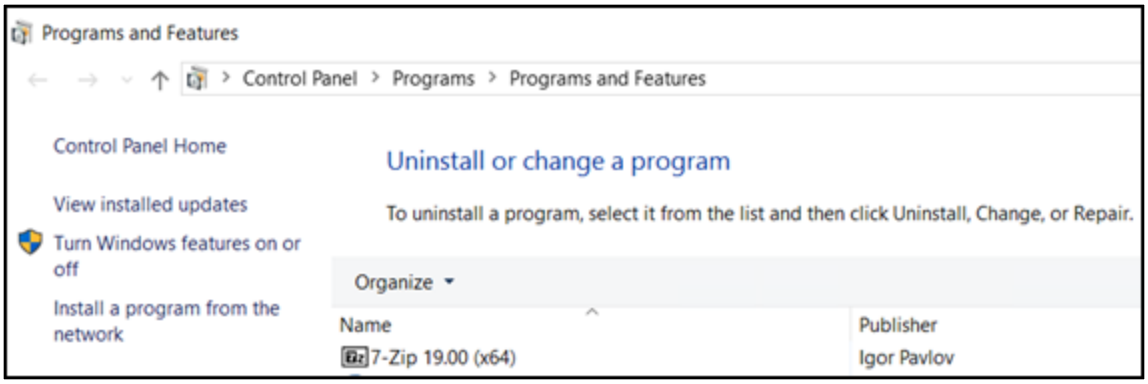
WellSky Corporation
Proprietary and confidential

5.8 IIS Installation

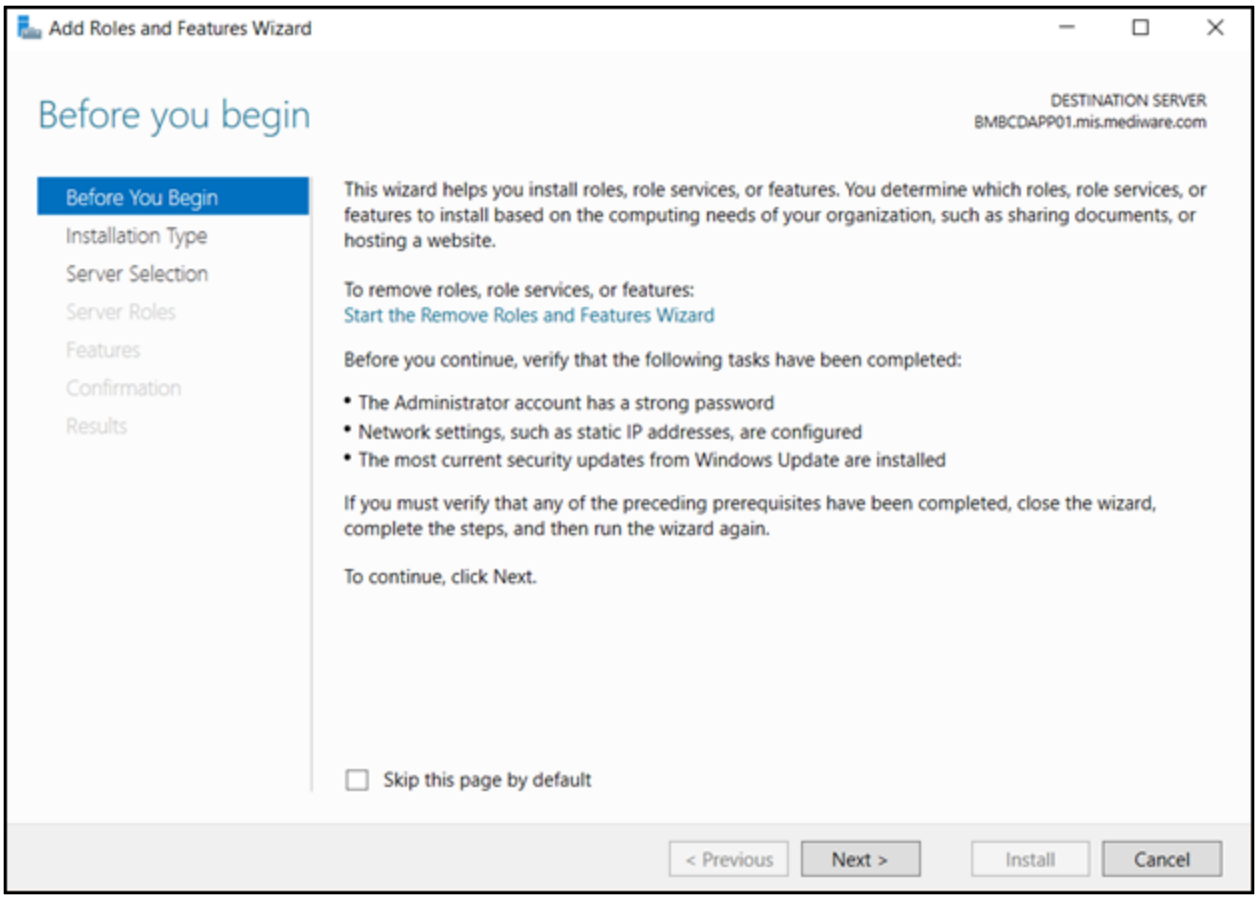
1. Press `Windows + R` to open the Run box.
2. Type `appwiz.cpl` in the Open field, and click **OK**.



This will open Programs and Features in the Control Panel:

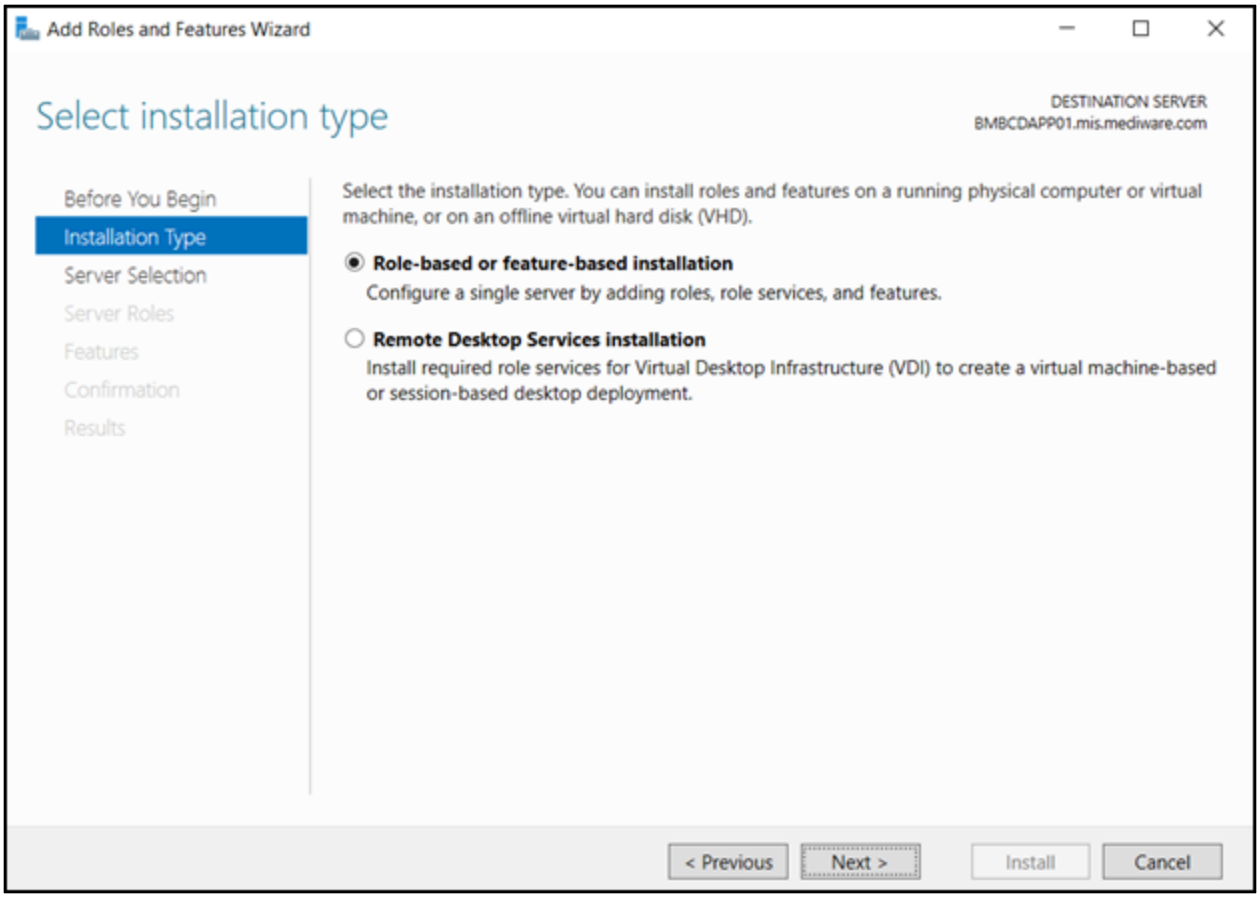


3. On the left side, click **Turn Windows features on or off**.

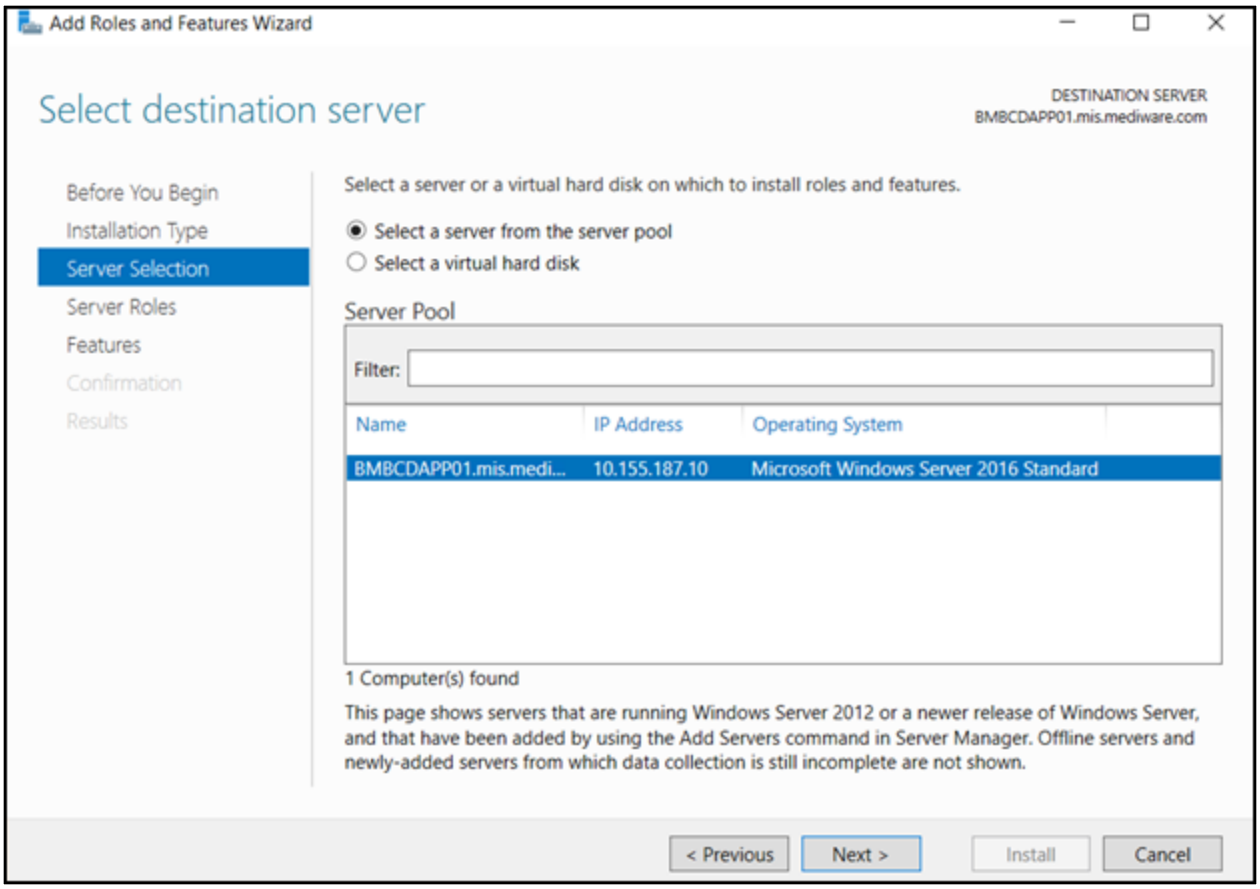


4. Click **Next**.

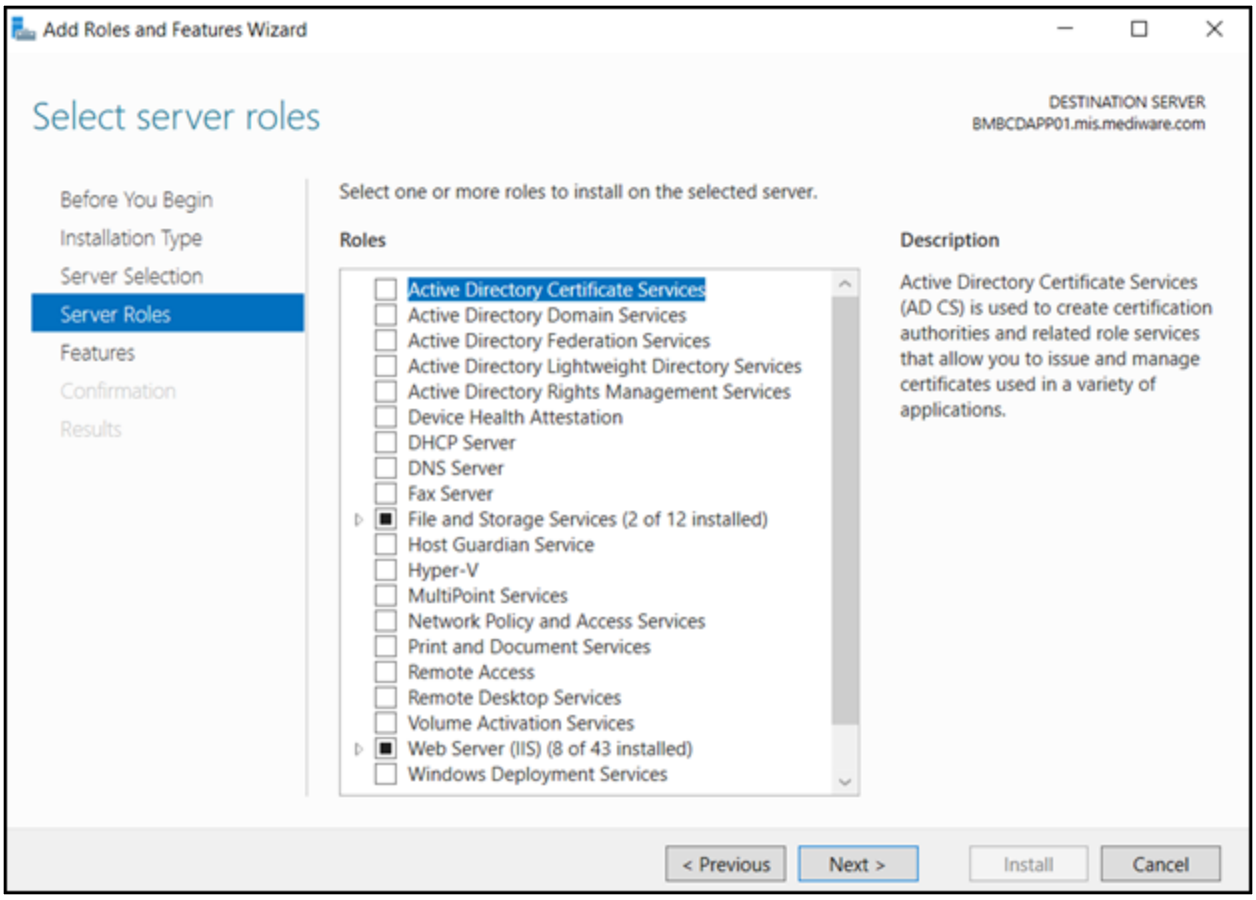
WellSky Corporation
Proprietary and confidential


Add Roles and Features Wizard
Select installation type
DESTINATION SERVER
BMBCDAPP01.mis.mediware.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select the installation type. You can install roles and features on a running physical computer or virtual machine, or on an offline virtual hard disk (VHD).
Role-based or feature-based installation
Configure a single server by adding roles, role services, and features.
Remote Desktop Services installation
Install required role services for Virtual Desktop Infrastructure (VDI) to create a virtual machine-based or session-based desktop deployment.
< Previous
Next >
Install
Cancel

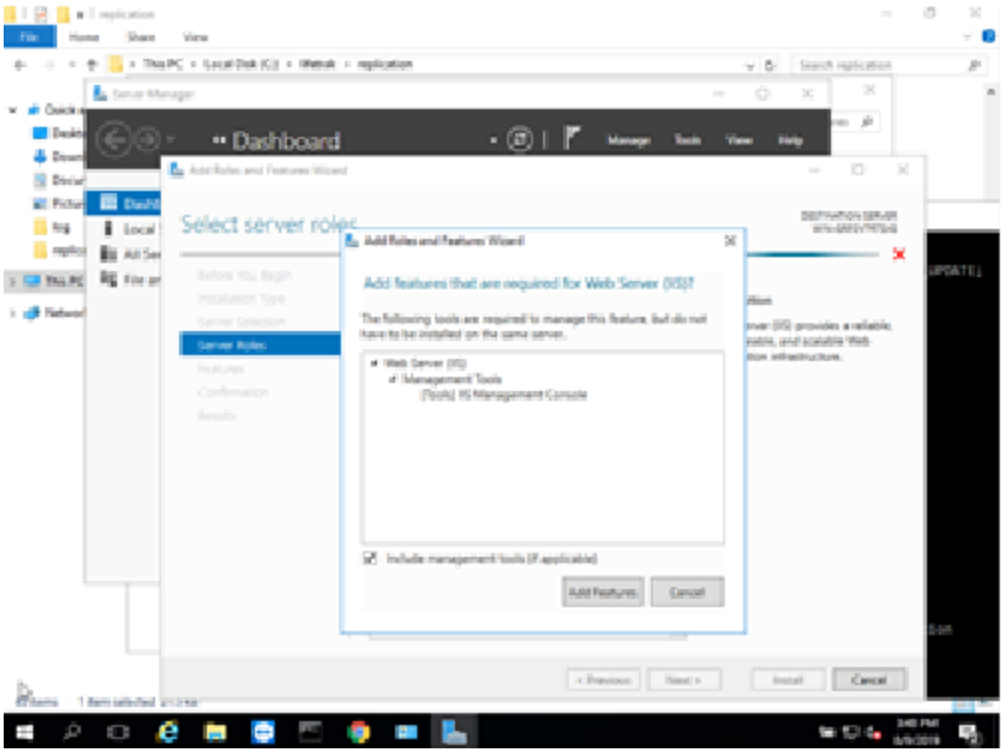
5. Click **Next**.


Add Roles and Features Wizard
Select destination server
DESTINATION SERVER
BMBCDAPP01.mis.mediware.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select a server or a virtual hard disk on which to install roles and features.
Select a server from the server pool
Select a virtual hard disk
Server Pool
Filter:
Name
IP Address
Operating System
BMBCDAPP01.mis.medi...
10.155.187.10
Microsoft Windows Server 2016 Standard
1 Computer(s) found
This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.
< Previous
Next >
Install
Cancel

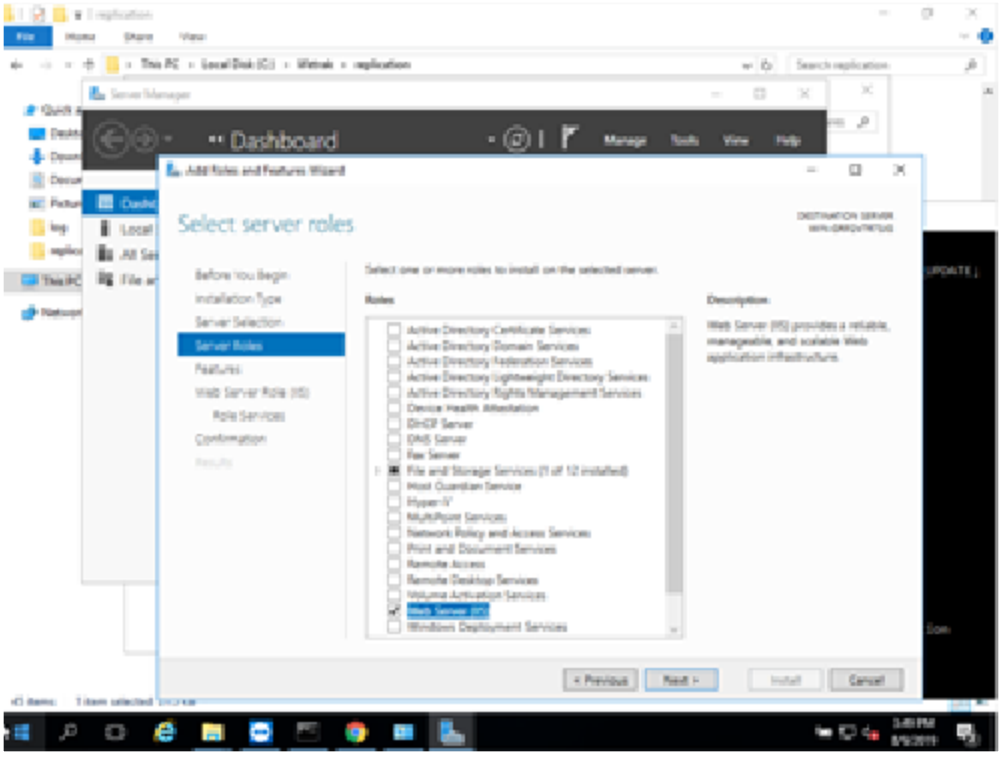
6. Click **Next**.



7. Select **Web Server (IIS)**, and click **Next**.

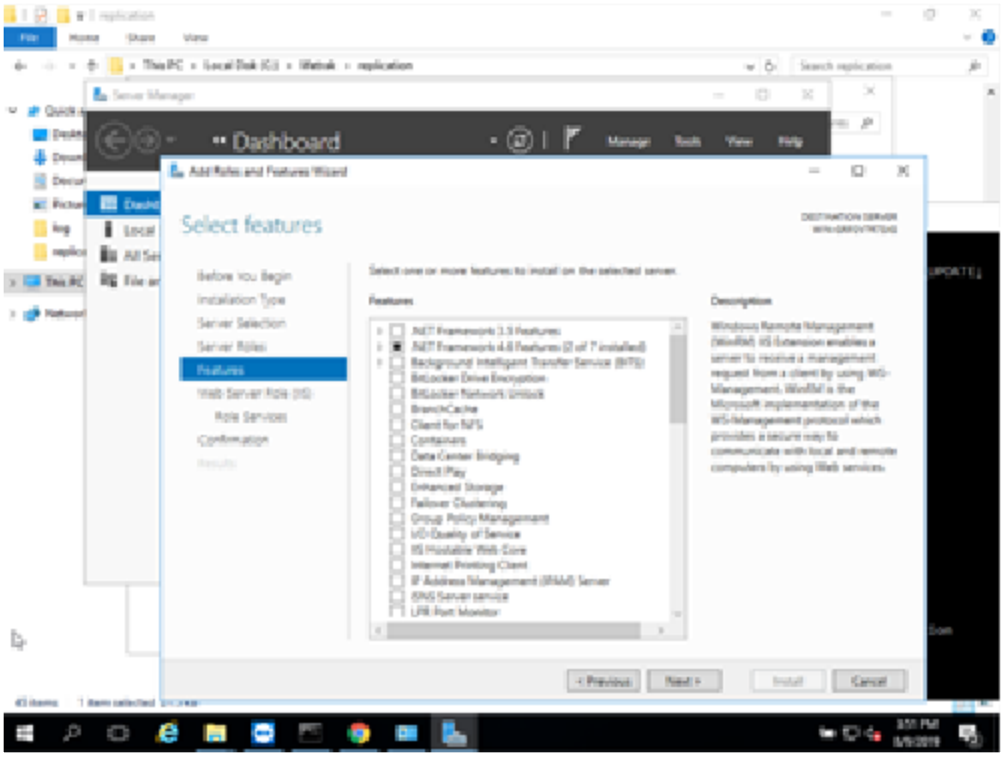


8. Click **Add Features**.

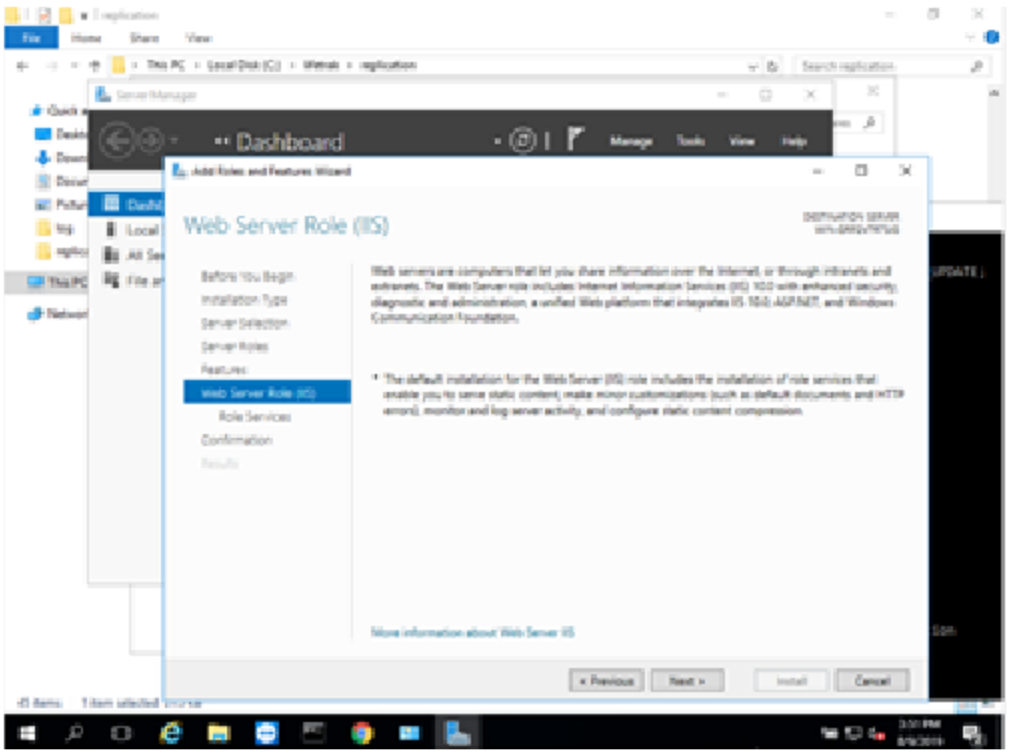


9. Click **Next**.

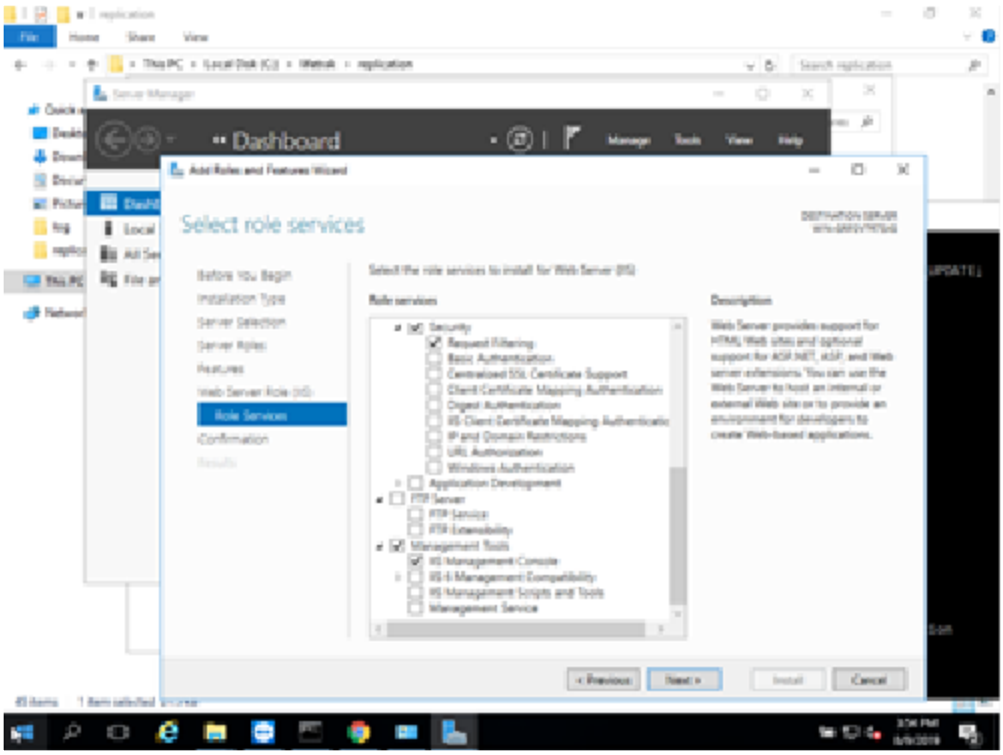
WellSky Corporation
Proprietary and confidential



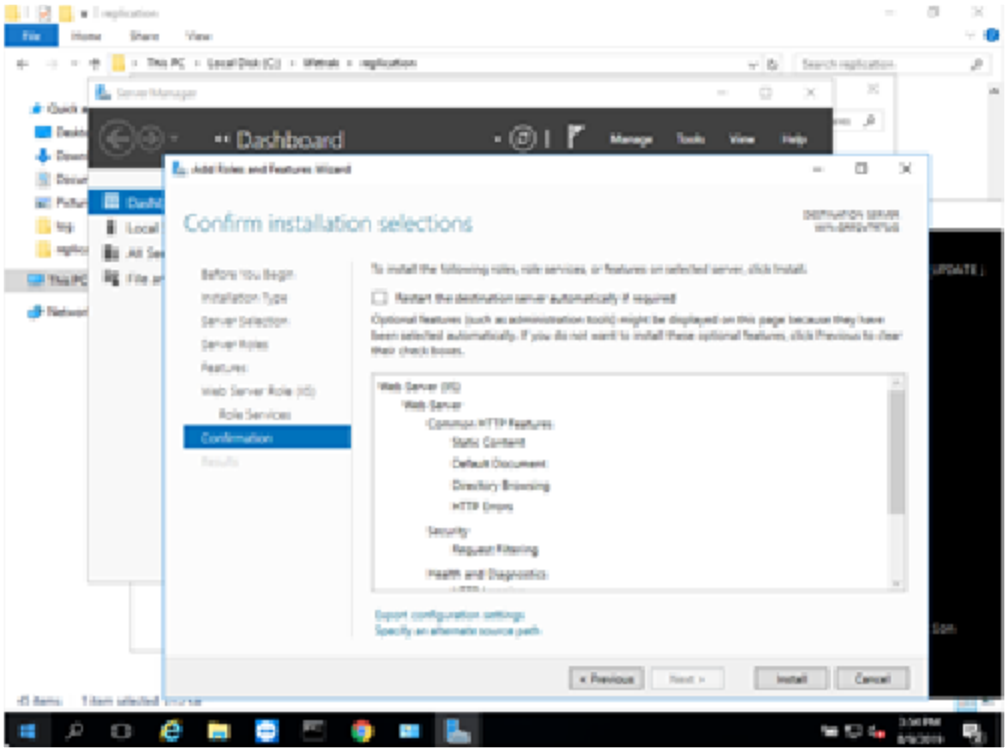
10. Click **Next**.



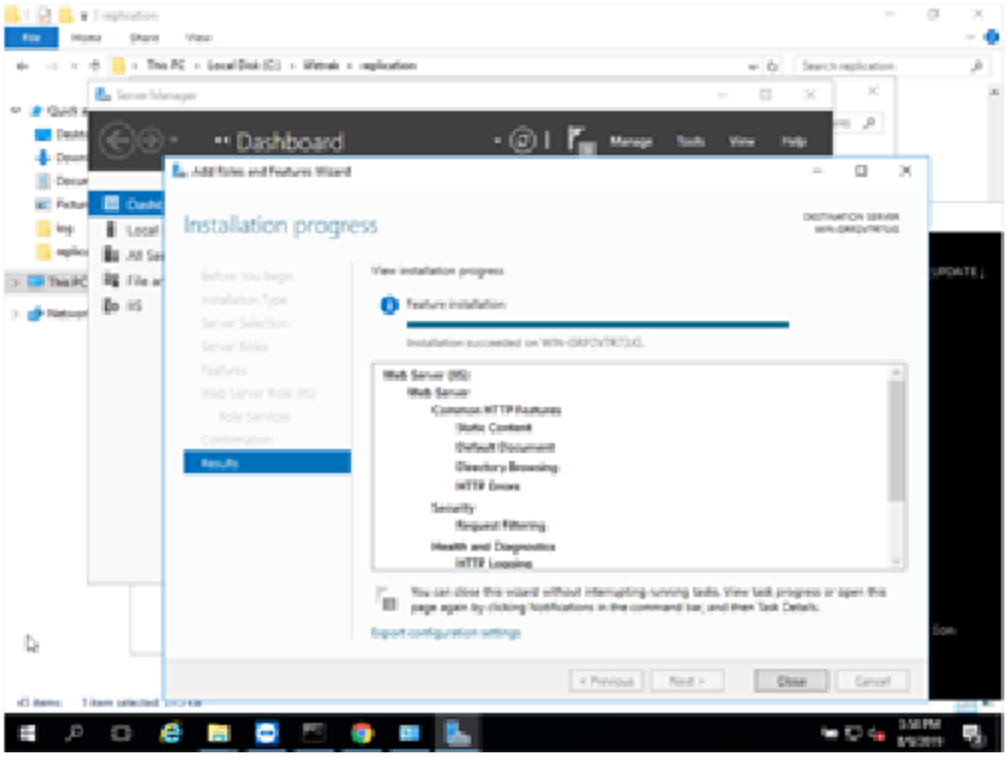
11. Click **Next**.



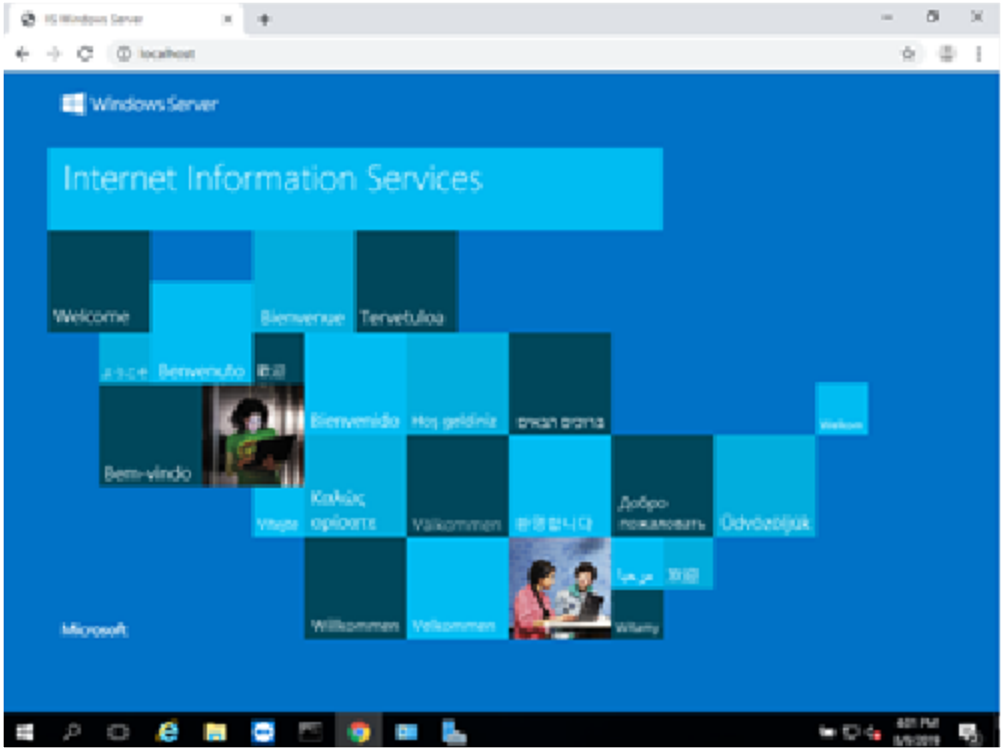
12. Click **Next**.



13. Click **Install**.

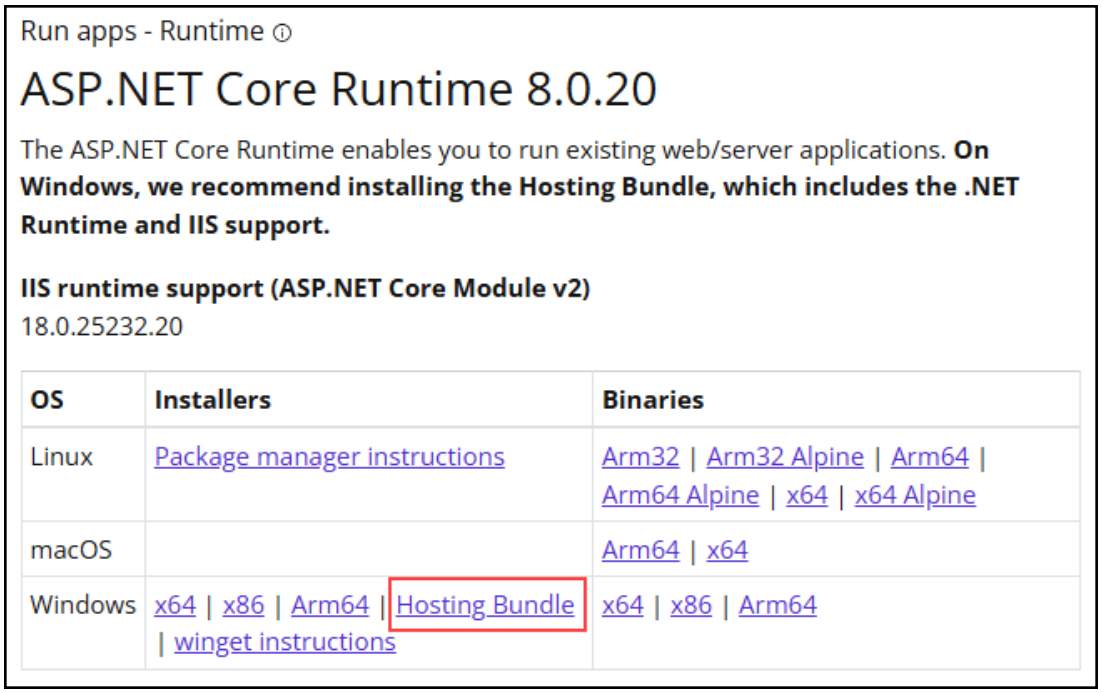


14. Click **Close**.
15. Open the Chrome browser, and enter the URL `http://localhost`. The IIS web page will display.



5.8.1 Microsoft .NET Core Installation

1. Go to `https://dotnet.microsoft.com/en-us/download/dotnet/8.0`, and download the .NET core software.
2. Click **Hosting Bundle**.


Run apps - Runtime

ASP.NET Core Runtime 8.0.20

The ASP.NET Core Runtime enables you to run existing web/server applications. **On Windows, we recommend installing the Hosting Bundle, which includes the .NET Runtime and IIS support.**

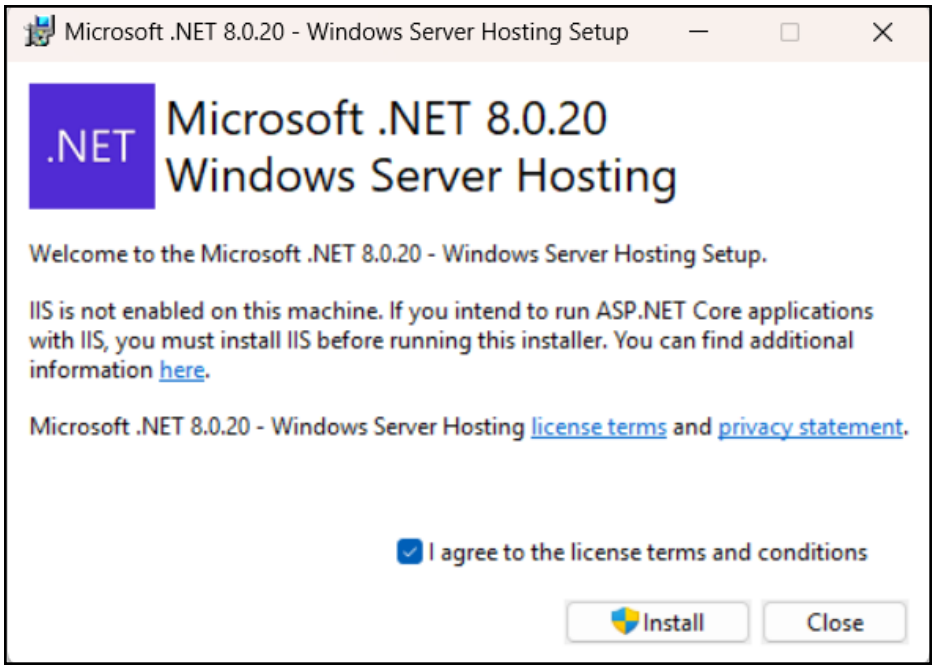
IIS runtime support (ASP.NET Core Module v2)
18.0.25232.20

OS	Installers	Binaries
Linux	Package manager instructions	Arm32 | Arm32 Alpine | Arm64 | Arm64 Alpine | x64 | x64 Alpine
macOS		Arm64 | x64
Windows	x64 | x86 | Arm64 | Hosting Bundle | winget instructions	x64 | x86 | Arm64

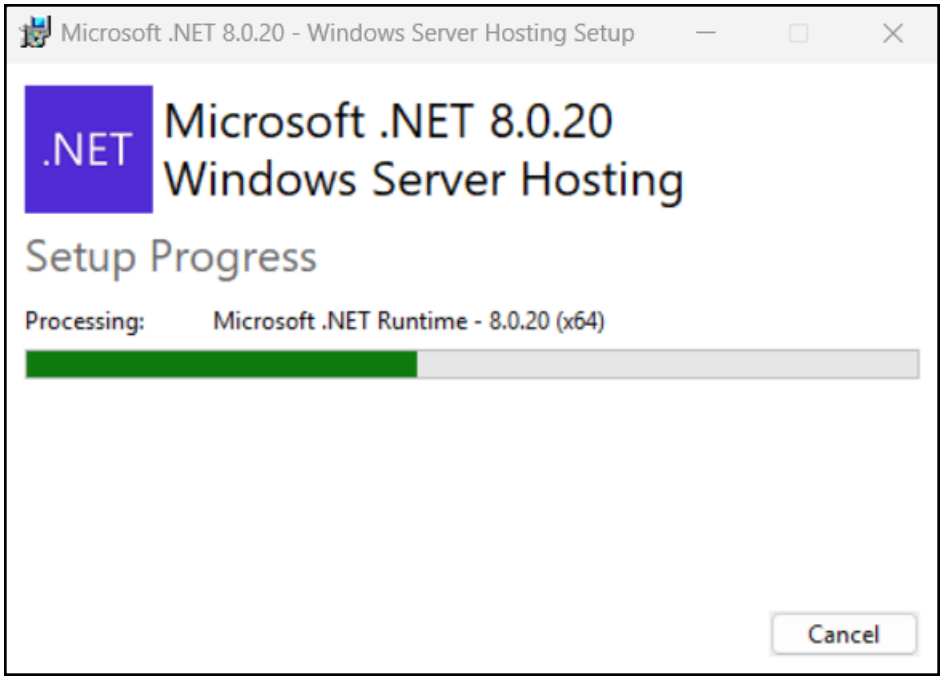
3. The **dotnet-hosting-8.0.20-win.exe** file will download.

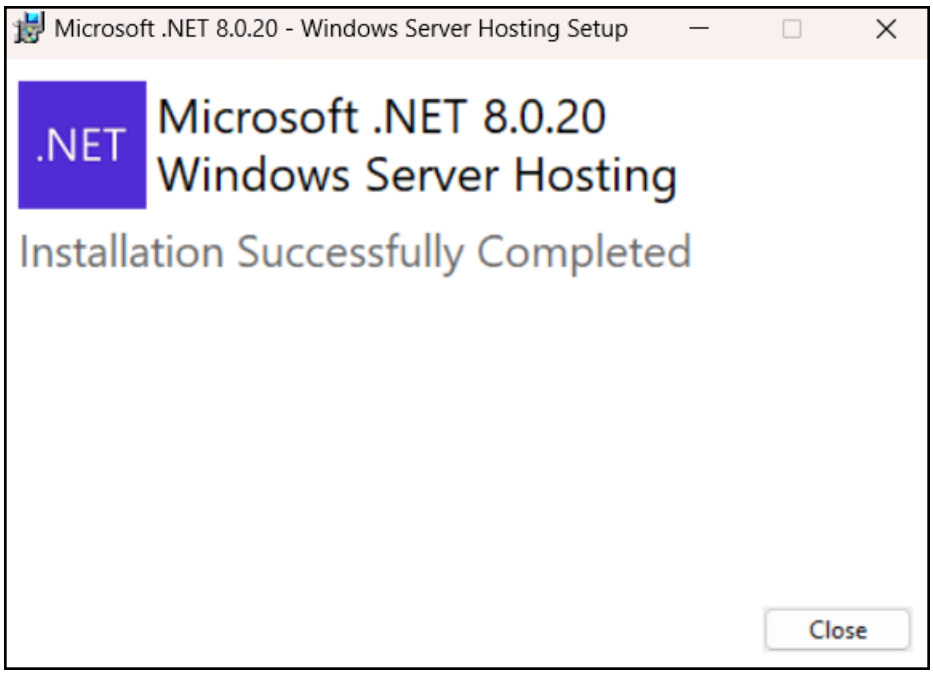
WellSky Corporation
Proprietary and confidential

4. Click **Run** then select the **I agree to the license terms and conditions** option.

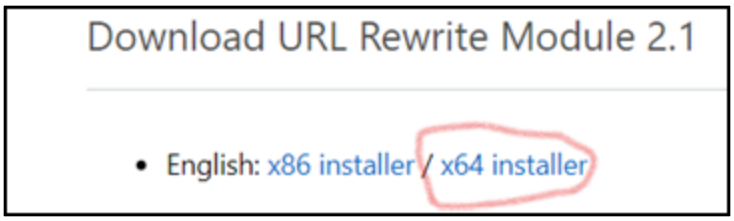


5. Click **Install**.



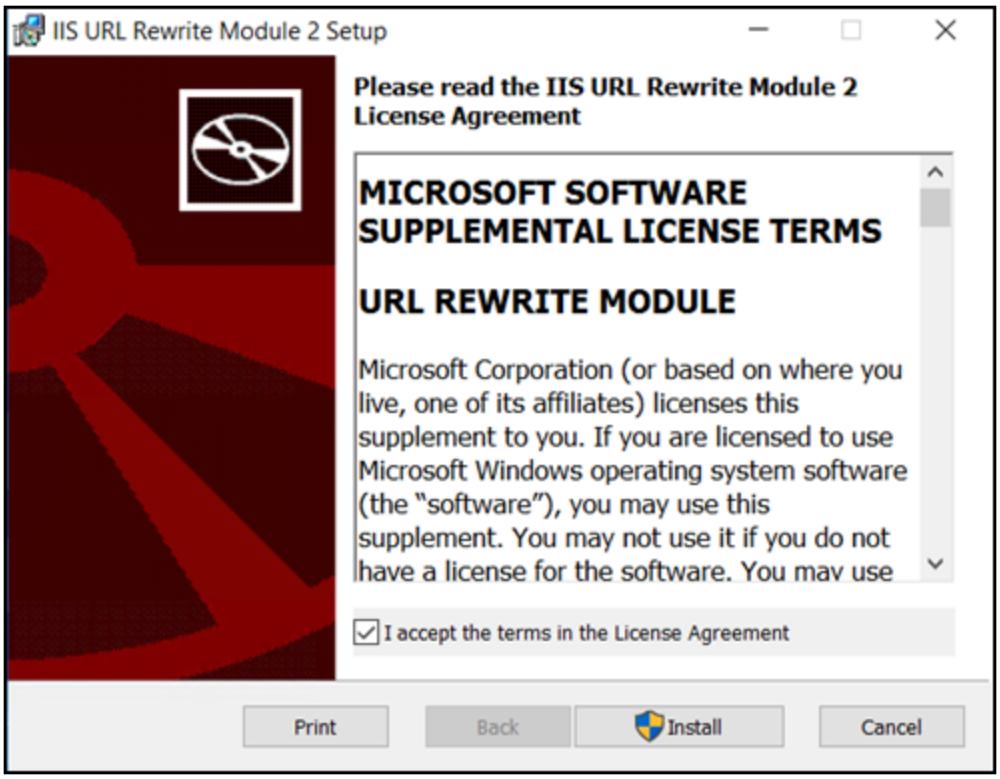


6. Click **Close**.
7. Go to `https://www.iis.net/downloads/microsoft/url-rewrite`, and click **Install this extension** to download the URLREWRITE extension.
8. Click **x64 installer** under Module 2.1.

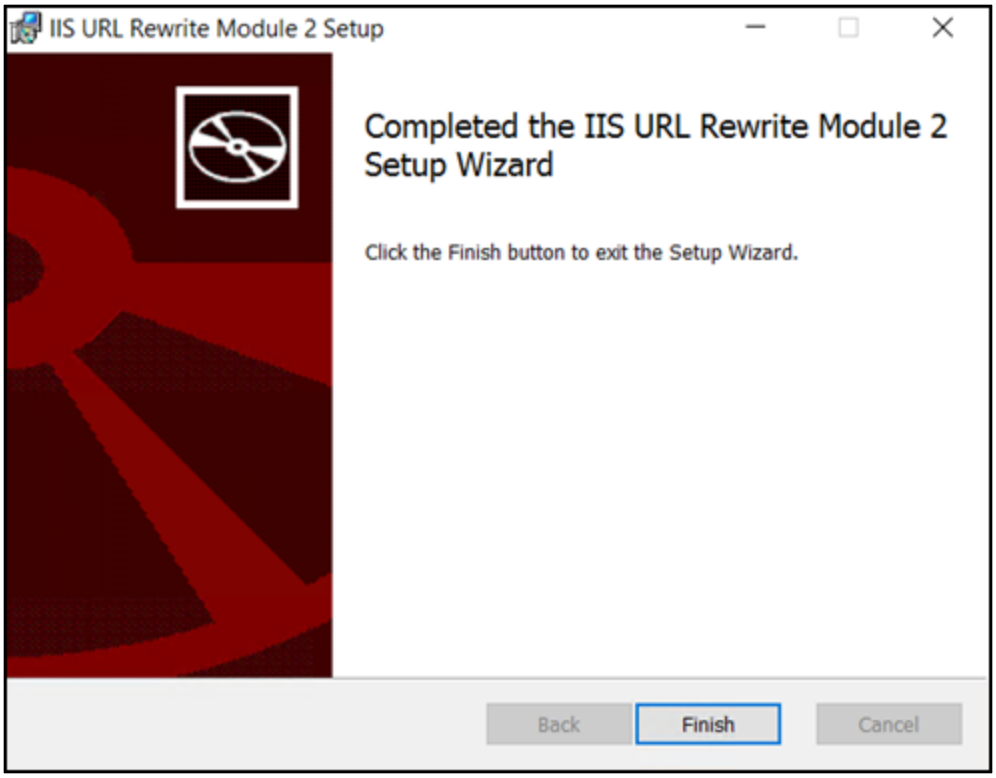


9. Double-click on the downloaded **rewrite_amd64_en-US.msi** file.

WellSky Corporation
Proprietary and confidential



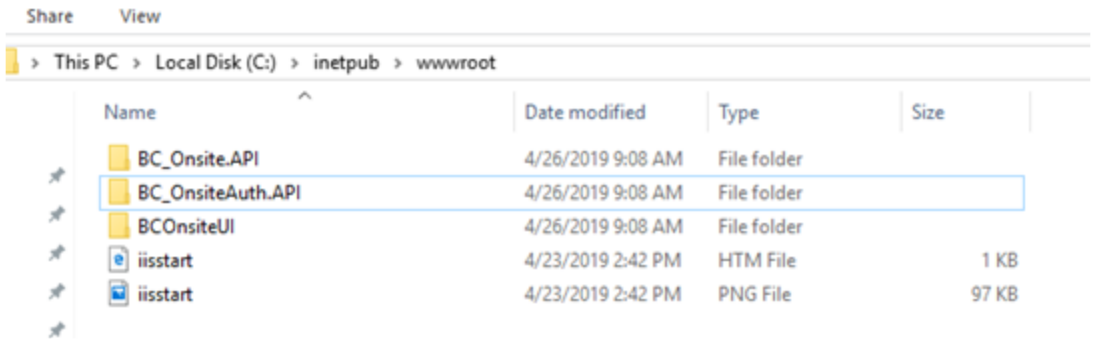
10. Click **I accept the terms in the License Agreement**, and click **Install**.



11. Click **Finish**.

5.8.2 Deploy the BC Onsite Application on the Application Server

1. Copy the following folders from the C:\bcmpc\BC_Deploy to the C:\inetpub\wwwroot directory.
   - BCOnsiteUI
   - BC_Onsite.API
   - BC_OnsiteAuth.API



2. Using a text editor, update the error-logging file path and DB connection in the following appsettings.json files.
   a. In Line 15, the default settings for path section and custom directory format are listed below. You can change them if you want the information in a different location.

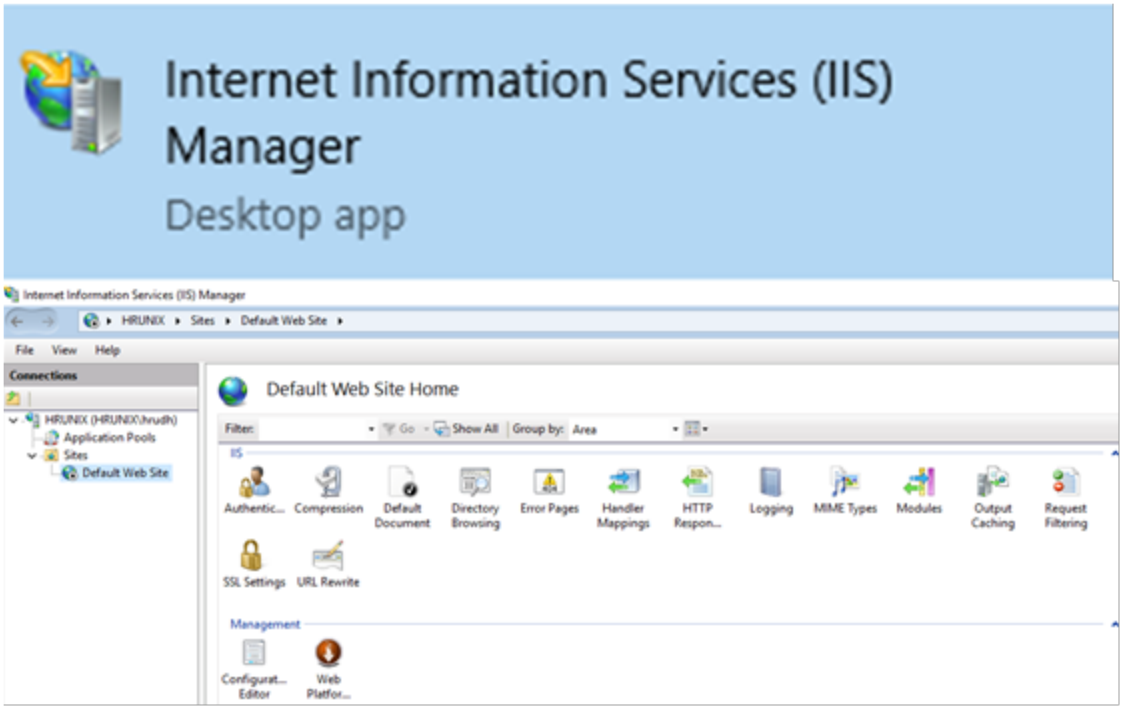
      BC_Onsite.api/appsettings.json
      "path": "C:\\OnsiteLogs\\Logs/OnsiteApp_.json"

      BC_OnsiteAuth.api/appsettings.json
      "path":"C:\\OnsiteLogs\\Logs/OnsiteAuthApp_.json"

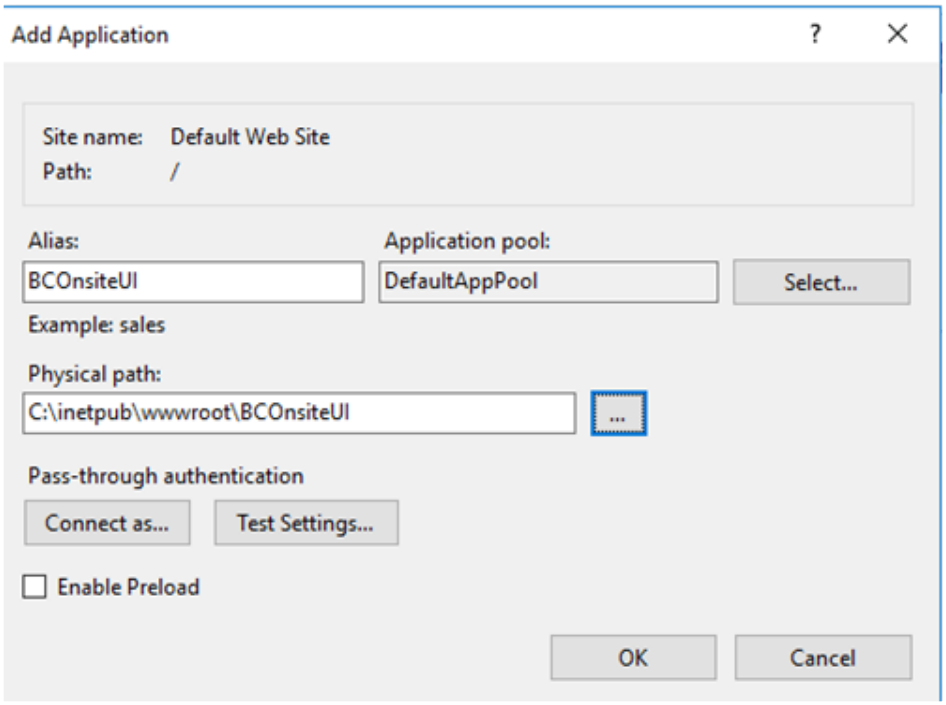
      The Logging file will generate in the Logs directory with json extension format (for example, OnsiteApp_**20210601**.json). In this example, the appended number is year, month, and date. Every day it will generate a new file with the appended date format.

   b. In Line 18, "retainedFileCountLimit": 7
      As per the retainedFileCountLimit setting in the appsettings.json configuration, only the latest 7 log files are retained on the server at any time.
   c. In Line 29, update the DB parameters in Default connection.
3. Type `IIS` in the search box to locate the Internet Information Services (IIS) Manager. When it is found, click on it.

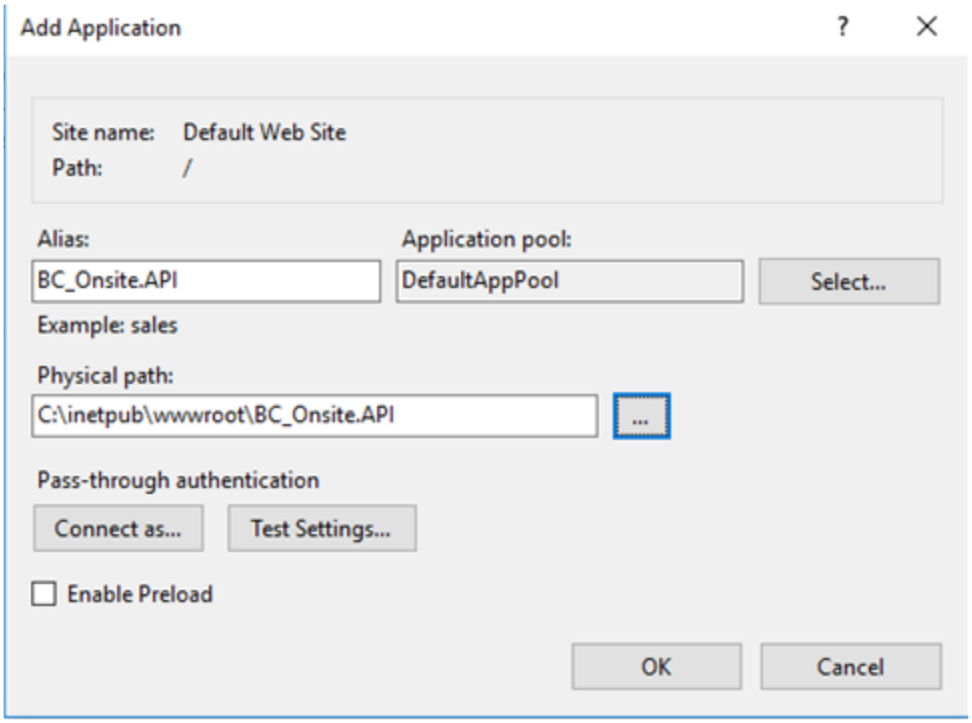
WellSky Corporation
Proprietary and confidential



4. Right-click on **Default Web Site**, and select **Add Application** from the menu.
5. On the Add Application form, enter the following:
   - Alias: `BCOnsiteUI`
   - Physical path: `C:\inetpub\wwwroot\BCOnsiteUI`



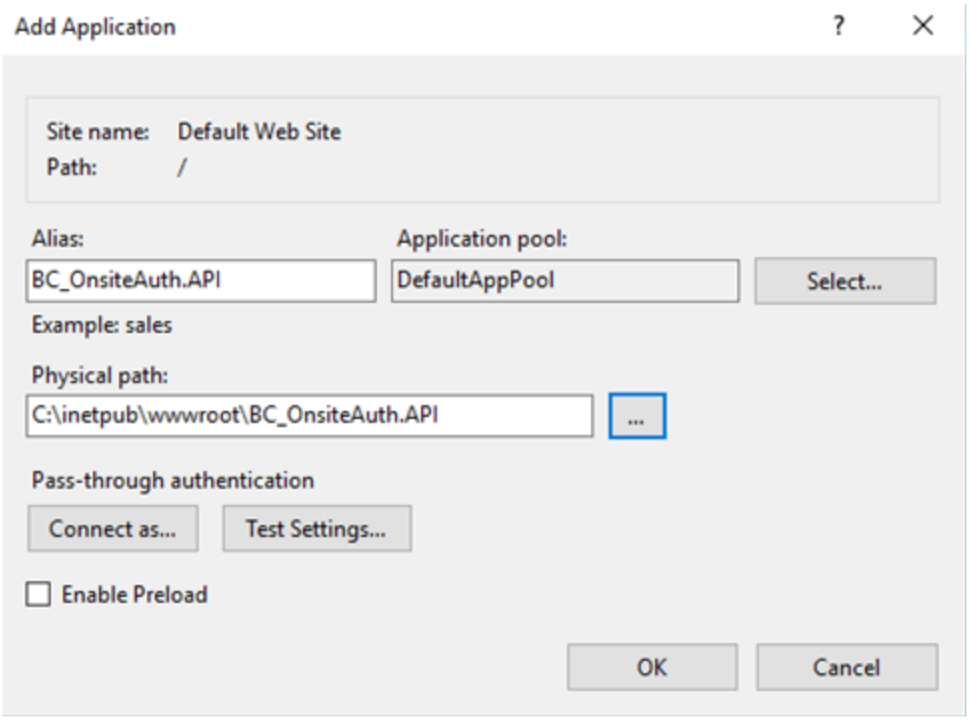
6. Click **OK**.
7. Right-click on **Default Web Site**, and select **Add Application** from the menu.
8. On the Add Application form, enter the following:
    - Alias: `BC_Onsite.API`
    - Physical path: `C:\inetpub\wwwroot\BC_Onsite.API`



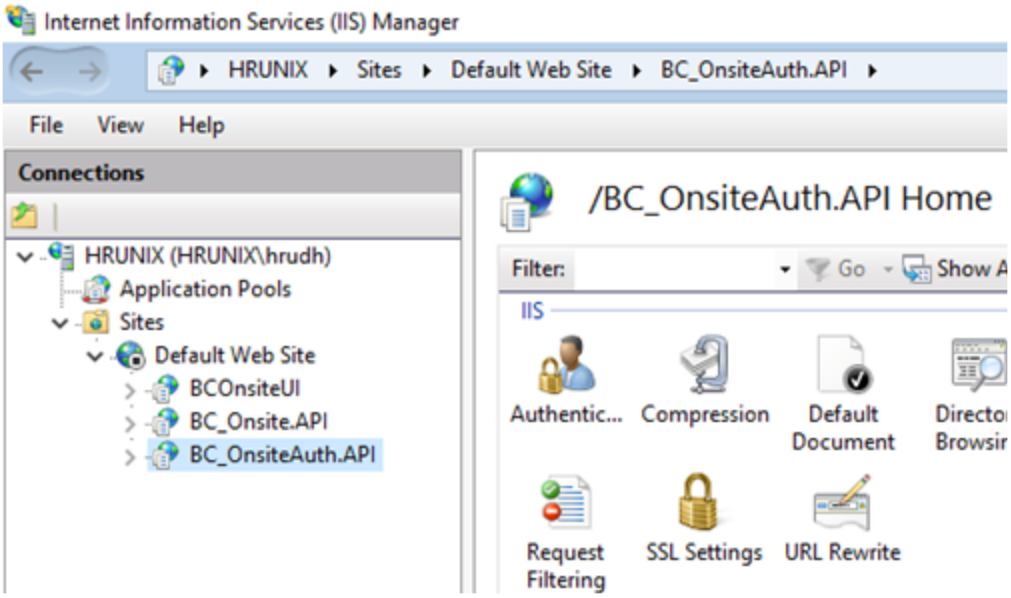
9. Click **OK**.
10. Right-click on **Default Web Site**, and select **Add Application** from the menu.

WellSky Corporation
Proprietary and confidential

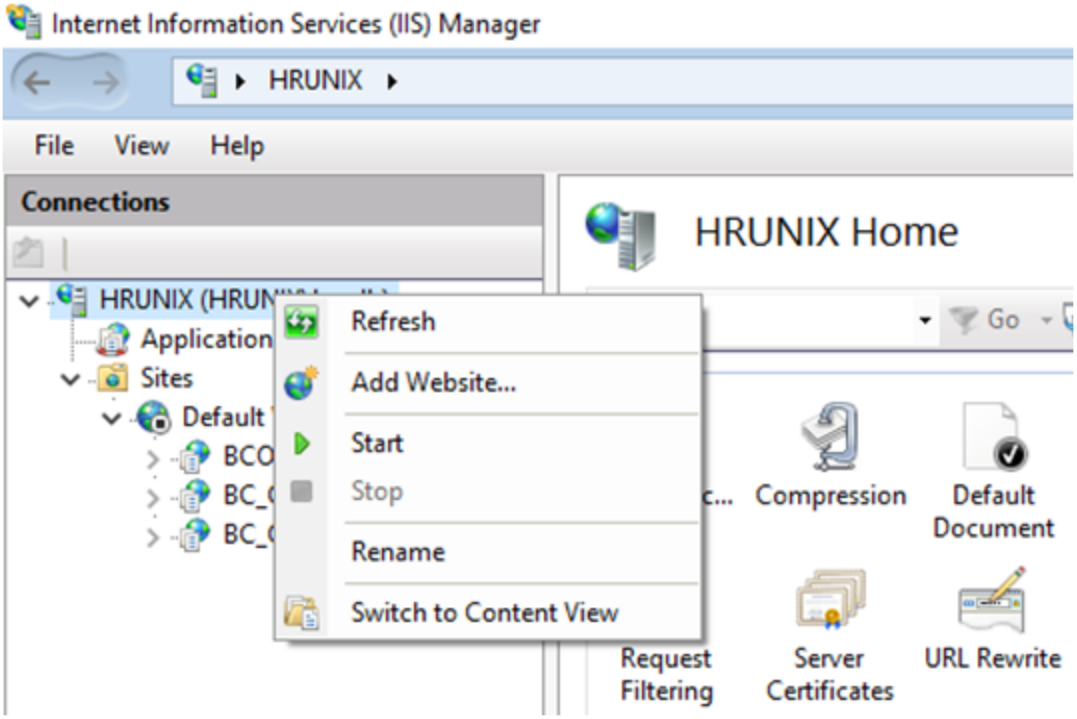
11. On the Add Application form, enter the following:
    - Alias: `BC_OnsiteAuth.API`
    - Physical path: `C:\inetpub\wwwroot\BC_OnsiteAuth.API`



12. Click **OK**. The BC Onsite applications will be listed in the IIS Manager.

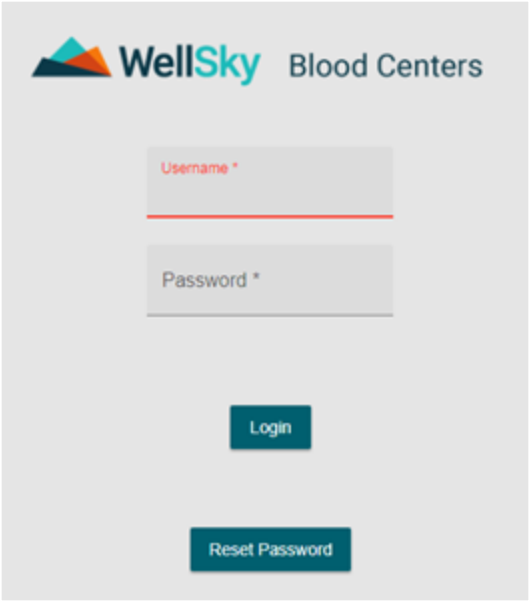


13. Right-click on the server (e.g. HRUNIX), and click **Start** to start the IIS Service.

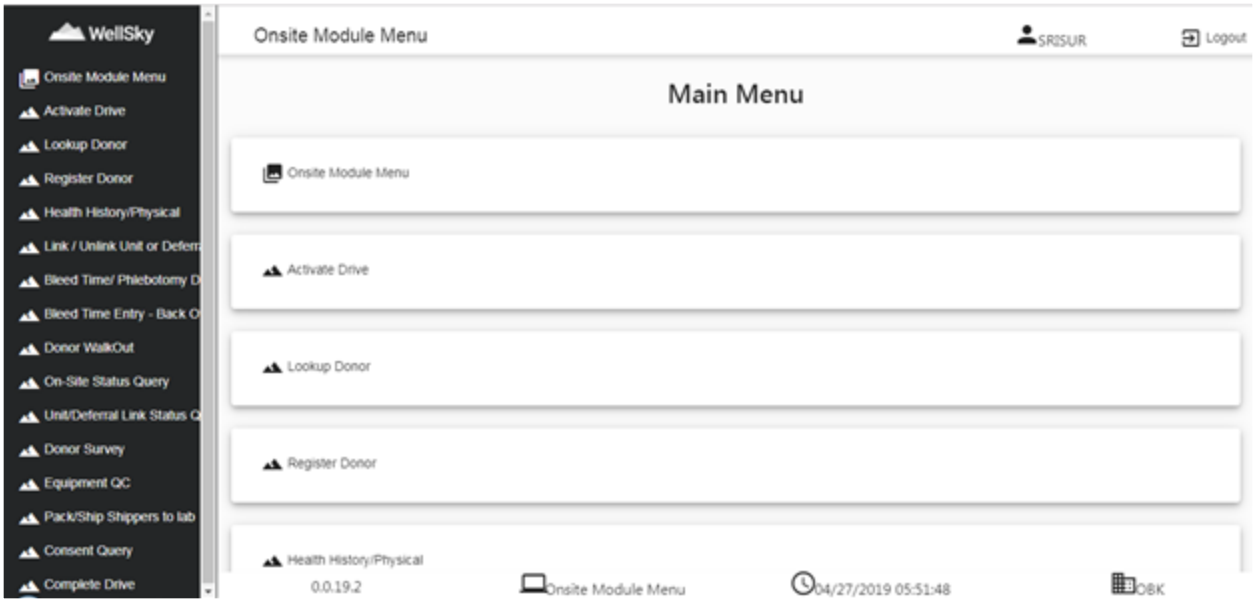


14. Launch the BC Onsite Application using the following URL: `http://HRUNIX/BCOnsiteUI/index.html`

> **NOTE –** Replace 'HRUNIX' with the actual IIS application server name.



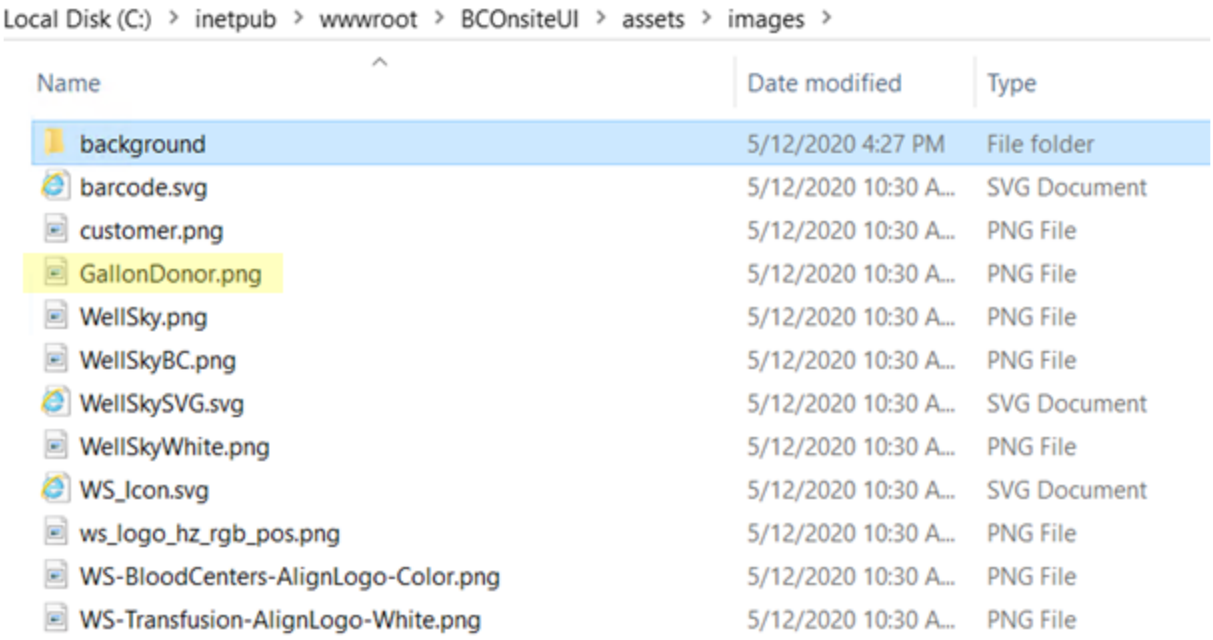
15. Enter your username and password in the respective fields, and click **Login**.



Gallon Donor Image File Location

The gallon donor image for the .NET application does not use the system parameter GALLON_IMG. The .NET application has the following requirements:

- The file must be named GallonDonor.
- The image type can be a .png, .jpg, or .svg image.
- The file must be placed in the following location: C:\inetpub\wwwroot\BCOnsiteUI\assets\images.



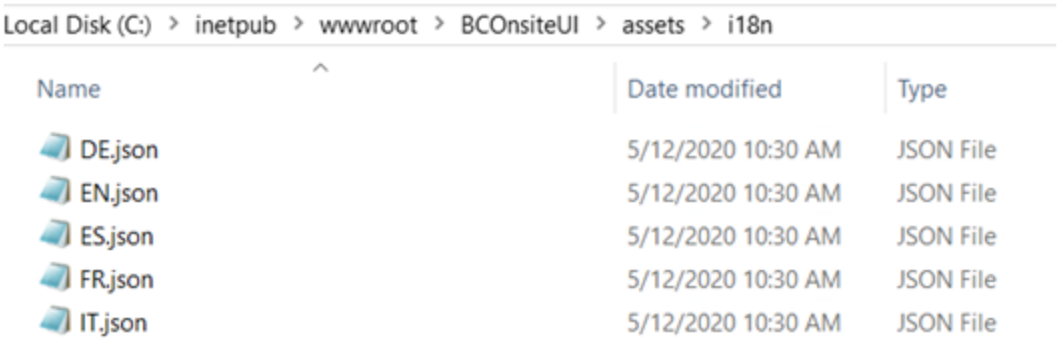
NOTE – The size of the file has not been an issue, but after loading the file you should view it and make sure it fits within the location in the Donor Header.

Multi-Language Files

The multi-language file is used for the health history questions, donor survey questions and answers, Donor Consents, FastTrak Login configuration, and the Donor Response initial displays information.

The files require the following:

- File has to be JSON.
- JSON file has to be labeled based on the "List of ISO 639-1 codes" (https://en.wikipedia.org/wiki/List_of_ISO_639-1_codes) for languages.
  - ES = Spanish
  - FR = French
  - IT = Italian
  - DE = German
- The JSON files MUST BE defined in upper case.
- The files must follow a specific format. Example files can be supplied by your Clinical Consultant, or by contacting Technical Support. Below is an example of a JSON file for Spanish.
- A file must be created for each language that you wish to support.
  - EN.json is an empty json file. Define the file, but information does not need to be present in the file.
- The files must be placed in the following location: C:\inetpub\wwwroot\BCOnsiteUI\assets\i18n.



WellSky Corporation
Proprietary and confidential

Example of JSON File for Spanish:

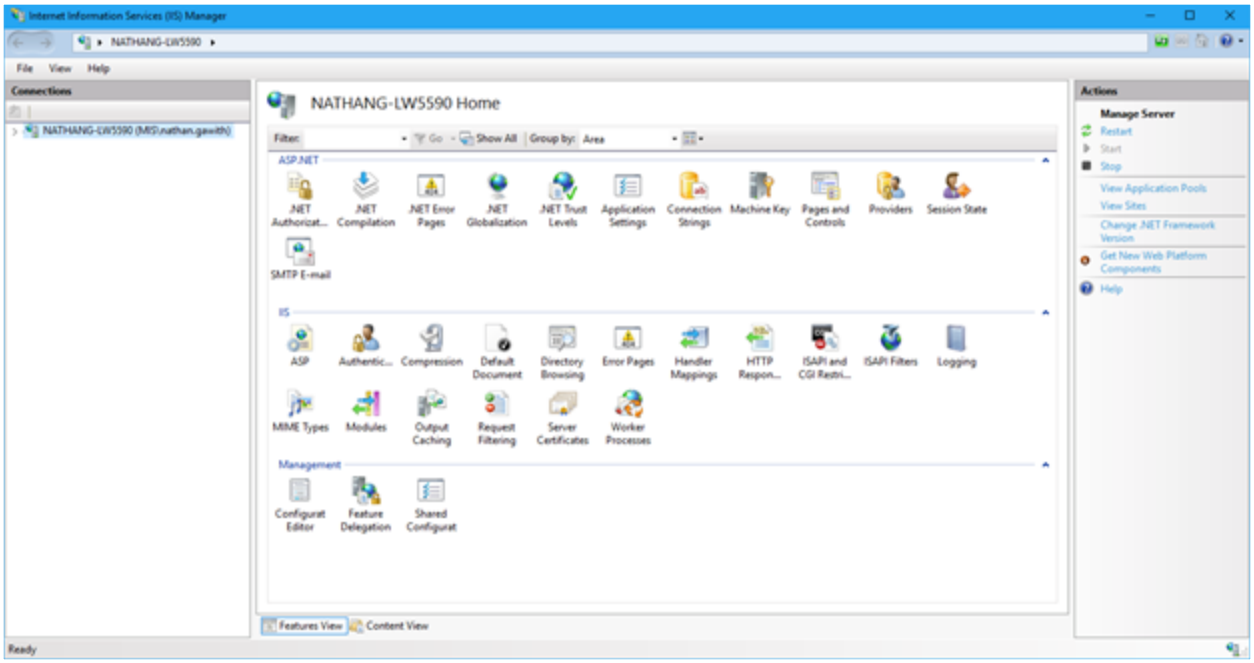
ES - Notepad

File Edit Format View Help

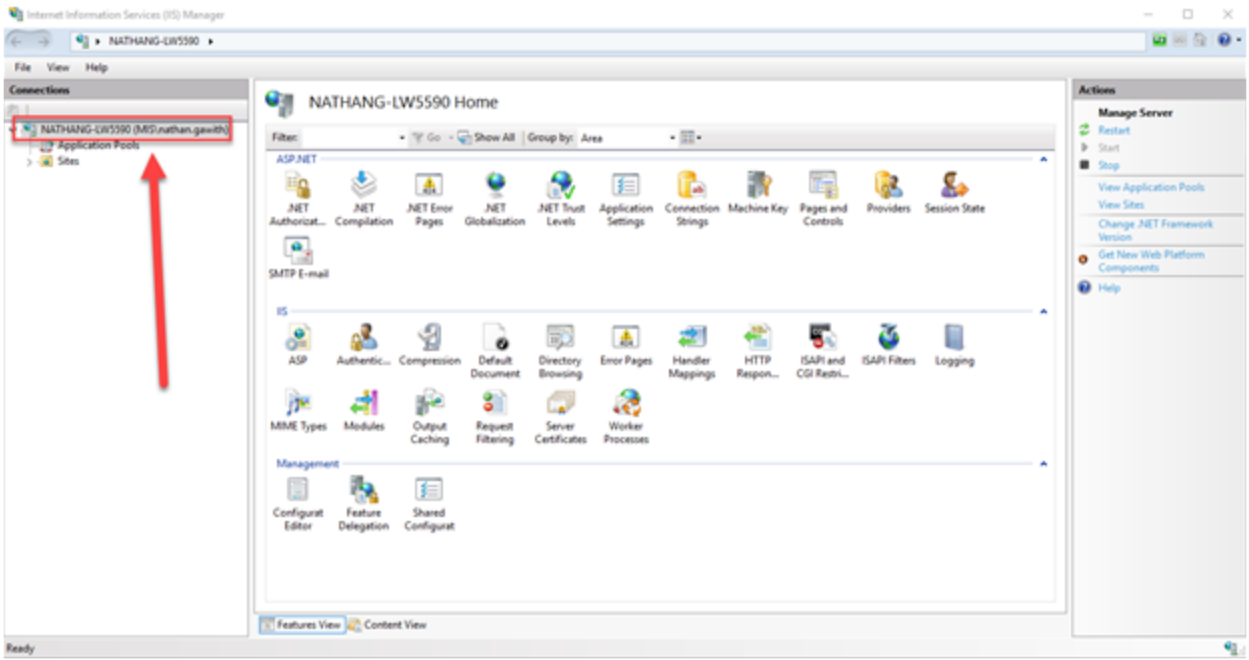

```
{
  "46-59": "46 a 59",
  "BackToQuestions": "Volver a las preguntas",
  "CANCEL": "CANCELAR",
  "CLEAR": "LIMPIAR",
  "CONTINUE": "CONTINUAR",
  "Chocolate Chip": "Chispas de chocolate",
  "DECLINE": "DECLINAR",
  "DECLINED": "DECLINADO",
  "DS0001": {
    "FIRST": "Donante por primera vez",
    "YES": "Sí",
    "text": "¿Eres un donante recurrente?"
  },
  "DS0002": {
    "NO": "No",
    "WARM": "Prefiero manta calentada",
    "YES": "Sí",
    "text": "¿Usas una manta cuando donas?"
  },
  "DS0003": {
    "1": "1-5 veces",
    "2": "6-10 veces",
    "FIRST": "Donante por primera vez",
    "text": "¿Cuántas veces dona en un año?"
  },
  "DS0004": {
    "NO": "No",
    "YES": "Sí",
    "text": "¿Eres un donante de féresis?"
  },
  "DeclineConfirmMessage": "¿Está seguro de que quiere rechazar este consentimiento?",
  "DonorSignature": "Firma del Donante",
  "FINISH": "FINALIZAR",
  "Finish": "Terminar",
  "FinishedWithDRMessage": "Gracias por sus respuestas. Haga clic en FINALIZAR y devuelva el dispositivo al Personal.",
  "GVT0001": {
    "60": "60+",
    "1721": "17-21",
    "2235": "22-35",
    "3645": "36-45",
    "4659": "46-59",
    "text": "¿Qué grupo de edad tienes?"
  },
  "HQ0010": "¿Es usted alérgico al Iodo y/o Látex?",
  "HQ0010-A": "A Desde su última donación, ha adquirido una alergia al yodo y / o ¿látex? - ABREVIADO",
  "HQ0010-D": "¿Alguna vez ha tenido una reacción alérgica o irritación de la piel al látex o productos con látex como los vendajes? Si es así, responda SÍ y el investigador lo discutirá con usted",
  "HQ0020": "¿Te sientes bien y saludable hoy?",
  "HQ0020-D": "No tenga ningún problema, como dolores de cabeza, dolor de espalda o estrés por donar.",
  "HQ0030": "En las últimas 8 semanas, ¿ha dado sangre, plasma o plaquetas?",
  "HQ0040": "¿Alguna vez ha sido rechazado como donante de sangre o le han dicho que no done sangre?",
  "HQ0071": "En las últimas 4 semanas, ¿ha tomado alguna píldora o medicamento?",
  "DCONPP_v2": {
    "content": "Entiendo que la donación de sangre para una féresis plaquetaria es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento de féresis plaquetaria"
  },
  "DCONAUTO_v1": {
    "content": "Entiendo que la donación de sangre para mí (autóloga) es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo..",
    "title": "Consentimiento del donante Donación autóloga"
  },
  "DCONTST_v3": {
    "content": "Doy mi permiso para tipear mis células sanguíneas y para pruebas de laboratorio detalladas de mi sangre, incluidas las pruebas de VIH (el virus del SIDA), hepatitis, pruebas de anticuerpos de glóbulos blancos y pruebas de investigación relacionadas con la seguridad de la sangre. Si un componente de mi sangre se almacena en estado congelado, doy mi permiso para realizar cualquier prueba de laboratorio adicional que se requiera en el futuro para aclarar la elegibilidad para donaciones pasadas o futuras. Prueba de comillas dobles \"derecha e izquierda\", comillas simples 'derecha e izquierda ', y & y barra diagonal /<br>Colon :<br>Semi colon;<br>Corchetes [derecha e izquierda]<br>paréntesis (derecha e izquierda)<br>Punto de exclamación !<br>A la señal @<br>Hash #<br>Dólar $<br>Porcentaje%<br>Carot ^<br>Asteriscos *<br>guión menos<br>Más +<br>Igual SALIDA=<br>Guion bajo _<br>Mayor que y menor que <><br>Barra diagonal \\<br>Tilde ~<br>Marca de acento grave<br>llaves (derecha e izquierda)<br>barra vertical |",
    "title": "Consentimiento del donante"
  },
  "DCONGEN_v1": {
    "content": "Tengo entendido que la donación de sangre es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento general para donación"
  }
},
"donorHint": "Pida asistencia al encuestador si no comprende el proceso.",
"donorSurvey": {
  "back": "ATRÁS",
  "complete": "Completar",
  "describe": "Por favor describa",
  "deviceMsg": "Por favor devuelve el dispositivo a un empleado",
  "greetings": "Hola {{name}}, ¡Gracias por tu donación!",
  "next": "SIGUIENTE",
  "okay": "ACEPTAR",
  "thankYouMsg": "Gracias por tu retroalimentación!"
},
"greeting": "Estimado donante",
"letterContent": "Nuestro centro de donación de sangre ha introducido la funcionalidad de preguntas contestadas por el propio donante, para mantener la privacidad del donante.",
"of": "de",
"preferredLanguageTitle": "Idioma preferido",
"verifyCommunicationQuestion": "Es usted capaz de leer en Español?",
"incompleteQuestionnaireMessage": "No has revisado todas las preguntas. ¿Quieres finalizar el proceso?",
"completeQuestionnaireMessage": "Has revisado todas las preguntas. ¿Desea finalizar el proceso e imprimir el código de barras?",
"logoutWarningMessage": "Sus respuestas no se guardarán si no imprime o envía por correo electrónico el código QR que se muestra al final del cuestionario. ¿Desea continuar?",
"WEB_THANQU": "Gracias por sus respuestas. Imprima el código de barras QR y llévelo a su Centro de sangre dentro de las 24 horas.",
"SENDEMAIL": "Enviar correo electrónico",
"EXIT": "SALIDA"
}
```

5.8.3 HTTPS Setup for IIS Server – Windows

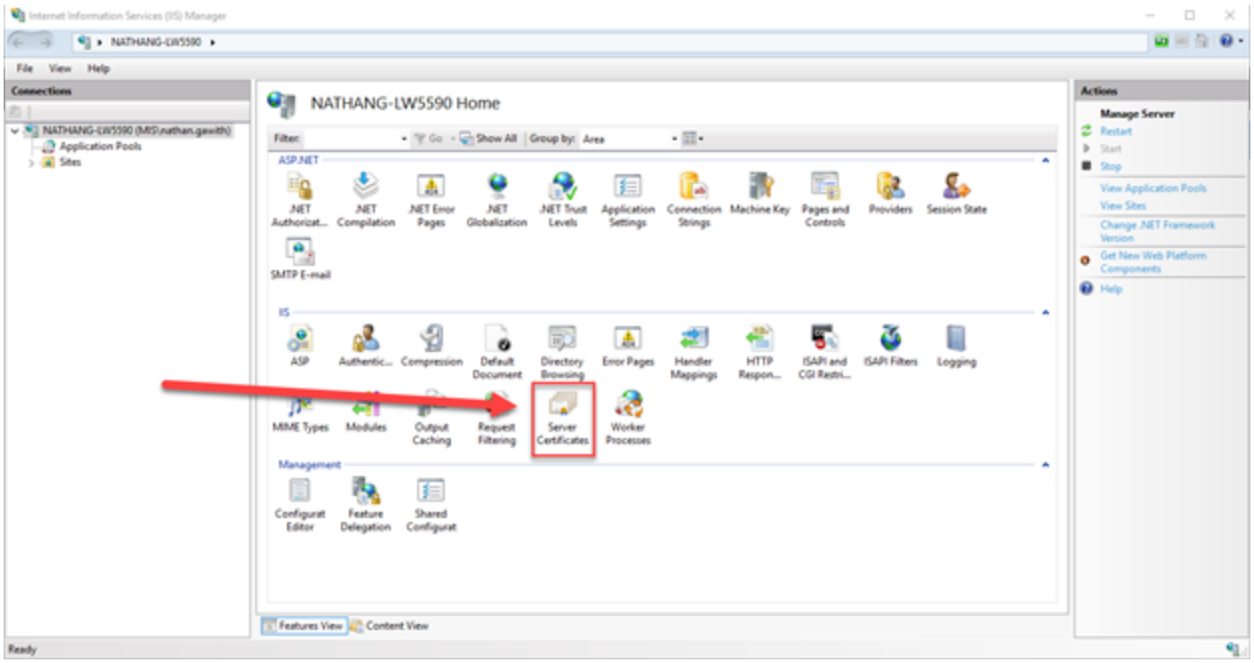
1. Open Internet Information Services (IIS) Manager.



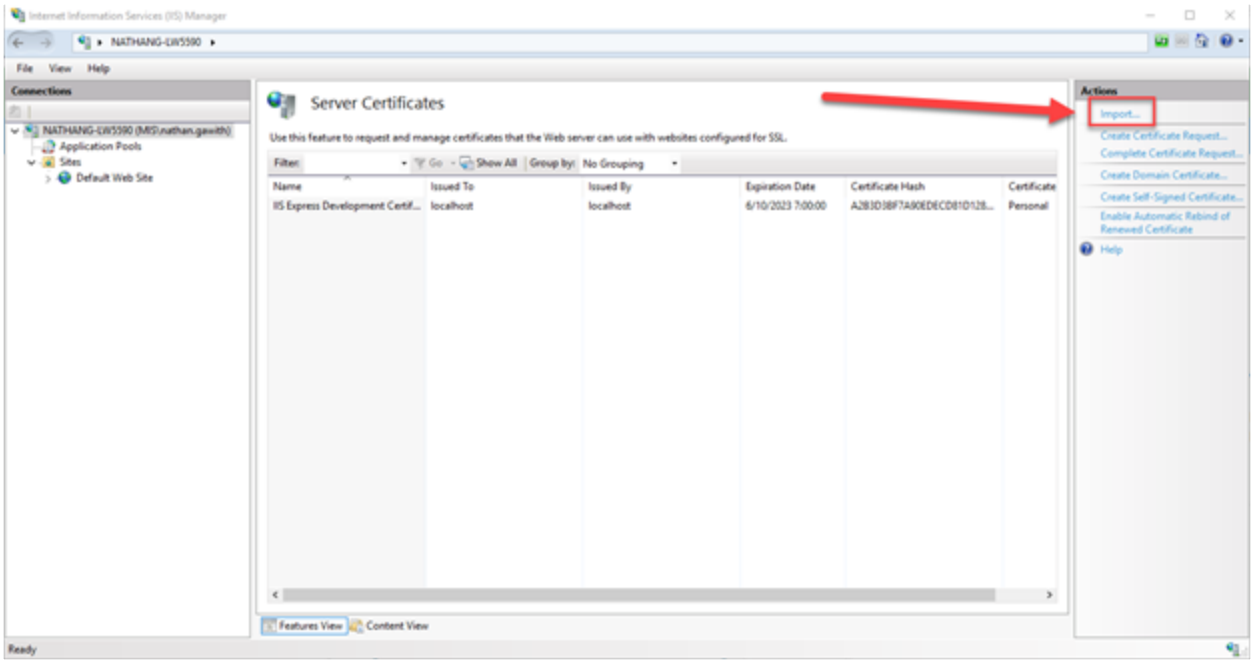
2. Click the server name.



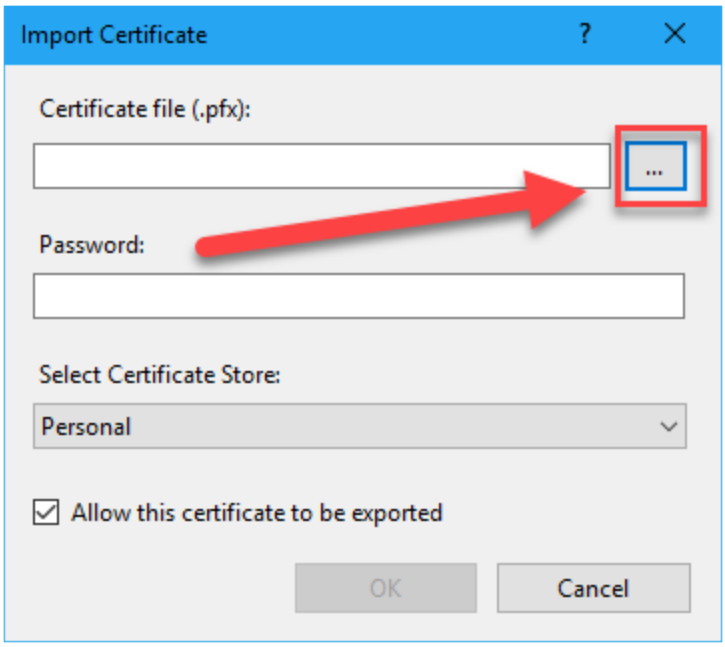
3. Double-click **Server Certificates**.



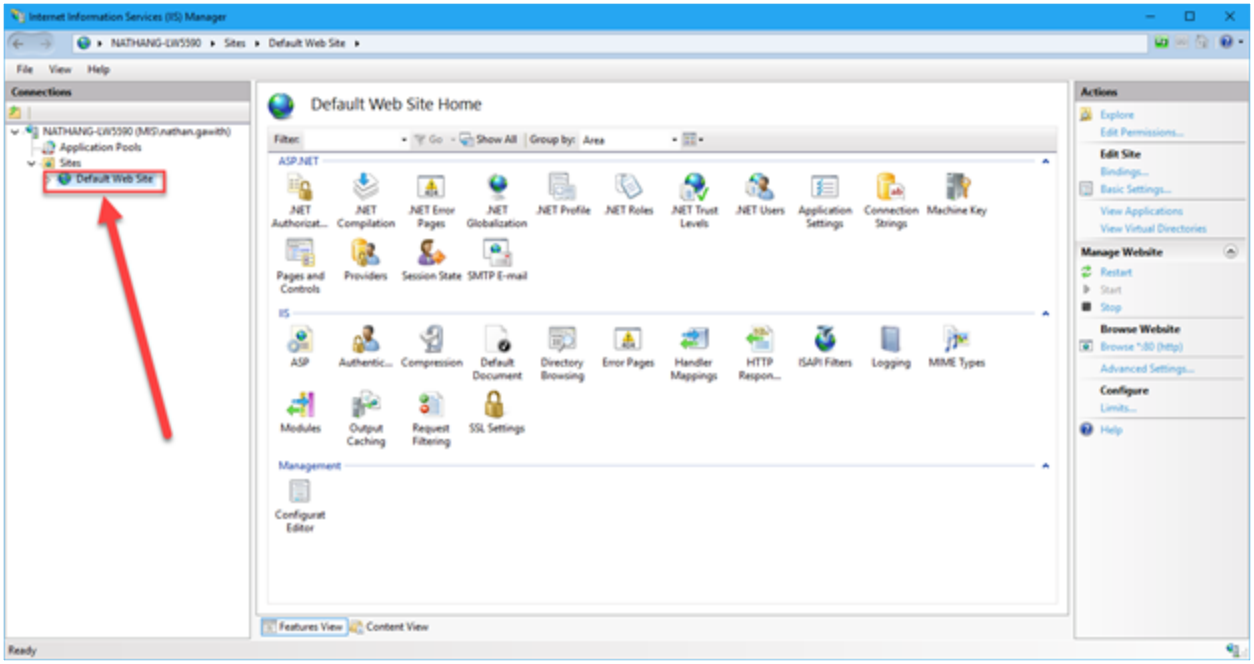
4. In the Actions menu on the right side of the screen, click **Import...**.



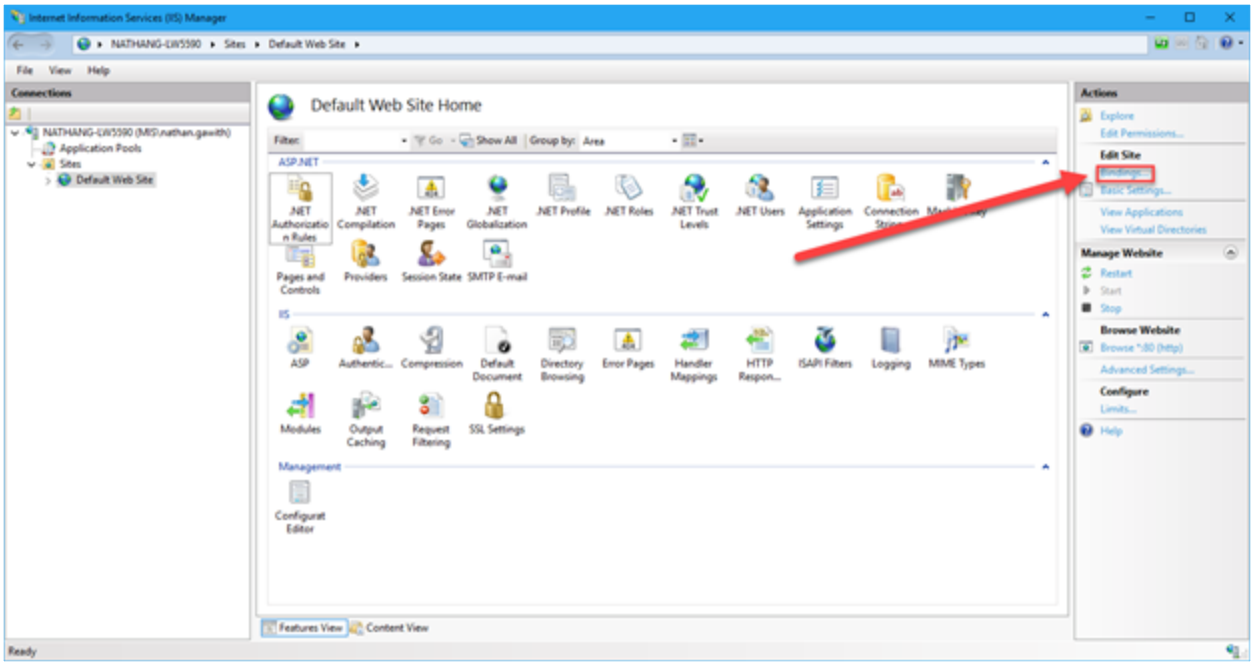
5. On the Import Certificate window, click the **...** button to browse for your SSL certificate to import. Your organization may already have an SSL certificate for you to import. If not, you can download one from an organization like digicert, which is one of many organizations that offer SSL certificates. Once you have selected the file path, click **OK** to import the selected SSL certificate.



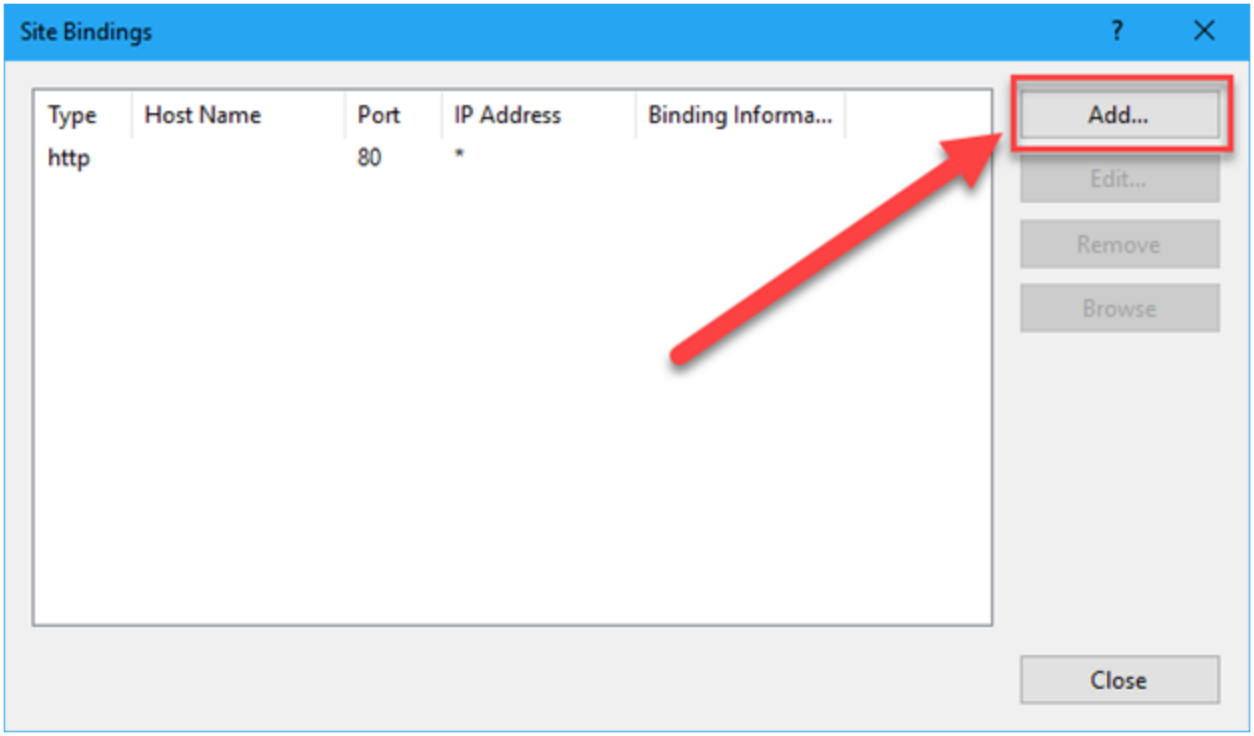
6. On the IIS Manager screen, click **Default Web Site**.



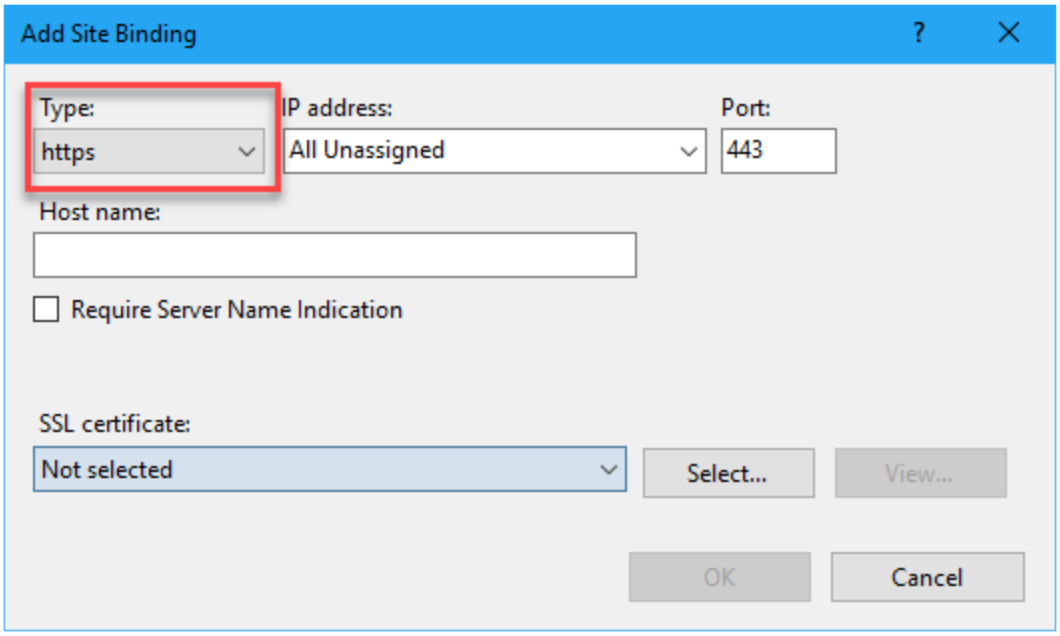
7. In the Actions menu on the right side of the screen, click **Bindings...**.



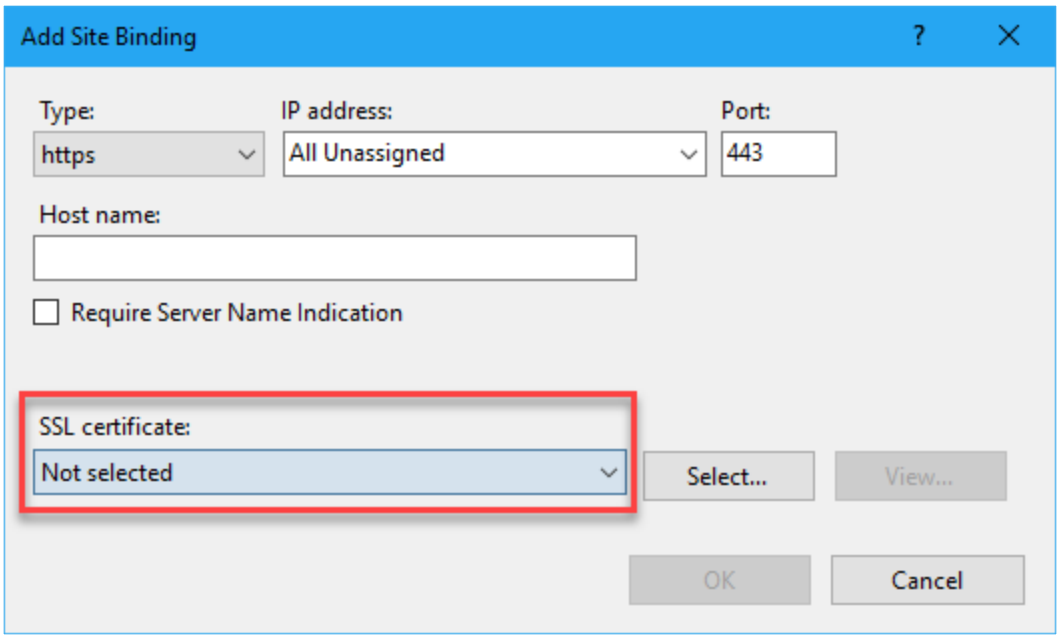
8. On the Site Bindings screen, click **Add...**.



9. On the Add Site Binding screen, select **https** from the *Type* drop-down.



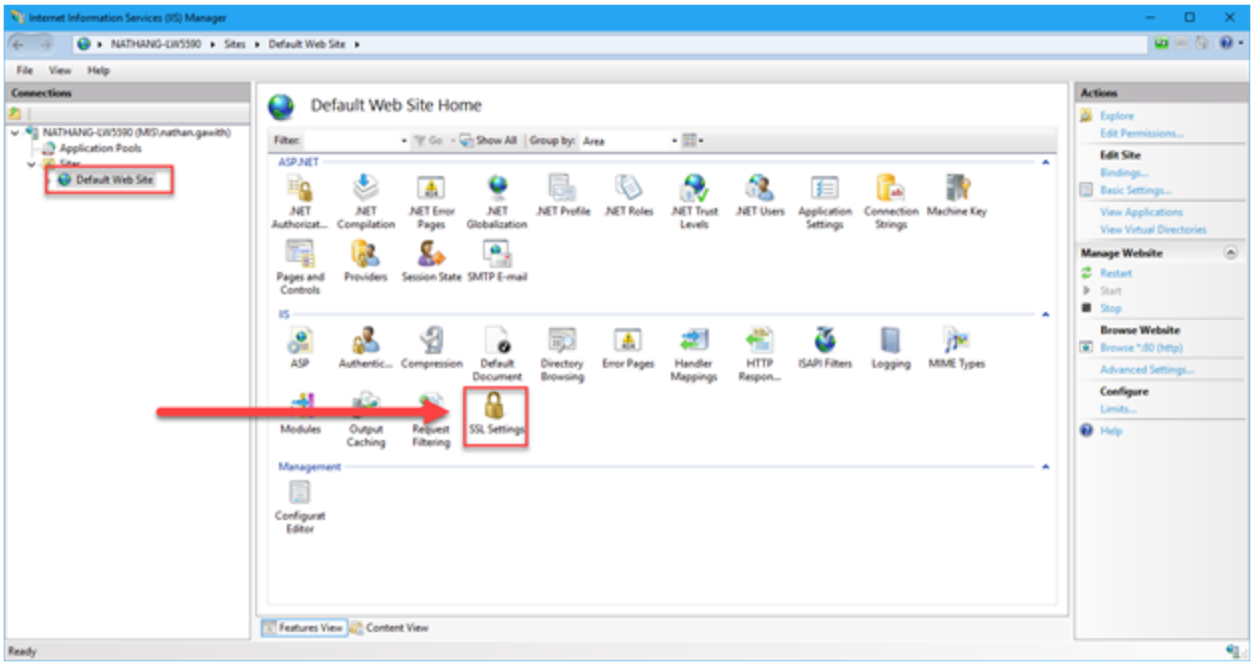
10. Select the SSL certificate you imported in step 5 from the *SSL certificate* drop-down.



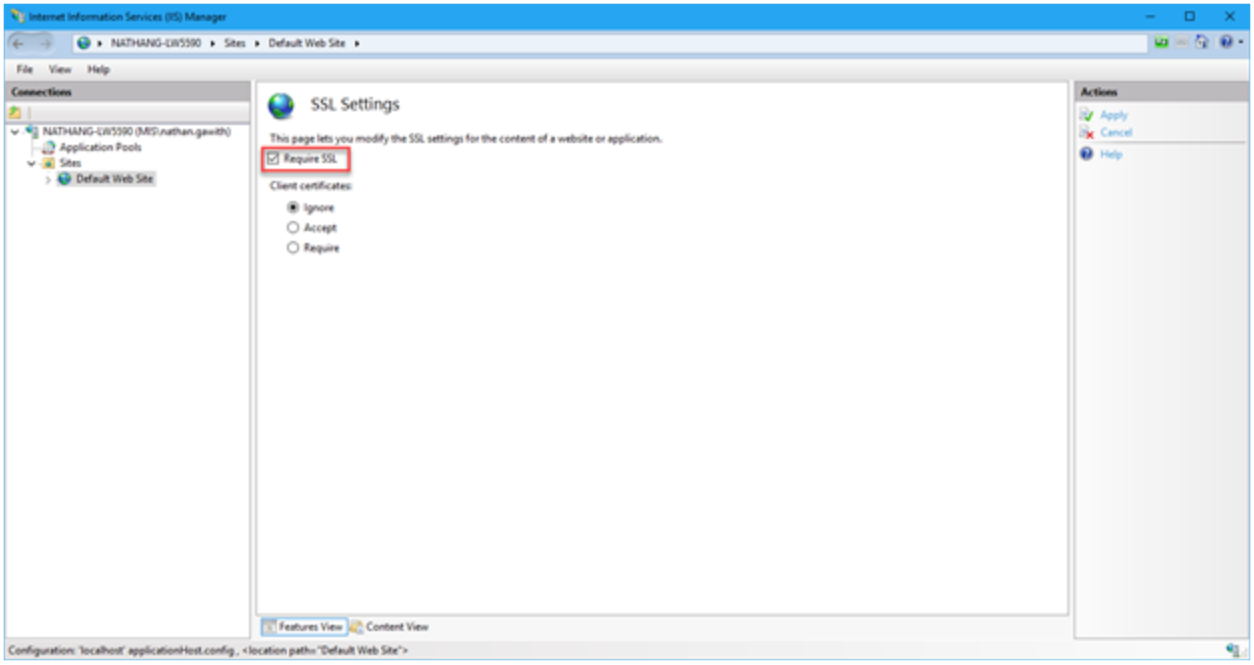
11. Click **OK**.

WellSky Corporation
Proprietary and confidential

12. On the IIS Manager screen, under **Default Web Site**, click **SSL Settings**.



13. On the SSL Settings screen, select the **Require SSL** option.

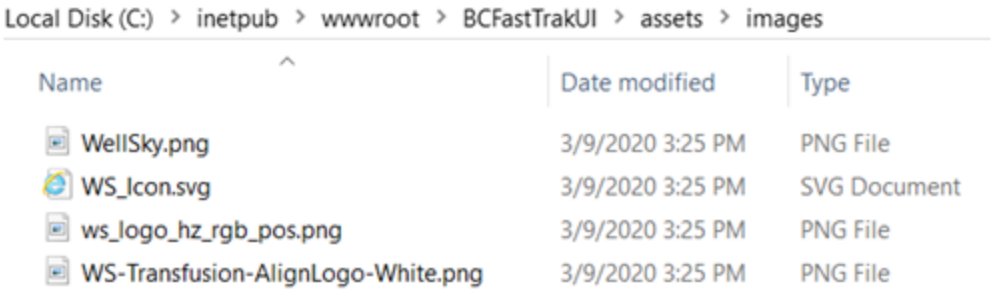


14. Launch the HTTPS application using the following URL: `https://HRUNIX/BCOnsiteUI/index.html`.

Gallon Donor Image File Location

The gallon donor image for the .NET application does not use the system parameter GALLON_IMG. The .NET application has the following requirements:

- The file must be named GallonDonor.
- The image type can be a .png, .jpg, or .svg image.
- The directory for the FastTrak application is: C:\inetpub\wwwroot\BCFastTrakUI\assets\images.



NOTE – The size of the file has not been an issue, but after loading the file you should view it and make sure it fits within the location in the Donor Header.

Multi-Language Files

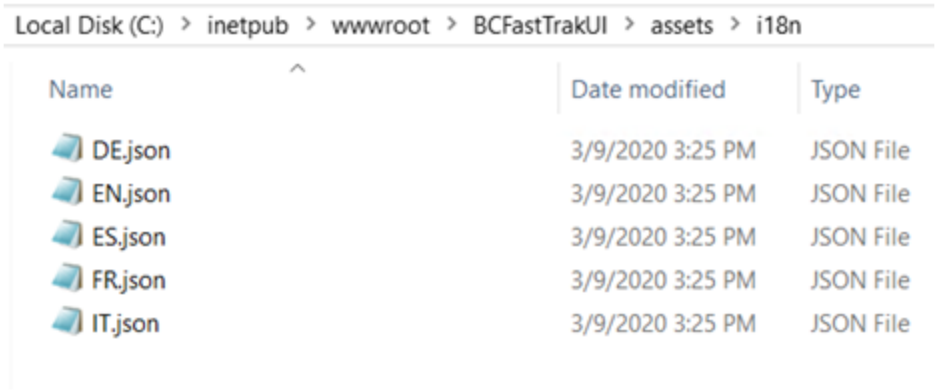
The multi-language file is used for the health history questions, donor survey questions and answers, Donor Consents, FastTrak Login configuration, and the Donor Response initial displays information.

The files require the following:

- File has to be JSON.
- JSON file has to be labeled based on the "List of ISO 639-1 codes" (https://en.wikipedia.org/wiki/List_of_ISO_639-1_codes) for languages.
  - ES = Spanish
  - FR = French
  - IT = Italian
  - DE = German
- The JSON files MUST BE defined in upper case.
- The files must follow a specific format. Example files can be supplied by your Clinical Consultant, or by contacting Technical Support. Below is an example of a JSON file for Spanish.
- A file must be created for each language that you wish to support.
  - EN.json is an empty JSON file. Define the file, but the information does not need to be present in the file.

WellSky Corporation
Proprietary and confidential

- Place the file for the FastTrak application in the following location: C:\inetpub\wwwroot\BCFastTrakUI\assets\i18n.



Example of JSON File for Spanish:

ES - Notepad

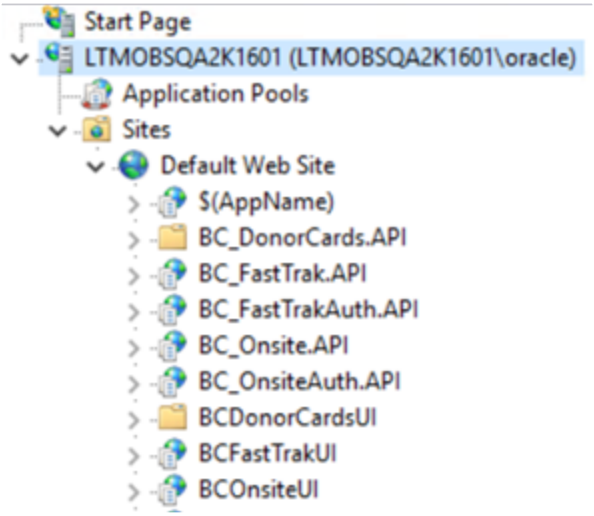
File Edit Format View Help


```
{
  "46-59": "46 a 59",
  "BackToQuestions": "Volver a las preguntas",
  "CANCEL": "CANCELAR",
  "CLEAR": "LIMPIAR",
  "CONTINUE": "CONTINUAR",
  "Chocolate Chip": "Chispas de chocolate",
  "DECLINE": "DECLINAR",
  "DECLINED": "DECLINADO",
  "DS0001": {
    "FIRST": "Donante por primera vez",
    "YES": "Sí",
    "text": "¿Eres un donante recurrente?"
  },
  "DS0002": {
    "NO": "No",
    "WARM": "Prefiero manta calentada",
    "YES": "Sí",
    "text": "¿Usas una manta cuando donas?"
  },
  "DS0003": {
    "1": "1-5 veces",
    "2": "6-10 veces",
    "FIRST": "Donante por primera vez",
    "text": "¿Cuántas veces dona en un año?"
  },
  "DS0004": {
    "NO": "No",
    "YES": "Sí",
    "text": "¿Eres un donante de féresis?"
  },
  "DeclineConfirmMessage": "¿Está seguro de que quiere rechazar este consentimiento?",
  "DonorSignature": "Firma del Donante",
  "FINISH": "FINALIZAR",
  "Finish": "Terminar",
  "FinishedWithDRMessage": "Gracias por sus respuestas. Haga clic en FINALIZAR y devuelva el dispositivo al Personal.",
  "GVT0001": {
    "60": "60+",
    "1721": "17-21",
    "2235": "22-35",
    "3645": "36-45",
    "4659": "46-59",
    "text": "¿Qué grupo de edad tienes?"
  },
  "HQ0010": "¿Es usted alérgico al Iodo y/o Látex?",
  "HQ0010-A": "A Desde su última donación, ha adquirido una alergia al yodo y / o ¿látex? - ABREVIADO",
  "HQ0010-D": "¿Alguna vez ha tenido una reacción alérgica o irritación de la piel al látex o productos con látex como los vendajes? Si es así, responda SÍ y el investigador lo discutirá con usted",
  "HQ0020": "¿Te sientes bien y saludable hoy?",
  "HQ0020-D": "No tenga ningún problema, como dolores de cabeza, dolor de espalda o estrés por donar.",
  "HQ0030": "En las últimas 8 semanas, ¿ha dado sangre, plasma o plaquetas?",
  "HQ0040": "¿Alguna vez ha sido rechazado como donante de sangre o le han dicho que no done sangre?",
  "HQ0071": "En las últimas 4 semanas, ¿ha tomado alguna píldora o medicamento?",
  "DCONPP_v2": {
    "content": "Entiendo que la donación de sangre para una féresis plaquetaria es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento de féresis plaquetaria"
  },
  "DCONAUTO_v1": {
    "content": "Entiendo que la donación de sangre para mí (autólogo) es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo..",
    "title": "Consentimiento del donante Donación autóloga"
  },
  "DCONTST_v3": {
    "content": "Doy mi permiso para tipear mis células sanguíneas y para pruebas de laboratorio detalladas de mi sangre, incluidas las pruebas de VIH (el virus del SIDA), hepatitis, pruebas de anticuerpos de glóbulos blancos y pruebas de investigación relacionadas con la seguridad de la sangre. Si un componente de mi sangre se almacena en estado congelado, doy mi permiso para realizar cualquier prueba de laboratorio adicional que se requiera en el futuro para aclarar la elegibilidad para donaciones pasadas o futuras. Prueba de comillas dobles \'derecha e izquierda\', comillas simples `derecha e izquierda ', y & y barra diagonal /<br>Colon :<br>Semi colon;<br>Corchetes [derecha e izquierda]<br>paréntesis (derecha e izquierda)<br>Punto de exclamación !<br>A la señal @<br>Hash #<br>Dólar $<br>Porcentaje%<br>Carot ^<br>Asteriscos *<br>guión menos<br>Más +<br>Igual SALIDA=<br>Guion bajo _<br>Mayor que y menor que <><br>Barra diagonal \\.<br>Tilde ~<br>Marca de acento grave<br>llaves (derecha e izquierda)<br>barra vertical |",
    "title": "Consentimiento del donante"
  },
  "DCONGEN_v1": {
    "content": "Tengo entendido que la donación de sangre es un acto totalmente voluntario y no se ha ofrecido ningún incentivo o remuneración. La donación de sangre / componentes es un procedimiento médico y que mediante la donación voluntaria. Acepto los riesgos asociados con este procedimiento. Se analizará mi sangre para detectar hepatitis B / C, parásito de la malaria, VIH / SIDA y enfermedades venéreas, además de cualquier otra prueba de detección requerida para garantizar la seguridad de la sangre. Prohíbo que cualquier información proporcionada por mí o sobre mi donación se divulgue a cualquier persona o agencia gubernamental sin ningún permiso previo.",
    "title": "Consentimiento general para donación"
  }
},
"donorHint": "Pida asistencia al encuestador si no comprende el proceso.",
"donorSurvey": {
  "back": "ATRÁS",
  "complete": "Completar",
  "describe": "Por favor describa",
  "deviceMsg": "Por favor devuelve el dispositivo a un empleado",
  "greetings": "Hola {{name}}, ¡Gracias por tu donación!",
  "next": "SIGUIENTE",
  "okay": "ACEPTAR",
  "thankYouMsg": "Gracias por tu retroalimentación!"
},
"greeting": "Estimado donante",
"letterContent": "Nuestro centro de donación de sangre ha introducido la funcionalidad de preguntas contestadas por el propio donante, para mantener la privacidad del donante.",
"of": "de",
"preferredLanguageTitle": "Idioma preferido",
"verifyCommunicationQuestion": "Es usted capaz de leer en Español?",
"incompleteQuestionnaireMessage": "No has revisado todas las preguntas. ¿Quieres finalizar el proceso?",
"completeQuestionnaireMessage": "Has revisado todas las preguntas. ¿Desea finalizar el proceso e imprimir el código de barras?",
"logoutWarningMessage": "Sus respuestas no se guardarán si no imprime o envía por correo electrónico el código QR que se muestra al final del cuestionario. ¿Desea continuar?",
"WEB_THANQU": "Gracias por sus respuestas. Imprima el código de barras QR y llévelo a su Centro de sangre dentro de las 24 horas.",
"SENDEMAIL": "Enviar correo electrónico",
"EXIT": "SALIDA"
}
```

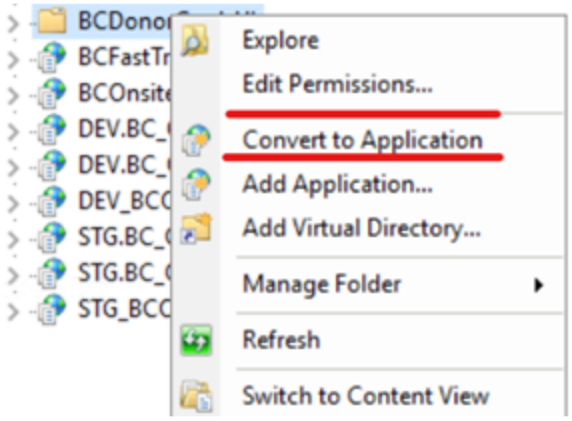

WellSky Corporation
Proprietary and confidential

5.8.4 Deploy the BC Donor Card Application on the Application Server

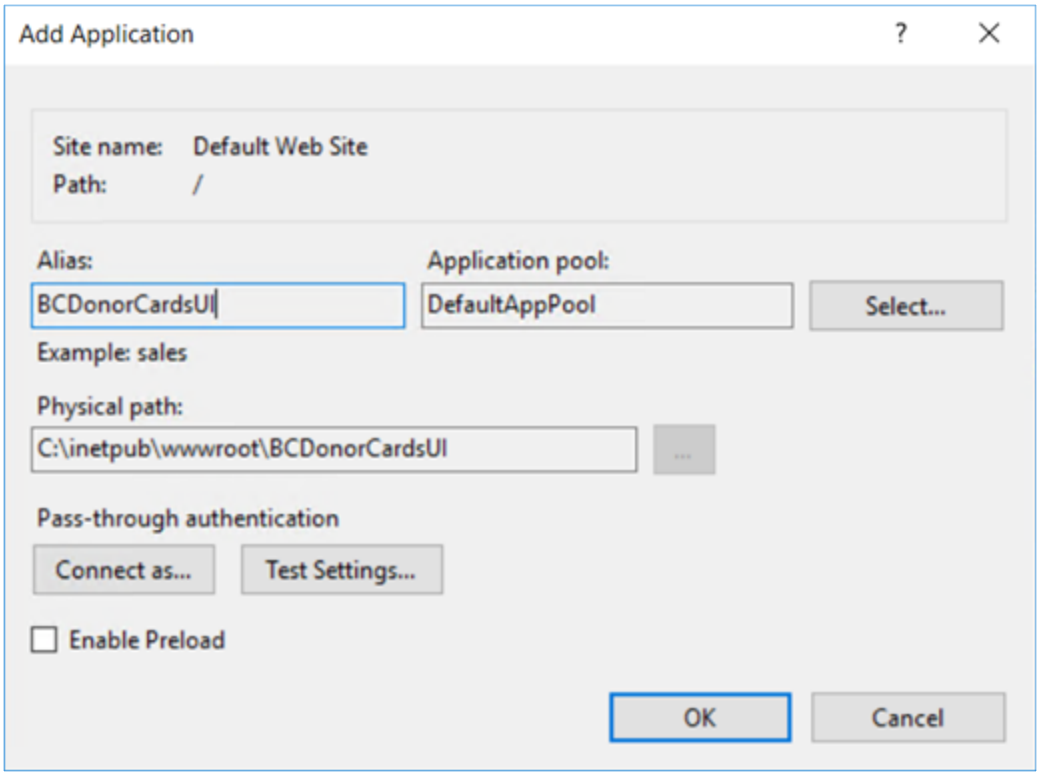
1. Copy the following folders from **C:\bcmpc\BC_DC_Deploy** to the **C:\inetpub\wwwroot**.
    - BCDonorCardsUI
    - BC_DonorCards.API
2. Edit the **BC_DonorCards.API\appsettings.json** file, and update it for DB Parameters in the DefaultConnection section, if needed.
3. Type `IIS` in the search box to locate the Internet Information Services (IIS) Manager.
4. When it is found, click on it to open it, and navigate to **Sites** > **Default Web Site**.



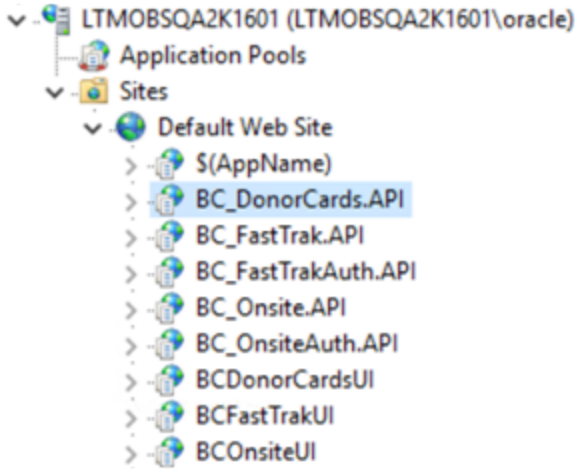
5. Right-click on **BCDonorCardsUI**, and select **Convert to Application**.



The Add Application screen displays.



6. Click **OK**.
7. Right-click on **BC_DonorCards.API**, and select **Convert to Application**.

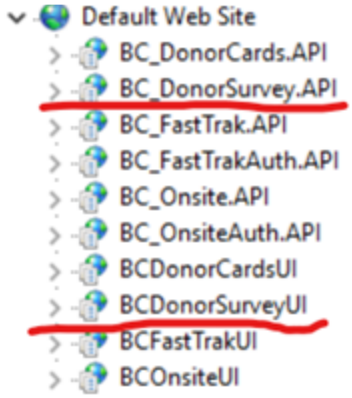


8. When the Add Application screen displays, click **OK**.
9. Restart the IIS server to apply the modifications.

WellSky Corporation
Proprietary and confidential

5.8.5 Deploy the BC Survey Application on the Application Server

1. Copy the following folders from **C:\bcmpc\BC_Survey_Deploy** to **C:\inetpub\wwwroot**.
    - BCDonorSurveyUI
    - BC_DonorSurvey.API
2. Edit the **BC_DonorSurvey.API\appsettings.json** file, and update it for DB Parameters in the DefaultConnection section, if needed.
3. Type `IIS` in the search box to locate the Internet Information Services (IIS) Manager.
4. When it is found, click on it to open it, and navigate to **Sites** > **Default Web Site**.
5. Right-click on **BCDonorSurveyUI**, and select **Convert to Application**.
6. The Add applications screen displays. Click **OK**.
7. Right-click on **BC_DonorSurvey.API**, and select **Convert to Application**.
8. The Add applications screen displays. Click **OK**.

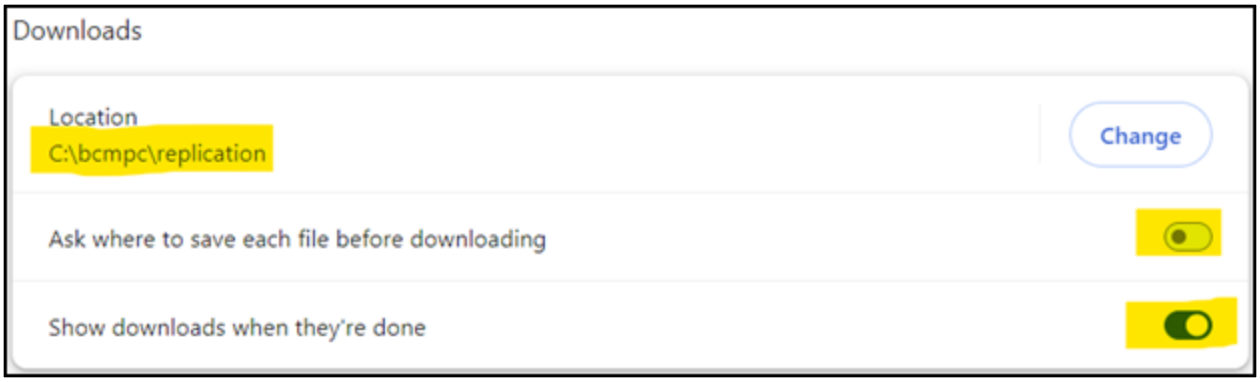


9. Restart the IIS server to apply the modifications.

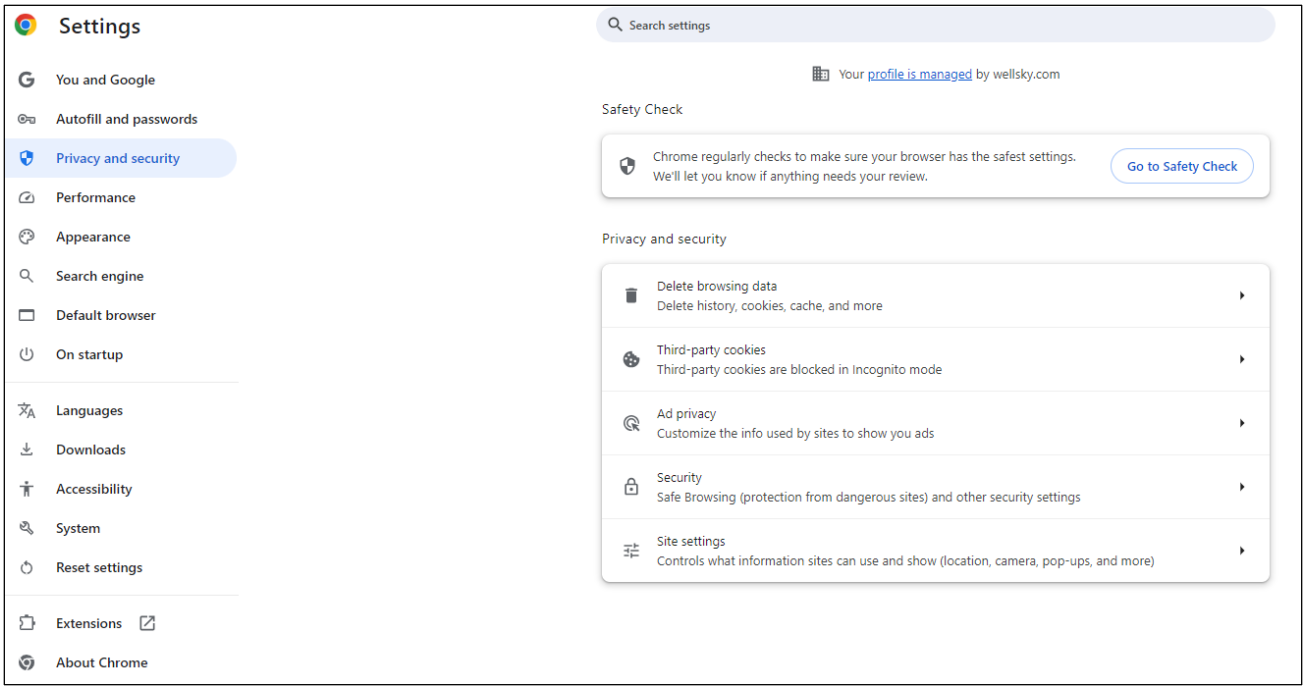
5.9 Google Chrome Setup for HTTP

Perform the following procedure to set up Google Chrome for HTTP.

1. Open Google Chrome.
2. Go to **Settings** > **Downloads**.
3. Click **Change** next to *Location*. Change the location to **`C:\bcmpc\replication`**.



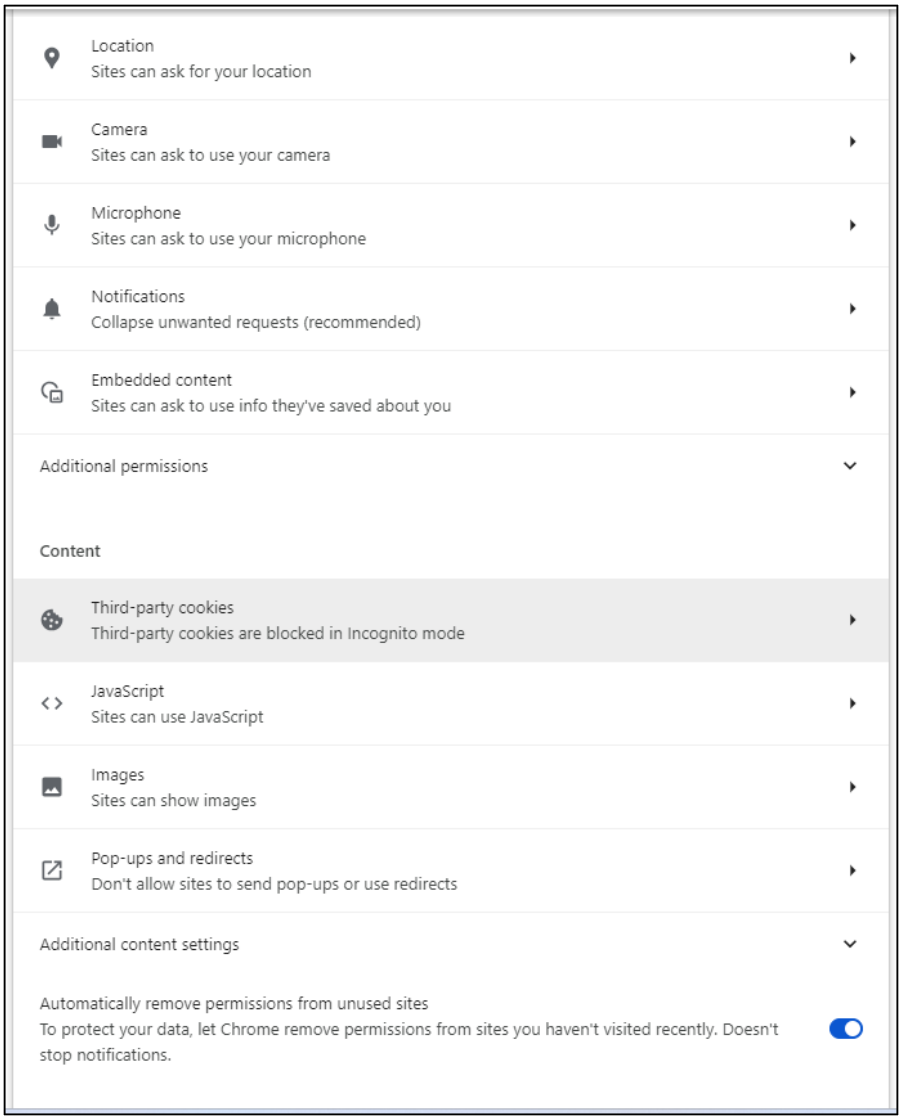
4. Click the **Privacy and security** option on the left side of the screen.



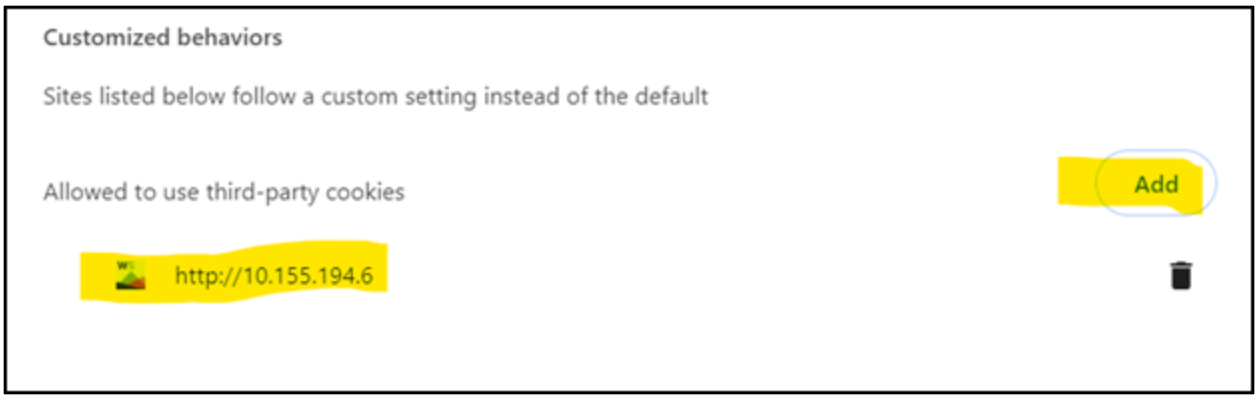
5. Click **Site Settings** at the bottom of the Privacy and security list.

WellSky Corporation
Proprietary and confidential

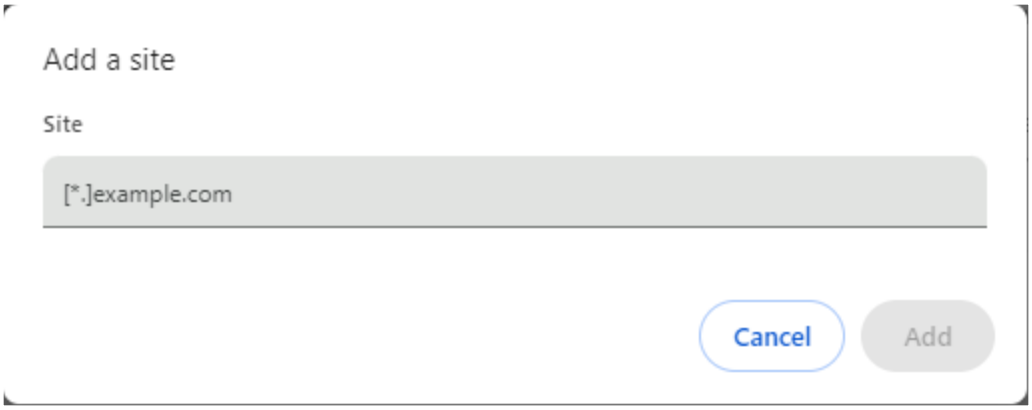
6. On the Site Settings menu, under *Content*, click **Third-party cookies**.



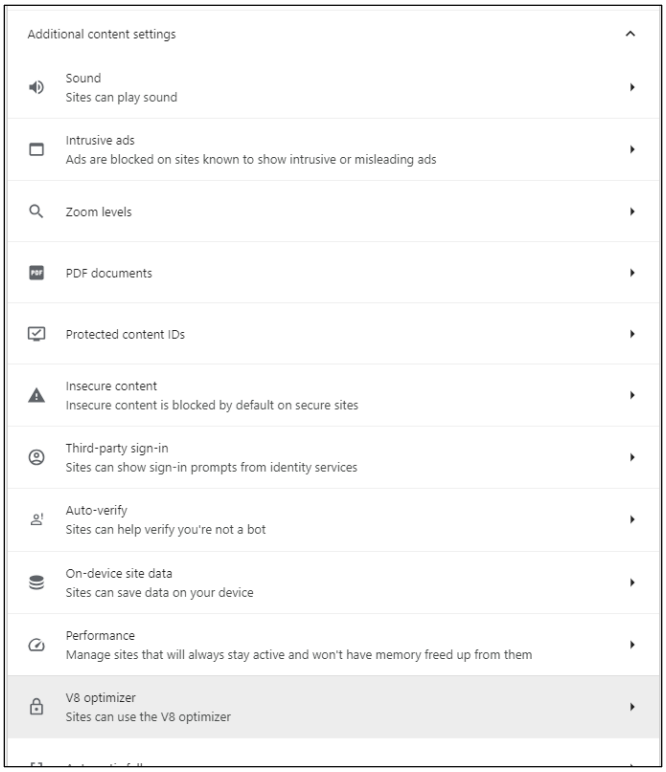
7. Under *Customized behaviors*, click **Add** next to *Allowed to use third-party cookies*.



8. In the Add a Site screen, add the Master server URL (**`http://10.155.194.6`**).

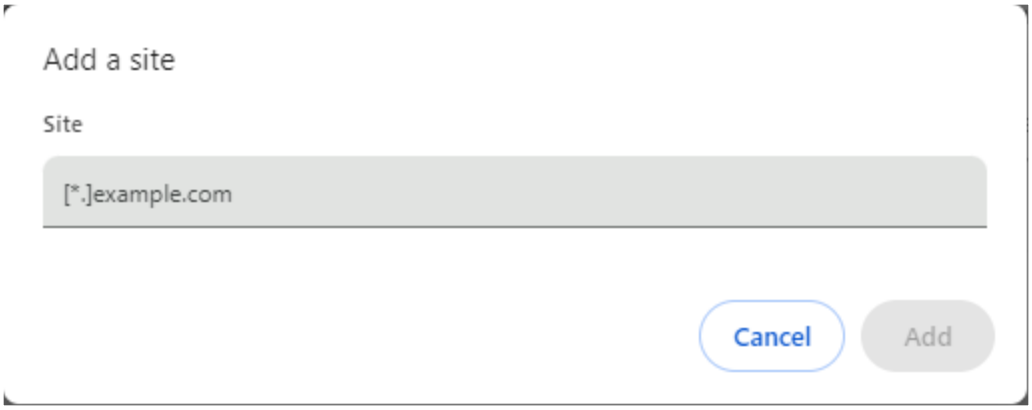


9. Click **Add**.
10. Go back to the Site Settings menu (**Privacy and security** > **Site settings**).
11. Expand the **Additional content settings** menu.



12. Click the **Insecure content** option.
13. Click **Add** next to the *Allowed to show insecure content* option.

14. In the Add a Site screen, add the Master server URL (**`http://10.155.194.6`**).



15. Click **Add**.



16. When you download the first time, a prompt will display for downloading multiple files. Click **Allow**.

This Page Intentionally Left Blank

6

Chapter 6 - Mobile PC Download, Extract, and Upload Process in PUSH Method

6.1 Mobile PC Download Process in PUSH Method 86
6.1.1 Set Up the Check_download Process in Windows Task Scheduler 86
6.1.2 Download Data 91
6.2 Mobile PC Data Extract Process in PUSH Method 96
6.2.1 Introduction 96
6.2.2 Process 96
6.2.3 Extract Data to File 96
6.2.4 Set Up the Extract Queue Process in Windows Task Scheduler 97
6.3 Create Data Extract File 103
6.4 Upload Data to the Main Server from the Mobile PC Server 107
6.4.1 Set Up the Upload Archive Process in Windows Task Scheduler 107
6.4.2 Upload Data to the Server from the Mobile PC 112

6.1 Mobile PC Download Process in PUSH Method

6.1.1 Set Up the Check_download Process in Windows Task Scheduler

NOTE – This setup needs to be performed only once.

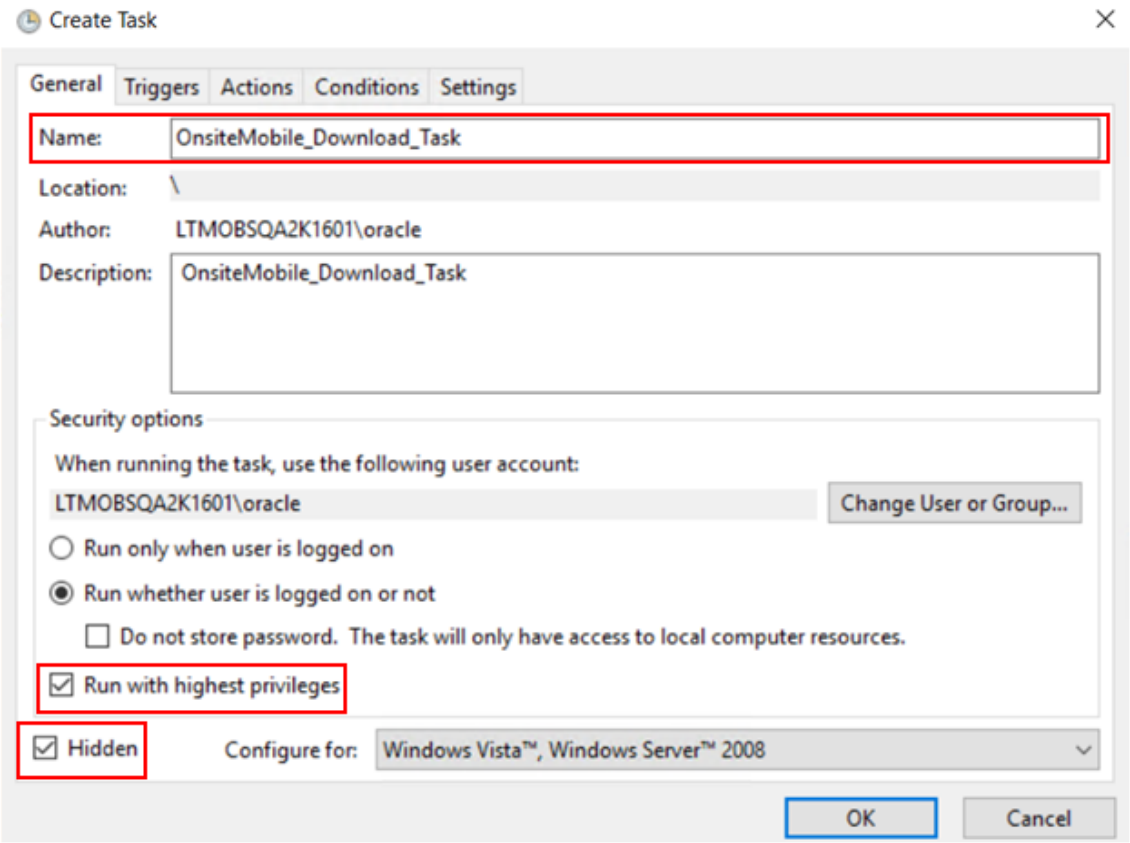
This procedure explains how to set up the C:\bcmpc\schedular_jobs\OnsiteMobile_Download_Task.bat script to run every five minutes in Task Scheduler.

1. Open the Task Scheduler application.

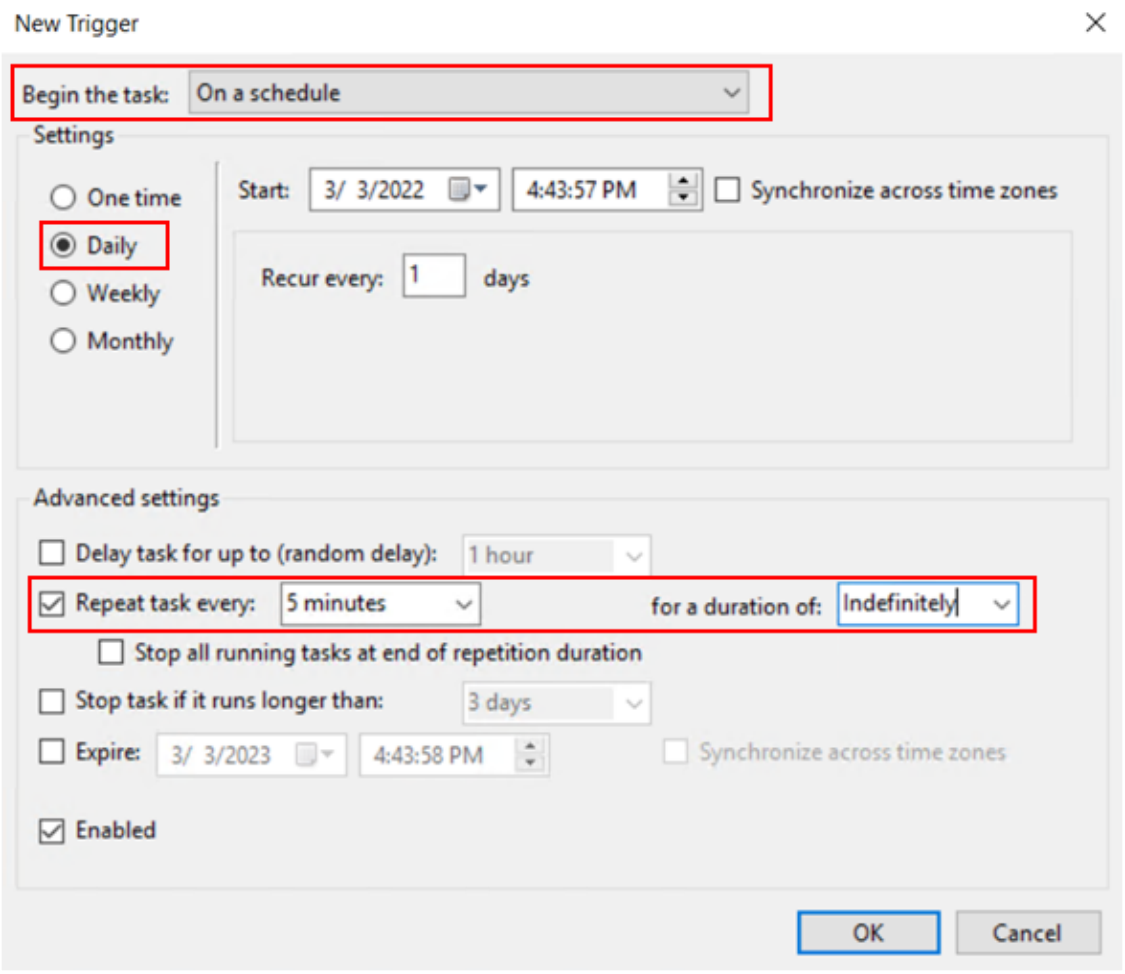
   NOTE – You can type `Task Scheduler` in the search box.

2. Click **Create Task** to open the Create Task window.
3. Name the task **`OnsiteMobile_Download_Task`**.
4. Select the **Run with highest privileges** option.
5. Select the **Hidden** option.

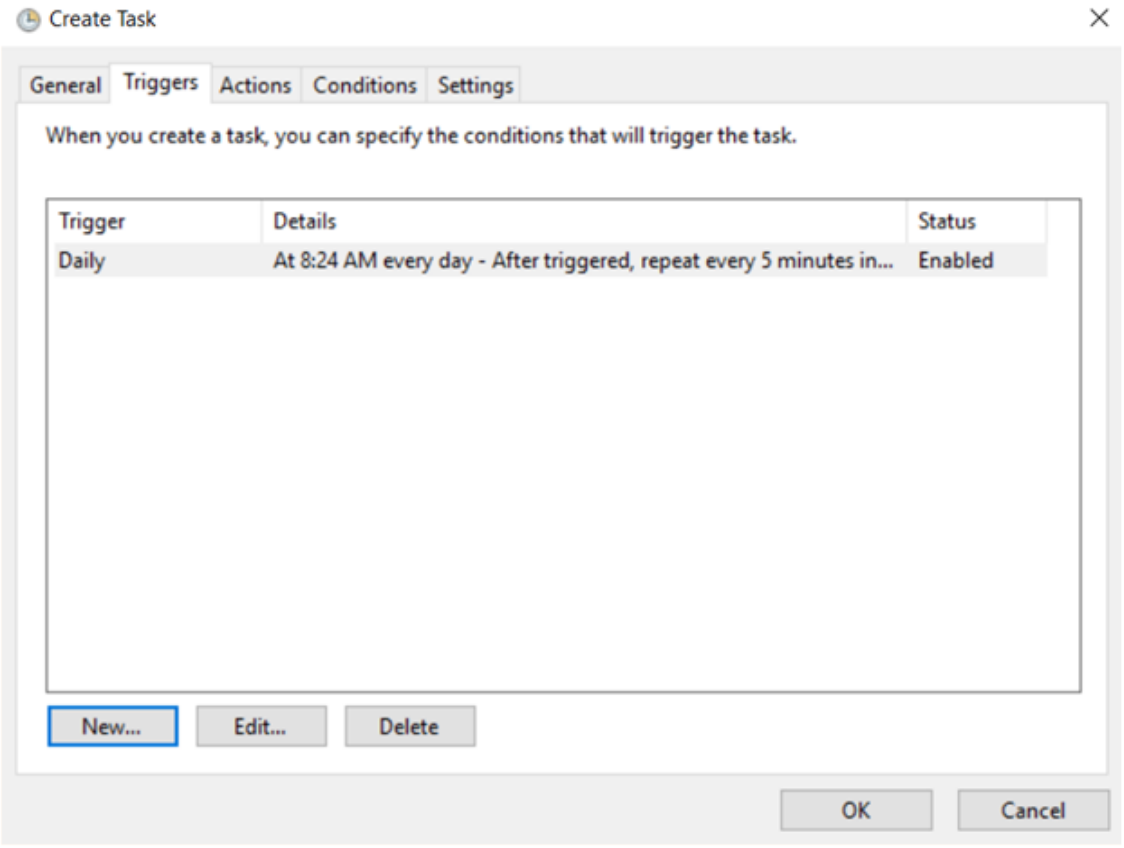
WellSky Corporation
Proprietary and confidential



6. Click the **Triggers** tab.
7. Click **New...**.
8. On the New Trigger screen, select **On a schedule** from the *Begin the task* drop-down menu.
9. Under Settings, select the **Daily** option.
10. Under Advanced Settings, select the **Repeat task every:** option.
    a. Select **5 minutes** from the drop-down menu.
    b. Select **Indefinitely** from the *for a duration of:* drop-down menu.

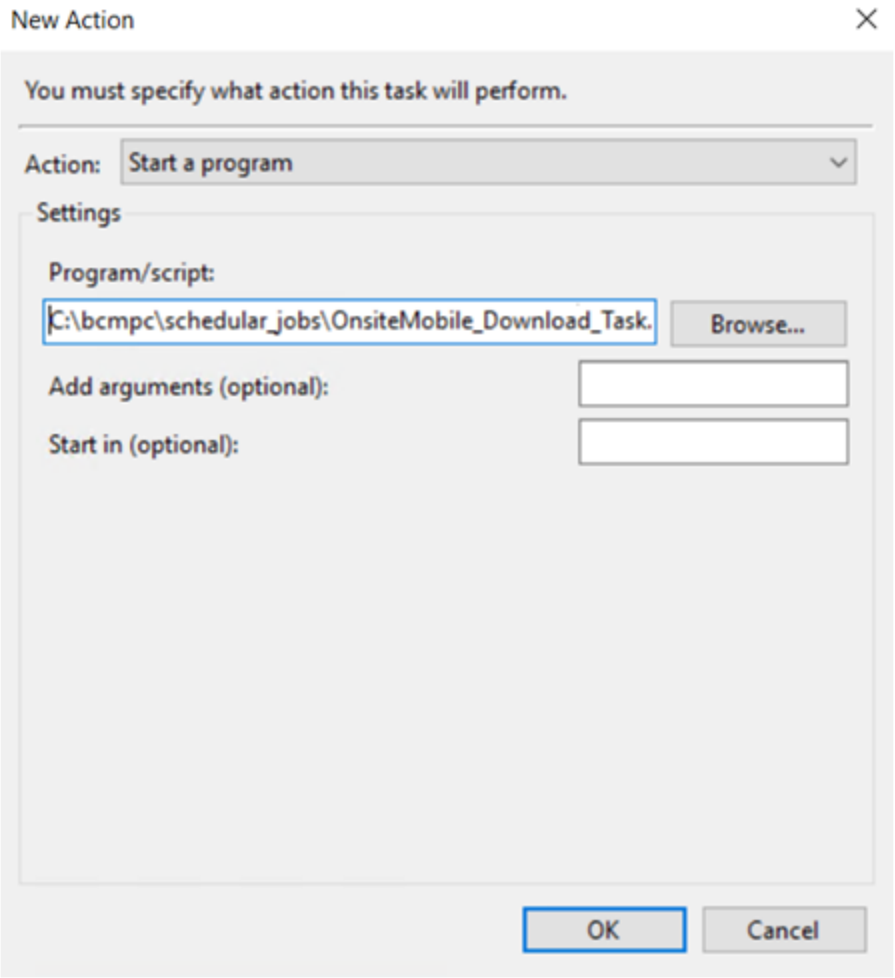

New Trigger
Begin the task: On a schedule
Settings
One time
Daily
Weekly
Monthly
Start: 3/ 3/2022 4:43:57 PM
Synchronize across time zones
Recur every: 1 days
Advanced settings
Delay task for up to (random delay): 1 hour
Repeat task every: 5 minutes
for a duration of: Indefinitely
Stop all running tasks at end of repetition duration
Stop task if it runs longer than: 3 days
Expire: 3/ 3/2023 4:43:58 PM
Synchronize across time zones
Enabled
OK
Cancel

11. Click **OK**. The new trigger displays on the Trigger tab, as shown in the following figure.



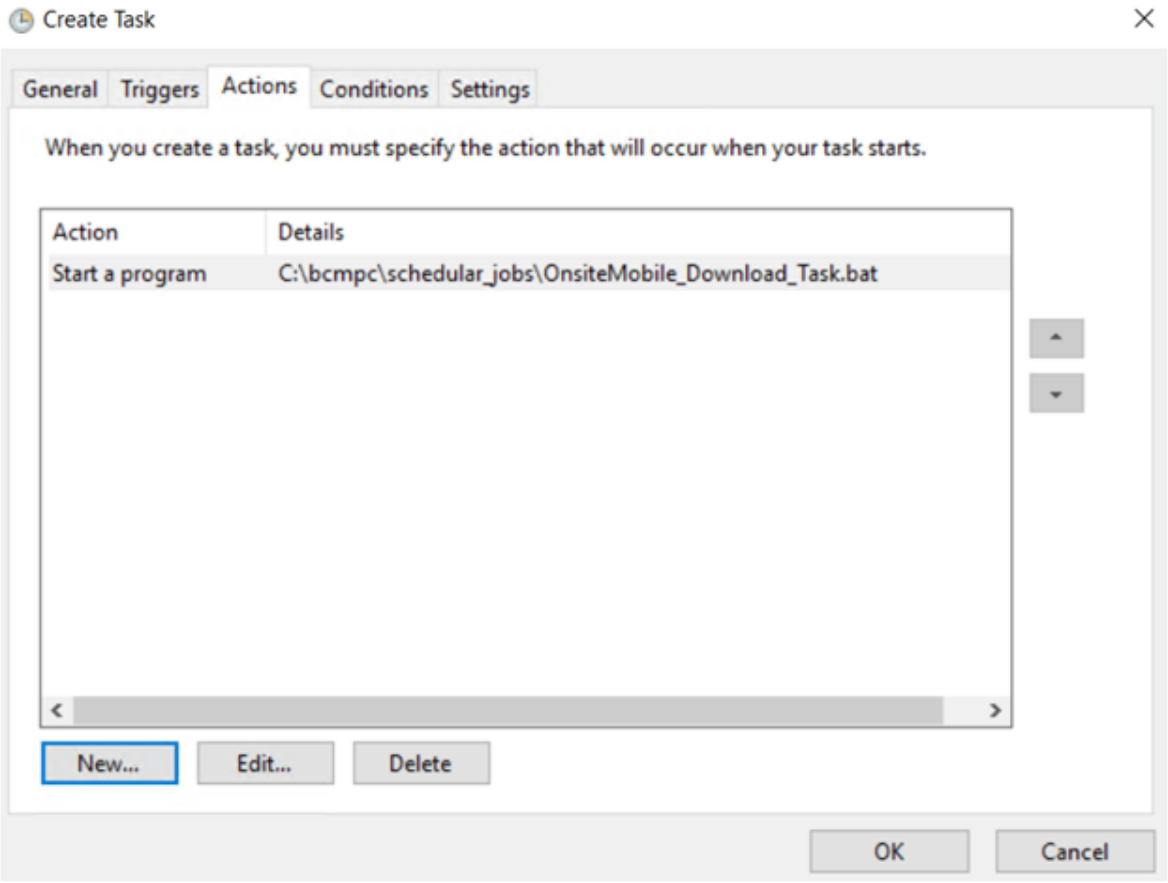
12. Click the **Actions** tab.
13. Click **New...**.

14. On the New Action screen, click **Browse...**and locate the OnsiteMobile_Download_Task.bat script, which will be here: C:\bcmpc\schedular_jobs\.



WellSky Corporation
Proprietary and confidential

15. Click OK. The new action displays on the Actions tab, as shown in the following figure.



16. Click OK to close the Create Task window.

6.1.2 Download Data

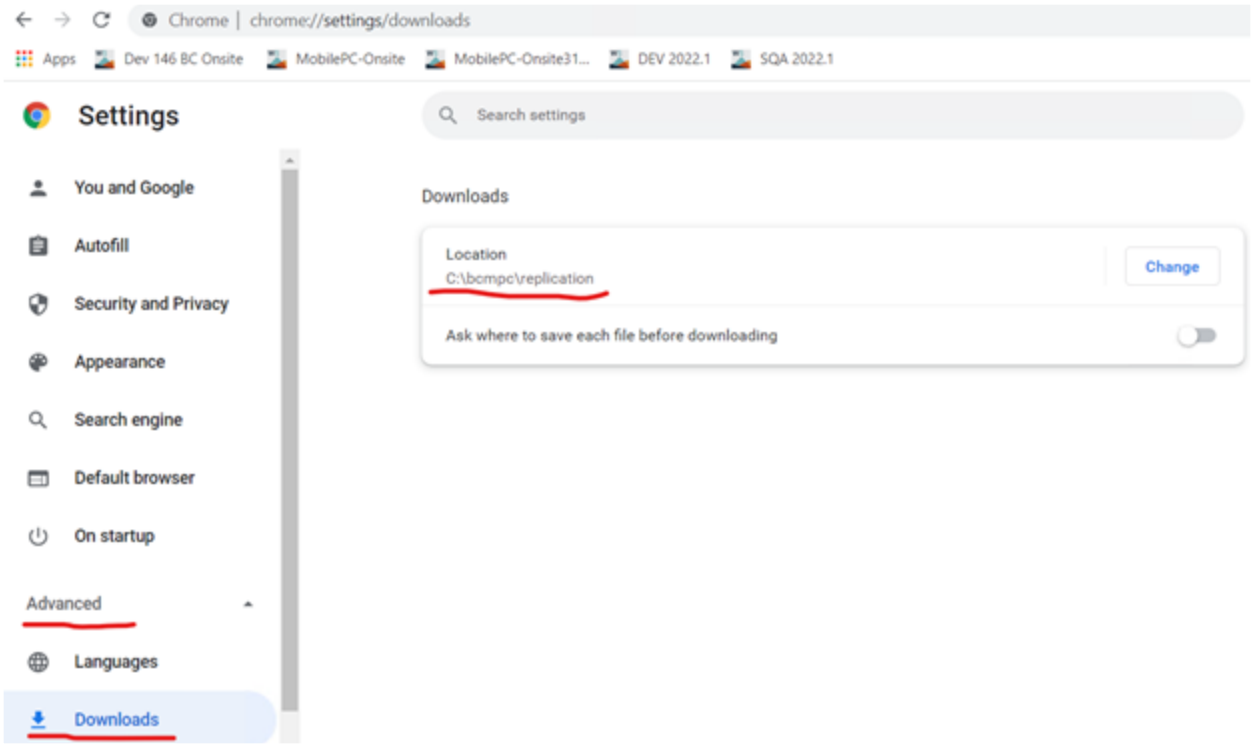
This process is repeated each time you need to refresh the data on the Mobile PC Laptop Server. Download the data for the laptop by opening the Production On-Site Application on the Mobile PC Laptop Server.

Important! This process loads all the donor data from the Blood Centers central database onto the Mobile PC Laptop Server.

If this laptop has been deployed to a mobile drive, it is recommended that you perform a data extract before downloading data to the Mobile laptop.

NOTE – WellSky recommends the download process takes place during non-peak times for optimal performance.

1. Open Google Chrome.
2. Access the Google Chrome settings by either clicking the vertical three dots next to your profile picture (), or type `chrome://settings` in the URL field.
3. Expand the **Advanced** menu, and click **Downloads**.
4. Click **Change** to update the C:\bcmpc\replication directory, as shown in the following figure.



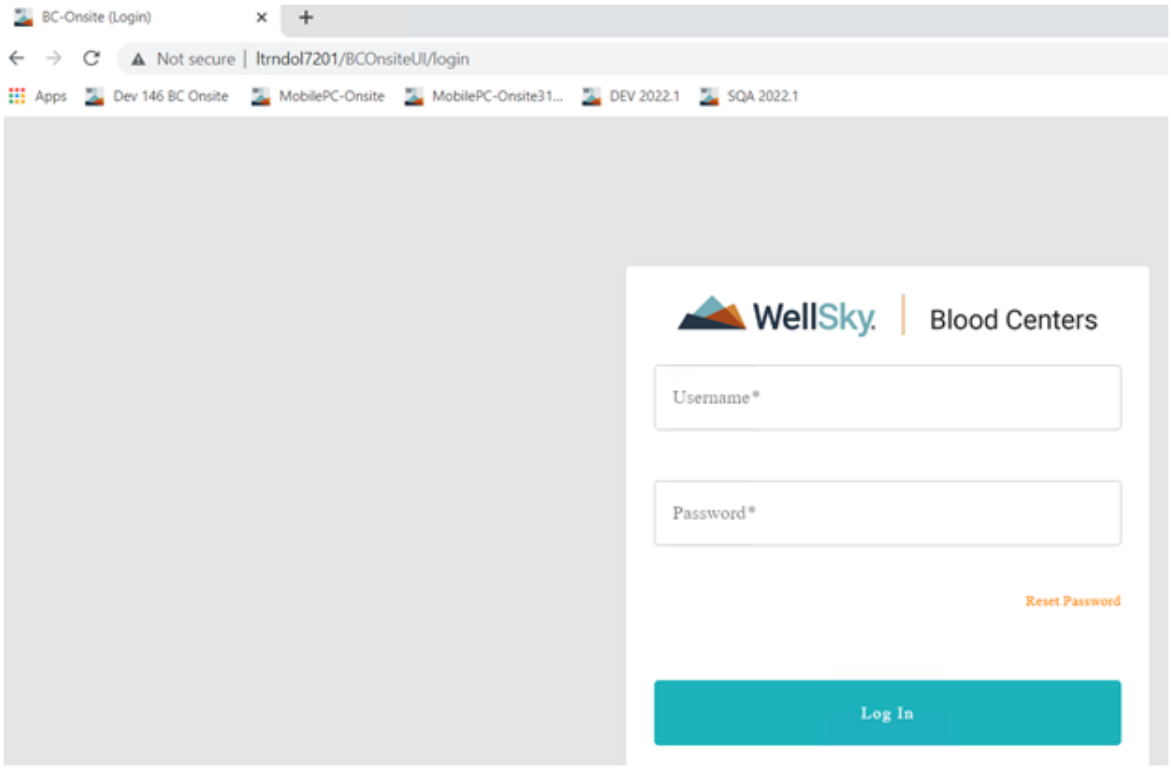
5. Enter the applicable URL for the Main Server's On-Site Application:

   Without SSL: `http://ltrndol7201/BCOnsiteUI/index.html`
   With SSL: `https://ltrndol7201/BCOnsiteUI/index.html`

> **NOTE –** Replace ltrndol7201 with the hostname/IP address of your Blood Centers production On-Site application server.

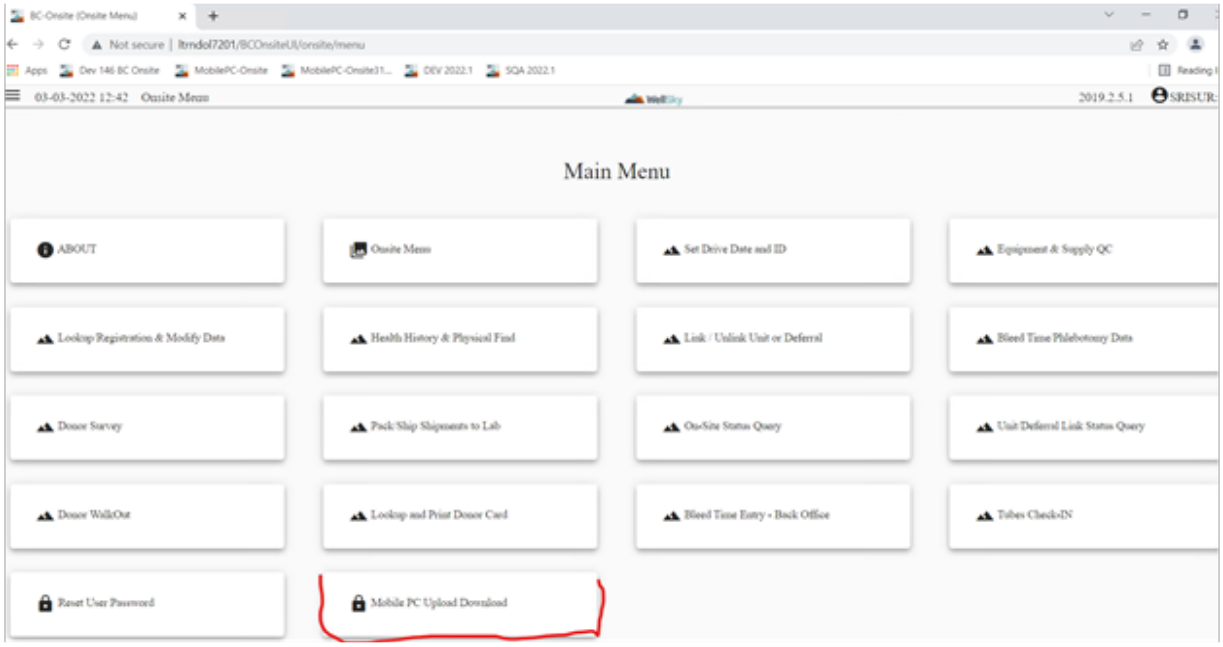
WellSky Corporation
Proprietary and confidential

6. The WellSky Blood Centers login window appears, as shown in the following figure.



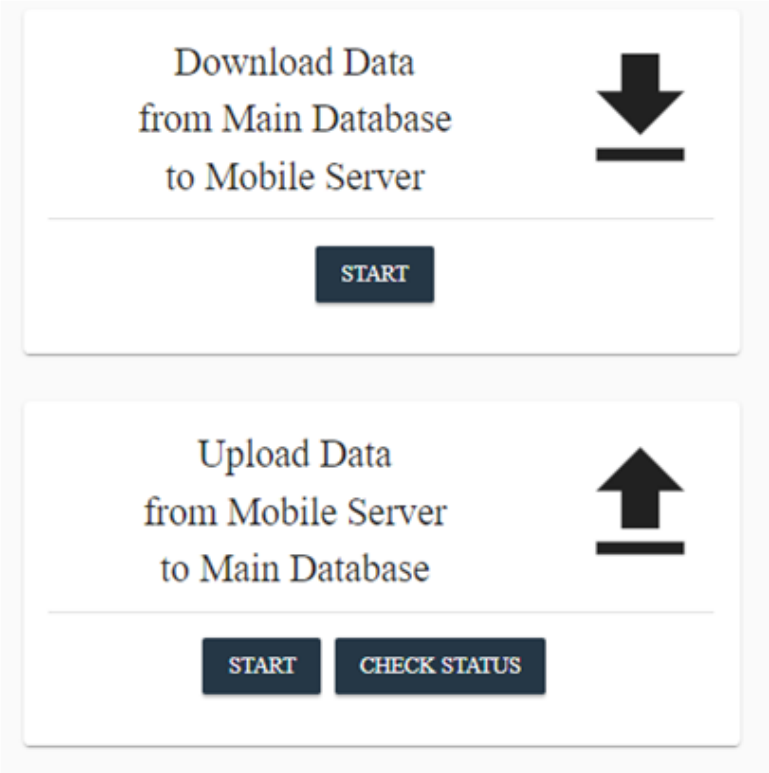
Enter your **username** and **password** in the respective fields.

7. Click **Log In**. The main menu of the On-Site application appears, as shown in the following figure.



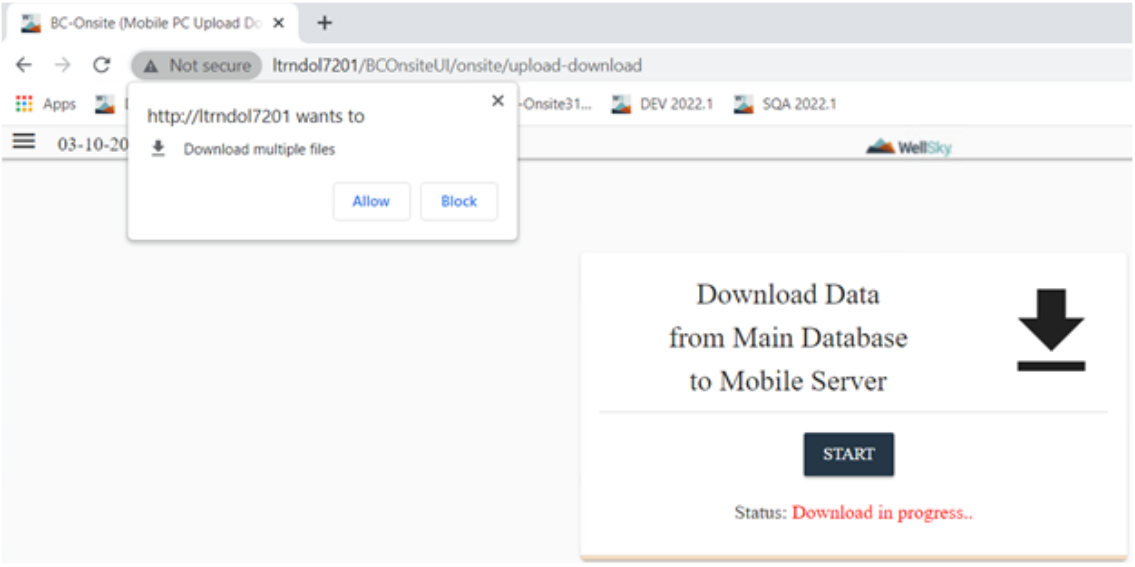
The Mobile PC Upload Download option appears at the bottom of the menu.

8. Click the **Mobile PC Upload Download** option to open it.



The screen shows two separate boxes, one for Upload and one for Download.

9. Click **START** in the *Download Data from Main Database to Mobile Server* box.
10. The first time you download data, you will be prompted whether or not to download multiple files, as shown in the following figure.

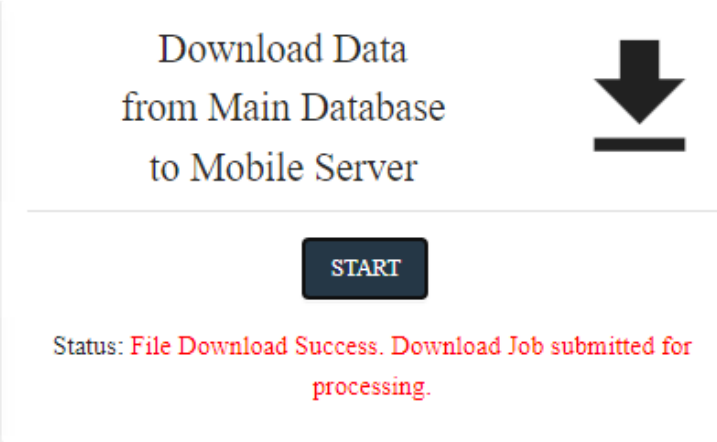


Click **Allow**.

WellSky Corporation
Proprietary and confidential

NOTE – The download time depends on your network bandwidth and the size of the files you are downloading.

When the download is complete, the following status message displays, indicating the download completed successfully.



11. Close the application and check the log files located at D:\lifetrak\replication\log to make sure that no errors occurred.

NOTE – You need to wait approximately 10-15 minutes before accessing the log files, based on the size of the data downloaded.

NOTE – Download is complete when all log files are updated with the current system date and time. Open the **import_error.log** file and verify no errors occurred during download. You should only see the following message: "*Download process Completed, Please check if any errors below*".

Recommendations

WellSky recommends the following:

- Downloads from the application server to laptops should be performed during non-peak times.
- To ensure that all current data is included in the cron job data file, make sure that all daily collections/deferral updates are committed to the database before the cron job's scheduled run time.
- To make sure that the information available at your mobile site is the most current, perform the download at the latest possible time before a drive. If you are uncertain whether or not the current data has been downloaded, perform a download.
- At download, a current copy of the Donor database is placed on a laptop server, enabling mobile drive staff to use existing donor data and collect new donor data for a drive without accessing the network. The download is date and time-stamped in the Donor database.

6.2 Mobile PC Data Extract Process in PUSH Method

6.2.1 Introduction

This feature provides the capability of extracting the data from the local database table to the corresponding text data files. The data is delimited by the configurable delimiter (|). User's can extract the data at any time during the process. Whatever point the data was captured into the table, the same amount of data will be extracted.

NOTE – This section describes the extraction steps for the Mobile Donor Upload/Download PC's database. The Mobile Donor Collections Lookup PC's database is for lookup only and these steps do not apply to that configuration.

6.2.2 Process

- Every new transaction or data modification is monitored by the database trigger and those transactions are recorded in the DOWNLOAD_MONITOR table.
- Regardless of the amount of data in the table, whatever data is marked in the monitoring table is the only data that will be extracted.

6.2.3 Extract Data to File

Before uploading the data from the Mobile PC to the Blood Centers production database, you must extract the data from the Mobile PC database.

NOTE – Perform these steps on the Mobile PC while it is offline from the Blood Centers production database.

One-time Setup

Before performing the extract process, verify the following items have been set correctly in the Blood Centers production database.

- The APIDR master code must be populated to specify the directory where the extracted files will be picked up by the upload process.

  NOTE – If you followed the instructions in this guide, the location for the extract files is c:\bcmpc\extract.

- The EXTDLT SYSPARM must be set to provide the delimiter used in the extract file for Mobile PC Donor Registration.
- The ENCRPT_REQ SYSPARM must be set to 'Y' to encrypt the data in the extract file; otherwise, the data in the extract file will not be encrypted.

WellSky Corporation
Proprietary and confidential

Push Extract/Script Extract

A facility can have data extracted automatically in the background at the Shipment Close screen. The SHIP_EXT parameter is used to configure this feature. Y turns the auto-extract feature ON, and N turns the auto-extract feature OFF.

- Automatic Extract ON: All the shipment units' related data is extracted in the background at the Shipment Close screen without any user intervention.

  NOTE – If auto-extract is turned ON, only one manual extract has to be performed at the end of the drive. This will extract all of the units or deferrals that are not extracted as part of the auto-extract.

  NOTE – WellSky recommends that manual extract be performed only at the close of the drive. Performing a manual extract during the drive will extract all of the data up until that point, and it may result in the auto-extract returning empty data.

- Automatic Extract OFF: The user must run manual extract in order to extract the data (i.e., run the script for SCRIPT method of extract OR navigate to the EXTRACT screen for the PUSH method of extract).

6.2.4 Set Up the Extract Queue Process in Windows Task Scheduler

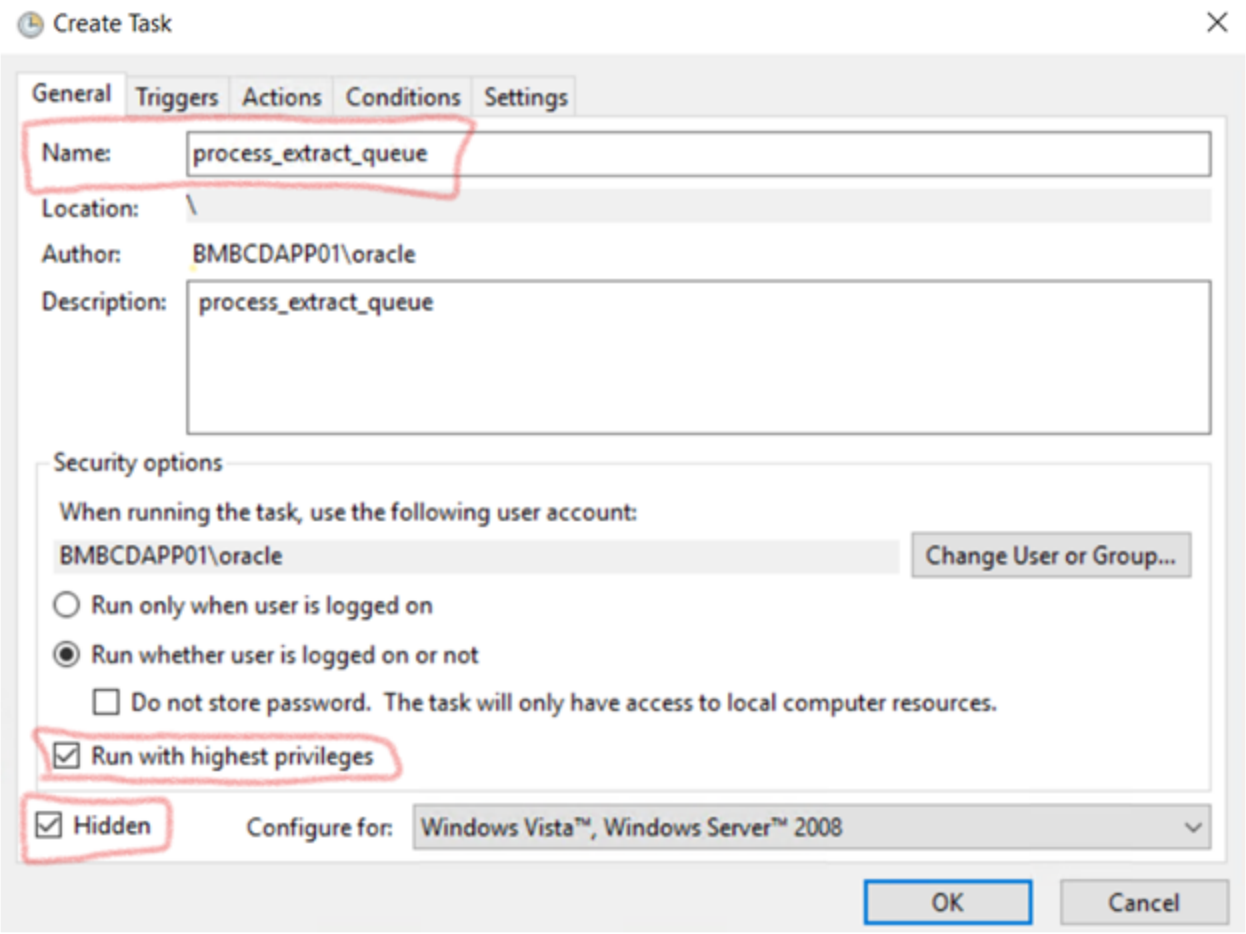
NOTE – This setup procedure needs to be performed only once.

This procedure explains how to set up the C:\bcmpc\replication\process_extract_queue.bat script to run every 5 minutes in Windows Task Scheduler.

1. Open the Task Scheduler application.

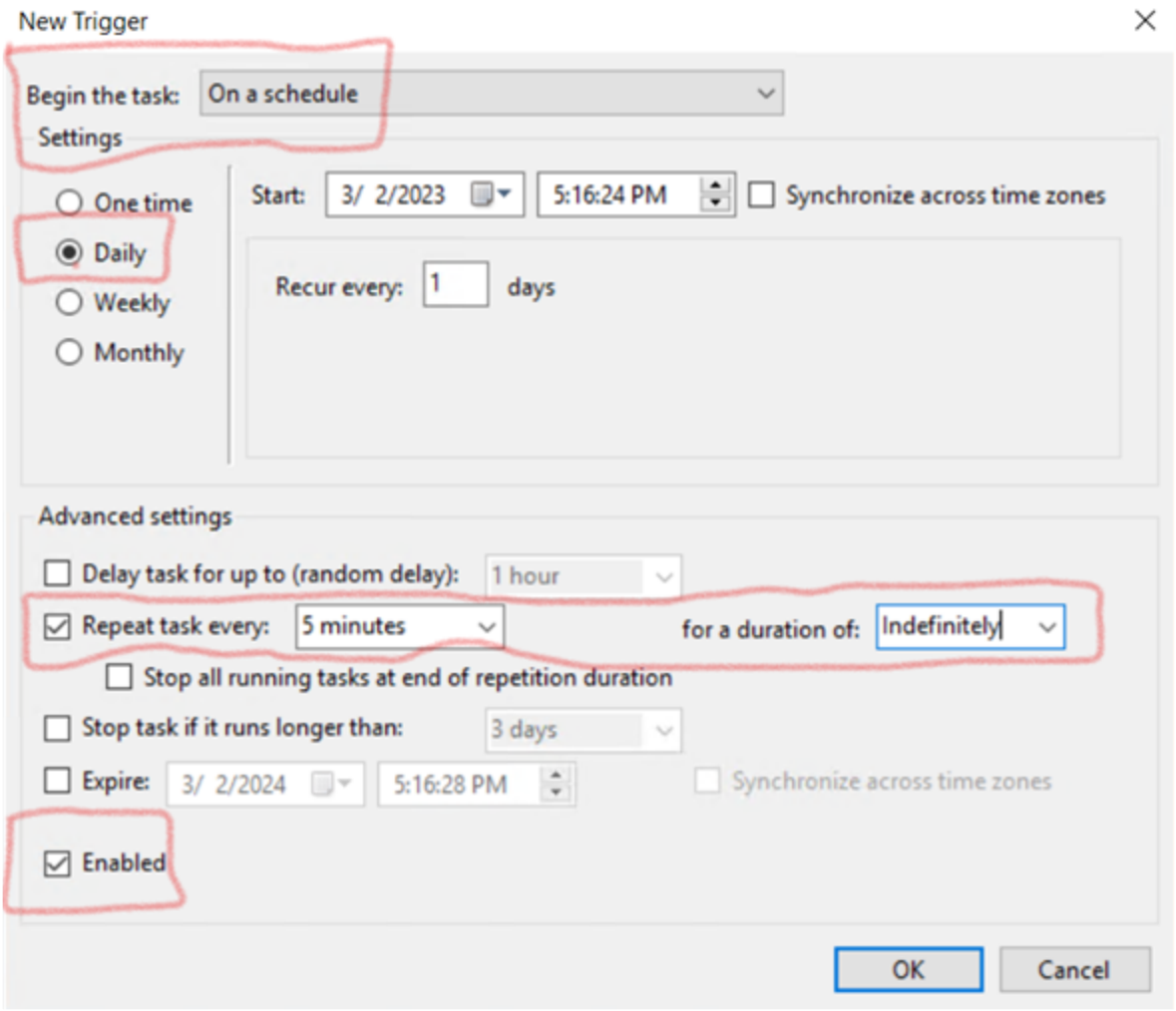
   NOTE – You can type **`Task Scheduler`** in the Search box.

2. Click Create Task to open the Create Task window.
3. Name the task **`process_extract_queue`**.
4. Select the Run with highest privileges option.
5. Select the Hidden option.

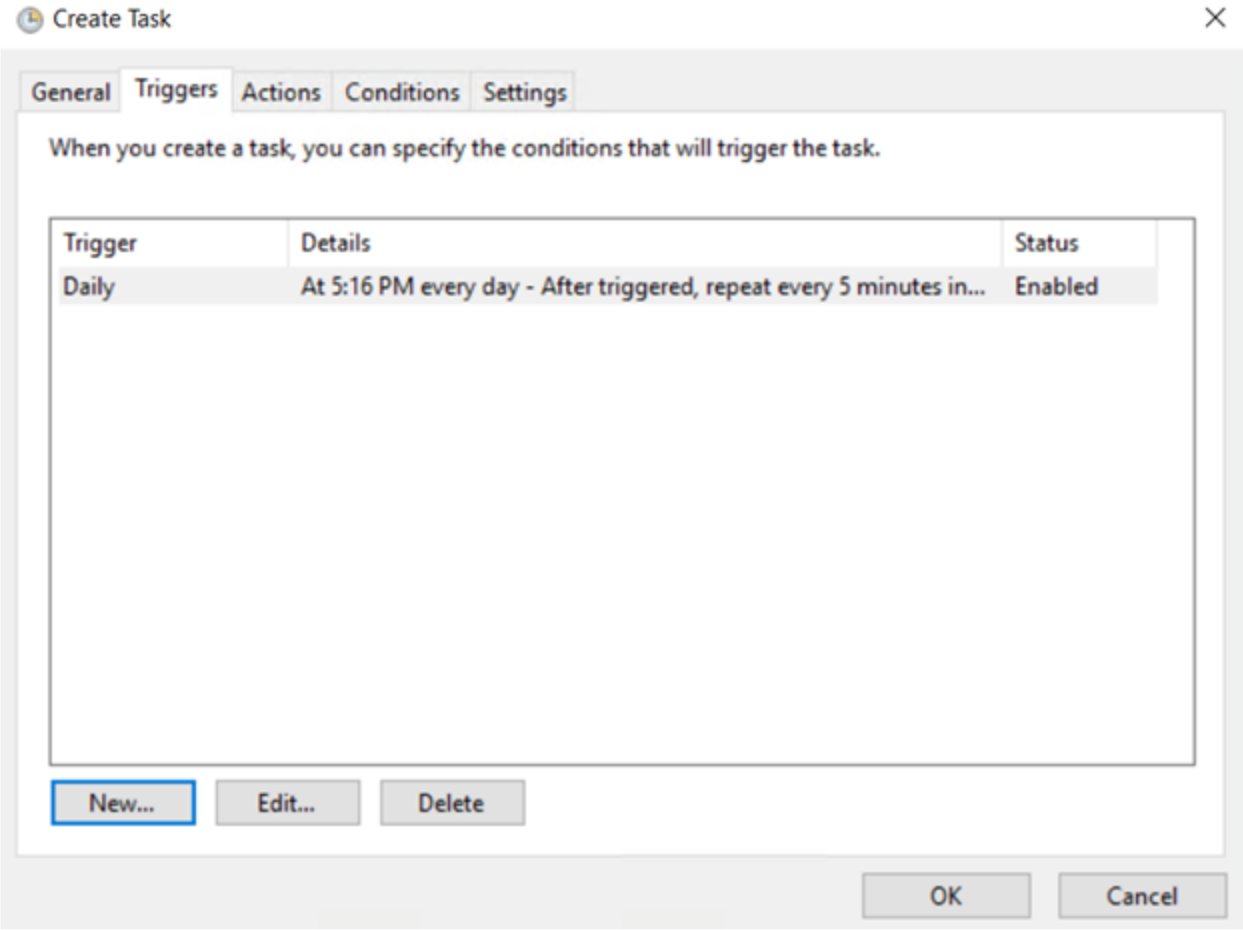


6. Click the **Triggers** tab.
7. Click **New**.
8. On the New Trigger screen, select **On a schedule** from the **Begin the task:** drop-down menu.
9. Under Settings, select the **Daily** option.
10. Under Advanced Settings, select the **Repeat task every:** option.
    a. Select **5 minutes** from the first drop-down menu.
    b. Select **Indefinitely** from the **for a duration of:** drop-down menu.

WellSky Corporation
Proprietary and confidential


New Trigger
Begin the task: On a schedule
Settings
One time
Daily
Weekly
Monthly
Start: 3/ 2/2023 5:16:24 PM
Synchronize across time zones
Recur every: 1 days
Advanced settings
Delay task for up to (random delay): 1 hour
Repeat task every: 5 minutes
for a duration of: Indefinitely
Stop all running tasks at end of repetition duration
Stop task if it runs longer than: 3 days
Expire: 3/ 2/2024 5:16:28 PM
Synchronize across time zones
Enabled
OK
Cancel

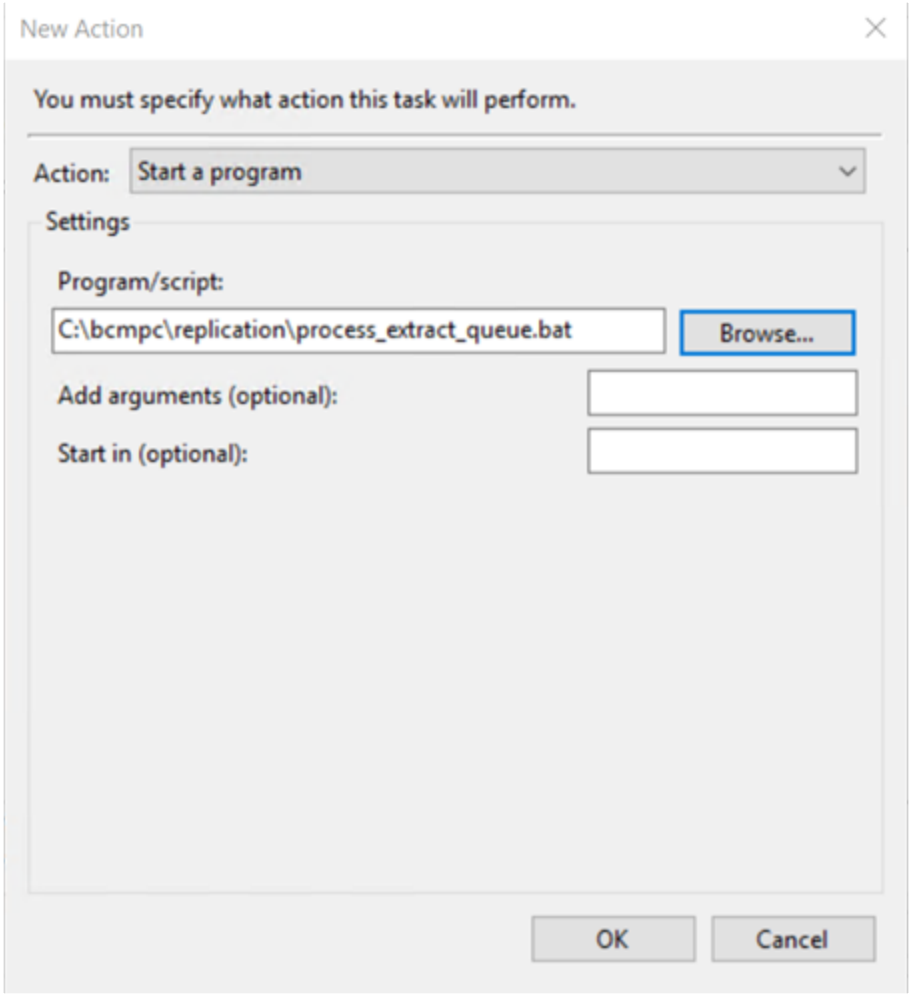
11. Click **OK**. The new trigger displays on the Trigger tab, as shown in the following figure.



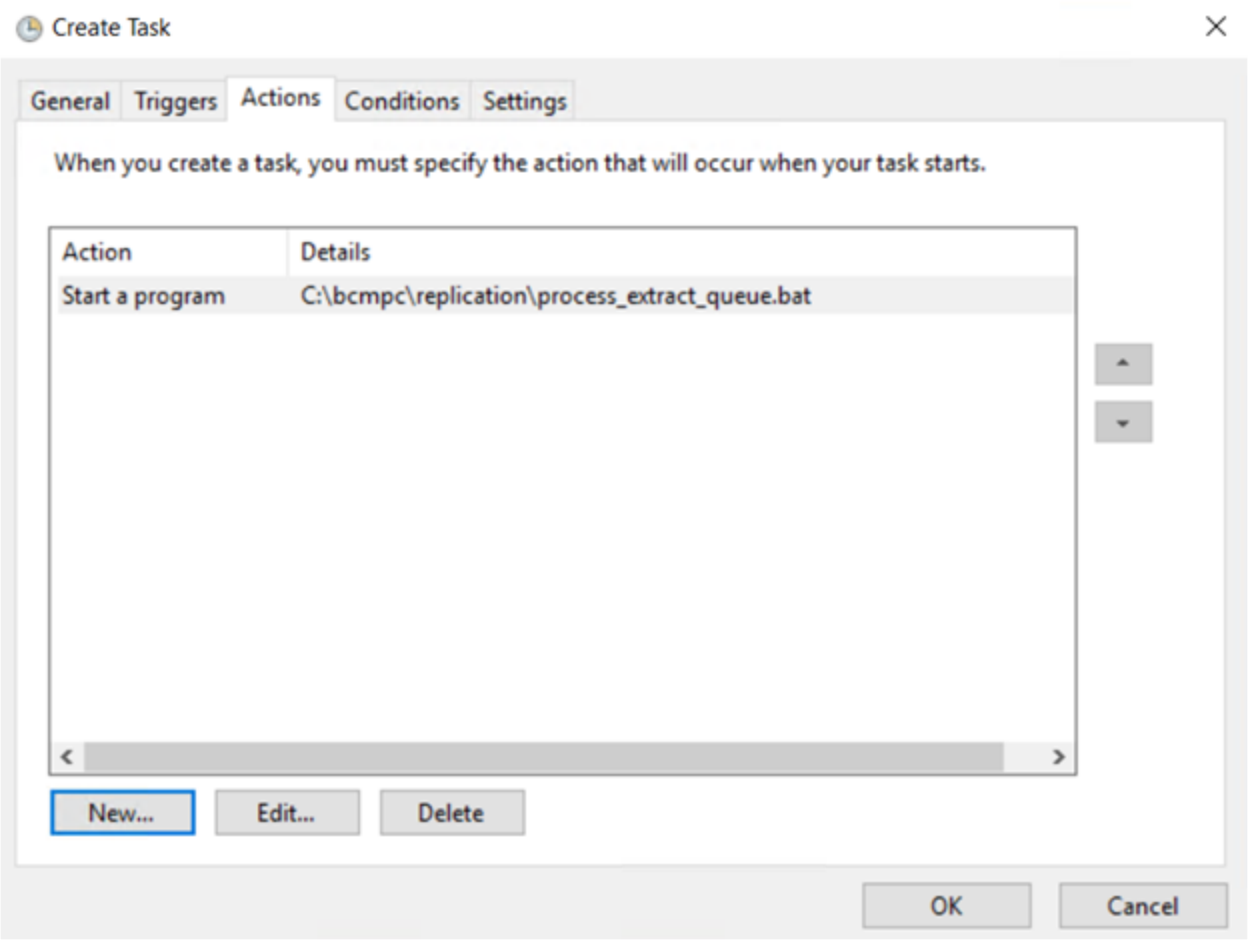
12. Click the **Actions** tab.
13. Click **New...**.
14. On the New Action screen, click **Browse...**.

WellSky Corporation
Proprietary and confidential

15. Locate the **process_extract_queue.bat** script, which will be here: C:\bcmpc\replication.



16. Click **OK**. The new action displays on the Actions tab, as shown in the following figure.



17. Click **OK** to close the Create Task window.

WellSky Corporation
Proprietary and confidential

6.3 Create Data Extract File

The data extraction process can be repeated as often as desired. The extract file that is created is stamped with the date and time of file creation.

NOTE – If the data is extracted in the middle of the drive, perform the extract, and use a flash drive to store the extracted file.

NOTE – For additional information regarding this screen, see the Blood Centers User Guide - Extract Data to File (EXTRACT) screen.

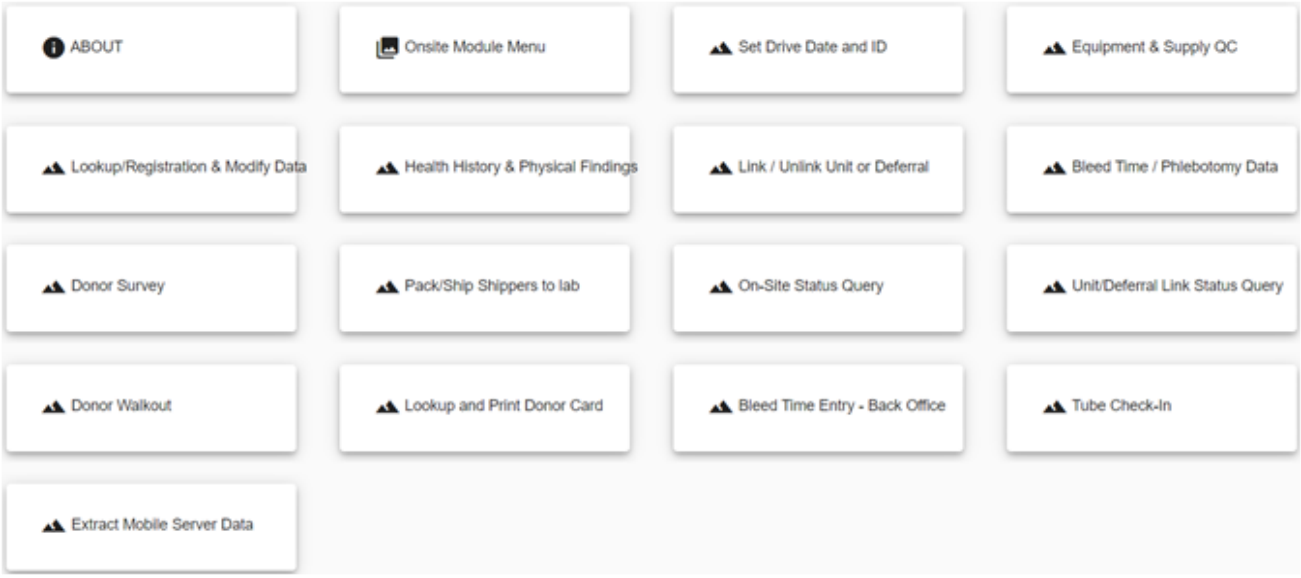
1. Launch the BC Onsite Application using the following URL: http://HRUNIX/BCOnsiteUI/index.html.

   The Login screen appears.



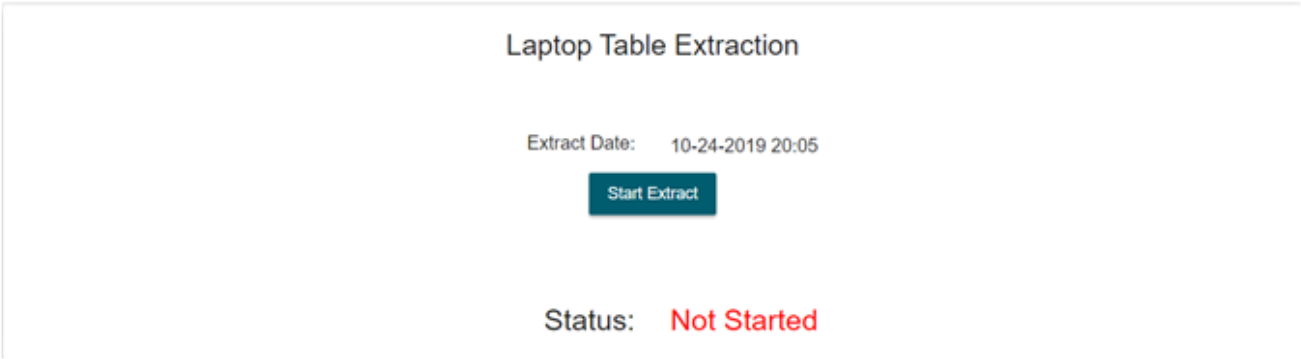
2. Enter your username and password in the respective fields, and click **Login**.
3. The Onsite Module Main Menu screen appears.





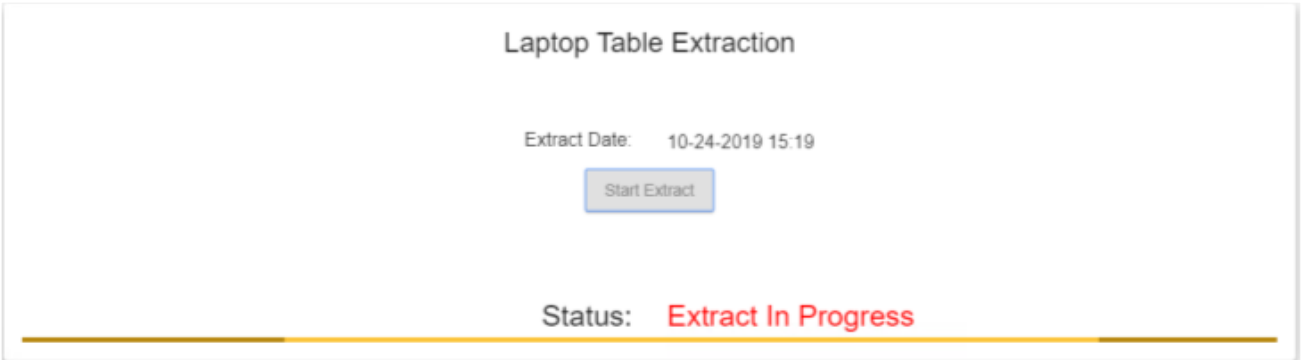
Click **Extract Mobile Server Data**.

4. The Laptop Table Extraction screen appears.



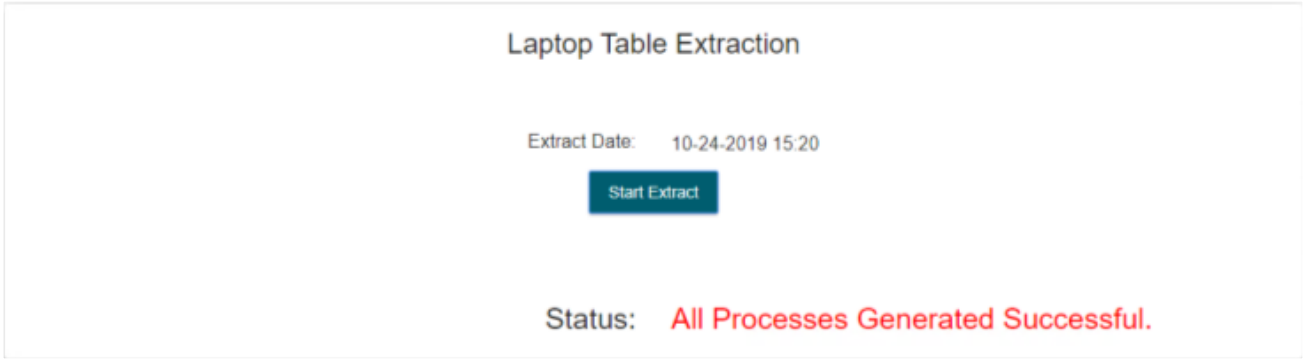
Click **Start Extract**.

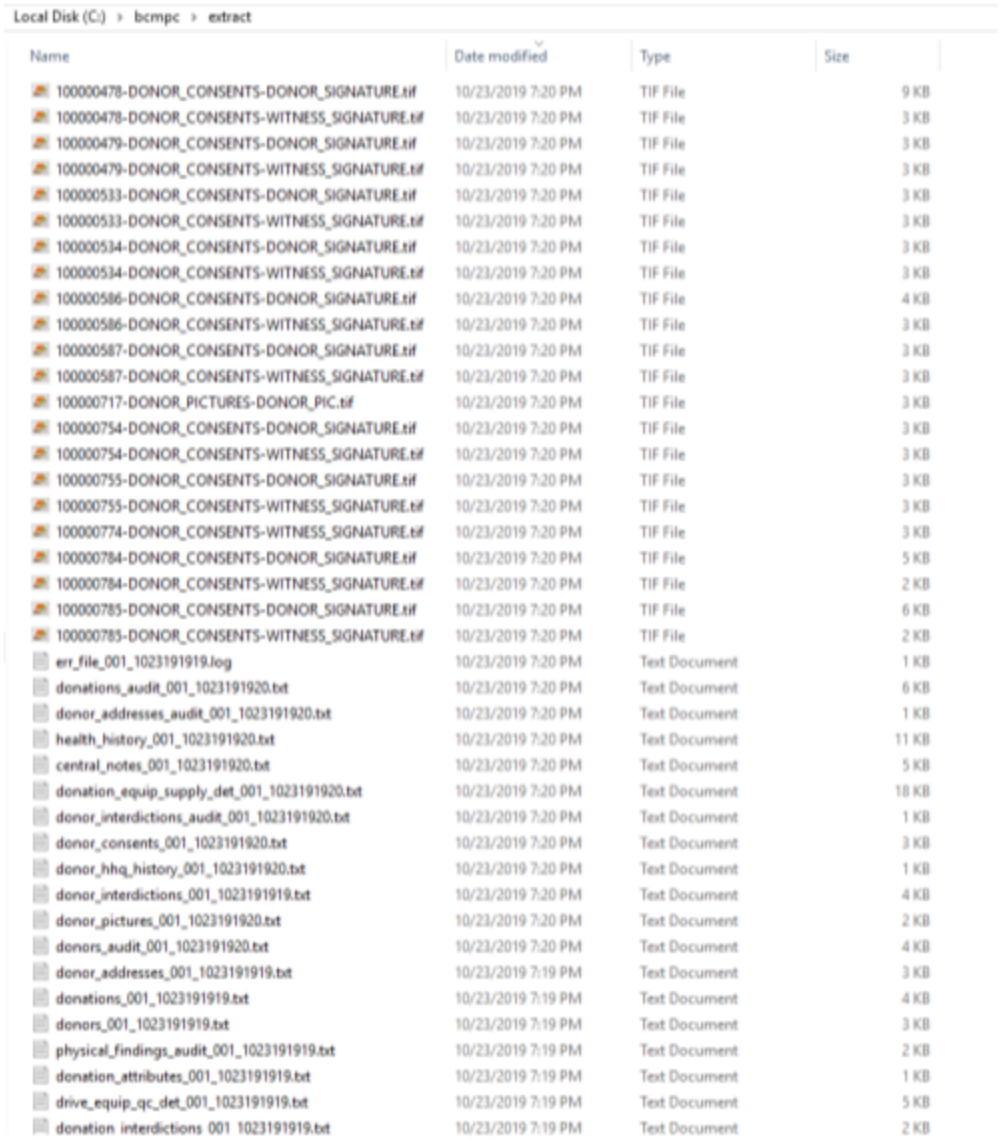
The process will generate an extract file and save it to the c:\bcmpc\extract directory on the laptop with the date and time as part of the file name.



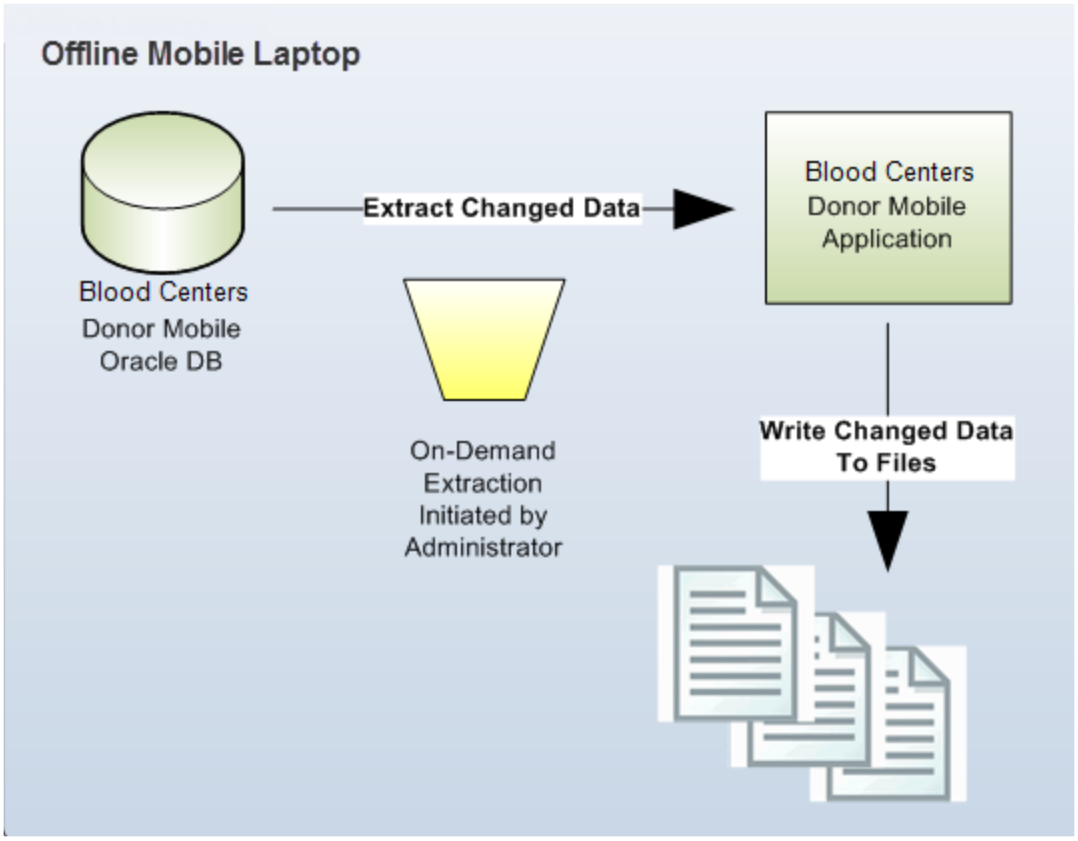
WellSky Corporation
Proprietary and confidential

When the process is complete, a status message appears.





Extract Process Illustration:



6.4 Upload Data to the Main Server from the Mobile PC Server

6.4.1 Set Up the Upload Archive Process in Windows Task Scheduler

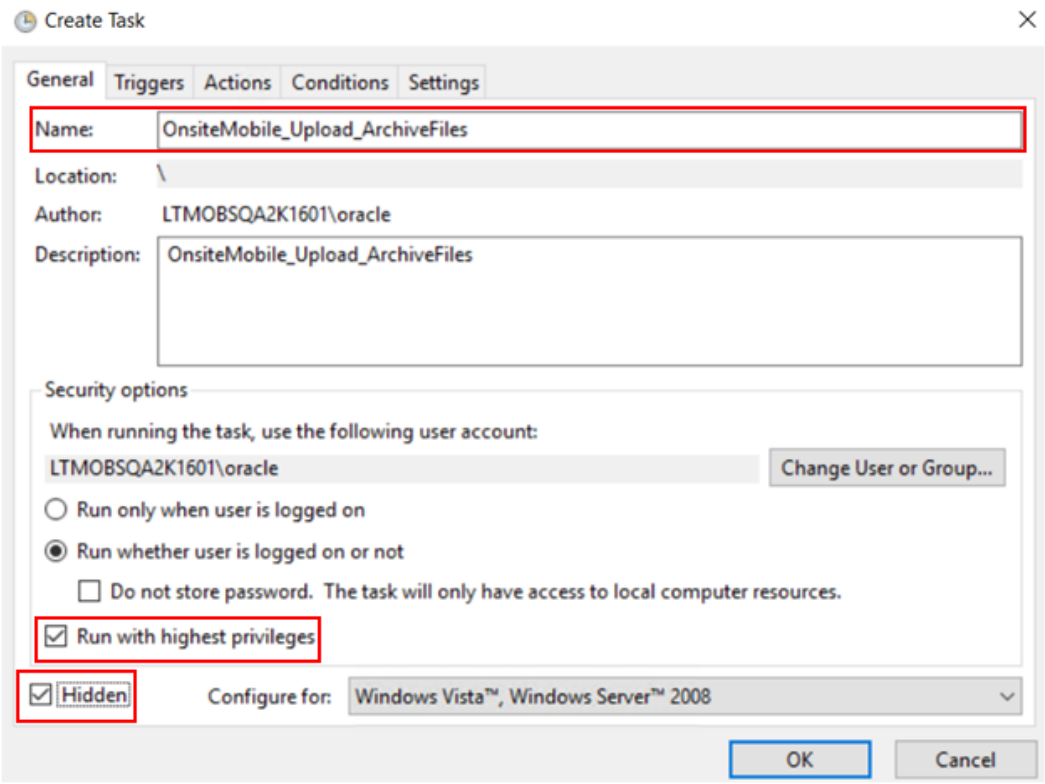
NOTE – This setup needs to be performed only once.

This procedure explains how to set up the C:\bcmpc\schedular_jobs\OnsiteMobile_Upload_ArchiveFiles.bat script to run every five minutes in Task Scheduler.

1. Open the Task Scheduler application.

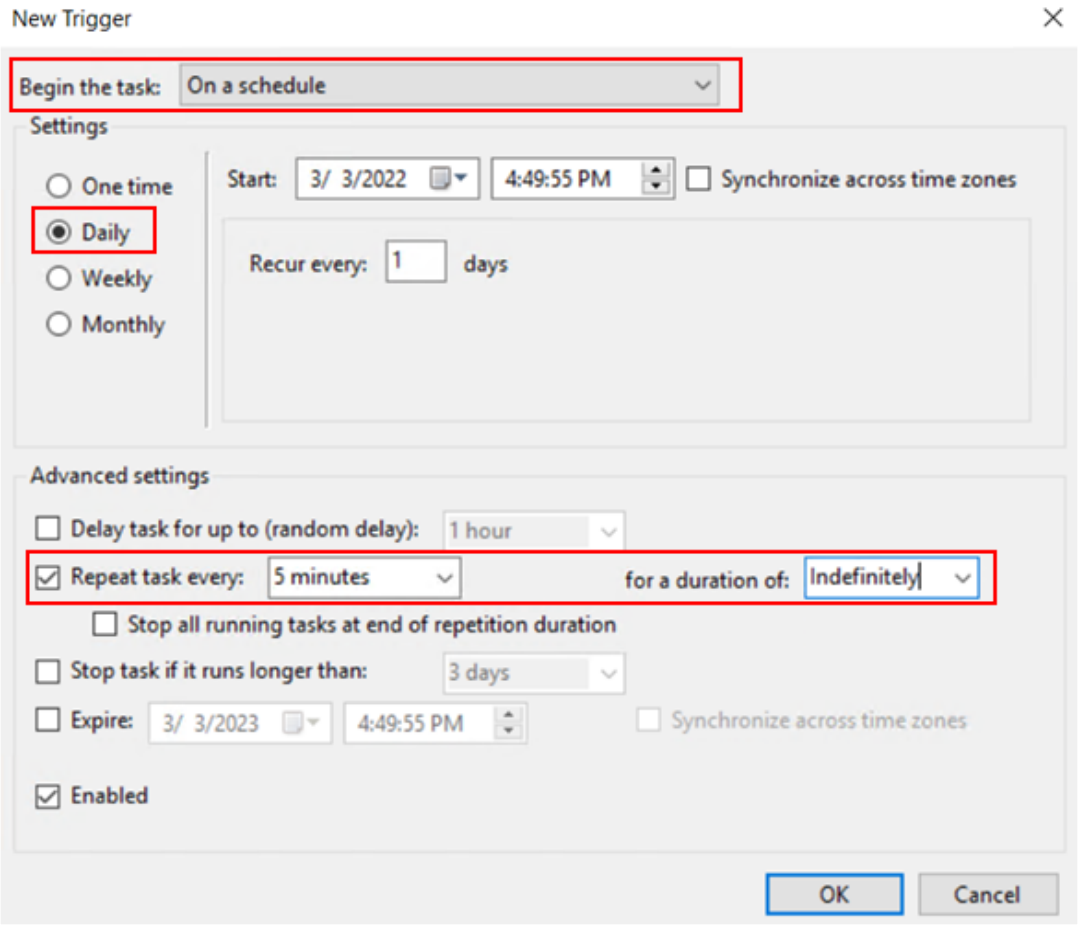
   NOTE – You can type `Task Scheduler` in the search box.

2. Click **Create Task** to open the Create Task window.
3. Name the task **`OnsiteMobile_Upload_ArchiveFiles`**.
4. Select the **Run with highest privileges** option.
5. Select the **Hidden** option.

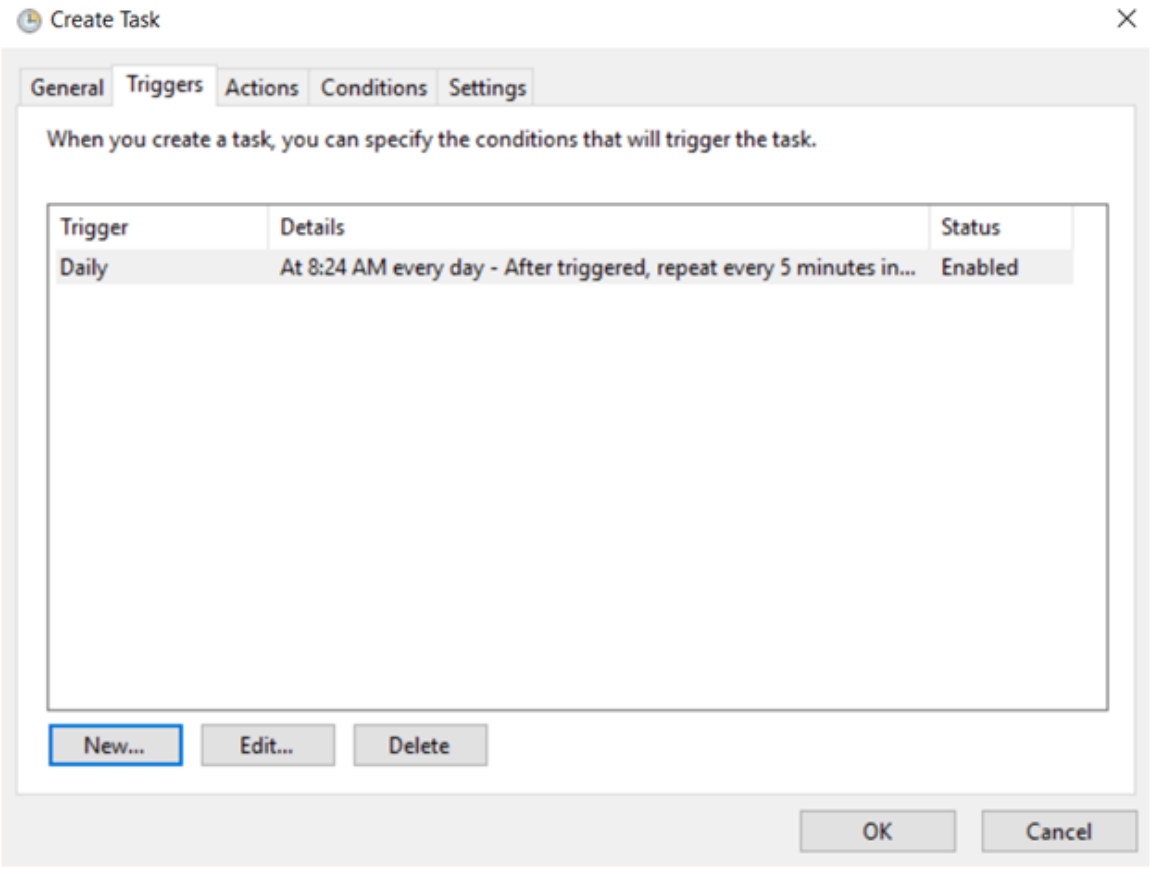


6. Click the **Triggers** tab.
7. Click **New...**.
8. On the New Trigger screen, select **On a schedule** from the *Begin the task* drop-down menu.
9. Under Settings, select the **Daily** option.
10. Under Advanced Settings, select the **Repeat task every:** option.
    a. Select **5 minutes** from the drop-down menu.
    b. Select **Indefinitely** from the *for a duration of:* drop-down menu.

WellSky Corporation
Proprietary and confidential

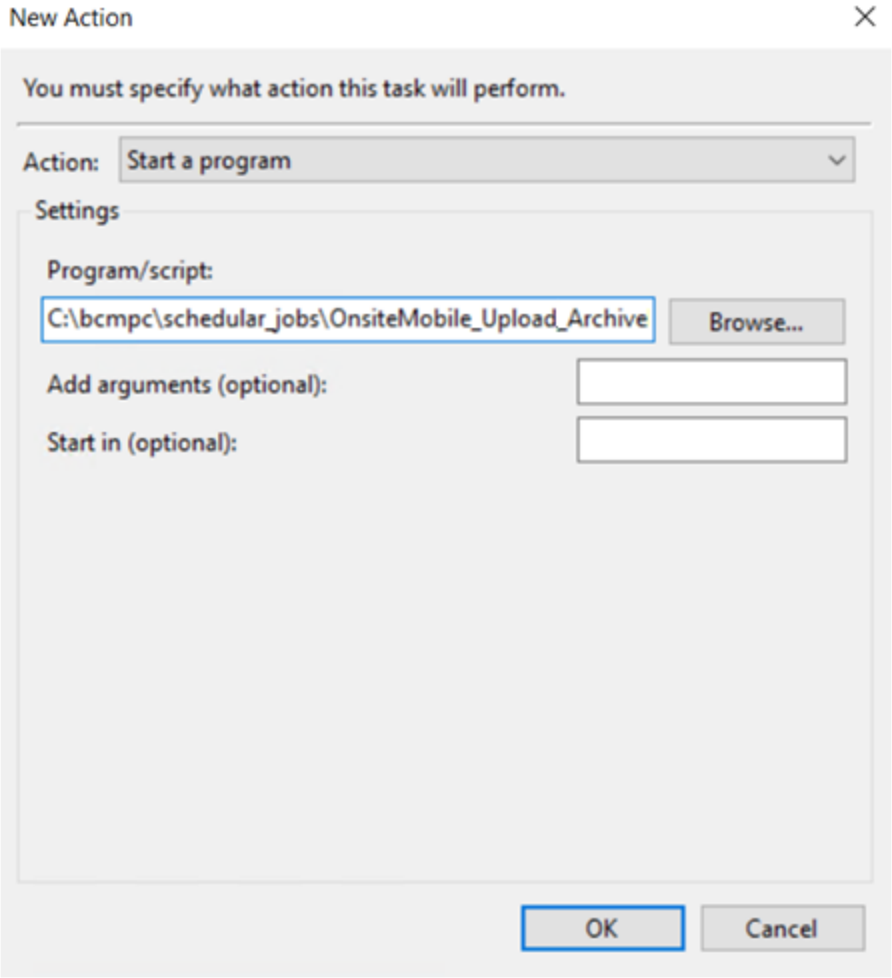

New Trigger
Begin the task: On a schedule
Settings
One time
Daily
Weekly
Monthly
Start: 3/ 3/2022 4:49:55 PM
Synchronize across time zones
Recur every: 1 days
Advanced settings
Delay task for up to (random delay): 1 hour
Repeat task every: 5 minutes
for a duration of: Indefinitely
Stop all running tasks at end of repetition duration
Stop task if it runs longer than: 3 days
Expire: 3/ 3/2023 4:49:55 PM
Synchronize across time zones
Enabled
OK
Cancel

11. Click **OK**. The new trigger displays on the Trigger tab, as shown in the following figure.

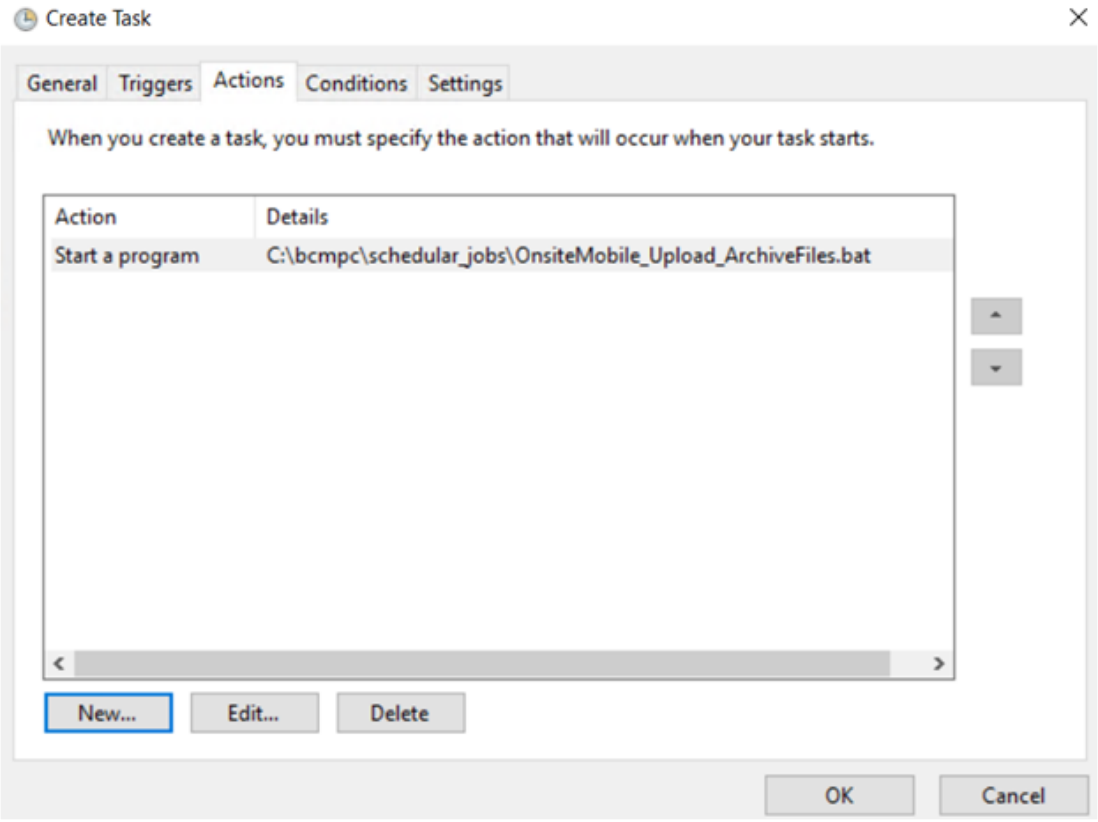


12. Click the **Actions** tab.
13. Click **New...**.

14. On the New Action screen, click **Browse...**and locate the OnsiteMobile_Upload_ArchiveFiles.bat script, which will be here: C:\bcmpc\schedular_jobs\.



15. Click **OK**. The new action displays on the Actions tab, as shown in the following figure.



16. Click **OK** to close the Create Task window.

6.4.2 Upload Data to the Server from the Mobile PC

The following steps are the preferred method of uploading data from the Mobile laptop to the Blood Centers database. However, if you prefer to use the older file copy method, refer to Section 6.5, Upload Data to Server.

NOTE – After collecting donor data during the mobile drive, you must use the offline menu function to generate the files and extract the data that is then stored on the hard drive of the mobile laptop.

For more information on extracting donor data, refer to the 'Extract Data to a File' (EXTRACTD) section in the *WellSky® Blood Centers User Guide* for more information on extracting donor data.

NOTE – When an extract file is uploaded, the production database records the file uploaded in the Laptop_ID_Details table.

Before an extract file is uploaded, the process checks the Laptop_ID_Details table to see if the extract file has already been uploaded. If it has, that file is skipped and not processed a second time.

WellSky Corporation
Proprietary and confidential

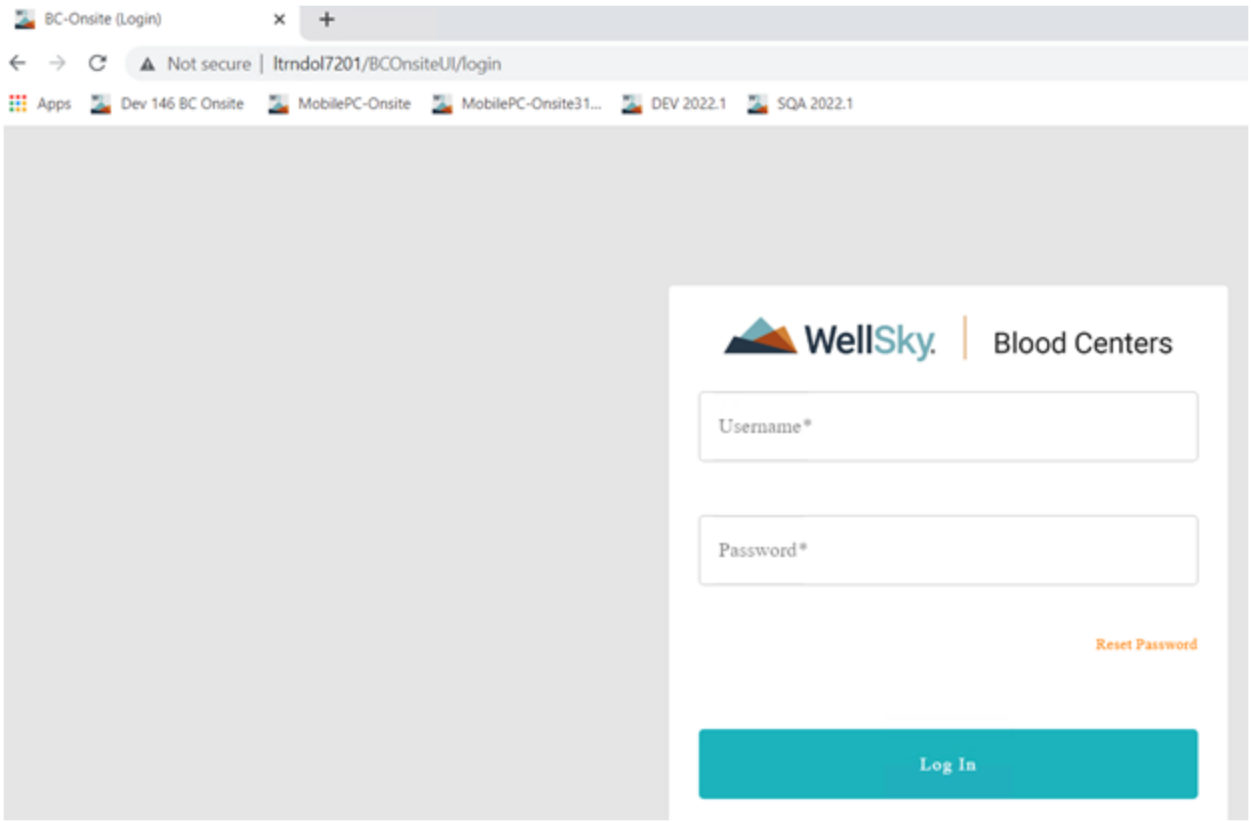
Upload the data from the Mobile laptop to the Blood Centers database by connecting to the Production On-Site application on the Mobile laptop.

1. On the Mobile laptop, open Google Chrome.
2. Enter the applicable URL:

   Without SSL: `http://ltrndol7201/BCOnsiteUI/index.html`
   With SSL: `https://ltrndol7201/BCOnsiteUI/index.html`

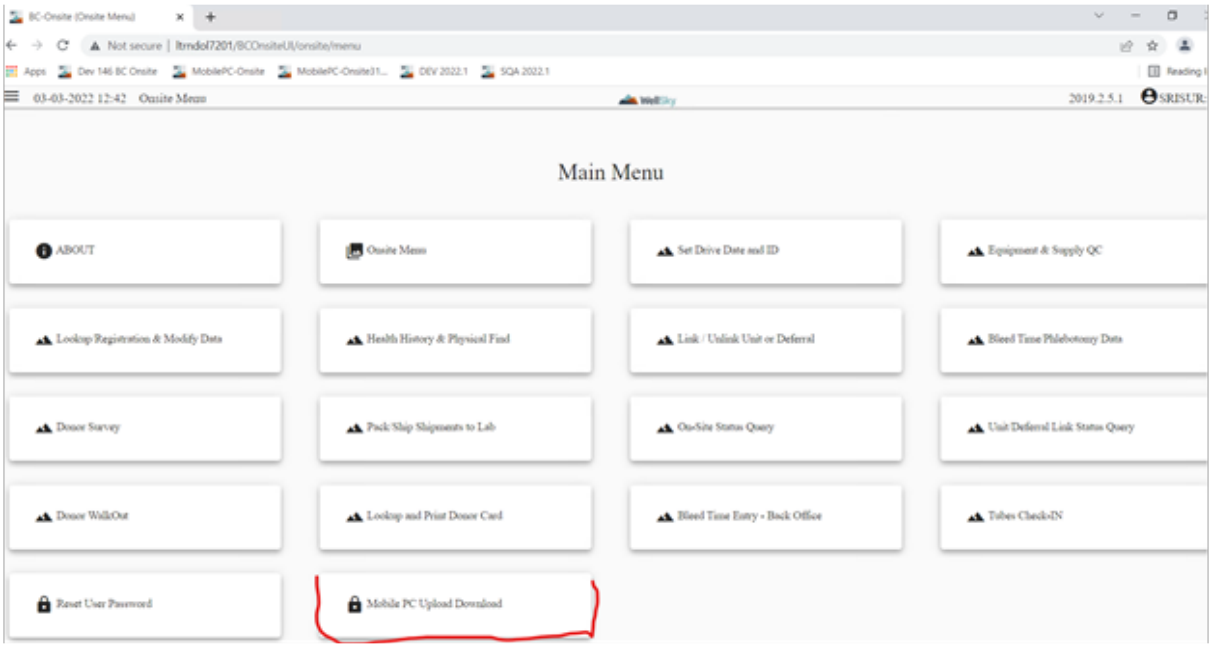
   NOTE – Replace ltrndol7201 with the hostname/IP address of your Blood Centers production On-Site application server.

3. The WellSky Blood Centers login window appears, as shown in the following figure.



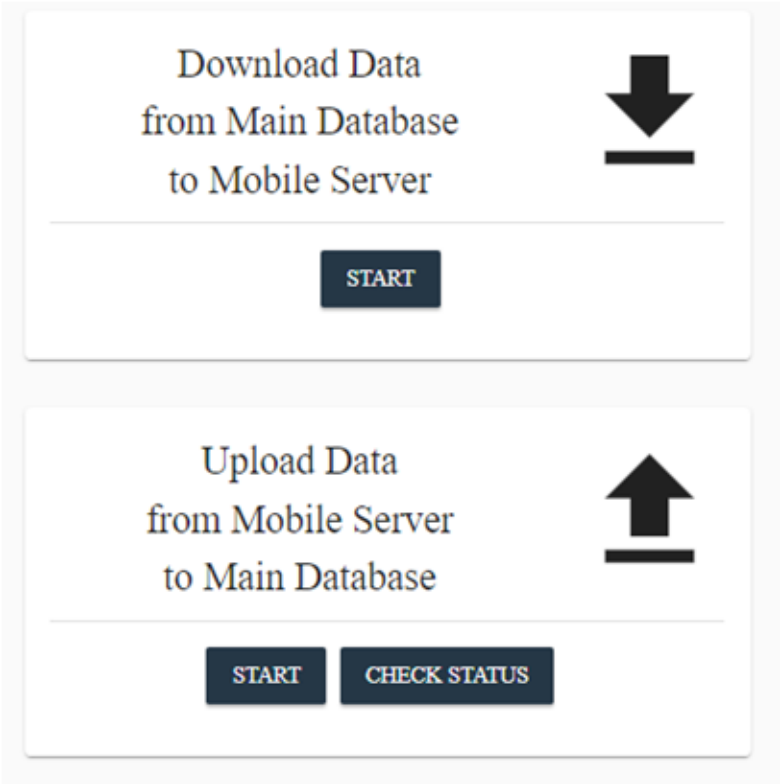
   Enter your **username** and **password** in the respective fields.

4. Click Log In. The main menu of the On-Site application appears, as shown in the following figure.



The Mobile PC Upload Download option appears at the bottom of the menu.

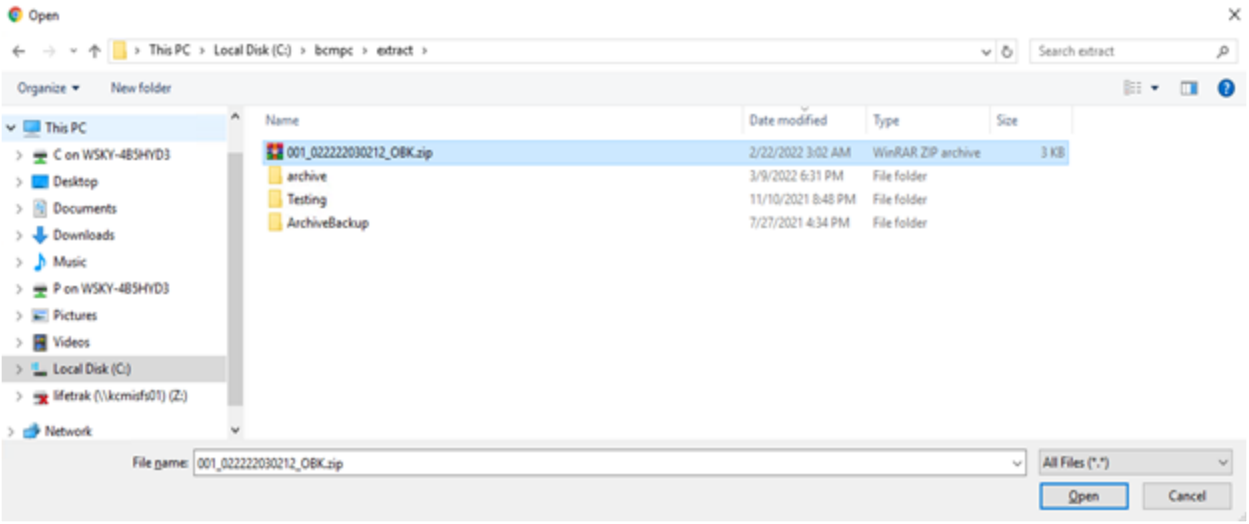
5. Click the Mobile PC Upload Download option to open it.



The screen shows two separate boxes, one for Upload and one for Download.

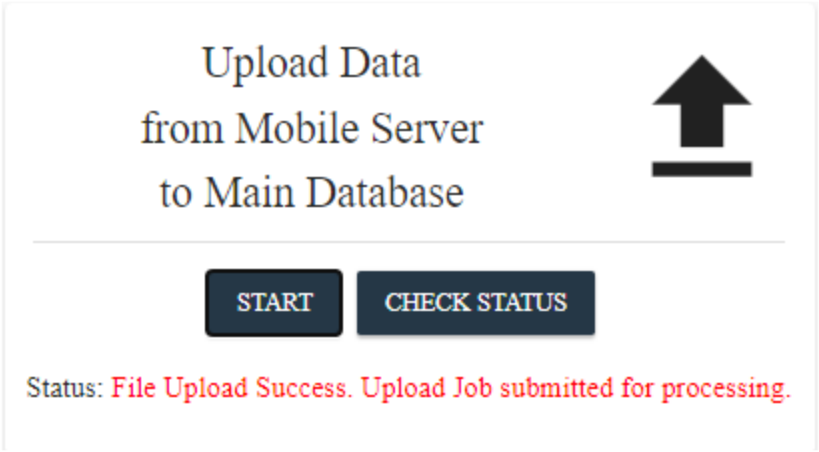
WellSky Corporation
Proprietary and confidential

6. Click **START** in the *Upload Data from Mobile Server to Main Database* box. The File Open screen displays, as shown in the following figure.



7. Locate and select the .zip file that was extracted.
8. Click **Open** to start the upload.

   When the upload is complete, the following message displays, indicating the upload process completed successfully.



   The Donor production database server runs the MPC_MASTER_UPLOAD.sh cron job at established intervals (WellSky recommends once per day) to upload the data from the mobile database to the master laptop.

NOTE – For the SFTP method of uploading, the Donor production database server needs the MPC_API_UPLOAD.sh cron job in addition to the MPC_MASTER_UPLOAD.sh cron job.

NOTE – An error message displays if the system cannot complete the upload.

NOTE – Perform an API Log Query to verify the upload completed without errors. Refer to the 'Blood Centers API Log Query (APILOG)' section in the *WellSky® Blood Centers User Guide* for more information.

The Check Status Button

- The Check Status button in the Upload Data from Mobile Server to Main Database box displays the status of the uploads for the current day and the day prior. The online ID is displayed in the status list after the .zip file is selected and uploaded successfully.
- In Queue Status – The extract file is displayed with a status of 'In Queue' after the .zip file is uploaded, but before the cron job is completed for the file.



List of Upload Jobs

Refresh Close

Laptop Id	Drive Id	Extract Seq	Extract Time	Upload Start Time	Upload End Time	Status
001	DRV0170103	3	03-10-2022 14:18	03-10-2022 14:31		In Queue
001	DRV0170103	2	03-10-2022 14:07	03-10-2022 14:28	03-10-2022 14:31	Completed
001	DRV0170103	1	03-10-2022 13:55	03-10-2022 14:30	03-10-2022 14:31	Completed
001	DRV0169800	2	03-08-2022 11:50	03-08-2022 11:50	03-08-2022 12:01	Completed
001	DRV0169800	1	03-08-2022 11:45	03-08-2022 11:45	03-08-2022 11:56	Completed

Upload Id: 001_030922181822_OBK

WellSky Corporation
Proprietary and confidential

- **Completed Status –** The extract file is displayed with a status of 'Completed' after the .zip file is uploaded, and the data is in the main database after the mobile upload cron job is completed.



List of Upload Jobs

Refresh Close

Laptop Id	Drive Id	Extract Seq	Extract Time	Upload Start Time	Upload End Time	Status
001	DRV0170103	3	03-10-2022 14:18	03-10-2022 14:31	03-10-2022 14:32	Completed
001	DRV0170103	2	03-10-2022 14:07	03-10-2022 14:28	03-10-2022 14:31	Completed
001	DRV0170103	1	03-10-2022 13:55	03-10-2022 14:30	03-10-2022 14:31	Completed
001	DRV0169800	2	03-08-2022 11:50	03-08-2022 11:50	03-08-2022 12:01	Completed
001	DRV0169800	1	03-08-2022 11:45	03-08-2022 11:45	03-08-2022 11:56	Completed

Upload Id: 001_030922181822_OBK

- If you are uncertain whether or not data has been uploaded from the mobile laptop server to the Blood Centers application server, check to see if there are any .zip files that were extracted in the extract folder (C:\bmcpc\extract). Upload any .zip files not already uploaded to the main database.

This Page Intentionally Left Blank

7

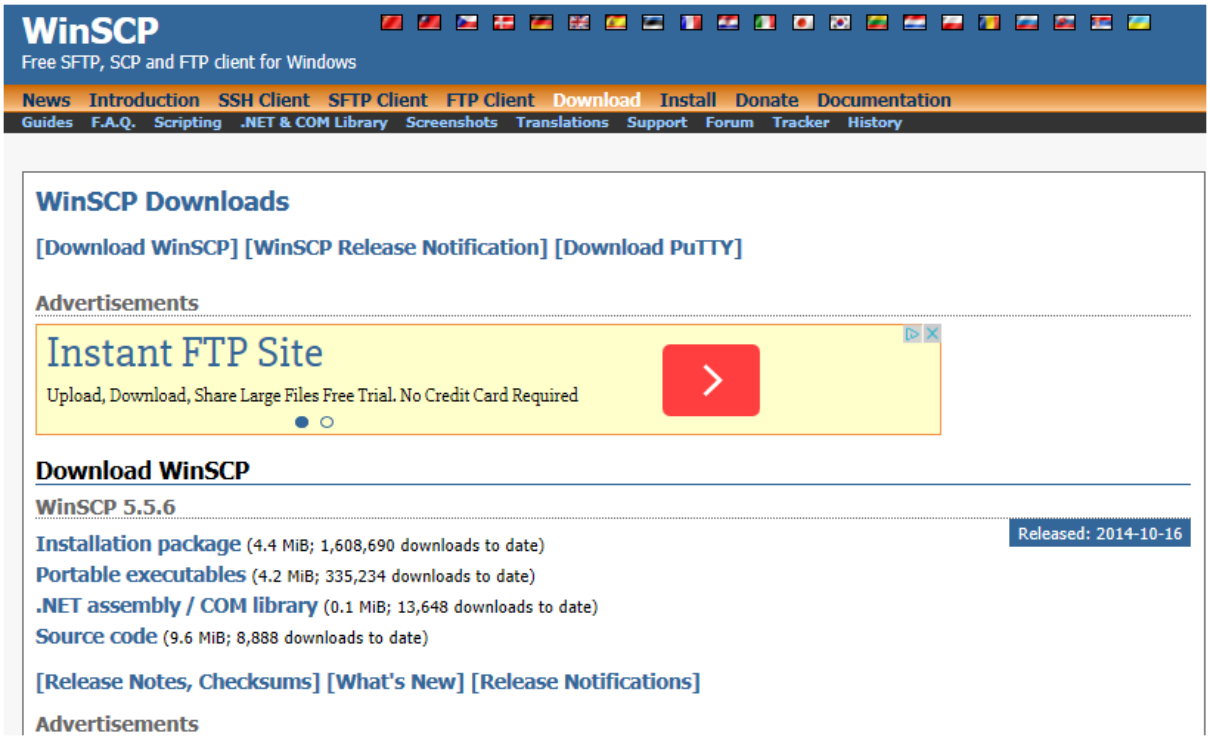
Chapter 7 - Mobile PC Download, Extract, and Upload Process in Script Method

7.1 Mobile PC Download Process in Script Method 120
7.1.1 WinSCP (SFTP) Install Process 120
7.1.2 Automating The Process Through Task Scheduler 131
7.2 Mobile PC Data Extract Process in Script Method 139
7.2.1 Introduction 139
7.2.2 Process 139
7.2.3 Extract Data to File 139
7.2.4 Set Up the Extract Queue Process in Windows Task Scheduler 140
7.3 Extract Data Using Batch Script 146
7.4 Upload Data to Server 147
7.5 Copy the Extracted Files into Database Server - File Copy Method 148

7.1 Mobile PC Download Process in Script Method

7.1.1 WinSCP (SFTP) Install Process

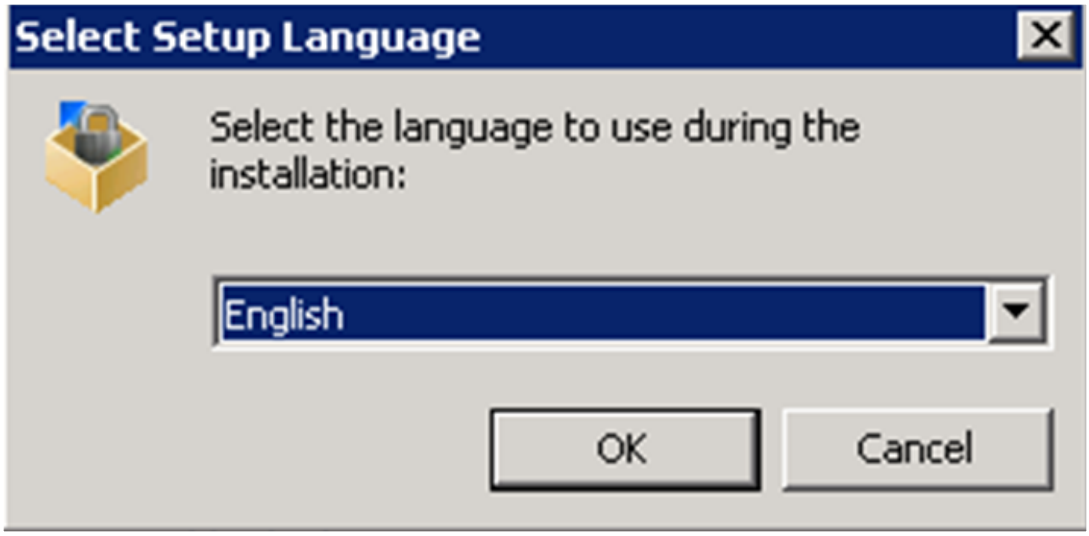
1. If you are downloading the Main Server database via the SFTP method, comment the last two lines of the script listed below, based on the Mobile PC Server Database Version in the Main Database Server, so that it will not execute the **/home/lifetrak/mobile_dmp/load_dump_files_in_database.sh** file.
    - /home/lifetrak/mobile_dmp/export19c.sh
2. Modify the **C:\bcmpc\replication\start_download_dmp_sftp_refresh.txt** file to update the database server information and Oracle password.
3. Using Internet Explorer, enter http://winscp.net/eng/download.php. The WinSCP window opens, as shown:



4. Click **[Download WinSCP]**.
5. Click **Installation Package**. The Run/Save/Cancel window appears at the bottom of the screen, as shown:



6. Click **Run** to begin the installation. The Select Setup Language window opens, as shown:



   a. Leave setting as **English**.

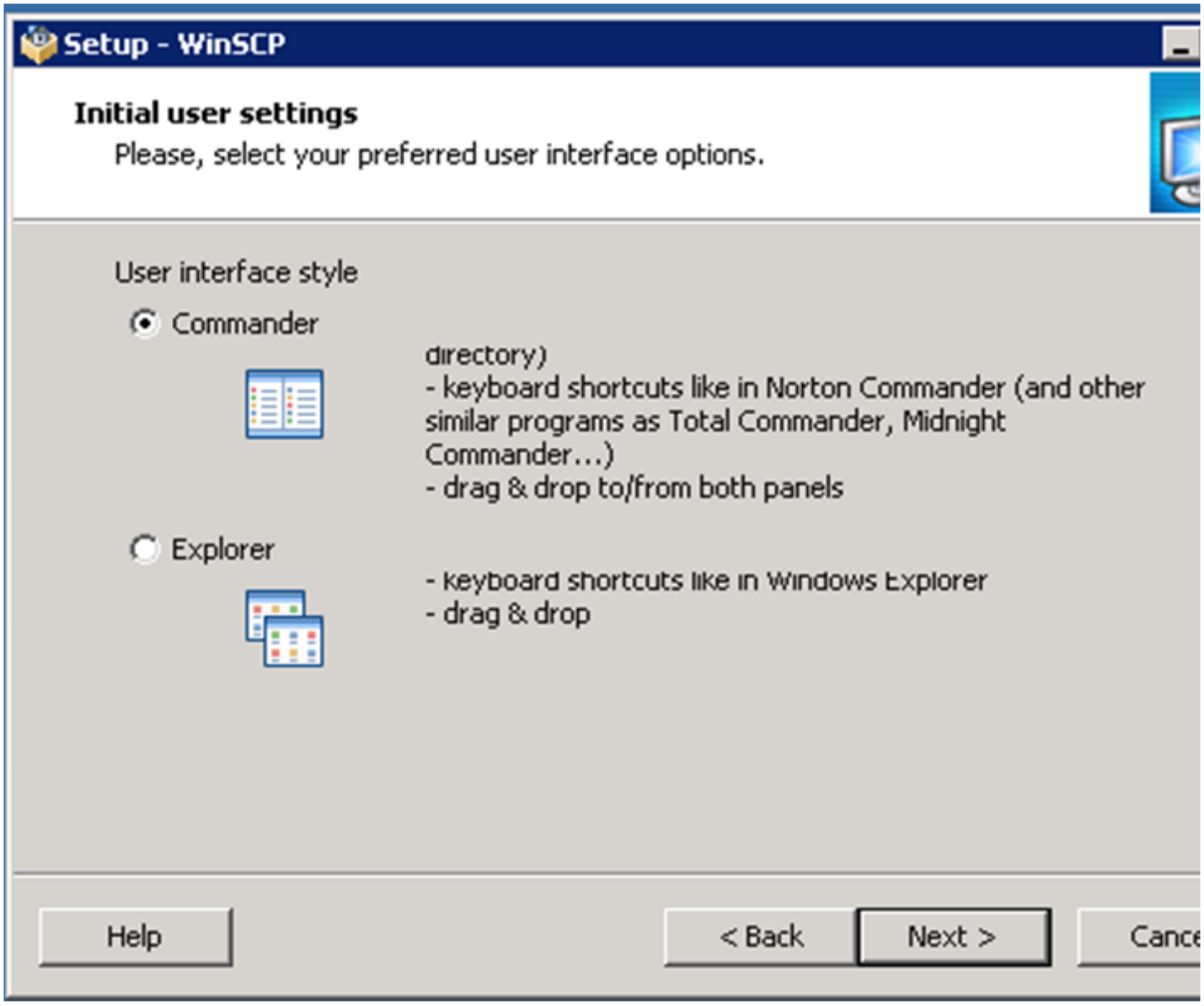
   b. Click **OK**.

7. The Setup - WinSCP window opens, as shown:



   a. Click **Next**.

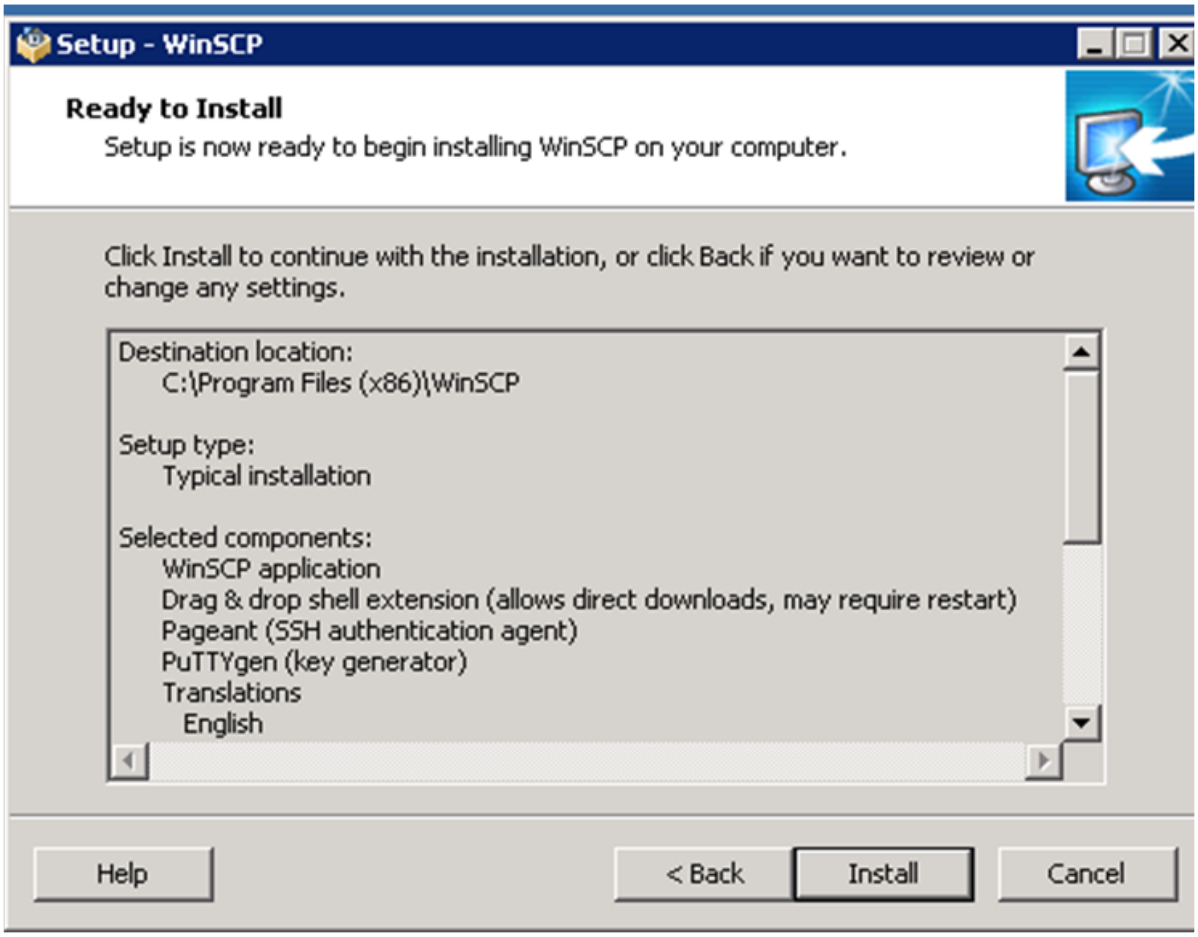
8. The Initial user settings window opens:



   a. Select **Commander**.

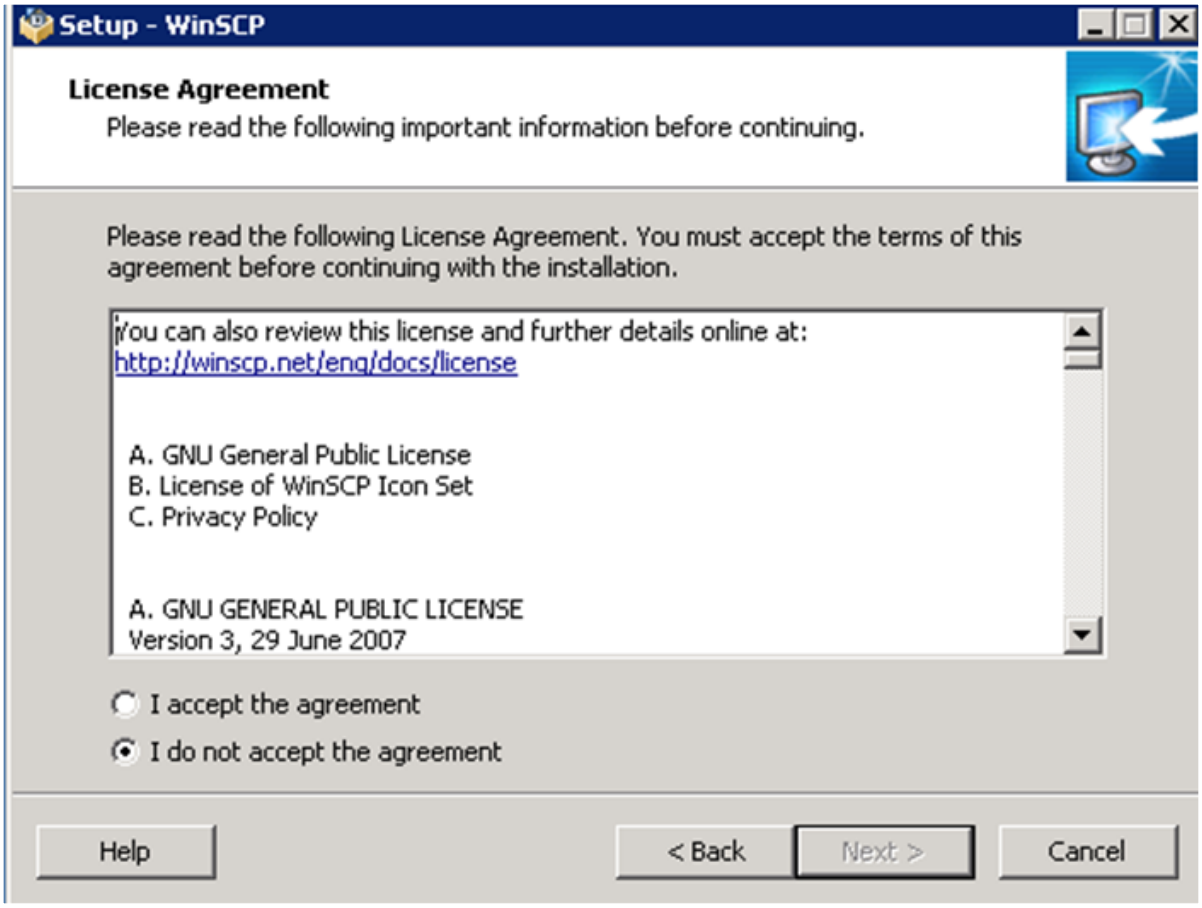
   b. Click **Next**.

9. The Ready to Install window opens:



a. Click **Install**.

10. The License Agreement window opens, as shown:

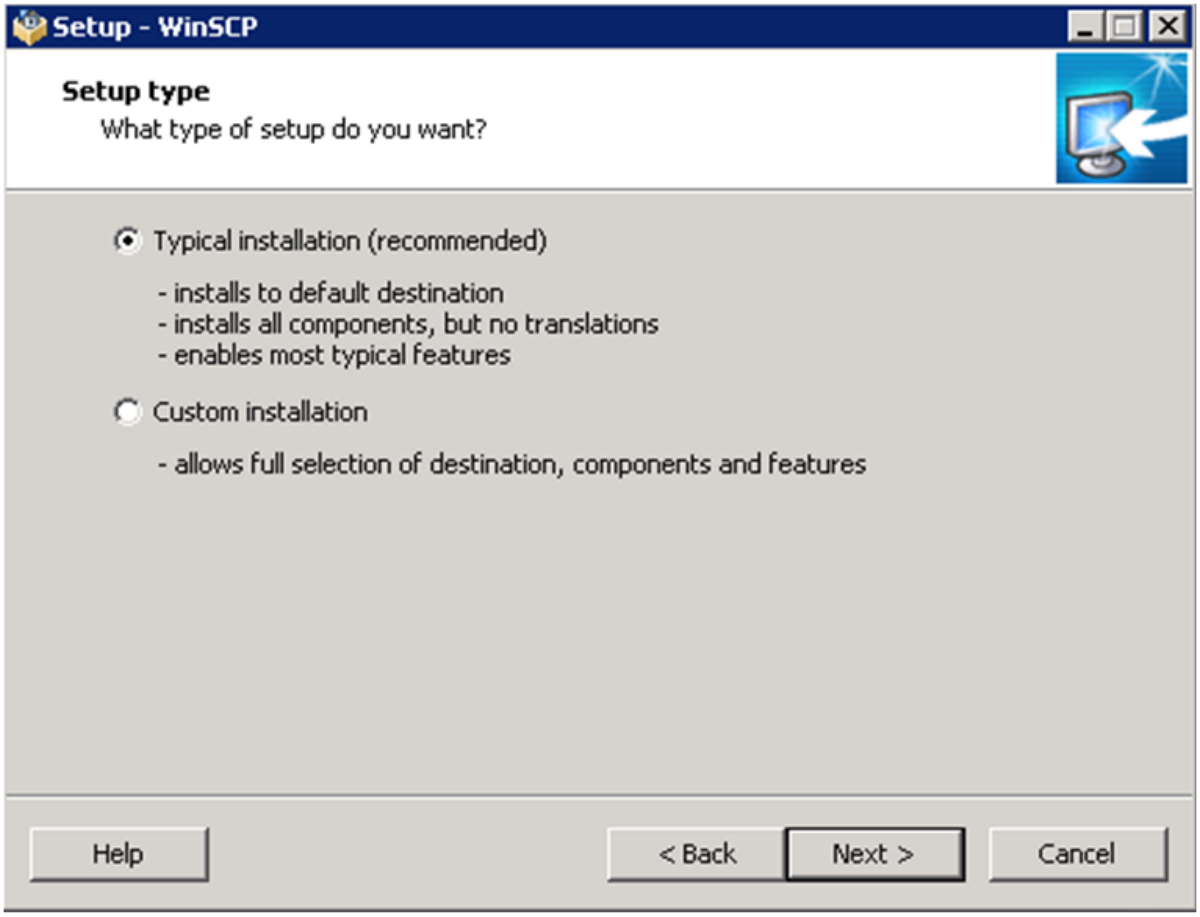


a. Select **I accept the agreement**.

b. Click **Next**.

WellSky Corporation
Proprietary and confidential

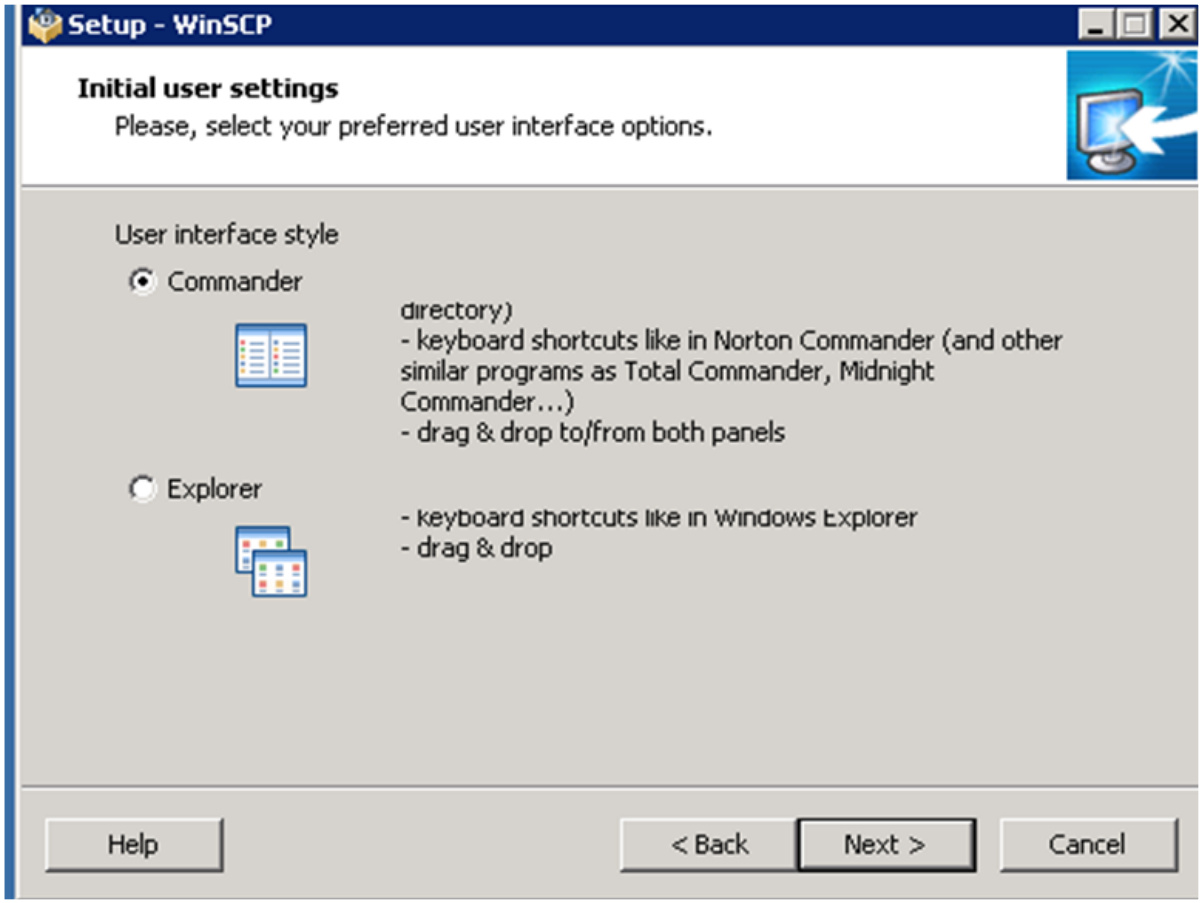
11. The Setup type window opens, as shown:



a. Select **Typical installation (recommended)**.

b. Click **Next**.

12. The Initial user settings window opens, as shown:

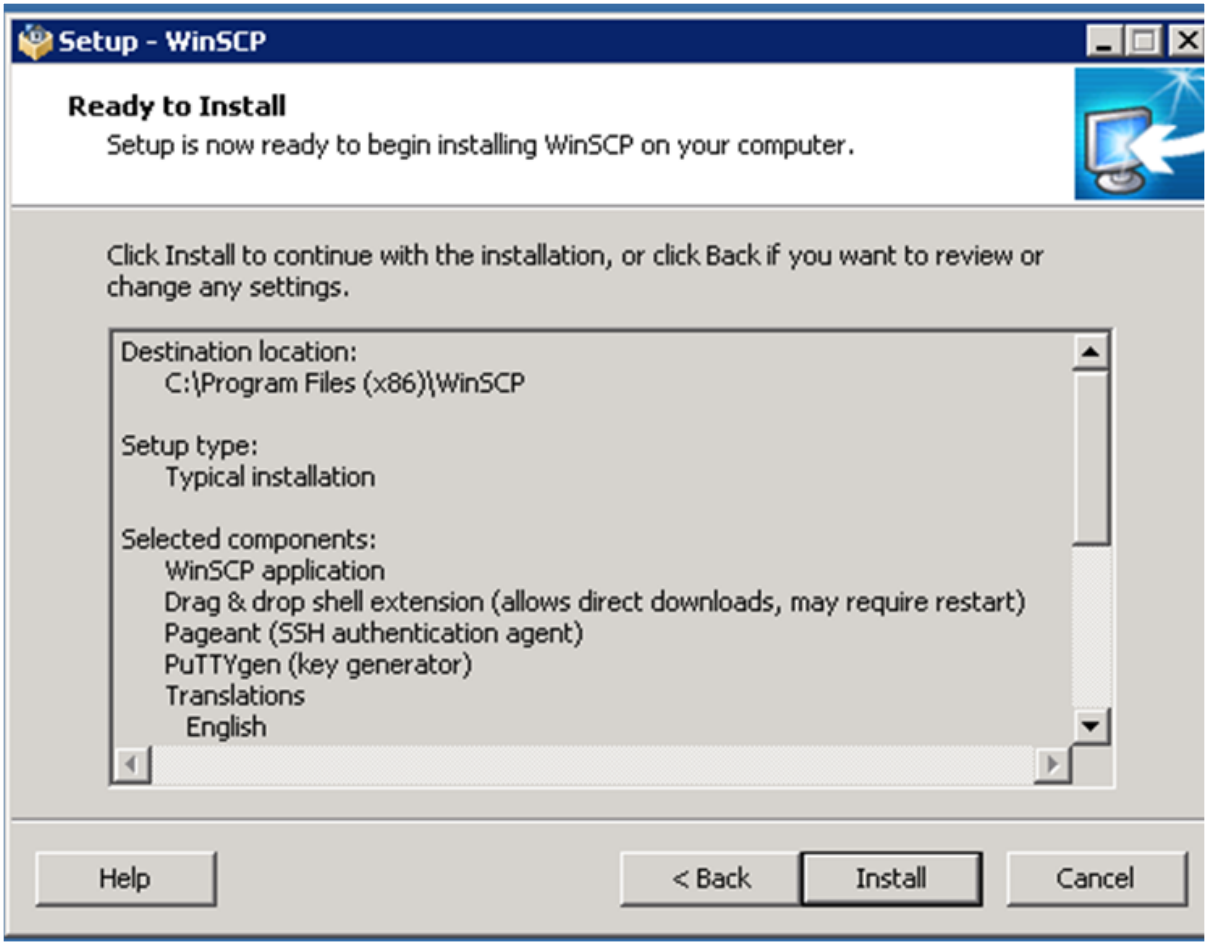


a. Select **Commander**.

b. Click **Next**.

WellSky Corporation
Proprietary and confidential

13. The Ready to install window opens, as shown:



a. Click **Install**.

14. The Completing the WinSCP Setup Wizzard window opens, as shown:



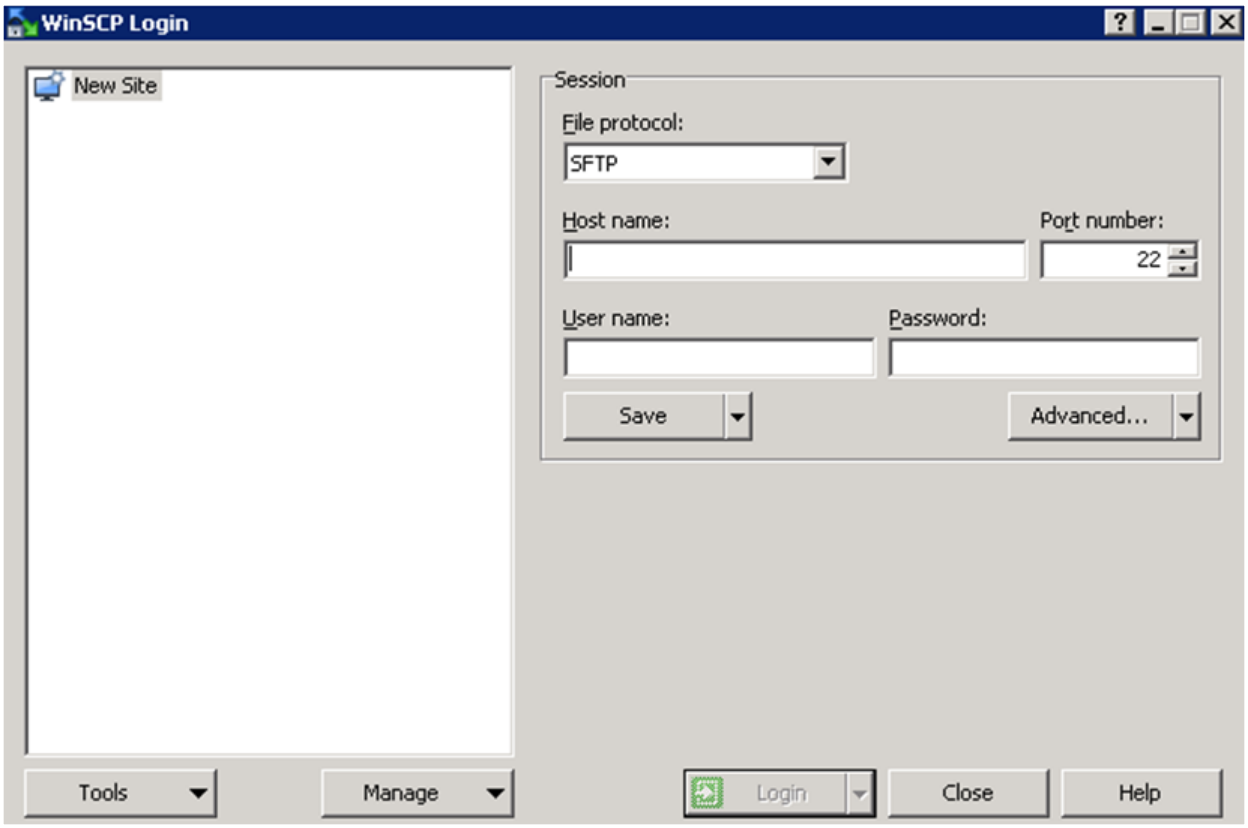
a. Select the **Launch WinSCP** check box.

b. Select the **Open Getting started page** check box.

c. Click **Finish**.

WellSky Corporation
Proprietary and confidential

15. The WinSCP Login window opens, as shown:



a. Select **SFTP** as the File protocol.

b. Enter your Blood Centers server's IP address in the *Host name* field.

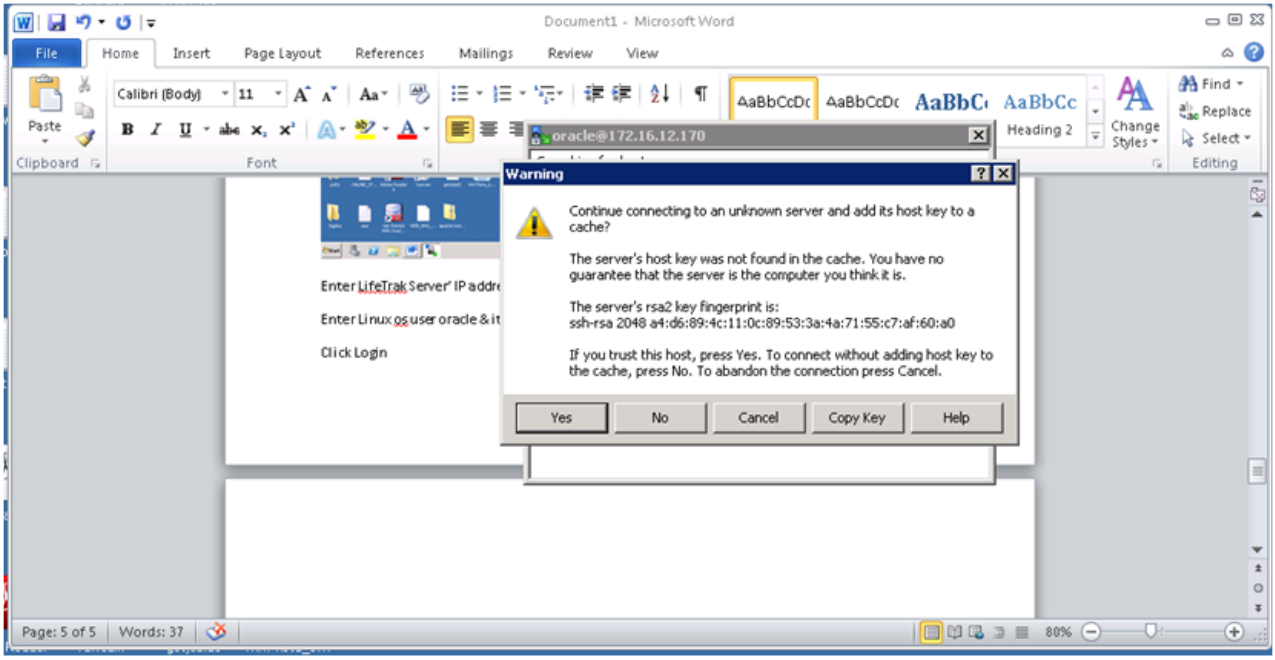
c. Leave the default Port number at 22, or enter your system's port number if it is different.

d. Enter `Oracle` in the *User name* field.

e. Enter your Oracle password in the *Password* field.

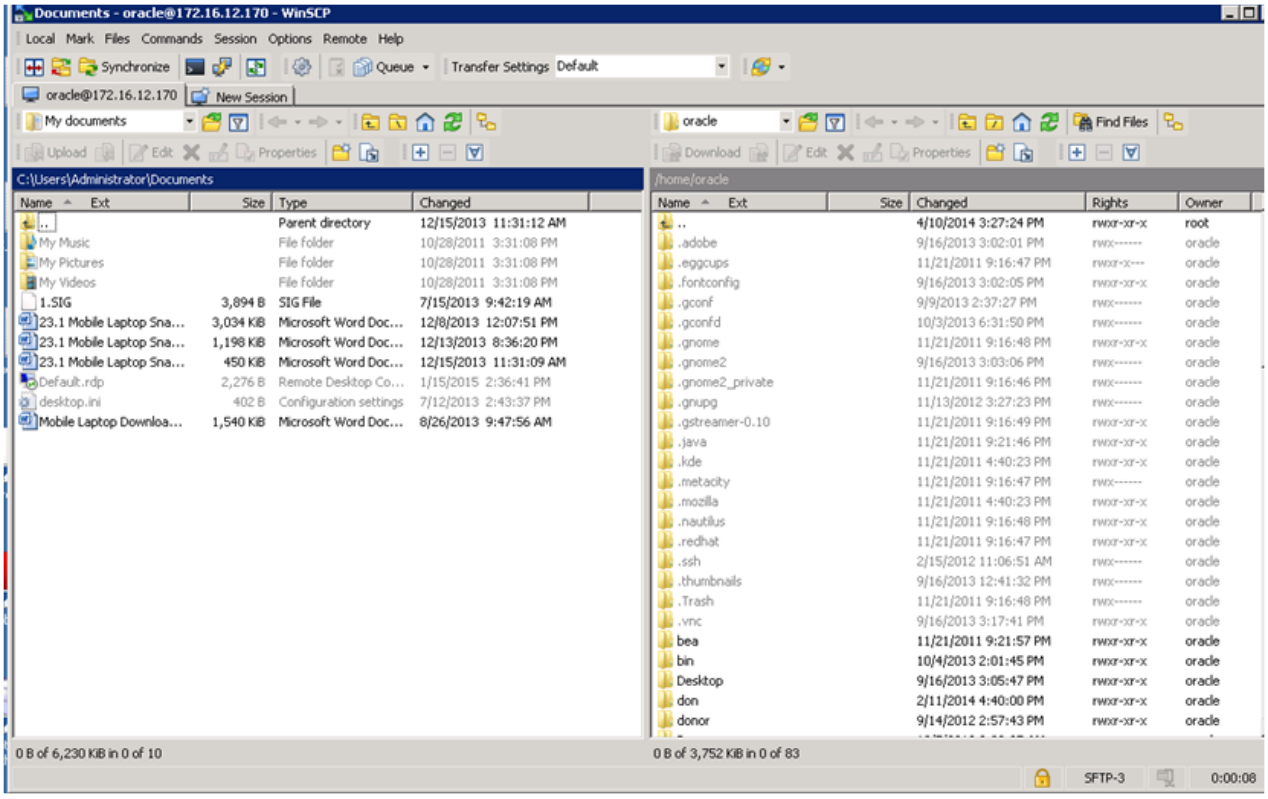
f. Click **Login**.

16. A Warning message displays, as shown:



a. Click **Yes** to close the message.

17. The Documents window opens, as shown:



a. Close the application.

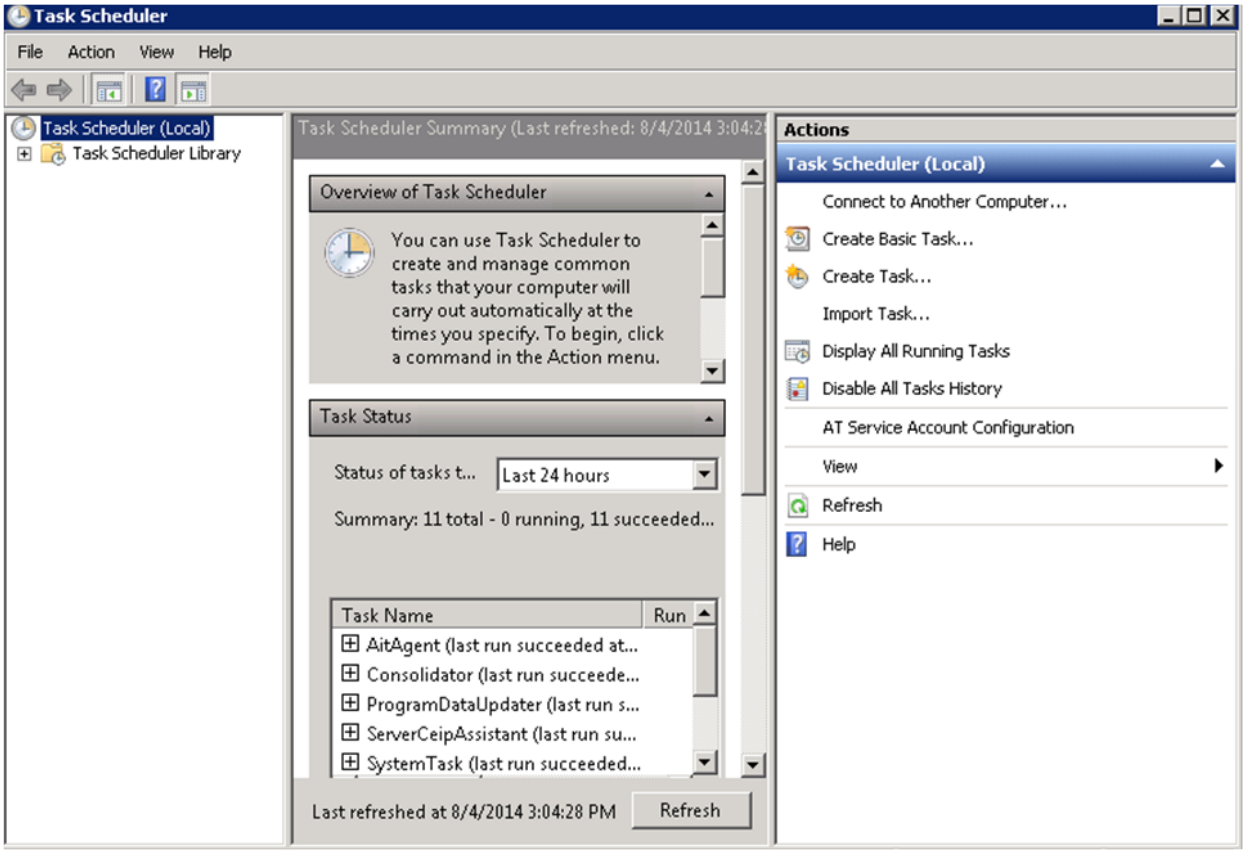
WellSky Corporation
Proprietary and confidential

18. Set up the `C:\bcmpc\replication\start_download_dmp_sftp_refresh.bat` file to run as an automatic process in the Task Scheduler. Refer to section Automating The Process Through Task Scheduler for details.

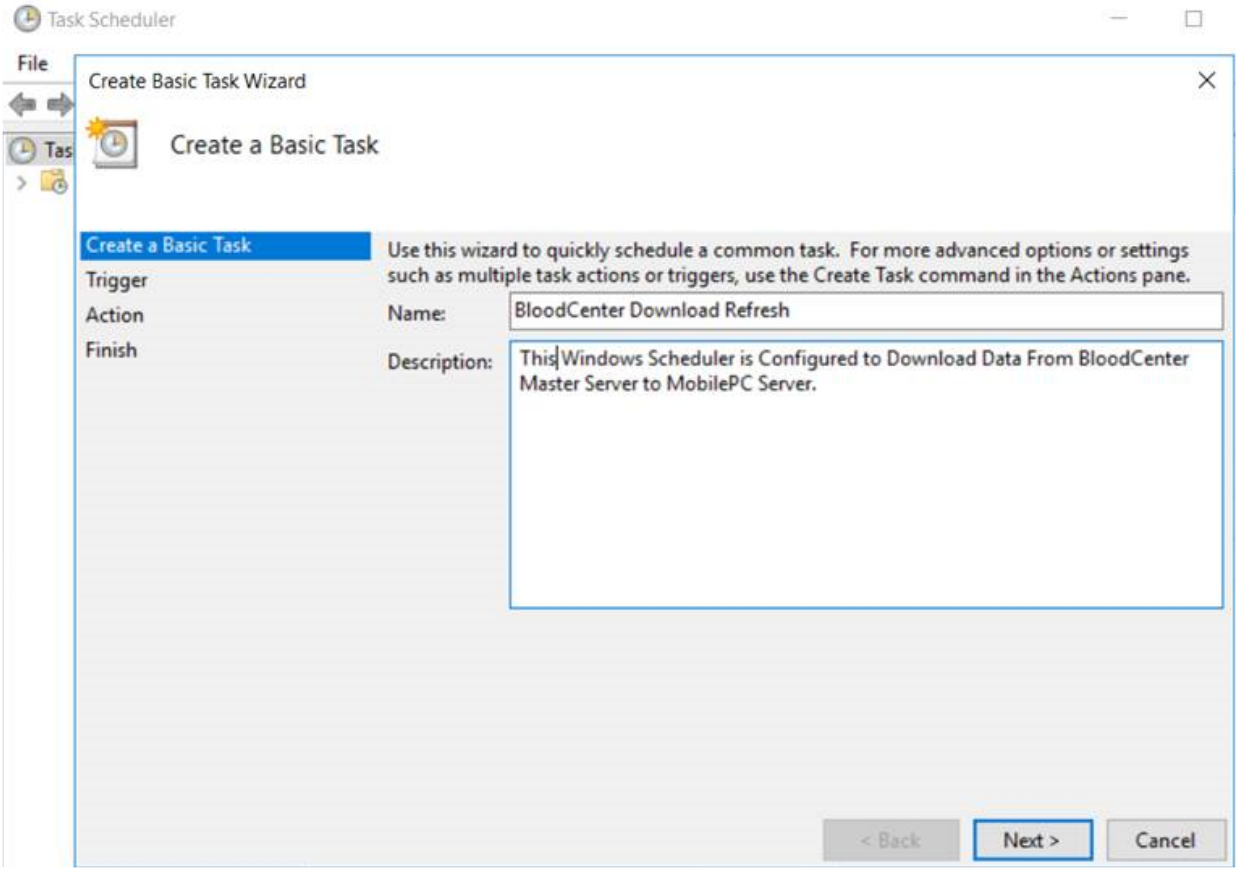
7.1.2 Automating The Process Through Task Scheduler

NOTE – Windows Task Scheduler to refresh the data in regular intervals

1. Click **Start** > **All Programs** > **Accessories** > **System Tools** > **Task Scheduler**. The Task Scheduler window opens, as shown:



2. Click **Action** > **Create Basic Task...**. The Create Basic Task Wizard opens, as shown:



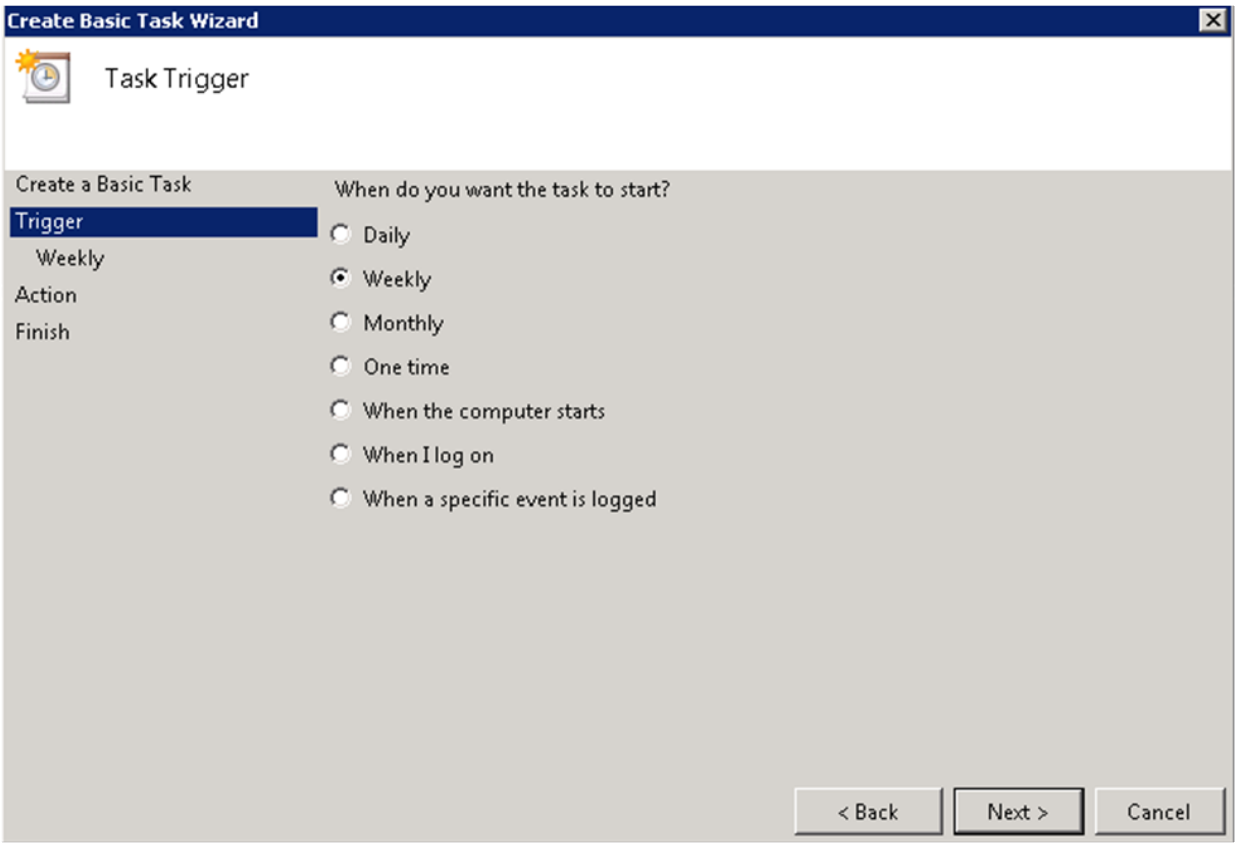
   a. Enter the task name in the *Name* field.

   b. Enter a description in the *Description* field. This description should include details regarding when the task will run.

   c. Click **Next**.

WellSky Corporation
Proprietary and confidential

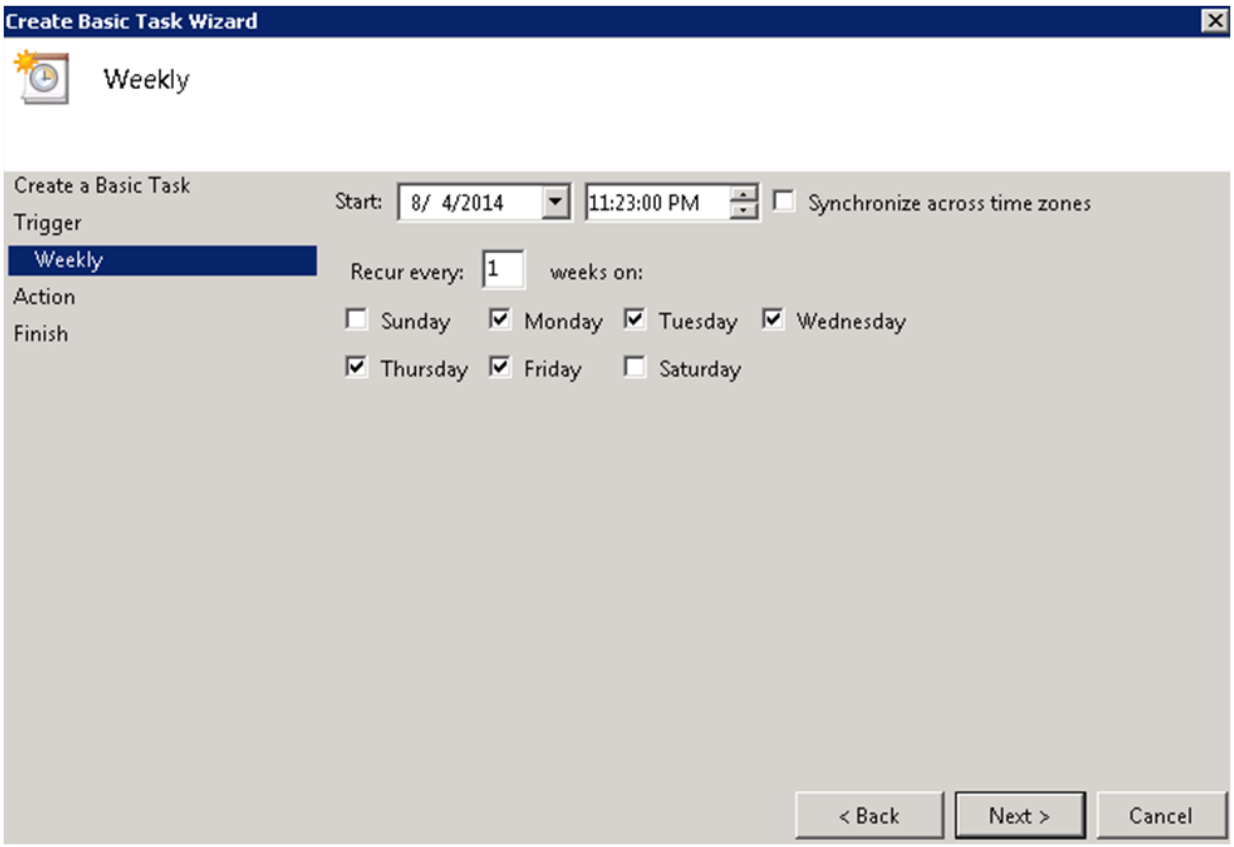
3. The Task Trigger window opens, as shown:



   a. Select **Weekly**.

   b. Click **Next**.

4. The Weekly window opens, as shown:



   a. Select the desired Start date and time.

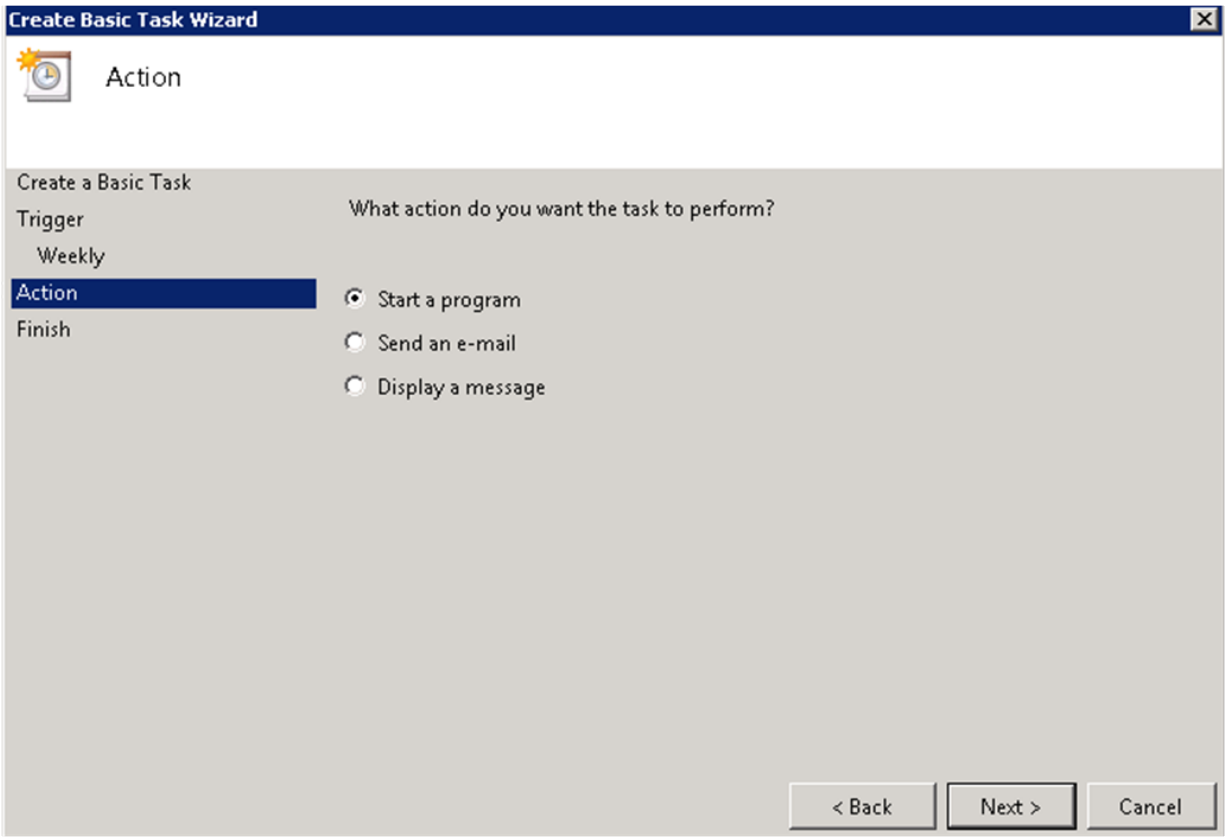
   b. Leave Recur every set to the default of 1.

   c. Select the desired days of the week when the process should run (for example, select Monday, Tuesday, Wednesday, Thursday, and Friday to have the process run once each weekday).

   d. Click **Next**.

WellSky Corporation
Proprietary and confidential

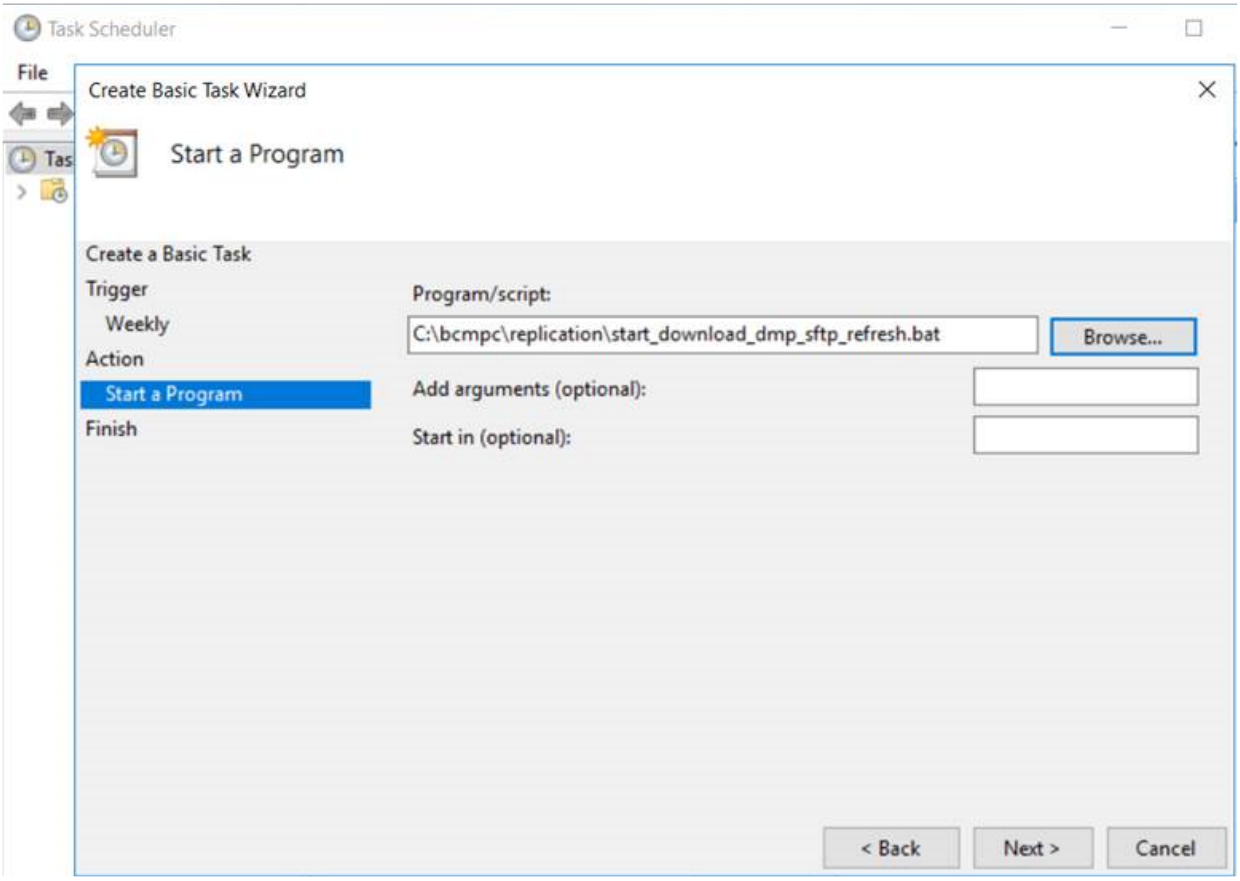
5. The Action window opens, as shown:



   a. Select Start a Program.

   b. Click Next.

6. The Start a Program window opens, as shown:



   a. Click **Browse**.

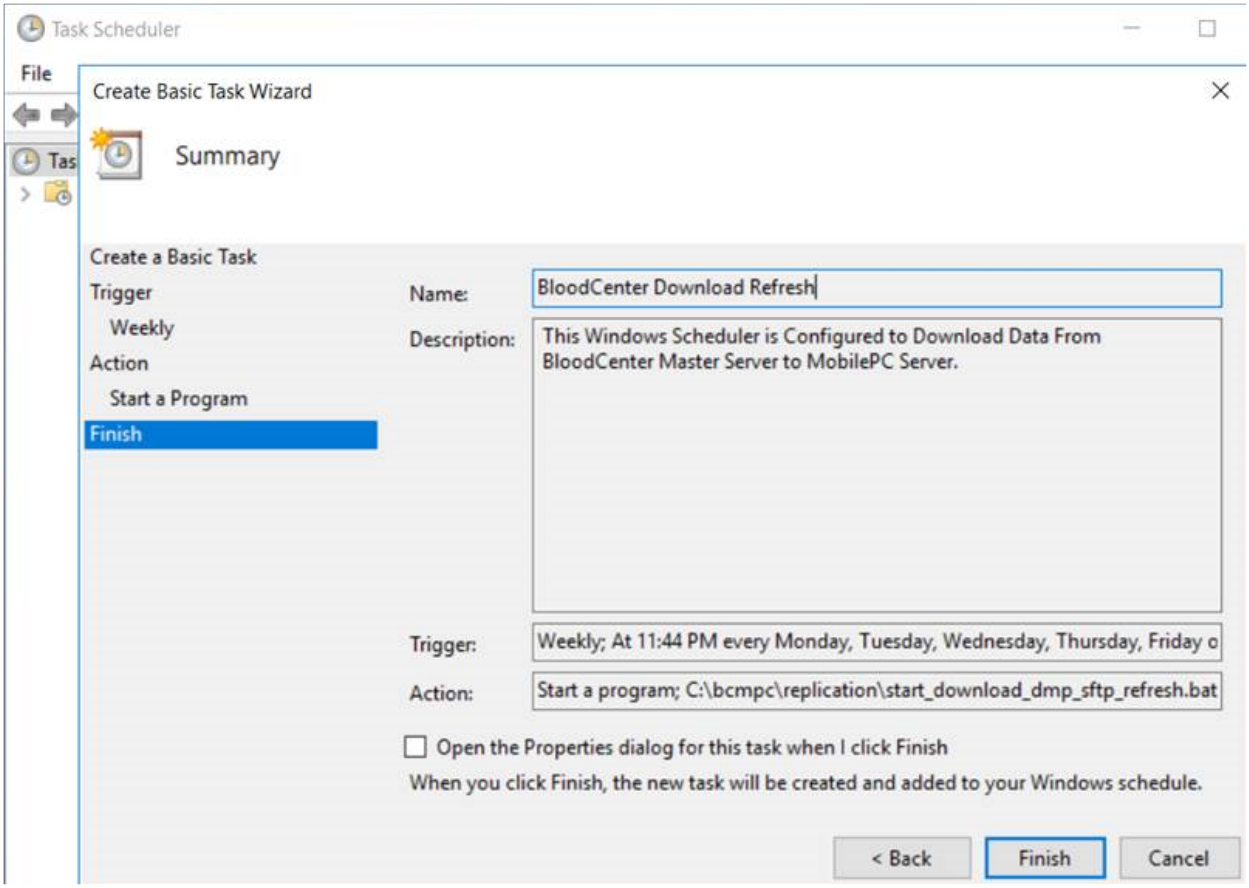
   b. Navigate to `C:\bcmpc\replication`.

   c. Double click the **start_download_dmp_sftp_refresh.bat** file.

   d. Click **Next**.

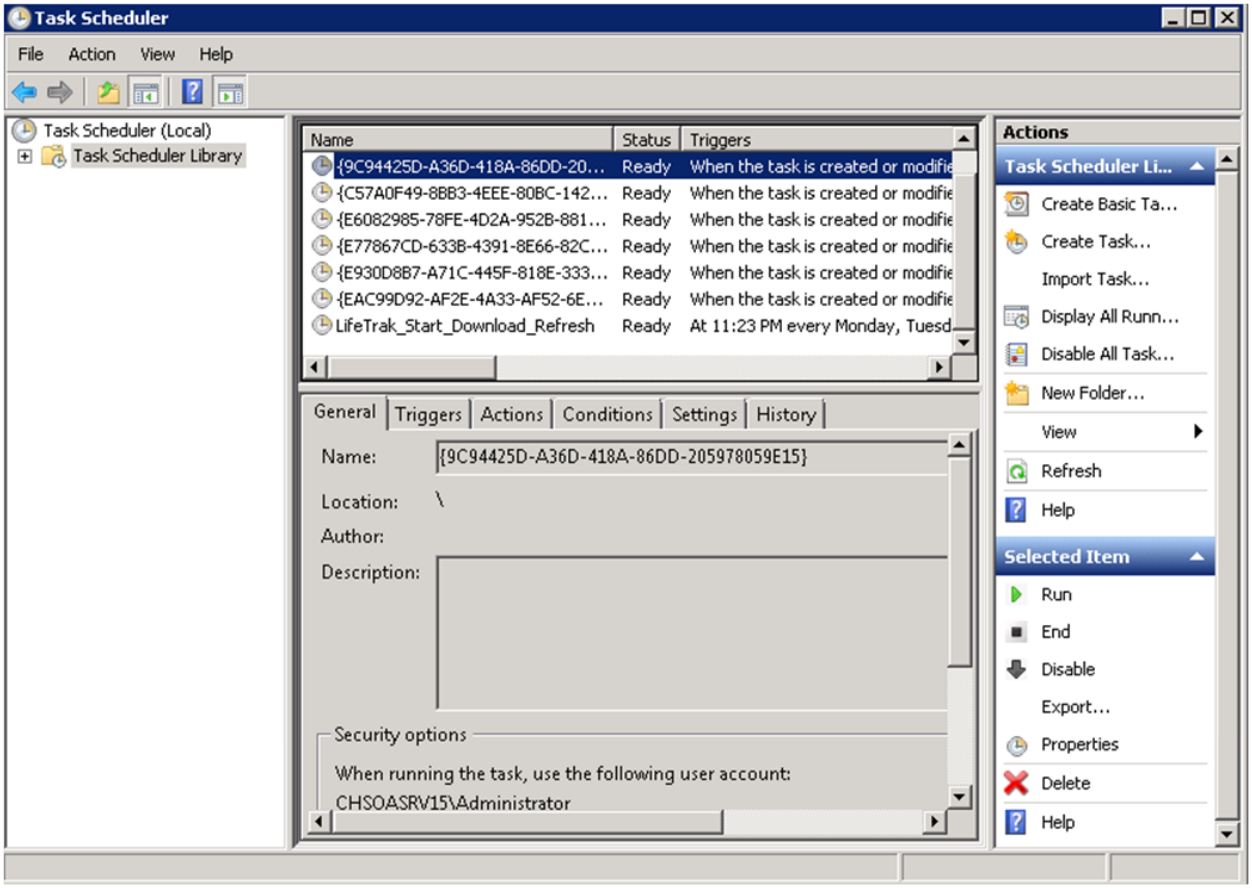
WellSky Corporation
Proprietary and confidential

7. The summary window opens, as shown:



   a. Click **Finish**.

8. If desired, check the task in the scheduler list, as shown:



NOTE – This LifeTrak_Start_Download_Refresh Scheduler will refresh the data from master database BCPRD to the replication database XE every weekday evening at 11:23 P.M.

WellSky Corporation
Proprietary and confidential

7.2 Mobile PC Data Extract Process in Script Method

7.2.1 Introduction

This feature provides the capability of extracting the data from the local database table to the corresponding text data files. The data is delimited by the configurable delimiter (|). User's can extract the data at any time during the process. Whatever point the data was captured into the table, the same amount of data will be extracted.

NOTE – This section describes the extraction steps for the Mobile Donor Upload/Download PC's database. The Mobile Donor Collections Lookup PC's database is for lookup only and these steps do not apply to that configuration.

7.2.2 Process

- Every new transaction or data modification is monitored by the database trigger and those transactions are recorded in the DOWNLOAD_MONITOR table.
- Regardless of the amount of data in the table, whatever data is marked in the monitoring table is the only data that will be extracted.

7.2.3 Extract Data to File

Before uploading the data from the Mobile PC to the Blood Centers production database, you must extract the data from the Mobile PC database.

NOTE – Perform these steps on the Mobile PC while it is offline from the Blood Centers production database.

One-time Setup

Before performing the extract process, verify the following items have been set correctly in the Blood Centers production database.

- The APIDR master code must be populated to specify the directory where the extracted files will be picked up by the upload process.

  NOTE – If you followed the instructions in this guide, the location for the extract files is c:\bcmpc\extract.

- The EXTDLT SYSPARM must be set to provide the delimiter used in the extract file for Mobile PC Donor Registration.
- The ENCRPT_REQ SYSPARM must be set to 'Y' to encrypt the data in the extract file; otherwise, the data in the extract file will not be encrypted.

Push Extract/Script Extract

A facility can have data extracted automatically in the background at the Shipment Close screen. The SHIP_EXT parameter is used to configure this feature. Y turns the auto-extract feature ON, and N turns the auto-extract feature OFF.

- Automatic Extract ON: All the shipment units' related data is extracted in the background at the Shipment Close screen without any user intervention.

  NOTE – If auto-extract is turned ON, only one manual extract has to be performed at the end of the drive. This will extract all of the units or deferrals that are not extracted as part of the auto-extract.

  NOTE – WellSky recommends that manual extract be performed only at the close of the drive. Performing a manual extract during the drive will extract all of the data up until that point, and it may result in the auto-extract returning empty data.

- Automatic Extract OFF: The user must run manual extract in order to extract the data (i.e., run the script for SCRIPT method of extract OR navigate to the EXTRACT screen for the PUSH method of extract).

7.2.4 Set Up the Extract Queue Process in Windows Task Scheduler

NOTE – This setup procedure needs to be performed only once.

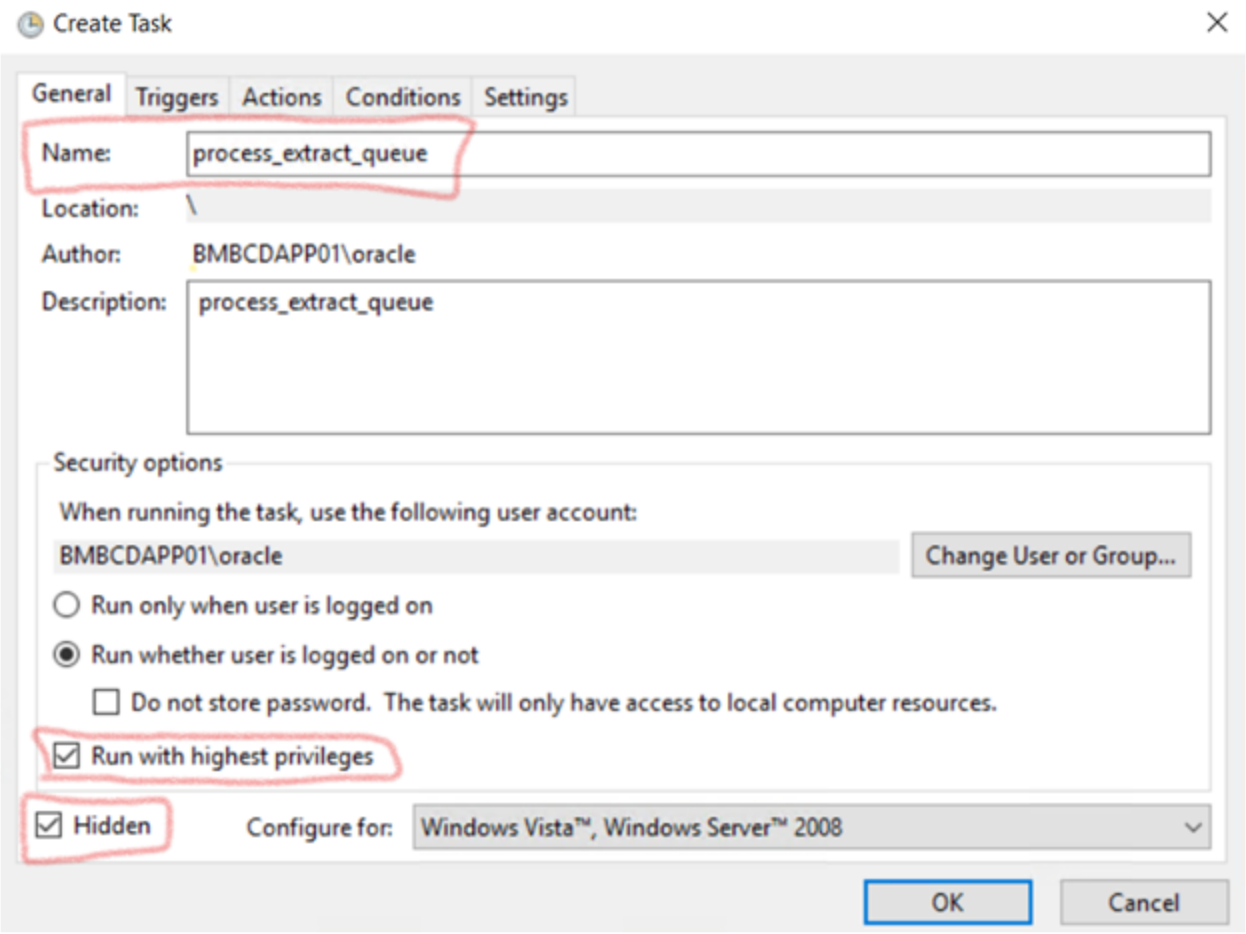
This procedure explains how to set up the C:\bcmpc\replication\process_extract_queue.bat script to run every 5 minutes in Windows Task Scheduler.

1. Open the Task Scheduler application.

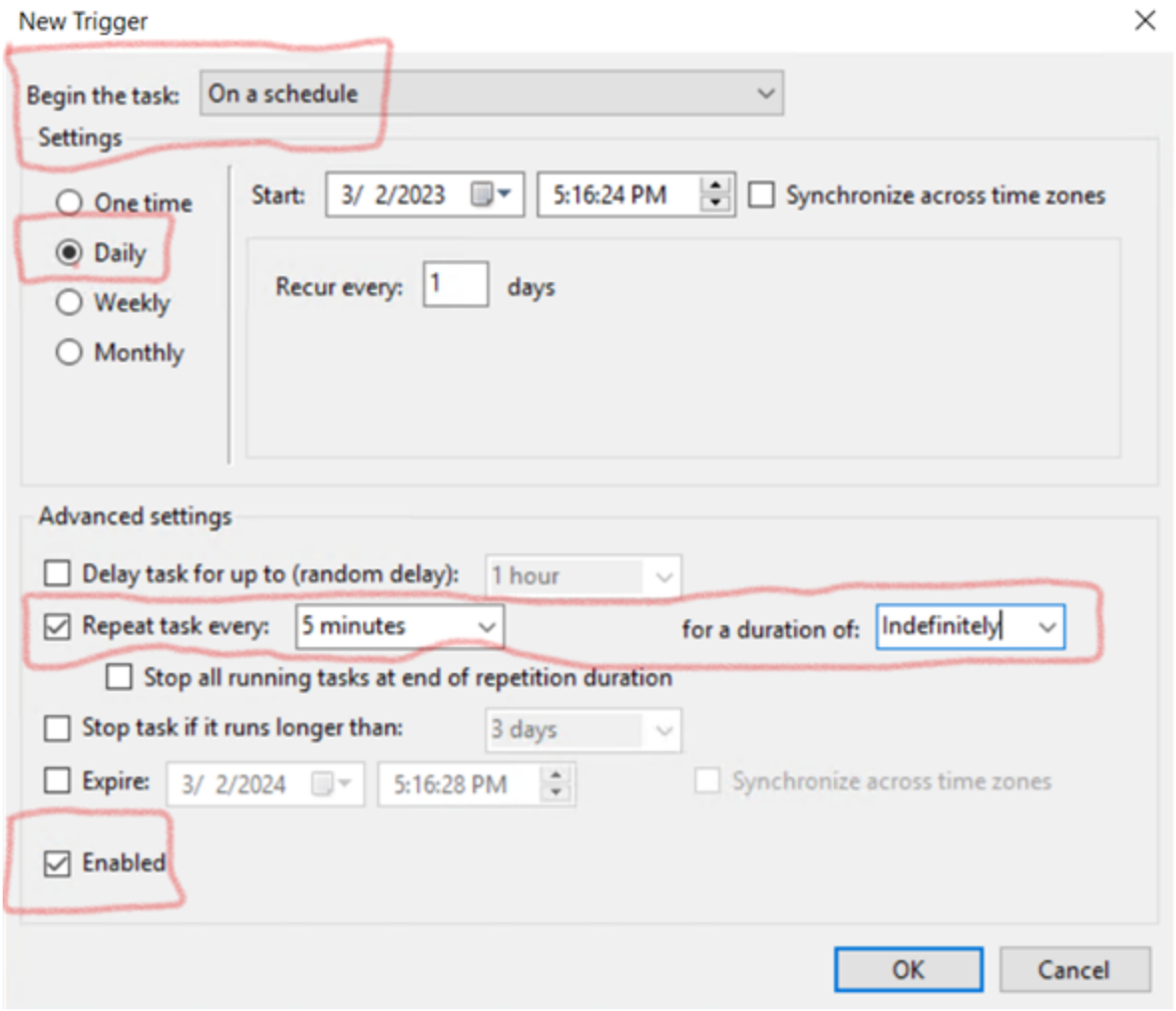
   NOTE – You can type **`Task Scheduler`** in the Search box.

2. Click Create Task to open the Create Task window.
3. Name the task **`process_extract_queue`**.
4. Select the Run with highest privileges option.
5. Select the Hidden option.

WellSky Corporation
Proprietary and confidential

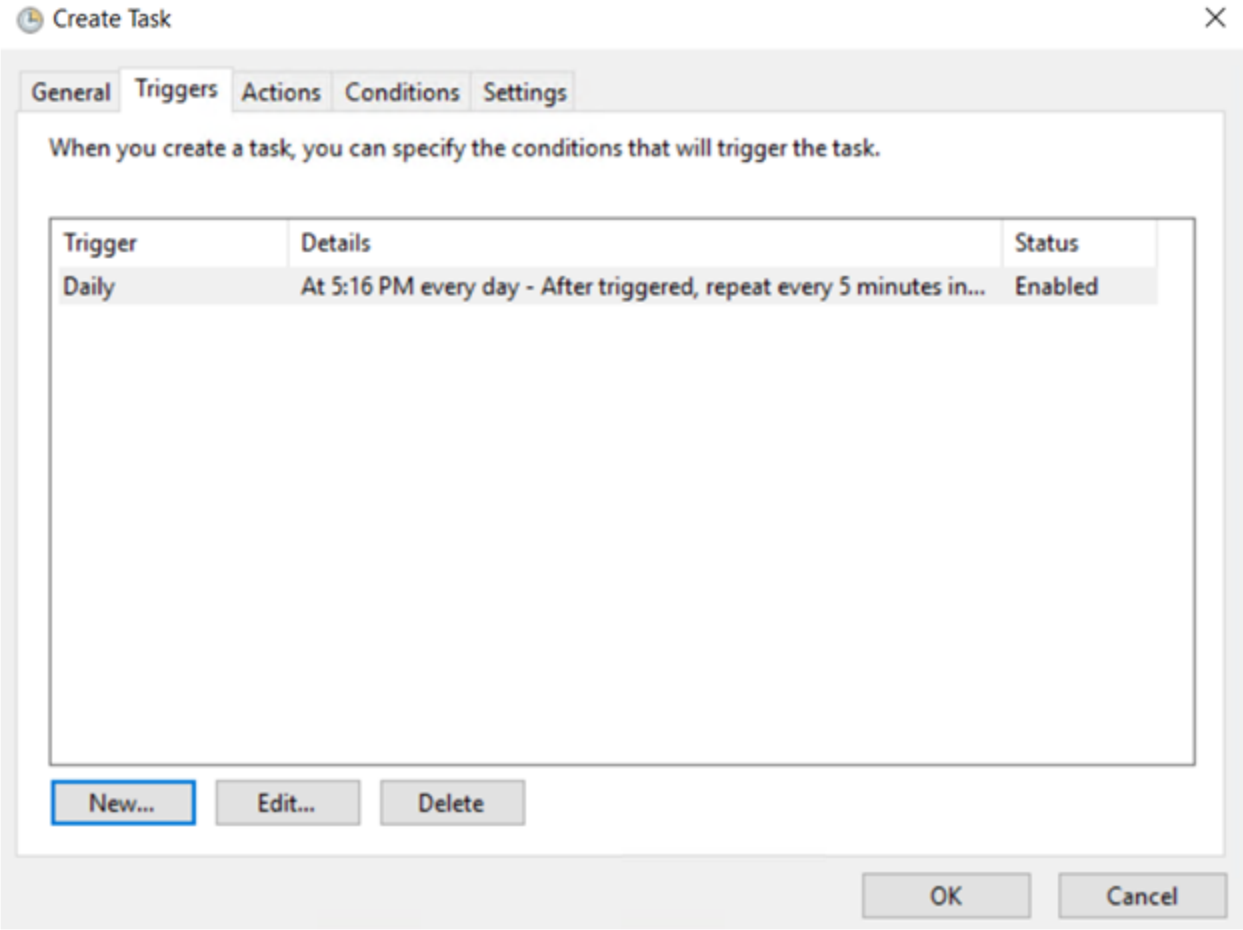


6. Click the **Triggers** tab.
7. Click **New**.
8. On the New Trigger screen, select **On a schedule** from the **Begin the task:** drop-down menu.
9. Under Settings, select the **Daily** option.
10. Under Advanced Settings, select the **Repeat task every:** option.
    a. Select **5 minutes** from the first drop-down menu.
    b. Select **Indefinitely** from the **for a duration of:** drop-down menu.



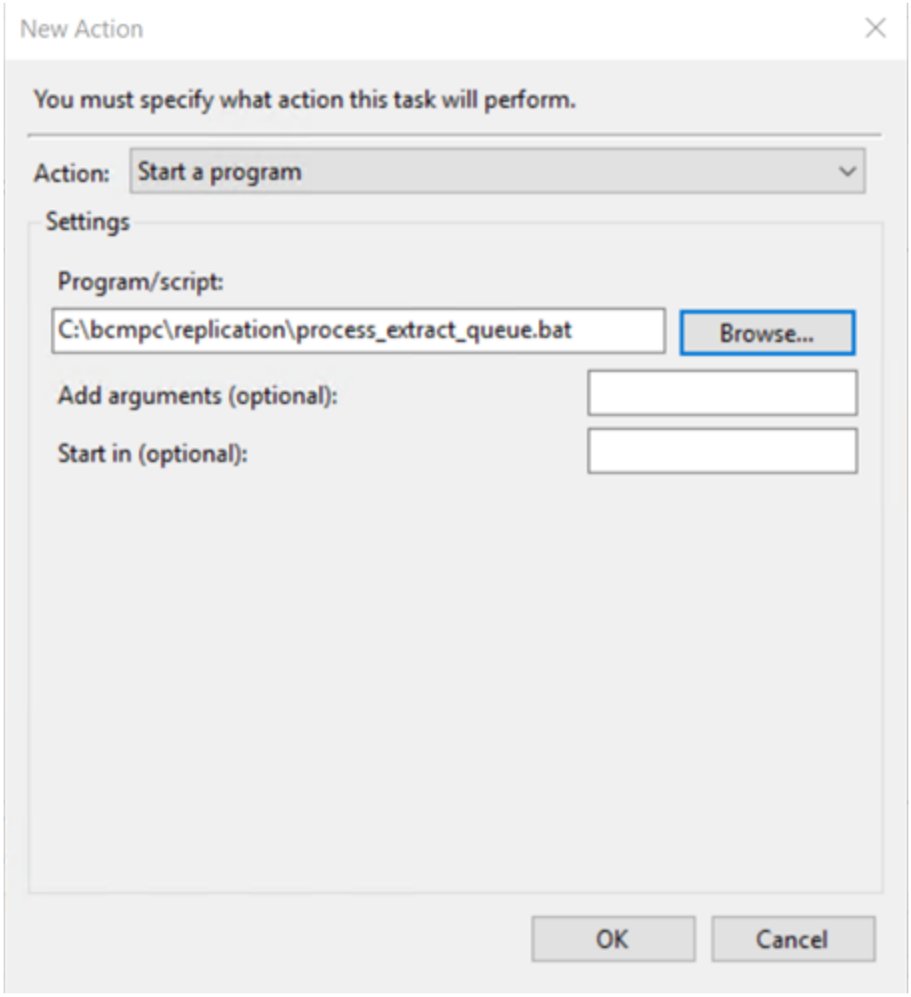
WellSky Corporation
Proprietary and confidential

11. Click **OK**. The new trigger displays on the Trigger tab, as shown in the following figure.

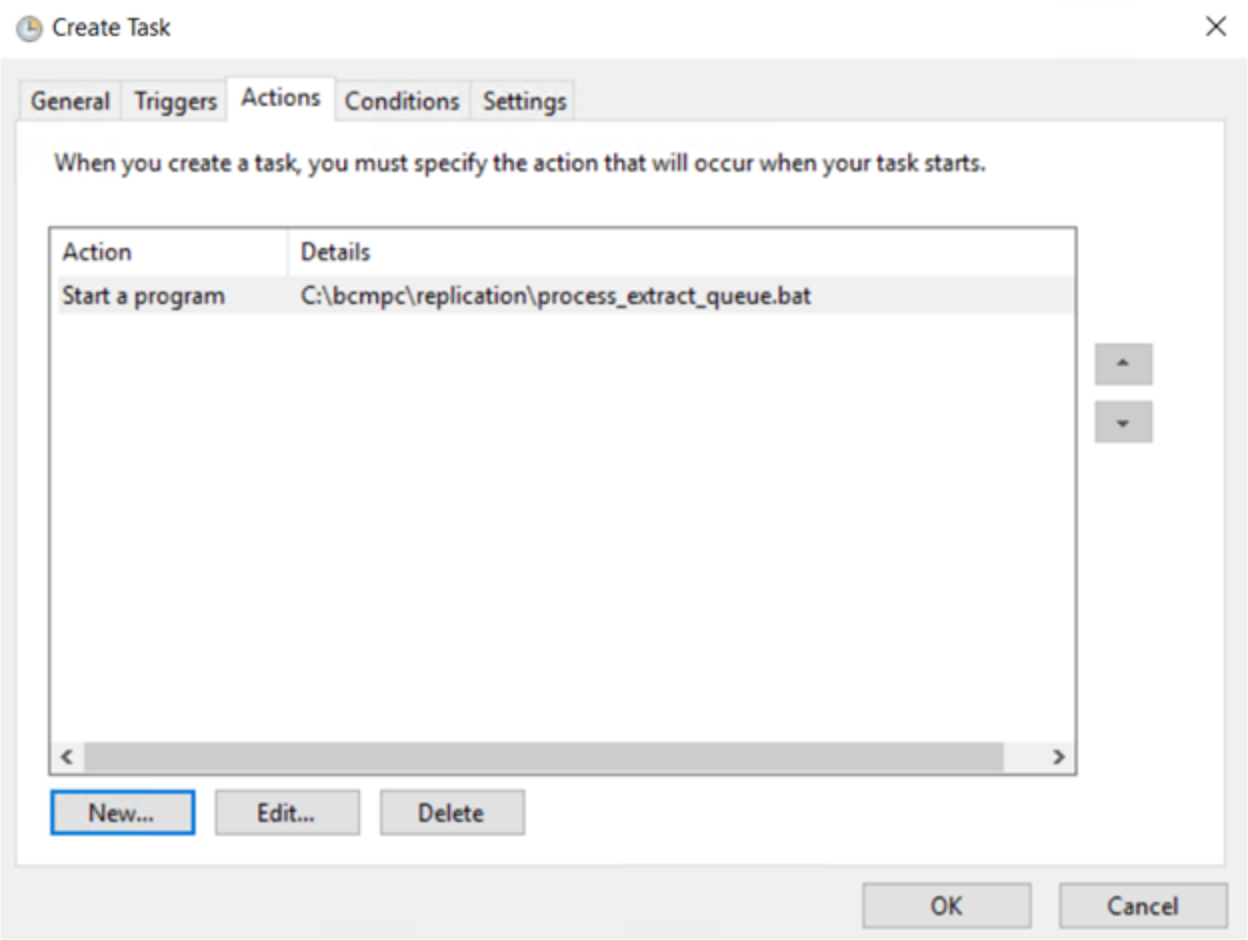


12. Click the **Actions** tab.
13. Click **New...**.
14. On the New Action screen, click **Browse...**.

15. Locate the **process_extract_queue.bat** script, which will be here: C:\bcmpc\replication.



16. Click **OK**. The new action displays on the Actions tab, as shown in the following figure.



17. Click **OK** to close the Create Task window.

7.3 Extract Data Using Batch Script

1. Extract the updated donors in mobile PC by running the batch file.
2. Create a desktop shortcut for **C:\bcmpc\replication\lt_table_extract.bat** file on the mobile PC.
3. Update any values, if needed, in **C:\bcmpc\replication\lt_table_extract.bat**.
4. Double-click the **C:\bcmpc\replication\lt_table_extract.bat** file before running the upload batch file.

   NOTE – You can either extract the modified data through the application screens or by using this script.

   This generates the lt_table_extract.log file under C:\bcmpc\replication\log directory.
5. You can set up this script in Windows Scheduler to automate it, instead of manually running it, but make sure this executes before the upload script runs.

WellSky Corporation
Proprietary and confidential

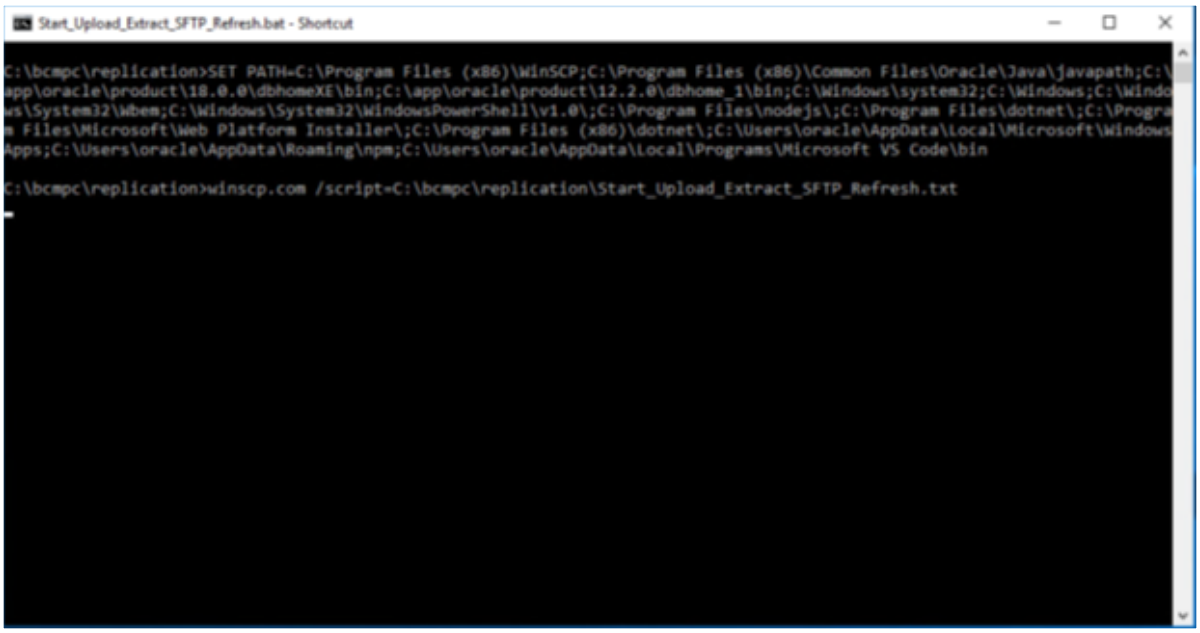
7.4 Upload Data to Server

The steps presented below are the preferred method of uploading data from the Mobile laptop to the Blood Centers database. However, if you prefer to use the older file copy method, refer to Copy the Extracted Files into Database Server.

NOTE – When an extract file is uploaded, the production database records the file uploaded in the Laptop_ID_Details table. Before an extract file is uploaded, the process checks the Laptop_ID_Details table to see if the extract file has already been uploaded. If it has been, that file is skipped and not processed a second time.

Upload the data from the Mobile laptop to the Blood Centers database by connecting to the **Production** Donor application on the Mobile laptop.

1. Make sure the Mobile PC Laptop server is in the Blood Center Database Server's network.
2. Edit the **C:\bcmpc\replication\Start_Upload_Extract_SFTP_Refresh.txt** file, and update the IP address of the Blood Center Database Server and password for **donsys** (Linux OS user).
3. Change the Blood Center database name in the Remote Directory field, if required, and save the file.
4. Set the Desktop shortcut for **C:\bmcpc\replication\Start_Upload_Extract_SFTP_Refresh.bat** file.
5. Double-click on the **C:\bcmpc\replication\Start_Upload_Extract_SFTP_Refresh.bat** file.



The files from the Mobile PC Laptop Server will be copied to the Blood Center's Database Server.

NOTE – There is a cron job running every minute to upload the data into the Database server from the extracted files.

7.5 Copy the Extracted Files into Database Server - File Copy Method

Although this file copy process is still a viable option, WellSky recommends you use the automated method of data transfer explained in section 6.4.

1. On the Mobile laptop, insert the desired media (USB jump drive or writable CD).
2. Open **Windows Explorer**.
3. Navigate to **c:\bcmpc\extract**.
4. Copy the extracted files (everything in the extract directory) to the media from step 1.
5. Take the media to the Blood Centers Database server and insert the media.
6. Log in as **donsys**.
7. Copy the files into the **/home/donsys/bcprd/LTAPI/MOBILE_PC/MPC_FILES** directory.



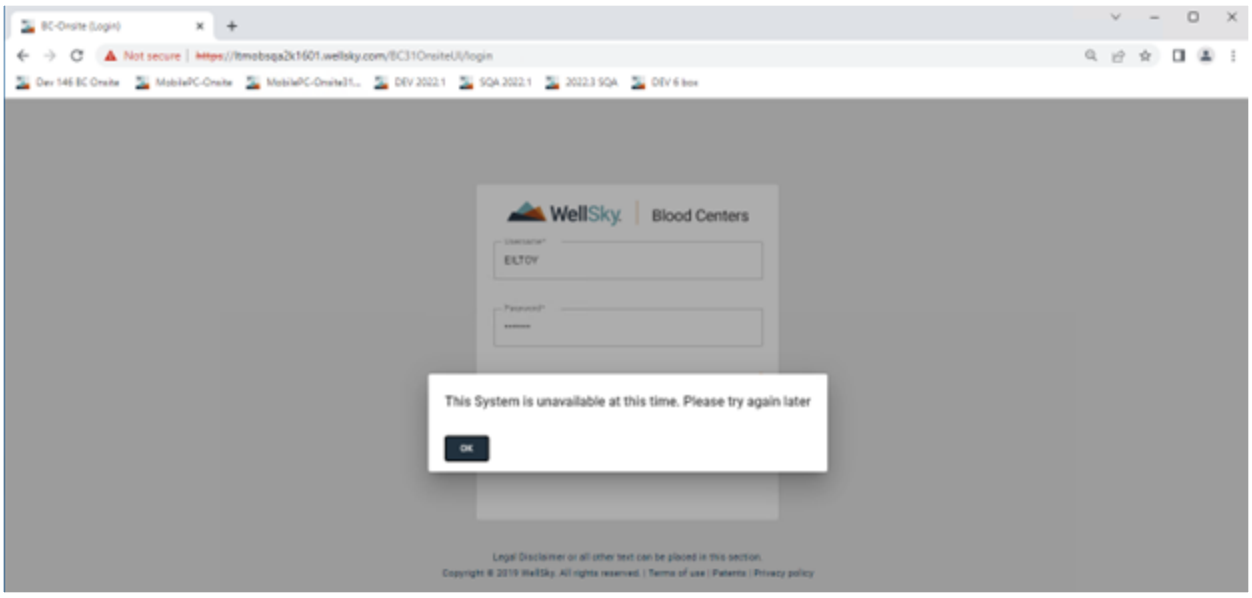
NOTE – There is a cron job running every minute to upload the data into the Database server from the extracted files.

WellSky Corporation
Proprietary and confidential

Appendix A - Recreate the Audit Sequence

After downloading the database from the main production server into the mobile PC server, if the LAPTOP_ REGISTER table does not have the laptop id entry, you will get an error message in the application stating, *"The System is unavailable at this time. Please try again later."*

NOTE – The above error message is configurable. Use Error Code: 16340.

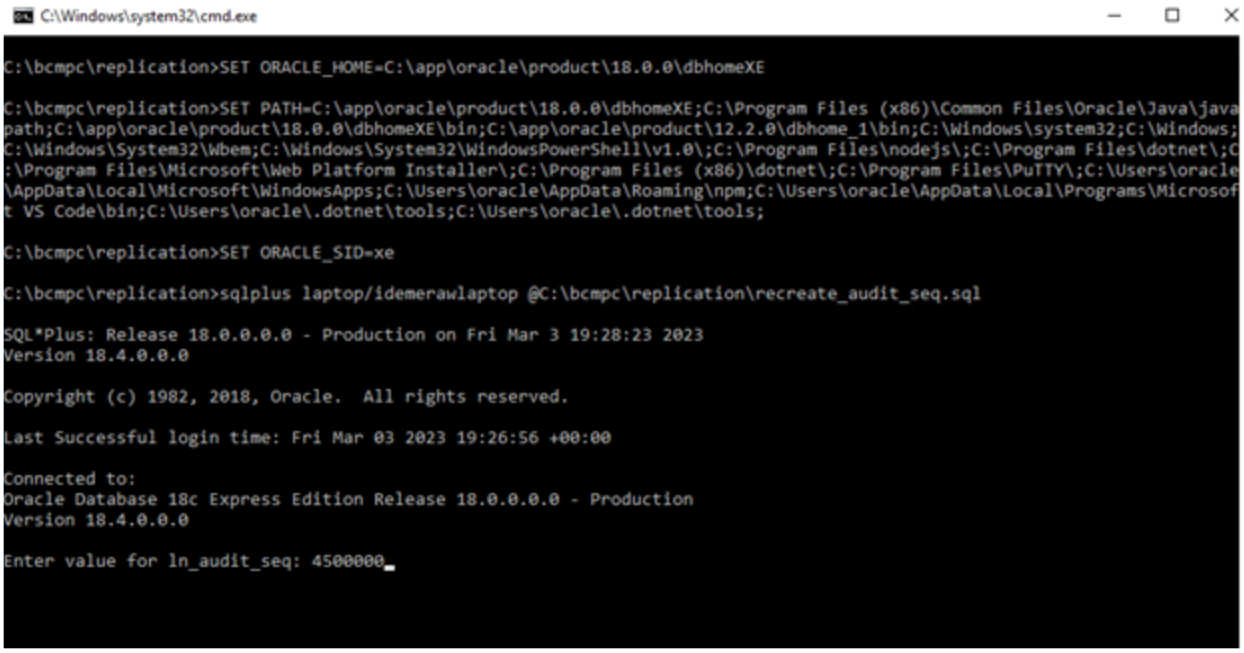


The Mobile PC Administrator must recreate the Audit Sequence in the Mobile PC database using a script (explained below).

Important! The administrator must ensure that the sequence number added to the script does not overlap with any other mobile server sequences. This is very important, and utmost caution must be used when determining the sequence number. Failure to use caution will result in data loss on the server.

1. After verifying the above requirements, at the command prompt, run the C:\bcmpc\replication\recreate_audit_seq.bat file.
2. Enter the Sequence value.

   NOTE – Write down the Sequence value, so you can use it in the next step to register the mobile server.



   Users will now be able to log in to the application without any issues.
3. Register the Mobile Laptop server in the LAPTOP_REGISTER table (MOBPCLR) so that this issue does not occur again.
4. Contact the WellSky Support team if you have any questions or issues.

Appendix B - Mobile PC Server Setup in Compress Mode

NOTE – This is an optional procedure.

Perform the following steps to set up the Mobile PC database in compress mode.

1. Open the command prompt, and change the directory to C:\bcmpc\replication\LAPTOP_DB_COMPRESS.
2. Run the LAPTOP_DB_COMPRESS_CONFIG.bat file. This script will do the following:
    - Restart the Laptop XE database.
    - Drop the laptop schema in the XE database.
    - Create the laptop schema and import the compressed laptop database objects.
3. Verify the laptop upload and download processes completed, and ensure there were no errors.
4. Contact the WellSky Technical Support team if there are any specific issues.

This Page Intentionally Left Blank